

AWI-ESM3 (v3.4, CORE3 / TCO95) PI-control tuning campaign

Tuning_test_06* — progress report

Evaluation via `release_evaluation_tool2` (part2 / part8 / part4)

2026-06-30 (rev. **2026-09-25**)

Abstract

AWI-ESM3 v3.4 (OpenIFS 48r1 / FESOM2 / LPJ-GUESS, TCO95L91) is being tuned for a pre-industrial spin-up against two faults: a positive TOA and surface radiative imbalance, and a NH high-latitude boreal-summer cold bias severe enough to lose the boreal forest. An AMIP testbed (§11) runs the atmosphere alone so the vegetation-albedo feedback is open and the root cause can be hunted without the coupled runaway closing over it.

Where we are (2026-09-25). Screening is over and the campaign is producing: **PICAL_ccnice** has run 1940–2100 on the v3.5 defaults with `cvmix_TKE+cvmix_IDEMIX`. It is not converged. It cools 0.21 K/decade at the surface while net TOA is positive, because total ocean heat content is flat and the upper 700 m is losing heat to below 700 m at 0.32 W m^{-2} equivalent (§57.2). The Antarctic sits at 62 % of observed winter volume, the ACC at 98.9 Sv against 134 ± 11.2 observed, and the boreal forest is gone to grass competition (§57). Five Southern Ocean levers were tried and excluded (§54); the GM setting that improved the CDW cost 16.7 Sv of Drake Passage (§57.4). Work is moving to albedo for queue time; see `HANDOFF_albedo_spinup_continuation.md` and §59.

What is left. Three routes are closed by measurement rather than opinion: transpiration→cloud (§58.2), snow-melt albedo (§27.2), and land surface albedo, which buys at most one sixth of the universal bias (§30.3). Skin conductivity is falsified for both the winter route (§28.1) and the boreal organic layer (§30.1). **Round 22 relocated the two remaining targets** (§38): the Southern Ocean is confirmed as the largest regional term and grows period-cleanly to $+7.46 \text{ W m}^{-2}$, but two thirds of it is missing cloud *area* that no lever — including the adopted INP branch — has ever moved; and the **global tropospheric cold bias** is now located, at the *extratropical tropopause* rather than in tropical convection, where the atmosphere is both 1.6 K too cold and 10 % too dry, two large errors that cancel in OLR and so cannot be fixed one at a time.

Read §58 for the causal web and what is still dark; §25 for what the observations can and cannot settle; §60 for every mechanism this campaign has killed, with the measurement that killed it. Parts I–III are the coupled work that preceded the AMIP testbed; Part IV is the atmosphere-only campaign in the order it happened.

Front-matter note — new-sea-ice branch Tuning_test_09A/09B/09C

The dated account of this branch is §12 in the timeline; what follows is the run-identity reference only.

This report now includes the new three-way coupled comparison branch: 09A (baseline + new sea ice), 09B (new sea ice + tuned stack), and 09C (new sea ice, 06V variant). All three completed the same 30-year segment and are compared on the same last-10-year window (model years 1370–1379).

Run identities

- 09A: `Tuning_test_09A_lpjguess_Baseline_coupled_fromCRUNCEP_newSeaIce`
- 09B: `Tuning_test_09B_06T_1hCPL_MOSPP_KPLOW_CRUNCEPinit_newSeaIce`
- 09C: `Tuning_test_09C_06V_CRUNCEPinit_newSeaIce`

09B tuning block documented on the figures (original → tuned)

- `oasis3mct.time_step`: 7200 s → 3600 s
- `use_momix`: `.false.` → `.true.`
- `kpp_av0`: 0.005 → 0.003, `kpp_kv0`: 0.005 → 0.003
- `kpp_avbckg`: $1\text{e-}4$ → $5\text{e-}5$, `kpp_kvbckg`: $1\text{e-}5$ → $5\text{e-}6$
- `pndaspect`: 0.8 → 1.3, `rfracmax`: 1.0 → 0.75, `albpnd`: 0.2 → 0.28

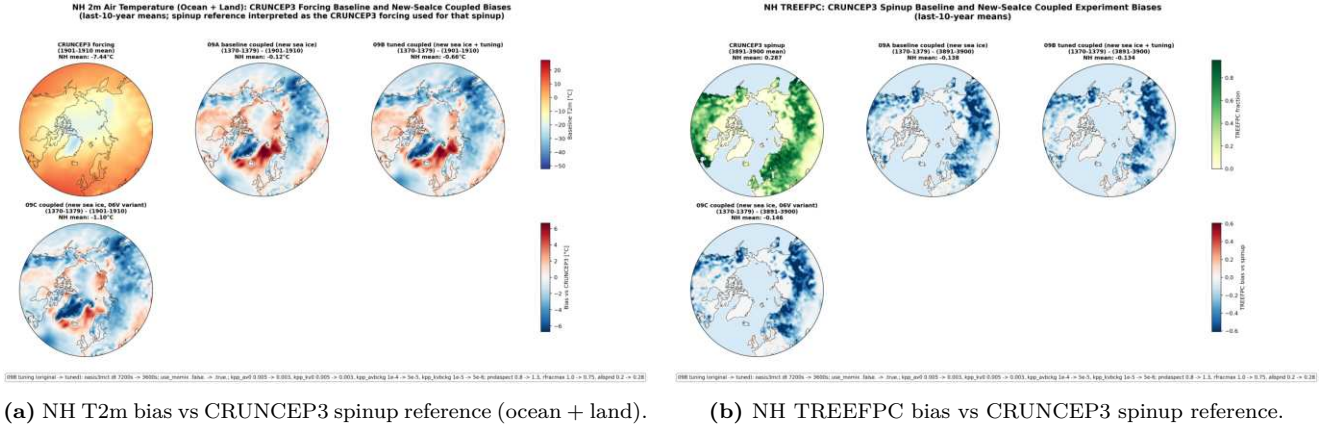


Figure 1: New-sea-ice branch update: common last-10-year means (1370–1379) for 09A/09B/09C.

Part I

Tuning away the high TOA imbalance

We set out to cut the AWI-ESM3 PI-control’s $+1.5 \text{ W/m}^2$ TOA imbalance. Every effective lever does it by *cooling* the model, and the aggressive coupling/ocean-mixing levers cut the imbalance hardest — but push the planet too cold against ERA5. This part documents that trade-off across 22 runs and traces the aggressive lever to a Southern-Ocean ocean-convection mechanism.

1 Goal & setup

Reduce the net top-of-atmosphere (TOA) radiative imbalance of the AWI-ESM3 pre-industrial control so it equilibrates faster, *without* degrading the CMIP6-normalised performance index (CMPI, [part4](#)) or inflating the 2 m-temperature RMSE vs ERA5 ([part8](#)). 22 experiments, each a 30-year segment (model years 1350–1379) of the CORE3/TCO95 configuration (FESOM2 CORE3 mesh, OpenIFS TCO95/A096, LPJ-GUESS), all branched from one **Baseline**. The baseline PI-control still drifts with a positive TOA imbalance of $\approx +1.5 \text{ W/m}^2$ (energy *into* the system — the signature of unequilibrated spin-up and/or ongoing ocean heat uptake). **A caution on the target.** The tuning levers change the model’s *equilibrium*, not its rate of approach, so “lower imbalance” is a useful *ranking* objective but not automatically “more equilibrated”: a low-imbalance-but-cold state is not obviously better than a high-imbalance-but-unbiased one. This is why the campaign’s real aim (§9) is a *warm-floor* state where the imbalance is cut by a *moderate* lever, not by cooling into a cold bias — i.e. we treat imbalance and 2 m-T bias as joint targets, not imbalance alone.

Structural note. Almost every non-baseline run also carries a *common melt-pond shift* (`albpcnd` $0.2 \rightarrow 0.28$, `rfracmax` $1.0 \rightarrow 0.75$, `pndaspect` $0.8 \rightarrow 1.3$). Differences below are each run’s *distinguishing* change on top of that.

Four parts. (I) evaluates the 22 imbalance-tuning runs and exposes the imbalance $\leftrightarrow$ cold-bias trade-off, tracing the aggressive lever to a Southern-Ocean mechanism. (II) shows that the cold bias which makes that lever costly is largely *not* the tuning but a **structural** LPJ-GUESS boreal-forest collapse (§6) — a *counter-lever*: fixing it warms the NH high latitudes, buying the RMSE headroom to spend the aggressive global cooling on the imbalance. Feedback-free spin-ups then quantify the sensitivity to forcing and isolate the comparatively small contribution from replacing net radiation alone. (III) records the original warm-floor work plan. (IV) supersedes its ordering where newer evidence requires it: first fix the atmospheric forcing on AMIP, then re-spin LPJG and tune only the residual (§15, §16).

2 Headline: the imbalance $\leftrightarrow$ cold-bias trade-off

Every lever that meaningfully lowers the TOA imbalance does so by *cooling* the model, which worsens the cold bias against ERA5. Two families separate cleanly: the sea-ice/albedo/melt-pond changes move gently down-left (small imbalance gain, small bias cost); the coupling + ocean-mixing changes cut the imbalance hard (30–57 %) but push the planet 1.7–2.1 K too cold.

The cold-bias axis is two things at once. It mixes (i) a *shared structural floor* — every run, including the baseline and the “clean” melt-pond runs, carries a strong Siberia/N. Canada cold bias that is *not* a tuning artefact but an LPJ-GUESS boreal-forest-cover bug (§6); and (ii) the *differential*, tuning-induced global cooling that the aggressive runs add on top. The structural floor is the same for all points and is independently fixable; the tuning only moves runs vertically relative to it. This is why the “warming lever” in §9 is the LPJG fix, not another atmospheric retune.

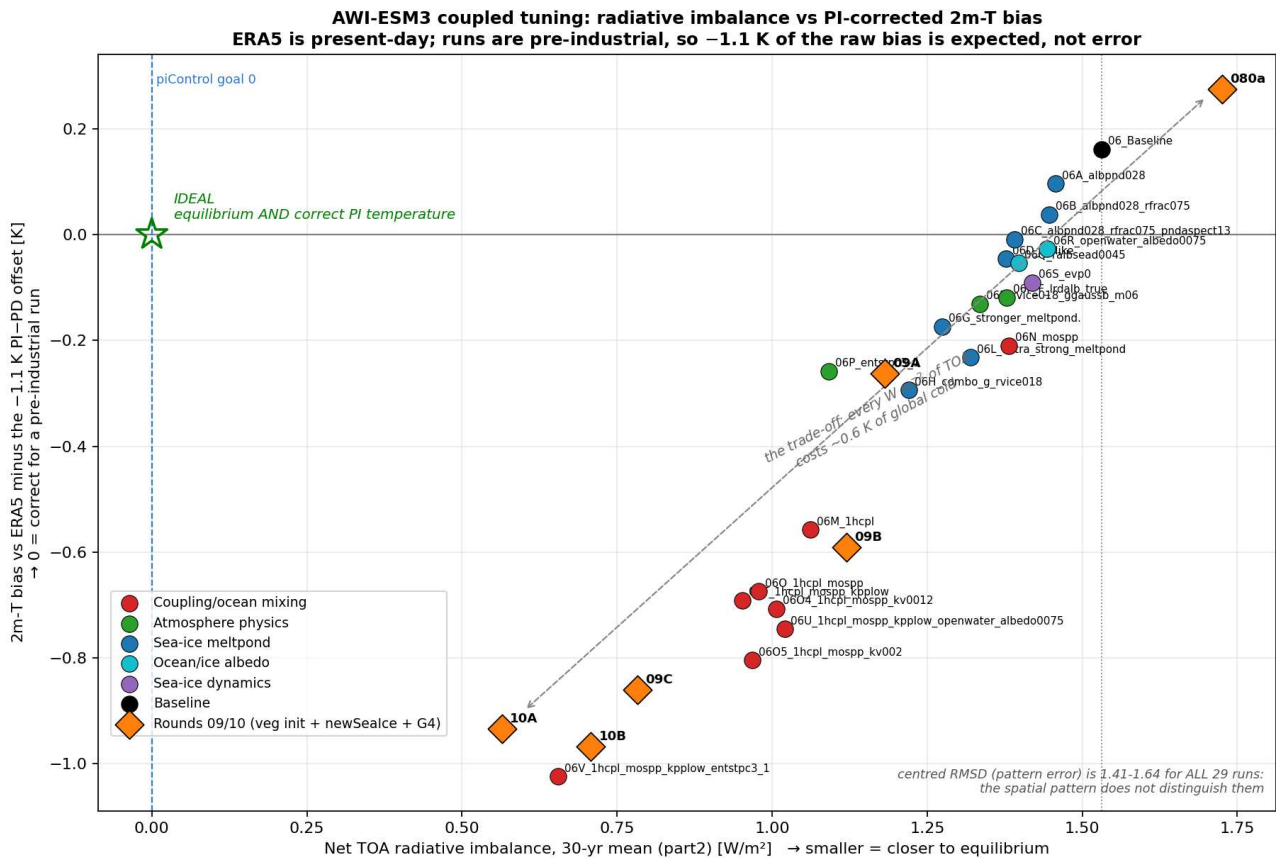


Figure 2: Net TOA imbalance (30-yr mean, part2) against the 2 m-T bias vs ERA5 *corrected for the reference period*, now including rounds 09 and 10 (orange diamonds, common window 1354–1379). **ERA5 is present-day and the runs are pre-industrial, about 1.1 K apart, so that much of the raw cold bias is expected rather than error;** the y axis subtracts it, and 0 is correct for a PI run. Ranking by raw RMSD penalises exactly the runs that are correctly cold, which is why the earlier version of this figure was misleading. Dashed blue = the corrected piControl goal of 0 (§37); the star marks the ideal, which nothing occupies. **The runs lie on a line, not in a cloud:** every W m^{-2} of TOA improvement costs ≈ 0.6 K of global cold, from 080a at (1.73, +0.28) to 10A at (0.57, -0.93). And the *pattern* error is not what distinguishes them — centred RMSD, $\sqrt{\text{RMSD}^2 - \text{bias}^2}$, which removes any uniform offset, is 1.41–1.64 K across all 29 runs, a $\pm 8\%$ spread against a raw-RMSD spread of 64%. Colour = parameter family.

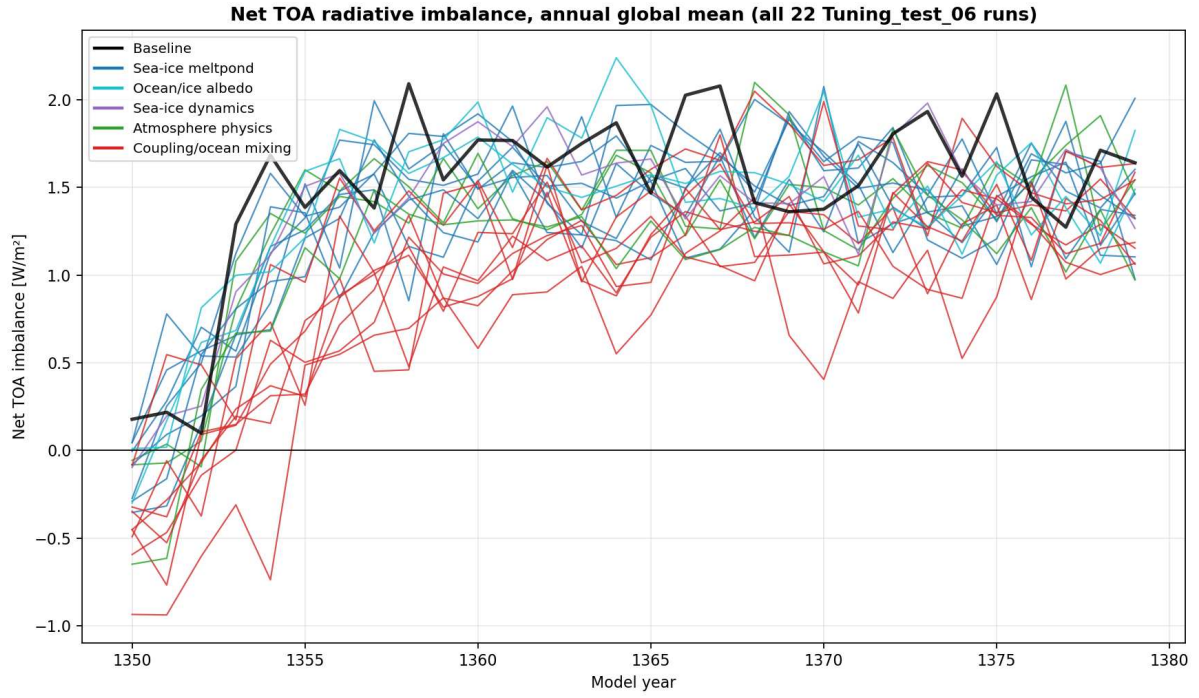


Figure 3: Annual global-mean net TOA imbalance, all 22 runs (independent CDO calculation; matches `part2`). None has flattened within 30 yr, so treat the mean level as a *ranking*, not a final equilibrium value.

3 Master metric table (coupled runs, sorted by TOA imbalance)

Run	TOA imb. [W/m^2]	vs base	2m RMSD [K]	2m bias [K]	CMPI
06V_1hcpl_mospp_kpplow_entstpc3_1	0.655	−57%	2.671	−2.124	0.803
06T_1hcpl_mospp_kpplow	0.952	−38%	2.385	−1.791	0.773
06O5_1hcpl_mospp_kv002	0.968	−37%	2.511	−1.904	0.788
06O_1hcpl_mospp	0.978	−36%	2.389	−1.774	0.772
06O4_1hcpl_mospp_kv0012	1.007	−34%	2.387	−1.807	0.781
06U_1hcpl_mospp_kpplow_openw0075	1.020	−33%	2.422	−1.845	0.776
06M_1hcpl	1.062	−31%	2.315	−1.658	0.773
06P_entstpc3_1	1.091	−29%	2.076	−1.358	0.759
06H_combo_g_rvice018	1.220	−20%	2.082	−1.394	0.757
06G_stronger_meltpond	1.274	−17%	1.990	−1.274	0.753
06L_extra_strong_meltpond	1.320	−14%	2.050	−1.332	0.751
06E_rvice018_ggaussb_m06	1.335	−13%	1.969	−1.231	0.754
06D_HRlike	1.377	−10%	1.892	−1.146	0.750
06F_lrdalb_true	1.378	−10%	1.962	−1.219	0.752
06N_mospp	1.382	−10%	2.011	−1.310	0.751
06C_albpnd028_rfrac075_pndaspect13	1.391	−9%	1.884	−1.109	0.751
06Q_ralbsead0045	1.397	−9%	1.891	−1.154	0.748
06S_evp0	1.419	−7%	1.935	−1.191	0.745
06R_openwater_albedo0075	1.443	−6%	1.860	−1.127	0.744
06B_albpnd028_rfrac075	1.447	−5%	1.847	−1.062	0.741
06A_albpnd028	1.457	−5%	1.811	−1.004	0.748
06_Baseline	1.531	—	1.717	−0.939	0.745

Rounds 09/10 (added 2026-08-07), same window 1354–1379.

Bias column also given PI-corrected (raw +1.1 K); cRMSD = pattern error.

10A (06V + G4)	0.565	−63%	2.522	−2.034 (−0.93)	cR 1.49
10B (10A + <code>tanh</code> depletion)	0.708	−54%	2.547	−2.067 (−0.97)	cR 1.49
09C (06V + newSeaIce)	0.783	−49%	2.475	−1.960 (−0.86)	cR 1.51
09B (06T + newSeaIce)	1.120	−27%	2.238	−1.691 (−0.59)	cR 1.47
09A (baseline + newSeaIce)	1.182	−23%	2.042	−1.363 (−0.26)	cR 1.52
080a (CRUNCEP veg init)	1.726	+13%	1.629	−0.825 (+0.28)	cR 1.41

The 09/10 block is the campaign’s central trade-off in one place, but read the bias column with the reference period in mind: ERA5 is present-day and these runs are pre-industrial, ≈ 1.1 K apart, so the PI-corrected value in brackets is the one that means

anything. On that basis 10A is 0.93 K too cold where 06_Baseline is 0.16 K too warm — a real gap, but a third of what the raw numbers suggest. **And the spatial pattern is not what changes:** centred RMSD is 1.41–1.64 across all 29 runs, so the entire raw-RMSD spread is the global mean, counted twice. Nothing yet occupies the low-TOA, correct-temperature corner. Over their own longer window (1370–1399) 10A reads 0.802/2.565 and 10B 0.916/2.383, so the ranking is window-dependent and only the common window is tabulated. CMPI is left blank for these rows rather than guessed — it needs `part4_cmpi.py` and its own reference set. TOA = 30-yr global-mean net TOA from `part2_rad_balance.py`. RMSD/bias from `part8` methodology (cos-lat weighted, model–ERA5, yrs 1354–1379); cross-checked vs the tool’s on-figure values (baseline 1.717 vs 1.721; 06V 2.671 vs 2.678). CMPI from `part4_cmpi.py`: < 1 = better than CMIP6 ensemble-mean error; lower is better.

The per-run distinguishing parameter change (which knob each 06* run alters on top of the common melt-pond shift) is catalogued in App. A.

4 Core diagnostics: radiation, 2 m temperature, CMPI

4.1 Radiation budget (part2)

Surface net flux tracks TOA to within $\approx 0.1 \text{ W/m}^2$ in every run — i.e. the atmospheric column stores essentially no energy on these multi-year means, as expected. Note this does *not* by itself localise the *cause*: a TOA $\approx$ surface match is equally consistent with a genuine TOA radiative error and with ocean heat uptake / unequilibrated spin-up drift passing cleanly through the surface (§5, §6.3). What it rules out is an atmosphere-only bookkeeping leak, so the imbalance and the surface energy budget can be read interchangeably. Baseline $\approx +1.53 \text{ W/m}^2$; the coupling+mixing stack drives it down to $+0.66 \text{ W/m}^2$ (06V).

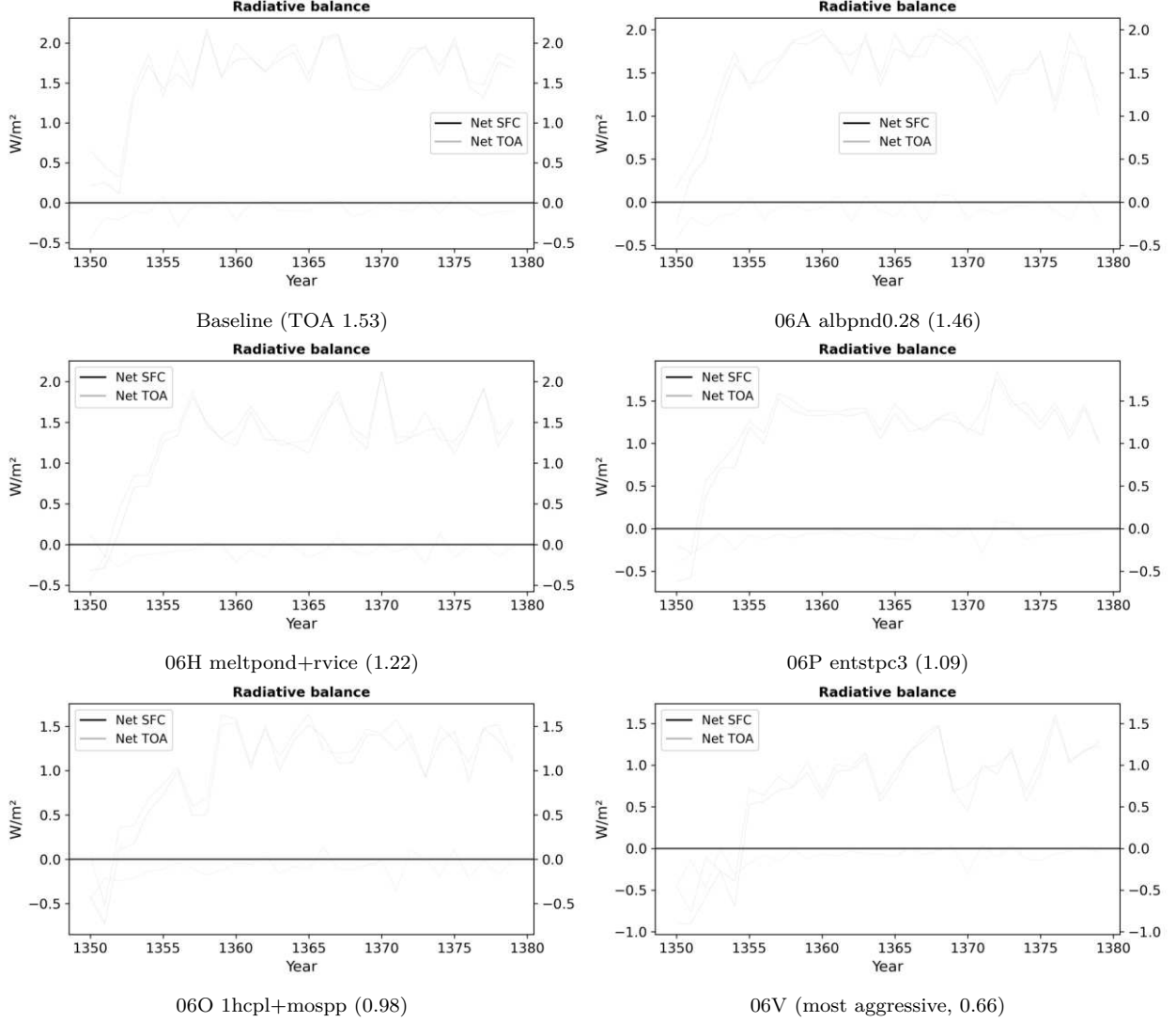


Figure 4: part2 radiation budget: grey = net TOA, blue = net surface, dark = their difference (W/m^2).

4.2 2 m temperature bias vs ERA5 (part8)

Baseline: cold over NH continents, warm over the Southern Ocean and tropical upwelling. The aggressive coupling runs remove the Southern-Ocean warm bias but turn the whole planet cold (06V is cold almost everywhere). *Caveat:* this is PI-control vs present-day ERA5, so $\approx 0.5\text{--}0.7\text{ K}$ of global cold offset is physically expected; the coupling runs' -1.7 to -2.1 K is clearly too much, the melt-pond family (-1.0 to -1.4 K) is more defensible.

The deep **Siberia / N. Canada cold lobe** is present in *every* panel — baseline and the clean melt-pond runs included — and barely changes between them: it is the structural LPJ-GUESS forest-cover signal (§6), not a product of the imbalance tuning. The aggressive coupling runs add a broad, near-global cooling *on top* of that fixed lobe.

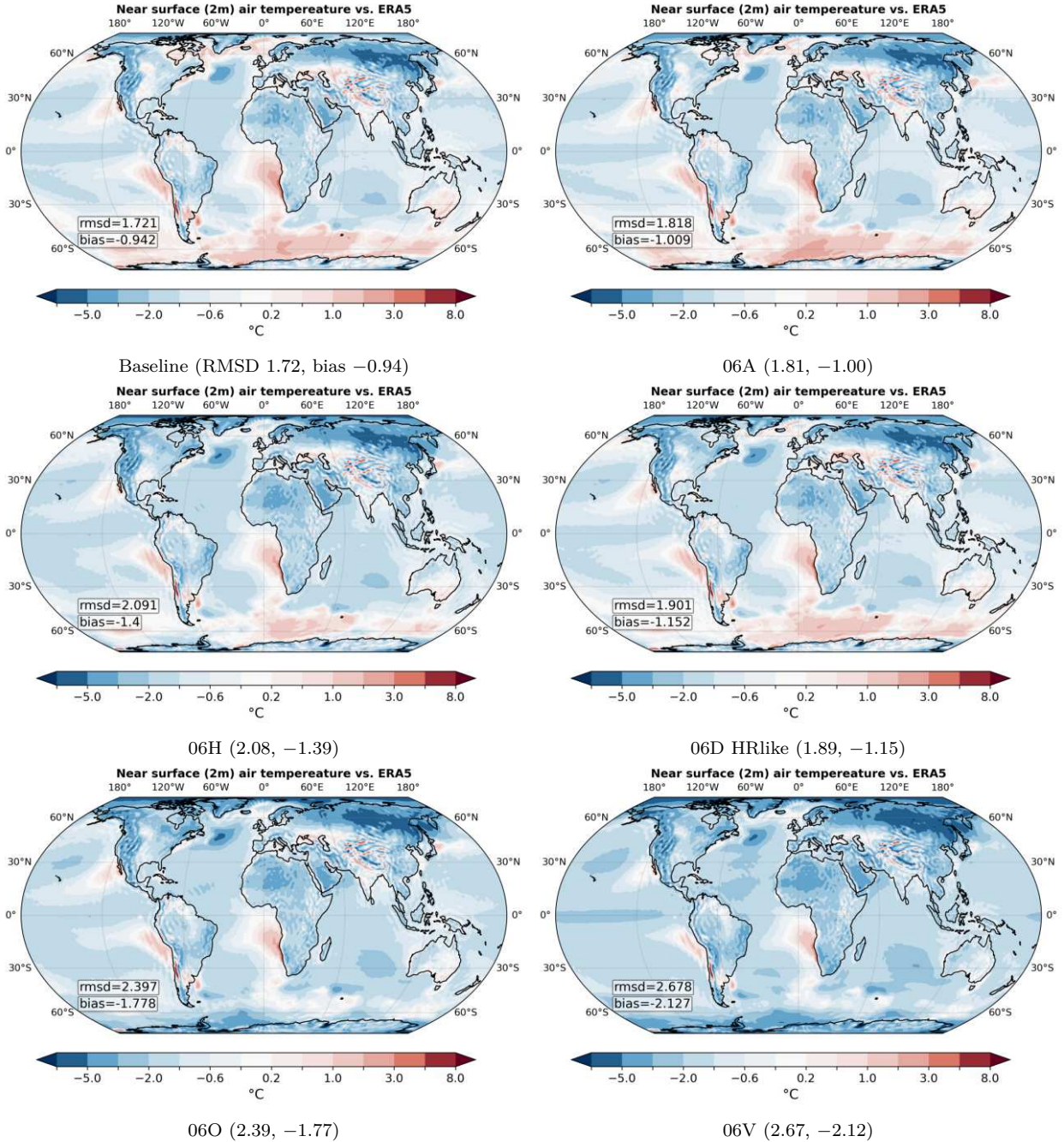


Figure 5: part8 2 m air-temperature bias (model – ERA5), EqualEarth, $\pm 8\text{ K}$ scale.

4.3 CMPI (part4)

CMPI integrates 13 fields (siconc, tas, clt, pr, rlut, uas, vas, ua@300, zg@500, zos, mlotst, thetao, so) over regions/seasons, normalised against 30 CMIP6 models; < 1 = **better than the CMIP6 ensemble-mean error**. Run on a subset spanning the trade-off front (now complete for all 22 runs).

How to read this (important). CMPI here compares a *PI-control* run to *present-day* observations, so a moderate global cold offset — and hence some CMPI rise — is *expected* and is not by itself a disqualifier. The informative question is not “did CMPI go up?” but “*where* did the error change, and is that consistent with the parameter that was altered?” A sea-ice-albedo change should move **siconc** in the polar regions and little else; if it instead degrades the tropics, that is a red flag. The per-variable / per-region breakdown below is the tool for that judgement; the global number is a summary, not a verdict.

Run	CMPI	TOA imb.	2m RMSD
06_Baseline	0.745	1.531	1.717
06A_albpnd028	0.748	1.457	1.811
06D_HRlike	0.750	1.377	1.892
06H_combo_g_rvice018	0.757	1.220	2.082
06O_1hcpl_mospp	0.772	0.978	2.389
06T_1hcpl_mospp_kpplow	0.773	0.952	2.385
06V__entstpc3_1	0.803	0.655	2.671

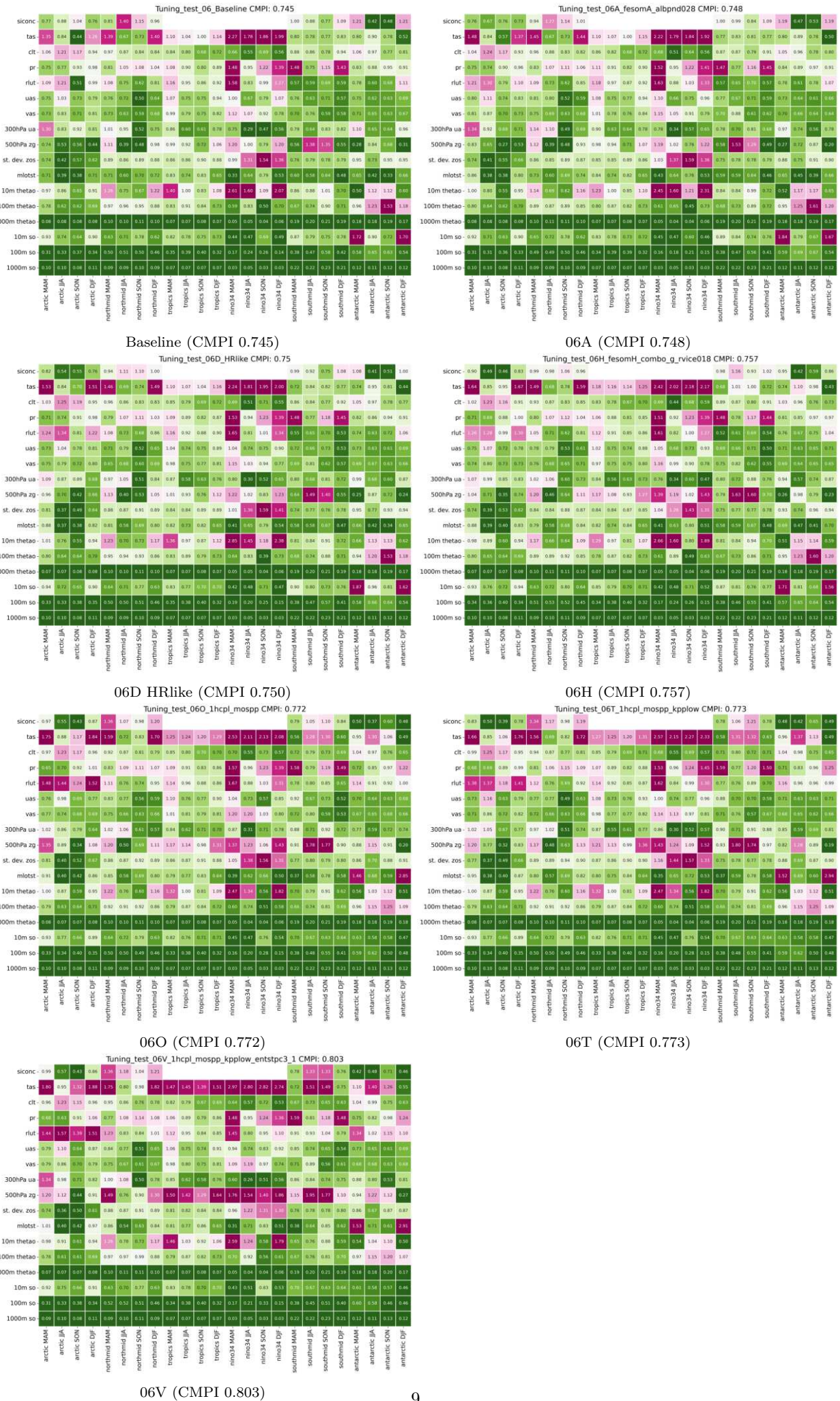


Figure 6: part4 CMPI heat maps (per variable/region/season, normalised to CMIP6).

4.3.1 Where does the error change? (per-variable / per-region)

The maps below show each run’s CMPI fractional error *minus baseline* (red = worse, blue = better). This is where the PI-vs-PD caveat earns its keep: the global CMPI rises monotonically ($0.745 \rightarrow 0.803$), but the *pattern* of change tells a much more favourable story for the targeted tunings.

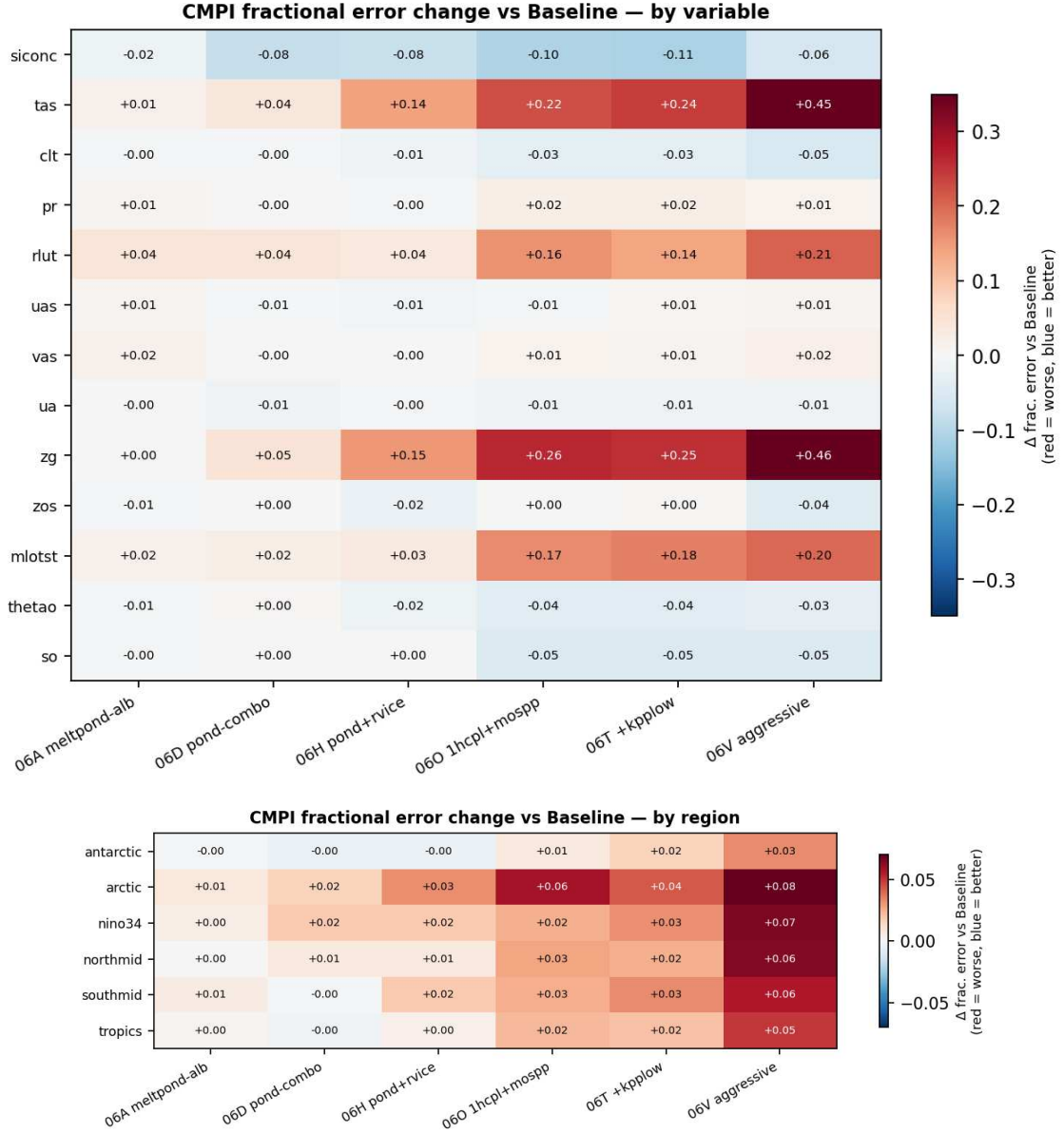


Figure 7: Change in CMPI fractional error vs Baseline, by variable (top) and region (bottom).

Reading the breakdown:

- **06A / 06D (melt-pond):** improve **siconc** ($-0.02/-0.08$) with essentially *zero* collateral elsewhere and a flat region map — a cleanly targeted change. These should be regarded as “no degradation where it matters,” despite the tiny global CMPI uptick.
- **06O / 06T (mospp + kpplow):** genuinely *improve* the ocean interior (**thetao**, **so**: $\approx -0.04/-0.05$), **siconc** and **clt** — exactly where ocean-mixing should act. But they *degrade* **tas/zg/rlut** ($+0.16$ to $+0.26$) and even their own target **m1otst** ($+0.17$), and the region map reddens *everywhere incl. tropics and Niño-3.4* — i.e. the cooling leaks well beyond where an ocean-mixing parameter has any business acting. *That* is the warning sign, more than the global number.
- **06V (+entstpc3):** largest, most widespread atmospheric degradation (**tas** $+0.45$, **zg** $+0.46$); the deep-convection change destabilises the whole atmospheric column.

5 Southern-Ocean mechanism (adapted from a270234’s notebooks)

To see *why* the ocean-mixing/coupling levers act as they do, two of a270234’s Weddell-Sea diagnostics (`mld_sic_sit_compare`, `temp_fesom_core3_WWS`) were re-pointed at the coupled `Tuning_test_06` runs (same CORE3 mesh). Panels show the Baseline absolute field plus each run *minus* Baseline, austral winter (September), yr1370–79. (The `temp_fesom_core3_WWS` tripyview transect could not decode these runs’ pre-1582 `gregorian` calendar, so an equivalent `pyfesom2` depth–latitude section was used.)

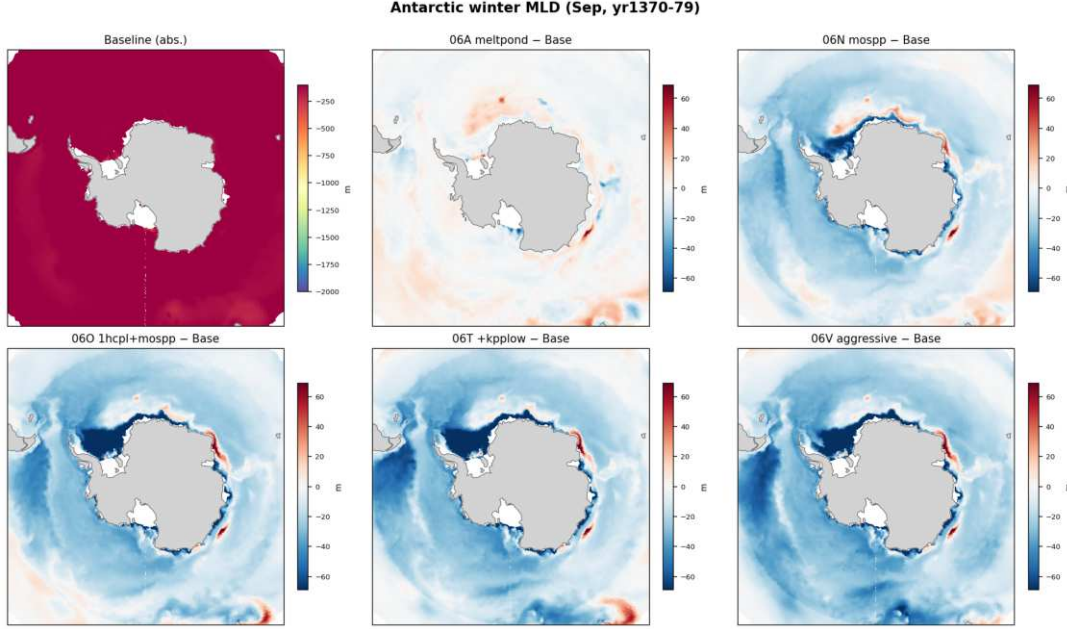


Figure 8: Antarctic September mixed-layer depth (MLD3): Baseline *absolute* (top-left, own scale), others = *anomaly* vs Baseline on a symmetric ± 70 m diverging scale (98th-ptile auto-scale; the intense Weddell/Ross convection core saturates it). The mixing runs (06N/O/T/V) restructure the Weddell/Ross winter mixing relative to Baseline, whereas melt-pond (06A) barely changes it. Deep open-ocean winter mixing there is FESOM “open-ocean convection” that ventilates warm deep water — a known driver of the Southern-Ocean warm bias (Heuzé et al. 2013; Heuzé 2021) and of TOA imbalance.

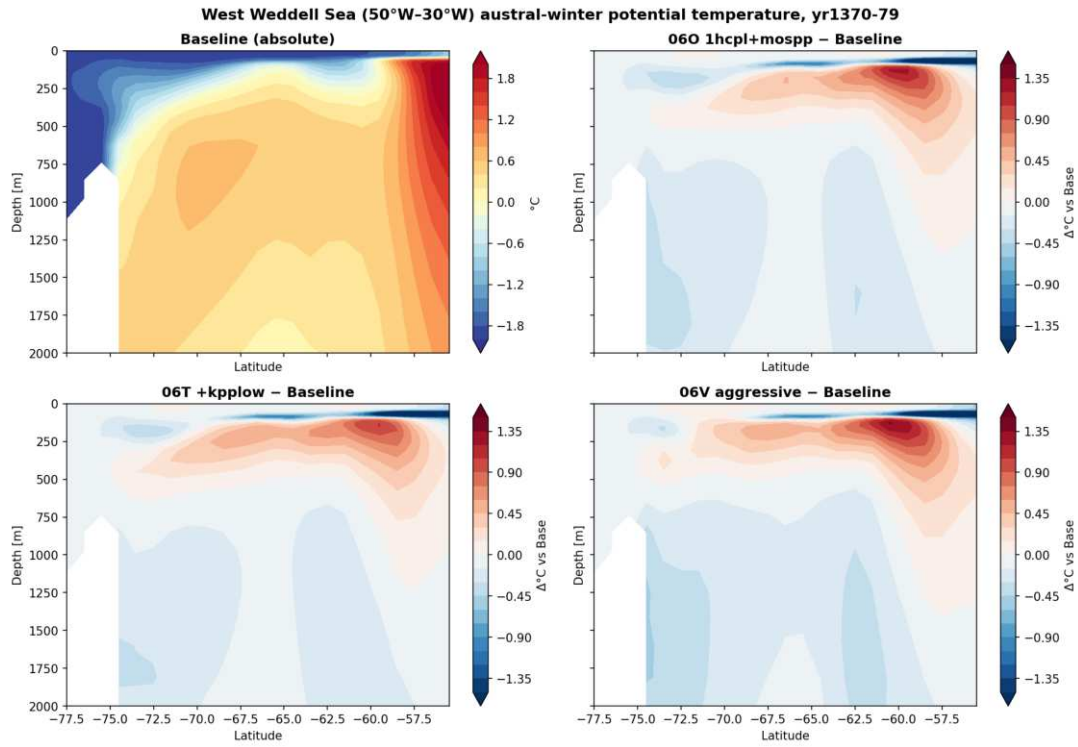


Figure 9: West Weddell Sea (50°W–30°W) September potential-temperature section: Baseline absolute and anomalies. The ventilated mid-column cools in the mixing runs (less convective entrainment of warm deep water to the surface), consistent with suppressed open-ocean convection; note that suppressing convection should leave the *deepest* water trapped/warmer rather than colder, and a near-surface edge warm anomaly is also present — i.e. the signal is a redistribution between layers, not uniform subsurface cooling.

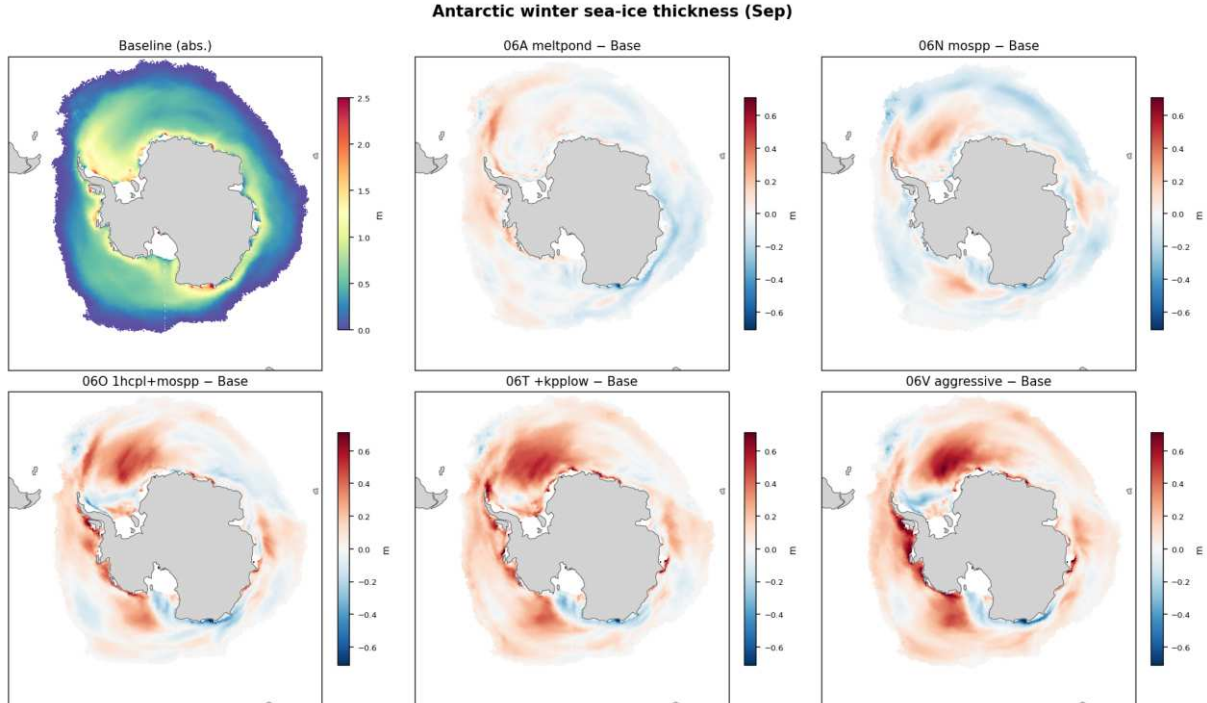


Figure 10: Antarctic September sea-ice *thickness* (Baseline absolute, top-left; others = anomaly vs Baseline). Thickness — not concentration — is the honest growth diagnostic here: winter concentration is already near 1 over most of the pack, so it *saturates* and cannot register added ice, whereas thickness is unbounded. The mixing runs (06O/06T/06V) thicken the pack broadly (positive/red anomaly over the Weddell and Ross sectors), consistent with suppressed open-ocean convection insulating and stratifying the surface; melt-pond (06A) barely changes it. (The concentration version, `sic_winter_antarctic.png`, shows only the ice-edge shifts for the reason above.)

Mechanism. The mospp / weaker-KPP changes reduce Weddell-Sea open-ocean winter convection, cooling the

subsurface and letting sea ice grow — improving `thetao/so/siconc` (consistent with the CMPI breakdown) and removing the Southern-Ocean surface warm bias. Spurious open-ocean deep convection in the Weddell/Ross gyres is the canonical CMIP Antarctic bias (Heuzé et al. 2013; Heuzé 2021, which evaluates AWI-CM in the multi-model set; Mohrmann et al. 2021 for the polynya surface expression), and open-ocean Weddell convection is a real-but-rare mode that models over-express (de Lavergne et al. 2014) — so damping it is physically the right direction. The *proximate radiative* closure is defensible: more winter ice → higher albedo → less absorbed SW → lower imbalance. We are more cautious about the “reduced ocean heat uptake” half of the chain: suppressing convection also *traps* heat at depth, which can *increase* deep ocean-heat-content drift, so the net uptake change is not established here — it should be pinned to an OHC-tendency diagnostic before being asserted. The cost is that the surface cooling propagates globally into the atmosphere (`tas/zg`), the broad cold bias seen in `part8`.

Counter-literature (a compensating-error risk this campaign must own). §5 attributes the SO warm bias and its removal almost entirely to *ocean* convection, but Hyder et al. (2018, *Nat. Commun.*) trace the leading 40–60°S warm-SST bias primarily to *atmospheric* net surface-heat-flux errors dominated by shortwave/cloud biases, and warn that a small *net* flux bias is often achieved by *compensating* component errors. The two coexist at different scales — broad atmospheric-SW warmth vs. localised Weddell/Ross deep-convection anomalies — and our mixing levers act only on the ocean branch. If part of the bias is atmospheric-SW in origin, the ocean-mixing lever is partly compensating for an atmospheric error, which is *exactly* the “cooling leaks beyond where the parameter should act” failure mode flagged for these runs in §4.3 (the region map reddens into the tropics/Niño-3.4). This tension is the report’s own evidence pointing at Hyder; it strengthens, not weakens, the case for treating aggressive ocean-mixing tunings as provisional.

Part II

Counter-tuning the NH cold bias — the structural floor

The cold bias that makes Part I’s aggressive lever expensive is largely **not** the tuning: it is a structural LPJ-GUESS boreal-forest collapse that every run inherits. That makes it a **counter-lever** — fixing it warms the NH high latitudes at essentially unchanged imbalance, buying the RMSE headroom to spend the aggressive global cooling on the imbalance. This part attributes the bias with a clean WITH-vs-WITHOUT-LPJG experiment, resolves its two mechanisms and a residual. Feedback-free spin-ups then show that the source of the low-tree state is strongly forcing-sensitive: the complete CRUNCEP–AMIP transfer matters greatly, whereas replacing CRUNCEP net radiation by CERES alone has only a small equilibrium effect.

6 The NH high-latitude cold bias is a structural LPJ-GUESS problem

The runs carry a severe near-surface **cold bias over Siberia and northern Canada** (visible in every part8 map). This chapter documents a working investigation into whether the coupled dynamic vegetation (LPJ-GUESS) is responsible — motivated by the idea that fixing it could be a *warming* lever to offset the imbalance-tuning cold bias, and by prior EC-Earth experience that LPJG drives such boreal biases.

6.1 Symptom: boreal needleleaf replaced by grass

Dominant-PFT maps (tool part26) for the HTESSSEL spin-up (the reference, run with the same atmosphere but *without* atmosphere↔LPJG feedback) vs the coupled runs show boreal needleleaf giving way to C3 grass over Siberia / N. Canada. Tree-dominant fraction north of 50°N: **20.1 % (spin-up) → 16.8 % (baseline) → 11.4 % (06V)** — progressively worse with the more aggressive (colder) runs. *Caveat on the prior*: DGVMs disagree substantially on boreal tree-vs-grass partitioning, and real boreal forest is genuinely open with substantial non-tree ground cover (Jiang et al. 2012), so “grass over Siberia” is not *automatically* a bug. What makes the establishment-failure reading defensible here is the *magnitude*: the coupled state reaches $cvh \approx 0.02\text{--}0.05$ with ~~no closed canopy anywhere~~ (FORESTFPC=0)¹ (§10), an outlier well below the $\sim 0.4\text{--}0.7$ closed-taiga range (MODIS VCF), not merely a grass-leaning partition.

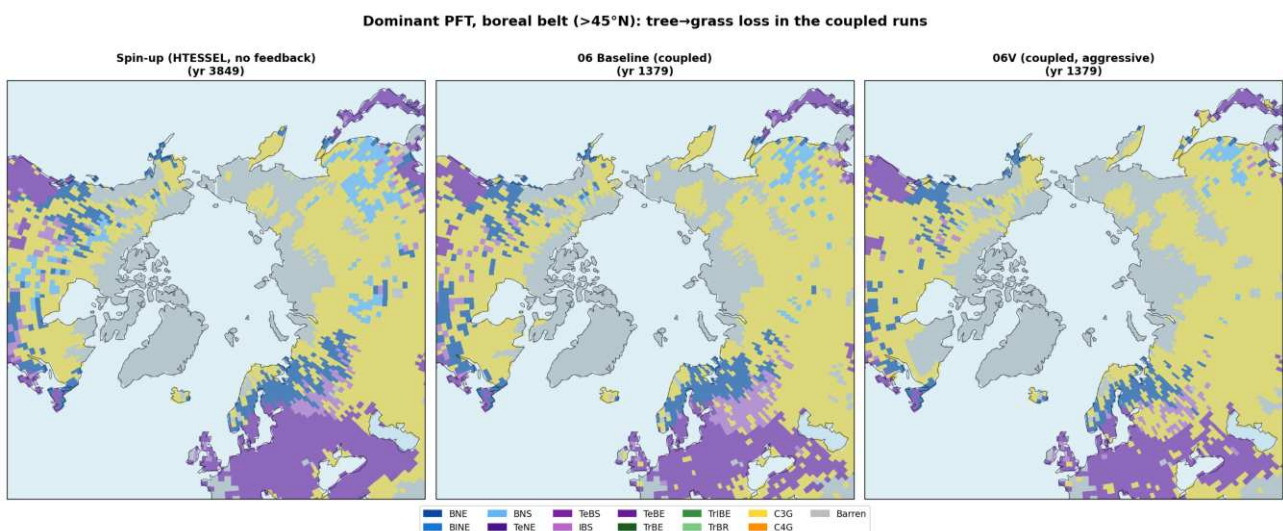


Figure 11: Dominant PFT, boreal belt: HTESSSEL spin-up (reference) vs coupled baseline vs 06V. Needleleaf (blues) thins and C3 grass (yellow) expands across central/eastern Siberia in the coupled runs.

¹**Void diagnostic.** FORESTFPC sums only PFTs with `landcover==FOREST` (`commonoutput.cpp:1143`), i.e. the *managed-forest* land-use class; natural trees are counted in TREEFPC via the `NATURAL` branch. In this PI configuration nothing is allocated to FOREST (`Natural_sum= 0.684`, `Total= 0.665`), so `FORESTFPC= 0` by construction while `TREEFPC= 0.323`. It is a land-use bookkeeping artifact and says nothing about canopy closure.

6.2 Is it grass growing, or trees dying? And is it a year-1 jump or a slow drift?

Belt-mean (55–75°N) needleleaf LAI is essentially *conserved* over the baseline run (0.40→0.37, trend ≈ 0), while C3 grass LAI *rises* (0.65→0.83, +0.10/dec): the signal is **grass growing**, with needleleaf loss only regional (E. Siberia) and offset by gains in the milder west. The coupled run is initialised directly from the spin-up’s year-3850 LPJG state, and baseline year-1350 boreal LAI (tree 0.407, grass 0.649) is *identical* to the spin-up end-state (0.401 / 0.669) — so there is **no wholesale year-1 collapse**; the degraded, grass-heavy state is *inherited from the spin-up*. Within 1350–1379 the boreal 2 m temperature is at its (cold) level from year one and does *not* drift down (trend +0.42 K/dec, noisy) — consistent with a memory-less atmosphere that equilibrates to the surface state within a year rather than a slow vegetation runaway. (These are belt-mean *LAI* diagnostics; as §6.3 shows, the quantity that actually matters is not LAI-per-tile but the forest *cover fraction*.)

6.3 The decisive test: a clean WITH-vs-WITHOUT-LPJG pair (CORE3, 30 yr)

The earlier control was confounded by the ocean (core2 vs core3). We removed that confound: a purpose-built **no-LPJG AWI-CM3-develop reference** — *same* CORE3 ocean, *same* TCO95 atmosphere and matched component branches, HTESSEL + prescribed satellite vegetation, a full 30-yr PI-control (model years 1350–1379, awicm3_noLPJG_CORE3_30y) — differs from the LPJG Baseline in **LPJG alone**, so the 2 m-T difference is now a clean attribution. Its net TOA imbalance is +1.73 W/m² (**part2**), near the warm/high-imbalance end of Fig. 1 — confirming the LPJG signal is a near-surface *land* effect, orthogonal to the TOA tuning of §2. **Crucially, it beats the LPJG Baseline on both skill metrics: global 2 m-T RMSD vs ERA5 1.514 K (bias −0.76 K, part8) vs 1.717/−0.94, and CMPI 0.738 (part4) vs 0.745** — at higher (warmer) imbalance. Removing LPJG improves the global temperature fit *and* the multivariate skill score *while* the imbalance changes by only +0.2 W/m² (+1.73 vs +1.53) — within the inter-run drift of these unequilibrated 30-yr segments (§8), so the warming lever is real *and costs essentially no imbalance to within that uncertainty* (not, strictly, “zero”). Over the final decade, boreal DJF:

- **Forest cover collapses.** Siberian high-veg cover cvh: **0.63 (no LPJG) → 0.02 (LPJG)** (pan-boreal 0.49→0.07). LPJG passes essentially no forest to the atmosphere.
- **And it is genuinely colder — confound gone.** With the ocean held fixed, the LPJG run is unambiguously **−3.1 K** colder over Siberia in DJF (−30.4 vs −27.4°C).
- **Snow-masking (spring albedo)** operates as before: forest removal brightens the snow season (MAM +0.077). This is the canonical boreal-forest climate mechanism — a masked snowpack lowers albedo and warms high latitudes, and its removal cools most in spring (Bonan, Pollard & Thompson 1992; Betts 2000; Bonan 2008), with tree/canopy cover setting the snow-albedo-feedback magnitude that many models misrepresent (Lorant et al. 2014; Chapin et al. 2005) — so a model that loses its forest gets exactly this feedback wrong. But it is a *sunlight* effect and cannot explain the equally-cold polar night (§6.4).

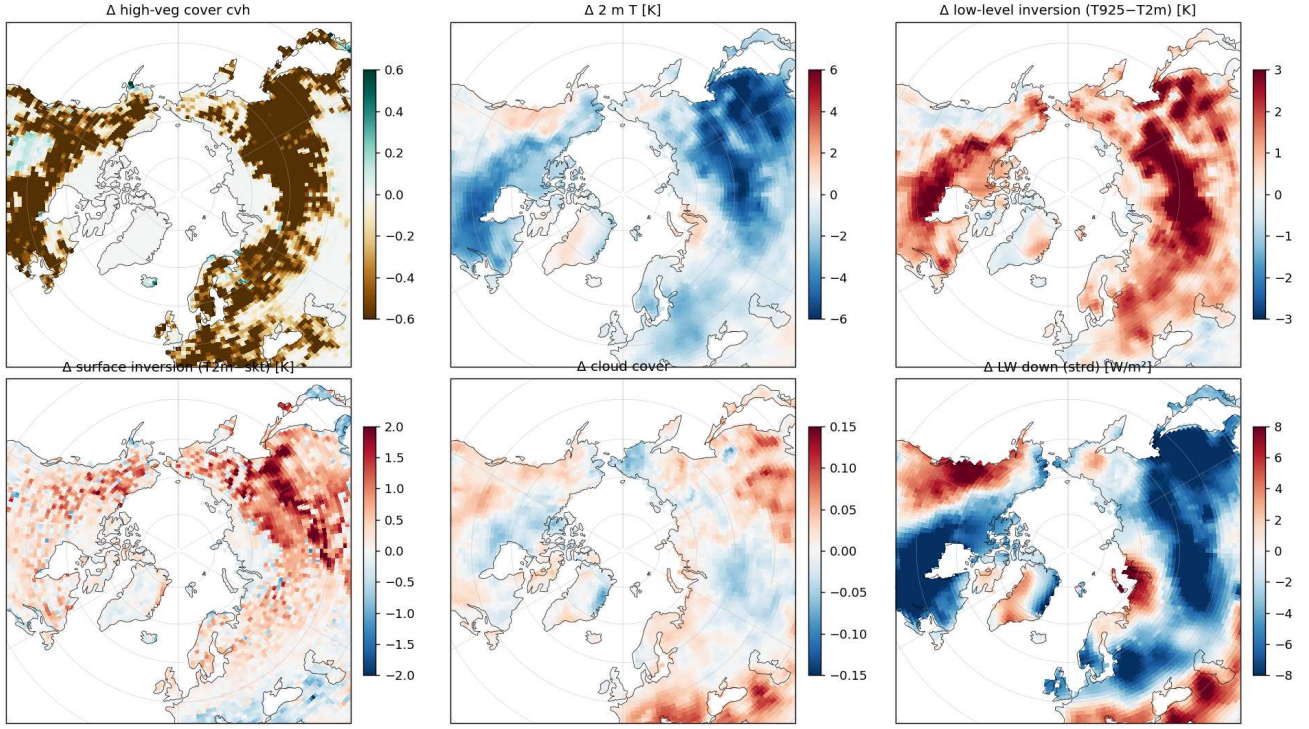


Figure 12: DJF WITH–WITHOUT LPJG, boreal NH (final decade, land-masked). Forest-cover collapse (cvh, top-left, brown) is spatially co-located with the cold anomaly (top-mid), a *stronger* low-level inversion ($T_{925}-T_{2m}$, top-right) and surface decoupling ($T_{2m}-T_{skt}$, bottom-left), at *unchanged* cloud (bottom-mid). LW-down (bottom-right) is modestly reduced over the cold interior.

6.4 Seasonality: which months carry the cold effect

Resolving the same clean CORE3 WITH–WITHOUT pair by *month* (boreal box 55–70°N, Siberia + N. America, final decade) shows the LPJG cold effect is **present in every month except high summer**. The WITH–WITHOUT 2 m-T difference is negative from September through June and only vanishes in July–August ($-0.1/-0.2$ K), peaking in the *shoulder/dark* seasons: **November –3.9 K, April –3.3 K**, with DJF -2.4 K and MAM -2.5 K (annual box-mean -2.0 K). **This is the key seasonal fingerprint:** a pure snow-albedo (sunlit) mechanism would confine the cold to the bright spring, yet the effect is *as large in near-sunless November/December* — which is exactly what motivates the winter roughness→inversion mechanism of §6.5. The near-zero July–August signal also confirms the growing-season atmosphere is unaffected (consistent with JJA gates being cleared, §6.3). Overlaid is the independent **OIFS–AMIP** run (§10.2: observed-SST, no-LPJG, PI 1870–79) — it tracks the *warm* WITHOUT curve (annual -7.6 vs -6.3 °C; DJF -25.6 vs -24.9 ; MAM even milder), confirming that a feedback-free, realistically-forced boreal climate is warm, and that the forcing we drive the decisive offline re-spin with (§10.2) is *not* itself cold-biased.

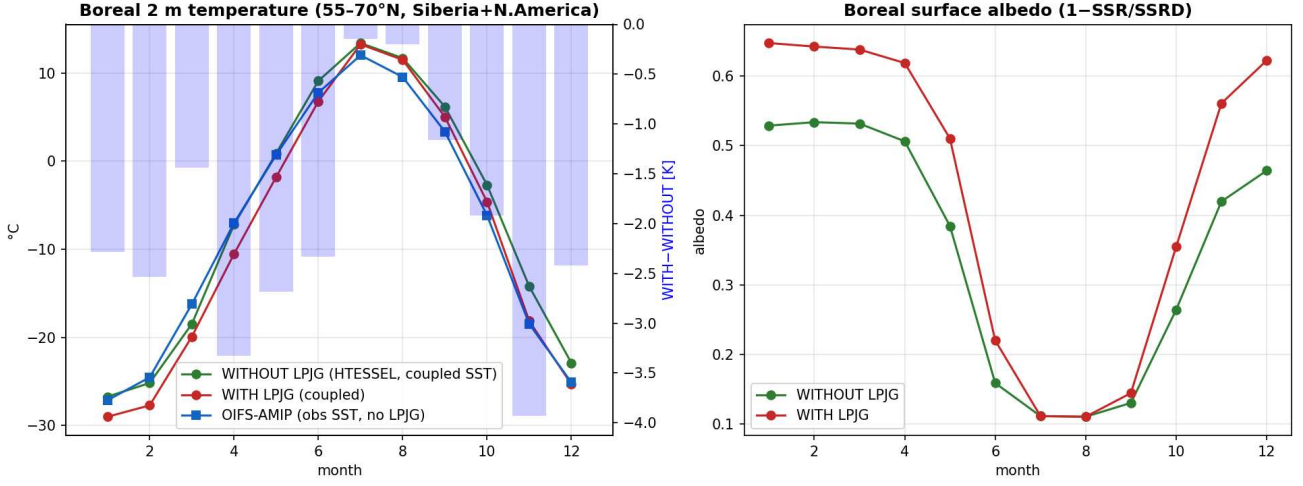


Figure 13: Boreal (55–70°N, Siberia+Canada) monthly climatology. *Left:* 2m-T annual cycle — WITHOUT LPJG (green, clean CORE3 reference), WITH LPJG (red, coupled), and the OIFS-AMIP obs-SST no-LPJG forcing run (blue); blue bars = WITH-WITHOUT [K, right axis], negative (LPJG colder) in every month bar July–August, strongest in Nov/Apr. *Right:* surface albedo (1-SSR/SSRD); LPJG brightens the snow season (spring/autumn), the sunlit part of the bias. (AMIP emits no down-welling SW, so it appears on the temperature panel only. WITH/WITHOUT are model years 1370–79; AMIP is obs-SST 1870–79 — both $\sim$ PI.)

6.5 A second, winter mechanism: roughness $\rightarrow$ inversion

Snow-albedo needs sunlight, yet the cold bias is as strong in DJF as in MAM. The missing forest acts a second way, in the dark. OpenIFS roughness length is cover-weighted by cvh , and needleleaf forest ($z_{0m} = 2.0$ m) is 20–60 $\times$ rougher than the tundra/grass (0.034–0.10 m, `susveg_mod.F90`) that replaces it. A smoother surface couples less efficiently to the atmosphere; in polar night the stable boundary layer (SBL) decouples and the surface radiatively cools, deepening the near-surface inversion. That surface-atmosphere coupling strength in the SBL controls screen temperature and inversion depth is well established (Holtslag et al. 2013; Sandu et al. 2013); we assemble the specific vegetation-roughness $\rightarrow$ SBL-decoupling $\rightarrow$ screen-T chain from the land-surface and SBL literatures rather than a single published result, so it is presented as a mechanism the data then test, not a settled citation. **The data show exactly this fingerprint** (DJF Siberia, LPJG – no-LPJG): a **+2.1 K stronger** low-level inversion ($T_{925} - T_{2m}$), a **+0.7 K stronger** surface inversion ($T_{2m} - T_{skt}$), **unchanged cloud** ($\Delta \approx 0.00$) and **-4.7 W/m² less** down-welling longwave — a decoupled, radiatively-cooling stable surface, not a cloud/radiation artefact. Spatially over boreal land the forest-loss pattern *is* the inversion pattern: area-weighted $r(\Delta cvh, \Delta inv) = -0.60$, $r(\Delta cvh, \Delta sfc\text{-decouple}) = -0.68$, $r(\Delta T_{2m}, \Delta inv) = -0.83$, while the negative control $r(\Delta cvh, \Delta cloud) = -0.05$ is ≈ 0 — a real mechanism, not a shared trend. Regression lever: **restoring cvh by +0.4 warms DJF screen-T by $\sim +1.2$ K** through roughness alone and weakens the inversion by ~ 0.7 K — additive to the spring/albedo warming. This slope-based estimate inherits the scatter of the underlying correlation ($r = 0.63$, so $< 40\%$ of grid-cell variance explained); read it as $\sim +1.2 \pm 0.5$ K, an order-of-magnitude lever rather than a precise prediction. The *direction* and existence of the effect rest on the WITH-WITHOUT experimental design (§6.3), not on this regression.

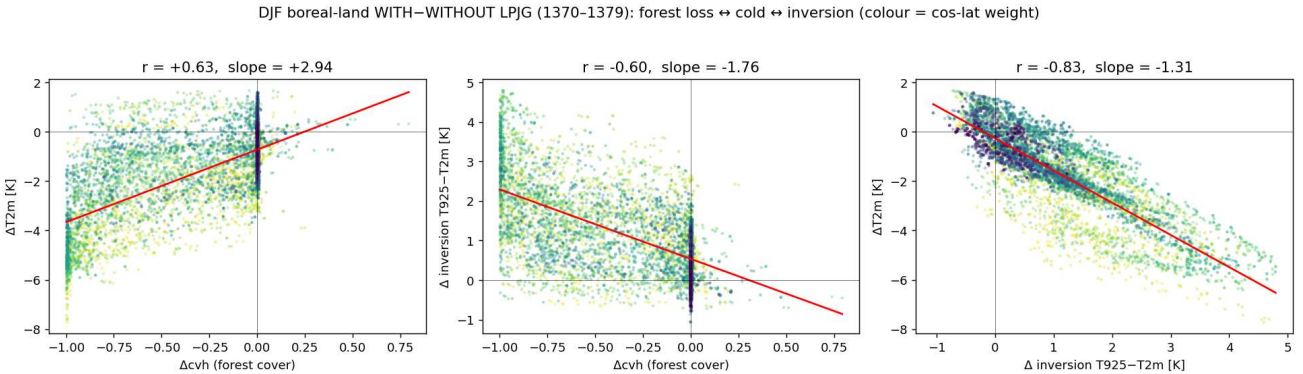


Figure 14: DJF boreal-land, WITH-WITHOUT LPJG (final decade). Forest loss (negative Δcvh) tracks the cold (ΔT_{2m} , left), the stronger inversion (middle) and, transitively, cold $\leftrightarrow$ inversion (right, $r = -0.83$). Colour = cos-lat weight.

6.6 How much does the LPJG fix buy? (vs ERA5)

Against ERA5 the current Baseline is -4.5 K too cold over Siberia in DJF. The lever helps but does not close it: a realistic partial fix ($+1\text{ K}$) leaves -3.5 K , and *full* forest restoration (the no-LPJG run itself — the ceiling) reaches -1.4 K . So vegetation/roughness accounts for $\sim 3\text{ K}$, roughly two-thirds of the Siberian DJF cold bias, and boreal-land RMSD falls $4.4 \rightarrow 3.3\text{ K}$ — but a residual $\sim 1.5\text{ K}$ survives *even with perfect forest*.

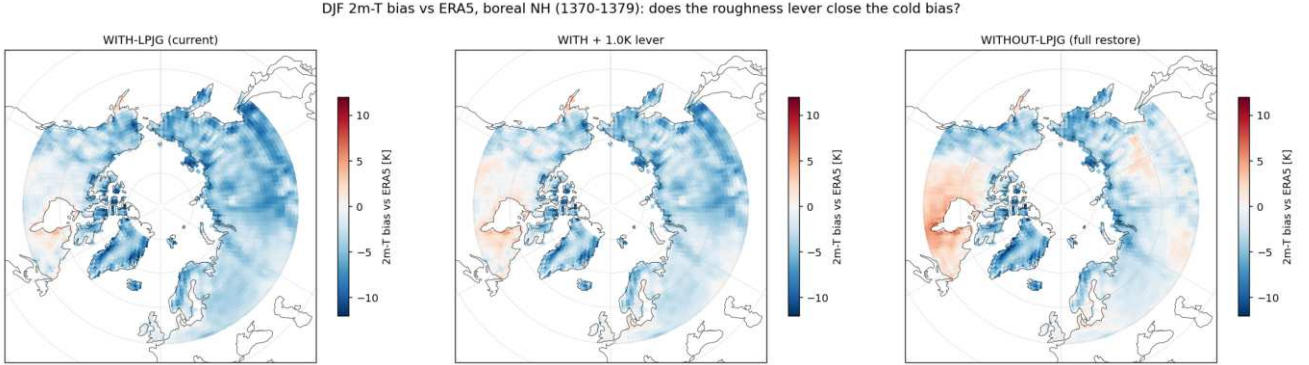


Figure 15: DJF 2m-T bias vs ERA5, boreal NH (final decade): current LPJG Baseline (left, model output), an **illustrative** $+1\text{ K}$ roughness lever (middle — a uniform $+1\text{ K}$ offset applied to the Baseline, *not* a simulation), and full forest restoration = the no-LPJG run (right, model output). The lever panel only visualises the magnitude of a plausible partial fix; the load-bearing comparison is left vs right, which shows a residual cold survives even full forest restoration.

6.7 The residual driver: an over-strong stable-boundary-layer inversion

Diagnosing that residual in the *full-forest* run (vegetation thereby ruled out) against ERA5 (DJFM, boreal land) points to a single process. The model’s low-level inversion is **$+1.4\text{ K}$ too strong** ($T_{925} - T_{2m}$: model $+5.6$ vs ERA5 $+4.2\text{ K}$), and the excess is **entirely surface-trapped** — the 925–1000 hPa inversion bias is $+0.07\text{ K}$ (≈ 0), so all the extra cooling sits in the lowest layer. Spatially this inversion-excess explains $r = +0.76$ of *where* the model is too cold; the Siberian High (MSLP bias $+0.15\text{ hPa}$, $r = 0.27$) and cloud-fraction (bias $+0.07$, $r = 0.00$) are ruled out. A too-strong wintertime polar inversion is a real, observationally-benchmarked model bias (Medeiros et al. 2011; Zhang et al. 2011), which supports reading the “ $+1.4\text{ K}$ too strong” residual as genuine. That surface-trapped signature, with circulation and cloud-fraction eliminated, is also the textbook fingerprint of a **stable boundary layer that under-mixes** — the IFS-family “too-stable” winter cold bias (Sandu et al. 2013). Its lever is the OpenIFS stable-BL turbulent mixing — the “long-tail” stability functions / minimum stable diffusion whose lineage runs Louis (1979) $\rightarrow$ Beljaars & Viterbo (1998) — *orthogonal* to the vegetation fix.

Two counters this diagnosis must own (both align with the report’s own “compensating error” worry). (a) *The mixing fix is a documented compensating error*: enhanced long-tail stable mixing is stronger than LES/observations justify and is retained only because reducing it degrades the large-scale flow — it improves 2 m temperature but deepens cyclones and harms low cloud (Sandu et al. 2013; Holtslag et al. 2013), so Phase 1B must check the storm-track/low-cloud climatology (which CMPI/part8 will not catch cleanly). (b) *The inversion may be a cloud-phase problem*: Pithan et al. (2014) attribute much of the modelled Arctic wintertime cold bias to missing mixed-phase low cloud (a LW, cloud-phase effect) — our null control ruled out cloud fraction but not phase, so we scope the residual claim to “given cloud fraction is unbiased” and flag a cloud-phase / LW $\downarrow$ test as a prerequisite. Likewise the $r = +0.76$ residual-to-inversion match is suggestive, not proof that the “fourth lever” is truly orthogonal.

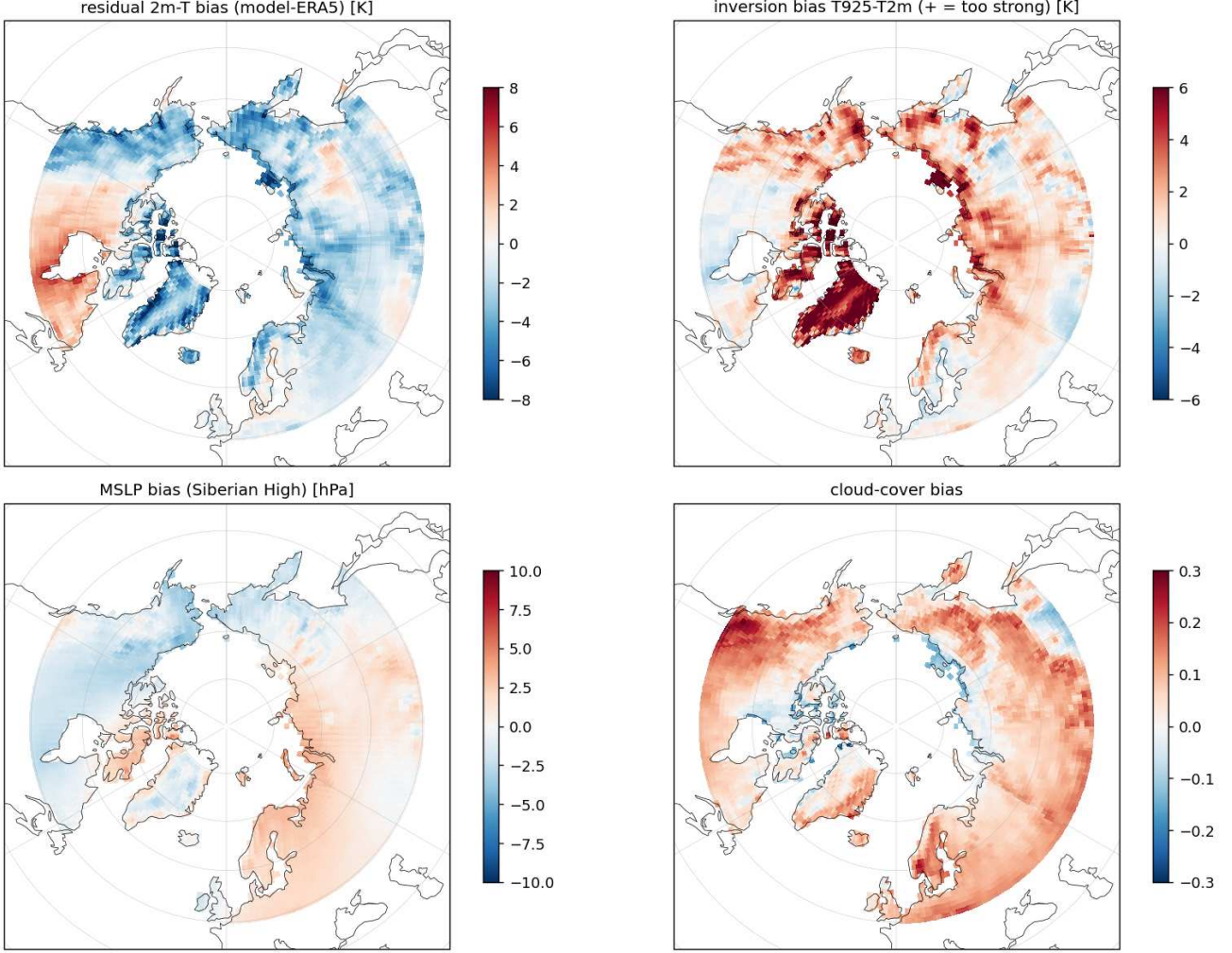


Figure 16: Second-driver search: full-forest (no-LPJG) run DJFM vs ERA5, boreal NH. Residual cold (top-left) matches the inversion-excess (top-right, $r = +0.76$); MSLP (bottom-left) and cloud (bottom-right) biases are diffuse and uncorrelated.

Near-complete attribution. The two mechanisms close the boreal DJF cold bias between them: of the ≈ -4.5 K Siberian bias vs ERA5, the LPJG forest-cover collapse (snow-albedo in spring, roughness in winter) contributes ≈ -3 K, and the over-strong stable-BL inversion the remaining ≈ -1.4 K. Two *orthogonal* levers — restoring a forcing-consistent forest state and enhanced stable-BL mixing — therefore target essentially the entire bias. (The veg cooling is also a *stable* offset, not a runaway: over 20 yr the LPJG run carries more/thicker Arctic ice as a bounded mean offset with no amplifying sea-ice drift.)

6.8 Working conclusion & lever

LPJ-GUESS sends essentially no boreal *forest cover* to OpenIFS ($cvh \sim 0.6 \rightarrow 0.02$ over Siberia, §6.3), so the canopy no longer masks snow (spring albedo, §6.3) *nor* roughens the surface (winter inversion, §6.5) — a physically robust driver of the Siberia/N. Canada cold bias, consistent with the EC-Earth family in which LPJ-GUESS is interactively coupled (Döscher et al. 2022) and in which dynamic vegetation-cover / effective-LAI improves high-latitude spring climate through the snow-albedo coupling (Alessandri et al. 2017). Source trace (verified): LPJG computes the high-veg fraction and exports it over OASIS as GUE_FRAH $\rightarrow$ cvh (framework.cpp::computeDailyLAIandFPCforIF namcouple line 120). With the AWI setting iftreefracca=1 (global.ins:151), cvh is the summed tree *crown area* $\times$ density (no 0.05 clamp — that constant is dead code; natural_frac= 1 over PI land), so $cvh \approx 0.05$ means boreal tree cover is genuinely $\sim 5\%$. The establishment gates ($gdd5min_est=500$, $twmin_est=5$) are *not* the blocker — they are cleared across the boreal — so the degraded state is **inherited from the LPJG spin-up**, not a coupled runaway. It was initially read as a tree-vs-grass competition failure, but the controlled forcing-transfer test (§14.3) reproduces most of the boreal loss without changing LPJG parameters; competition must therefore be diagnosed only after a forcing-consistent re-spin. Seasonally (§6.4) the LPJG cold effect spans September–June

(DJF -2.4 , MAM -2.5 K over the boreal box) and vanishes only in July–August — i.e. it is a cold-season, not a growing-season, effect. The co-equal *sunless* late-autumn/winter part — which pure snow-albedo cannot explain — is now resolved (§6.3–6.7): it is the roughness→inversion effect of the lost forest (§6.5) plus a model-intrinsic over-strong stable-BL inversion (§6.7).

Fixes: (1) correct the atmospheric forcing, regenerate the daily LPJG forcing and re-do the spin-up with one verified binary; only the residual against that forcing may justify tuning `greff_min`, C3-grass competitiveness or `pstemp_low` (§14.3, §16); (2) if a structural residual remains, revisit `iftreefracca/crown allometry`; (3) a woody stem-area floor on the snow-masking fraction in OpenIFS (`surfbc_ctl_mod.F90:389–391 / surfgrad_ctl_mod.F90:650–655`) — but this only bites once (1) raises `cvh`, since the masked-snow tile is weighted by `cvh` itself. Restoring forest cover is the physically-grounded *warming lever* of §9 and buys ~ 3 K (§6.6), but its cause must not be prejudged as a parameter error. **A fourth, orthogonal lever** attacks the residual ~ 1.4 K (§6.7): *enhanced stable-boundary-layer mixing* in OpenIFS (VDF stability functions / minimum stable diffusion), which would relax the over-strong surface inversion. Together (1)–(4) target essentially the entire boreal DJF cold bias.

6.9 Feedback-free AMIP spin-up — substantially qualified by §14.3 and §7

Read this section with the later controlled tests. The argument below turns on a standalone LPJG re-spin driven by the OIFS–AMIP forcing under-vegetating despite the establishment gates being cleared, which was read as an LPJG-internal competition failure. A later controlled experiment (§14.3) shows that *moving between the CRUNCEP and AMIP forcings alone*, with byte-identical parameters, costs -54% of Siberian and -66% of East-Siberian **TREEFPC**. The under-vegetation seen here is therefore largely what the forcing change predicts on its own, and cannot by itself demonstrate a competition failure. The four-arm radiation test (§7) further shows that replacing only CRUNCEP `rsns/rlns` by CERES removes only a small part of the NH contrast: the strong full-forcing effect is not a net-radiation effect alone. **FORESTFPC** is in fact a *void diagnostic* here: it counts only the managed-forest land-use class, which is empty in this PI configuration, so it is identically zero whatever the vegetation does (§14.3). There is no separate closed-canopy failure.

Is the boreal forest deficit a *climate* problem (fix the atmosphere) or an *LPJG-internal* tree-vs-grass competition problem (fix the vegetation)? We answered it directly with a **feedback-free** experiment: a clean OpenIFS–AMIP run (prescribed observed SST, no coupled ocean/LPJG cold feedback; $\sim$ PI 1870–79) generated a realistic daily forcing, which drove a 2000-yr standalone LPJG re-spin on a *dense* 11 538-cell gridlist (build detail in §10.2). Fig. 17 puts that forcing (top) beside the equilibrium vegetation it produced (bottom). **The forcing is *not* growing-season-cold:** over the Siberian box the warmest-month T2m is 14.7°C (100% of cells clear the `twmin_est` $= +5$ gate) and GDD5 is 730 (76% clear 500) — establishment is climatically permitted essentially everywhere trees should be. **Yet the spin-up under-vegetates and grass wins:** `cvh` $= 0.21$ (far below the 0.4–0.7 closed-taiga range), `GRASSFPC` $= 0.37 > \text{TREEFPC} = 0.21$, and ~~no closed canopy forms anywhere~~ (`FORESTFPC` $= 0$)², with grass filling exactly the East-Siberian interior. On a dense gridlist this cannot be dismissed as under-sampling. **Verdict: clearing the establishment gates does not prove that competition is the dominant error.** The controlled transfer test shows that the complete forcing change reproduces most of the regional tree loss (§14.3), while the new four-arm experiment shows that replacing net radiation alone explains only a small part of the NH contrast (§7). Competition remains a possible *residual* target, but it must be measured after a forcing-consistent re-spin rather than assumed from this AMIP arm. One climate co-limit visible here is the deep-winter cold (-28.3°C box mean, < -40 in NE Siberia) that gates out evergreens on $\sim 40\%$ of cells — but climatically-permitted *larch* (no cold gate) fails there too (BNS cover 0.035), so the winter warming of Phase 1B is only a *secondary* aid.

²**Void diagnostic.** `FORESTFPC` sums only PFTs with `landcover==FOREST` (`commonoutput.cpp:1143`), i.e. the *managed-forest* land-use class; natural trees are counted in `TREEFPC` via the `NATURAL` branch. In this PI configuration nothing is allocated to `FOREST` (`Natural_sum` $= 0.684$, `Total` $= 0.665$), so `FORESTFPC` $\equiv 0$ by construction while `TREEFPC` $= 0.323$. It is a land-use bookkeeping artifact and says nothing about canopy closure.

Decisive test: OIFS-AMIP (obs-SST, feedback-free ~PI) forcing vs the LPJG spin-up it drives (equilibrium yr 3900, NH boreal)

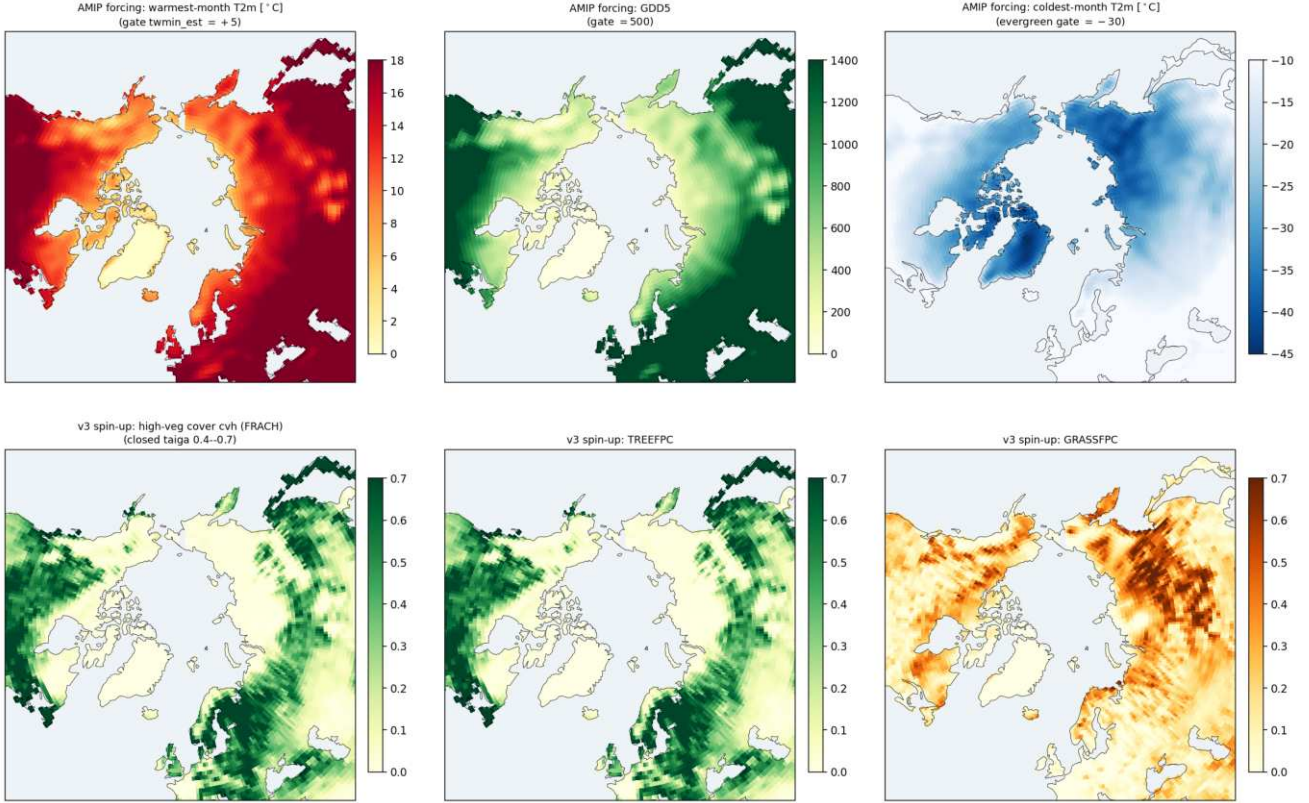


Figure 17: Decisive climate-vs-competition test (NH boreal, equilibrium yr 3900). *Top*: the feedback-free obs-SST ~PI OIFS-AMIP forcing that drove the standalone LPJG re-spin — warmest-month T2m (vs `twmin_est` = +5), annual GDD5 (vs 500), coldest-month T2m (vs evergreen `tcmin_est` = -30). *Bottom*: the equilibrium vegetation it produced — `cvh`, `TREEFPC`, `GRASSFPC`. The growing-season gates are largely cleared, yet the AMIP-forced equilibrium has `cvh` ≈ 0.2 and extensive East-Siberian grass. This panel diagnoses the AMIP-arm state; the controlled transfer test (§14.3) shows that it cannot by itself attribute the state to LPJG competition.

7 Forcing sensitivity: CRUNCEP radiation constrained by CERES EBAF

Question and controlled design. The full forcing-transfer test in Part IV shows that moving from CRUNCEP to AMIP strongly reduces equilibrium boreal tree cover, but it does not isolate which forcing fields cause that response. We therefore completed four 2000-yr standalone LPJ-GUESS spin-ups on TCO95 and compared their final year (model year 3900). The two additional arms isolate the narrower effect of replacing only CRUNCEP surface net shortwave (`rsns`) and net longwave (`rlns`) by CERES-constrained radiation:

experiment	daily forcing cycle	radiation treatment
AMIP CRUNCEP v3	OpenIFS-AMIP, 1870–1879 calibrated CRUNCEP v7, 1901–1910	corrected native OpenIFS <code>rsns/rlns</code> calibrated CRUNCEP net radiation
CRUNCEP + CERES direct	CRUNCEP v3 non-radiation fields	CERES EBAF Ed4.1 monthly 2001–2010, bilinear to TCO95 and linearly interpolated between month midpoints
CRUNCEP + CERES daily variability	CRUNCEP v3 non-radiation fields	CERES official 12-month climatological means, retaining CRUNCEP relative daily SW and additive daily LW anomalies

In both CERES experiments `tas`, `tasmin`, `tasmx`, precipitation, humidity and wind are byte-for-byte unchanged from CRUNCEP v3. Runtime files carry the generic name `AM04_atm_cmip6_1d_1990-1999_fast_lpjgforcing.nc`; checksums and full variable comparisons verified that they are exact runtime copies of the intended direct and daily-variability source files.

CERES provenance and what the comparison means. The observational reference is NASA Langley CERES EBAF Edition 4.1 on its native 1° grid, specifically `sfc_net_sw_all_clim` from `CERES_EBAF_Ed4.1_Subset_CLIM01-CLIM12.nc`. It is the official July 2005–June 2015 climatology (product DOI). Figure 18 puts all forcings on a common 1° land grid and shows JJA plus the Siberian seasonal cycle. AMIP–CERES and CRUNCEP–CERES are *forcing differences across products and epochs*, not strict pre-industrial biases. The small differences for the two CERES-derived forcings are *construction checks*, not independent validations: the direct case uses 2001–2010 monthly CERES data, whereas the daily-variability case is explicitly constrained to the same climatology used as the reference.

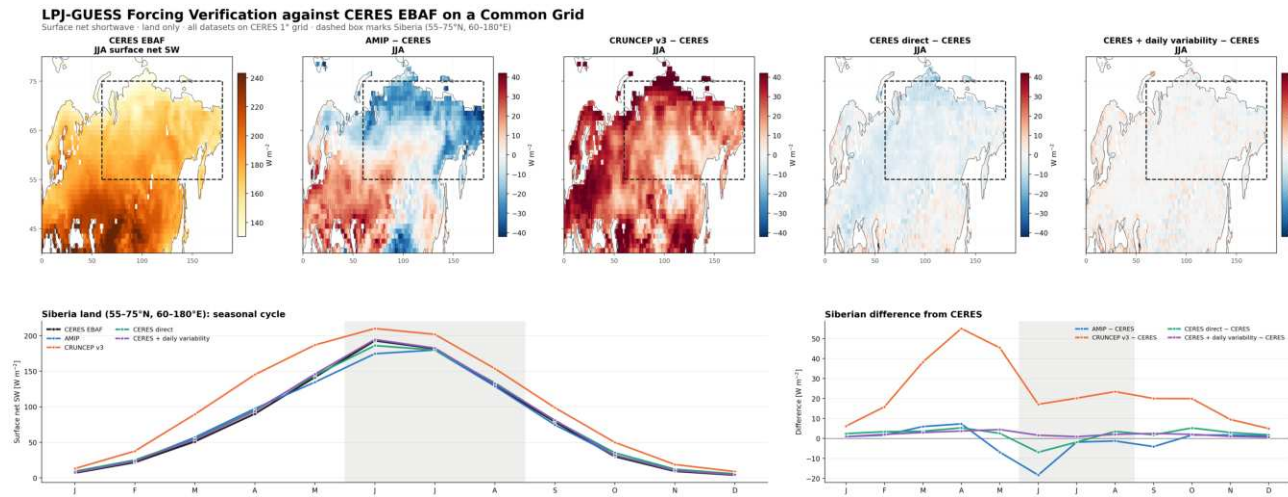


Figure 18: Surface net-shortwave forcing check against CERES EBAF Ed4.1. Top: CERES JJA net SW and each forcing minus CERES after common-grid remapping over land. Bottom: Siberian land seasonal cycles and their differences from CERES ($55\text{--}75^\circ\text{N}$, $60\text{--}180^\circ\text{E}$). Near-zero CERES-derived panels confirm the forcing construction; they are not an independent observational test.

Equilibrium vegetation response. In the annual `treeFrac` diagnostic, the NH-land display-grid mean is 0.186 under AMIP, 0.269 under CRUNCEP v3, 0.262 under direct CERES radiation and 0.263 when CRUNCEP daily variability is retained. Thus the full CRUNCEP–AMIP difference is +0.083, whereas the two CERES substitutions remain +0.076 above AMIP. Replacing CRUNCEP radiation reduces its mean tree fraction by only 0.006–0.007 and leaves about 92% of the CRUNCEP–AMIP difference. The almost identical direct and daily-variability results show that this choice of monthly-to-daily radiation treatment has negligible influence on the hemispheric mean at equilibrium, although regional differences remain visible.

Interpretation. Net radiation alone is not the dominant cause of the AMIP–CRUNCEP equilibrium tree-cover difference: changing only `rsns/rlns` removes roughly 8% of this hemispheric `treeFrac` contrast. Together with the regional `TREEFPC` transfer result of §14.3, this localises most of the full-forcing penalty to other meteorology (especially the documented temperature/GDD5 change), field covariability and nonlinear LPJG response. It does *not* establish a residual competition error. The appropriate conclusion is: *the vegetation equilibrium is strongly forcing-sensitive, but within the CRUNCEP family a CERES net-radiation correction alone is insufficient; competition parameters should be considered only after a forcing-consistent same-binary re-spin.*

Part III

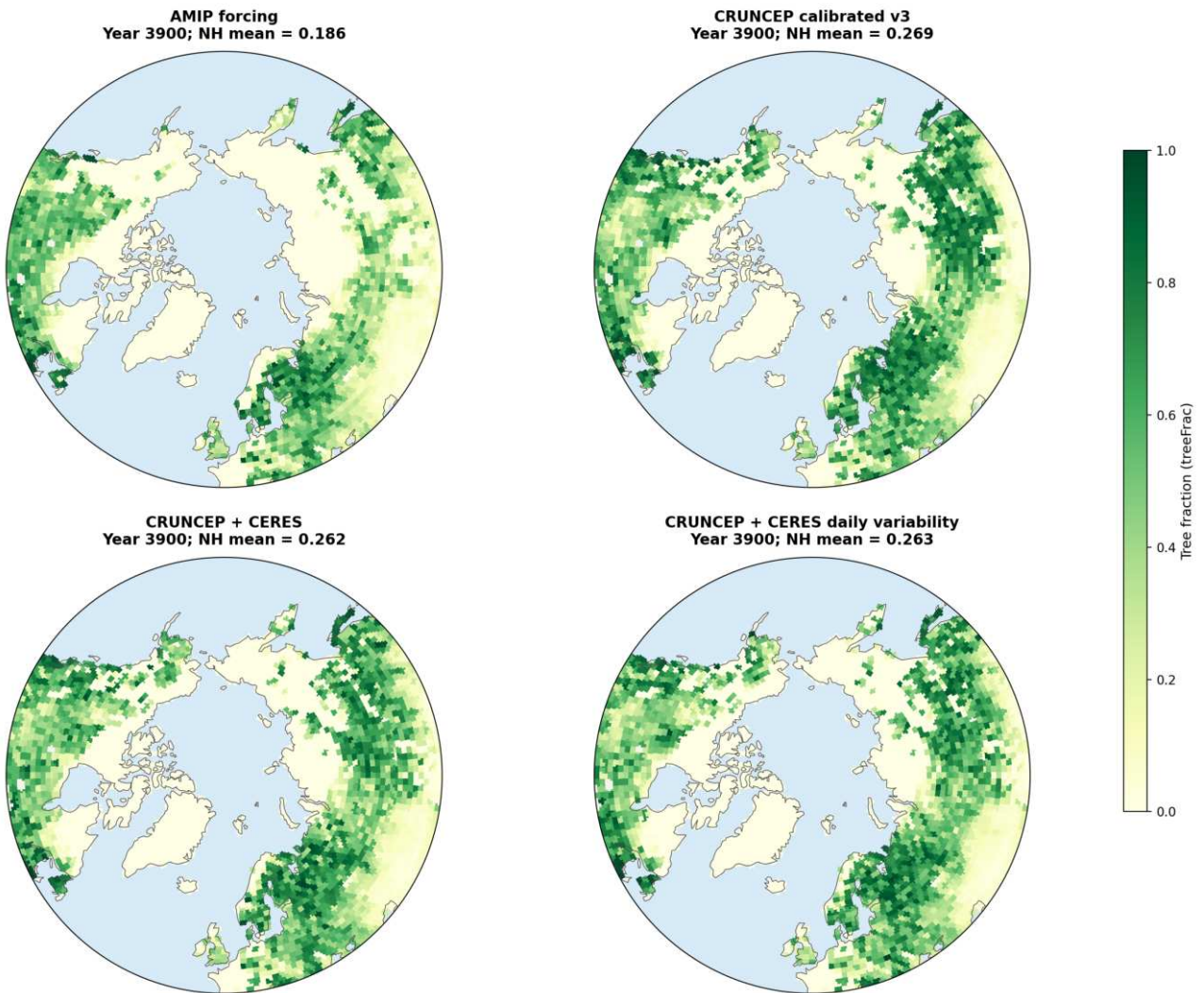
The path forward: raise the floor, then tune

Parts I–II give the strategy: the imbalance can only be cut by cooling, and cooling is only affordable once the structural NH cold floor is raised. So first fix the floor (vegetation competition + stable-BL mixing), then spend a moderate cooling lever to reach surface balance — now without the RMSE blow-up. First the caveats that condition this, then the recommendation and a phased plan.

8 Caveats

1. **06D “HRlike” has no HR-specific namelist change** — only the common pond shift (identical to 06C); its metrics confirm this. If a resolution/timestep/viscosity change was intended, it is not in the namelists.

Northern Hemisphere Tree Fraction in LPJ-GUESS Spinups (last simulation year)



Northern Hemisphere Tree-Fraction Differences from AMIP (last simulation year)

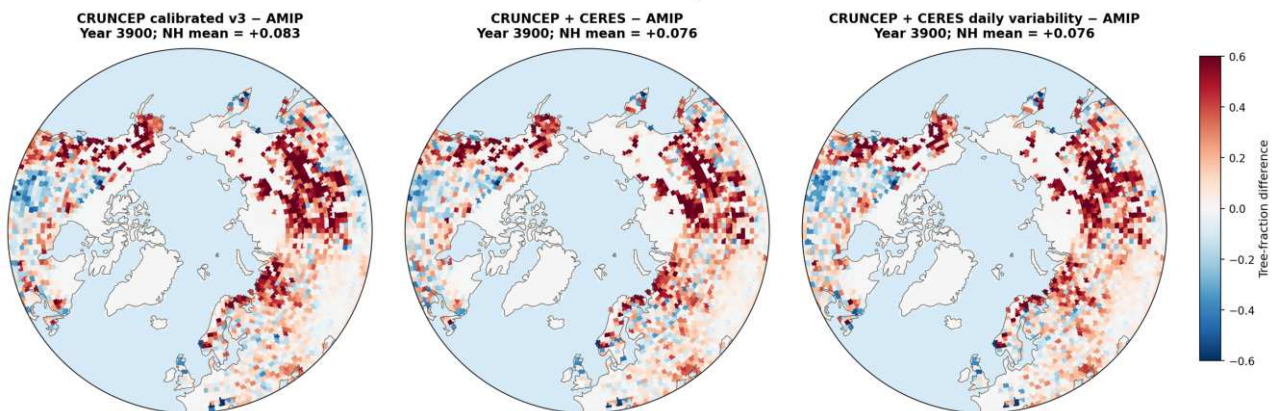


Figure 19: Final-year tree fraction in the four 2000-yr offline spin-ups. Top: absolute year-3900 tree fraction. Bottom: CRUNCEP-family differences from AMIP. CERES-constrained radiation modestly reduces CRUNCEP tree cover, but does not remove the much larger AMIP–CRUNCEP contrast.

2. **Filename convention.** These runs write `atm_remapped_1m_<var>_YYYY-YYYY.nc` (no `_1m_infix`). I added backward-compatible fallbacks to `part2`, `part8` and the CMPI preprocessing; existing configs are unaffected.
3. **Accumulation period** for these XIOS runs is **3600 s** (not the 21600 s of the old HR config); configs use `accumulation_period=3600` (verified: `tsr` $\rightarrow \approx 240 \text{ W/m}^2$).
4. None of the runs is at equilibrium within 30 yr — numbers are a ranking, not final values.
5. **CERES is a modern, monthly constraint**, not a pre-industrial daily observation (§7). The CERES-derived panels test that the forcing construction worked; only the AMIP–CERES and CRUNCEP–CERES panels compare independent products, and even those span different epochs.

9 Recommendation & synthesis

Reading global CMPI/RMSD as a guide rather than a gate (the PI-vs-PD offset is expected), the picture is:

- The **melt-pond / albedo family (06A–06D, etc.)** are *clean*: they improve sea ice where they should and leave everything else essentially untouched. Their imbalance gain is modest (5–20%) but it comes with no unexpected collateral — safe building blocks.
- The **ocean-mixing / coupling family (06O/T)** delivers the only large imbalance reductions (30–38%) *and* genuinely improves the ocean interior and sea ice — but the cooling leaks into the atmosphere globally (incl. tropics), which is degradation in places the parameter should not reach. That makes them promising but not yet “free.”
- **06V** pushes too far: the ENTSTPC3 convection change degrades `tas/zg` everywhere — not recommended as-is.

Best-balanced single steps: **06P** (–29%, well-localised) and **06H** (–20%).

Synthesis — and where the warming lever comes from. The two arcs join here. The imbalance can only be cut substantially by cooling levers (ocean mixing/coupling), so every combo tried so far overshoots into cold bias. But §6 shows the shared Siberia/N. Canada cold lobe is *not* a tuning artefact, and a purpose-built clean no-LPJG CORE3 reference (§6.3) now *quantifies* it: of the $\approx -4.5 \text{ K}$ Siberian DJF bias vs ERA5, $\approx -3 \text{ K}$ is the LPJG forest-cover collapse (via snow-albedo in spring *and* surface roughness in winter, §6.5) and $\approx -1.4 \text{ K}$ is a model-intrinsic over-strong stable-BL inversion (§6.7). The clean run is *better* than the Baseline globally (RMSD 1.514 vs 1.717 K) at *near-unchanged* imbalance ($+0.2 \text{ W/m}^2$, within drift) — indicating both are *warming levers largely orthogonal* to the TOA tuning. That reframes the path forward:

1. First correct the atmospheric forcing and re-spin LPJG with a verified common binary (§14.3, §16). The CERES experiment (§7) shows that net-radiation correction alone is insufficient; tune tree–grass competition only if a residual remains under the corrected forcing. Do not relax the already-adequate summer establishment gates.
2. Attack the residual with *enhanced stable-BL mixing* in OpenIFS (§6.7) — an independent knob that relaxes the surface-trapped winter inversion.
3. On that warmer baseline, apply a *moderate* mixing/coupling setting (06O-like, not 06V) for the imbalance, plus the clean melt-pond changes — the cooling it adds is then offset rather than additive.
4. Re-judge on the *pattern* of CMPI change (§6.3 method), not the global value, since the PI-vs-PD offset is expected.

In short: don’t chase the imbalance with ever-colder atmospheres; remove the structural cold bias first, then the moderate cooling needed for the imbalance becomes affordable.

9.1 2026–07–24 coupled follow-up from the CRUNCEP3-initialized branch: the year-50 land–ocean split refines the path

After the earlier report structure was drafted, the coupled CRUNCEP3-initialized branch (alias “080a”) reached 50 years, which adds an important constraint on the tuning strategy. The new result is *not* a simple hemisphere-wide cold drift. By year 50 the system shows a marked **warm-ocean / cold-land split**, while boreal tree cover continues to retreat — most strongly over Siberia / East Siberia.

Two points matter:

1. The older LPJG story remains necessary: the spin-up / competition state is still part of the problem, and the vegetation levers are real.
2. But the 50-year coupled adjustment shows that a *regional continental-land* climate failure also develops, so vegetation-only retuning is not sufficient.

The finished year-50 diagnostics make the asymmetry explicit. Relative to year 0, NH45+ TREEFPC declines from 0.336 to 0.215, but the strongest loss is concentrated over Siberia / East Siberia. In the Siberian box (55–75°N, 60–180°E) TREEFPC drops by -0.188 , and in East Siberia (55–75°N, 90–160°E) by -0.232 , accompanied by a strong drop in AGDD5 (-89 and -133 respectively). At the same time, NH45+ aggregate AGDD5 actually *increases*, and Scandinavia / Canada also warm in thermal-growing metrics. So the problem is not “the NH is too cold”; it is that **boreal continental land, especially East Siberia, remains too cold relative to the oceanic state**. This is exactly what one sees in the year-50 seasonal-temperature maps against the CRUNCEP3 reference: ocean-adjacent sectors are warm-biased while large boreal land sectors remain cold.

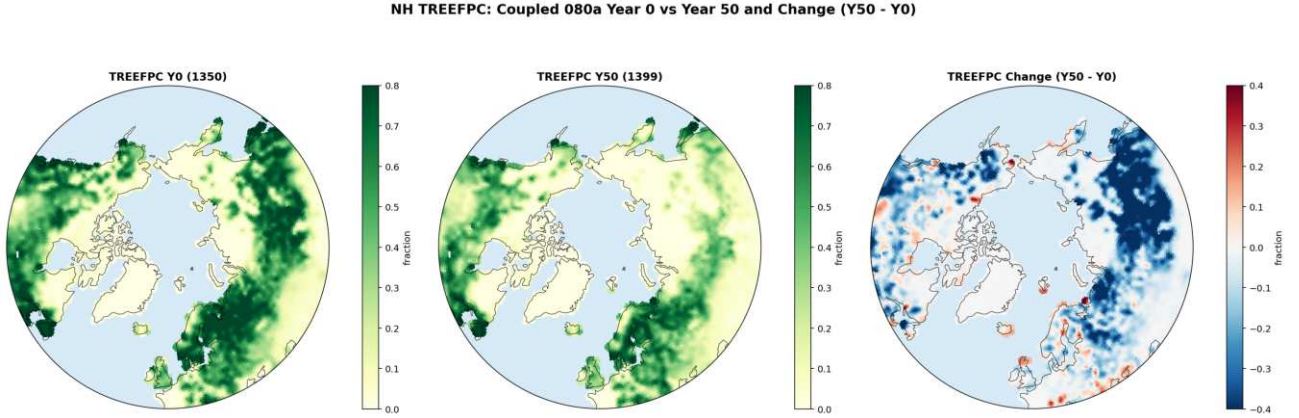


Figure 20: Coupled CRUNCEP3-initialized branch year-50 follow-up: NH TREEFPC at year 0 (1350), final year (1399) and change (Y50–Y0). The retreat is broad, but the largest loss is concentrated over Siberia / East Siberia.

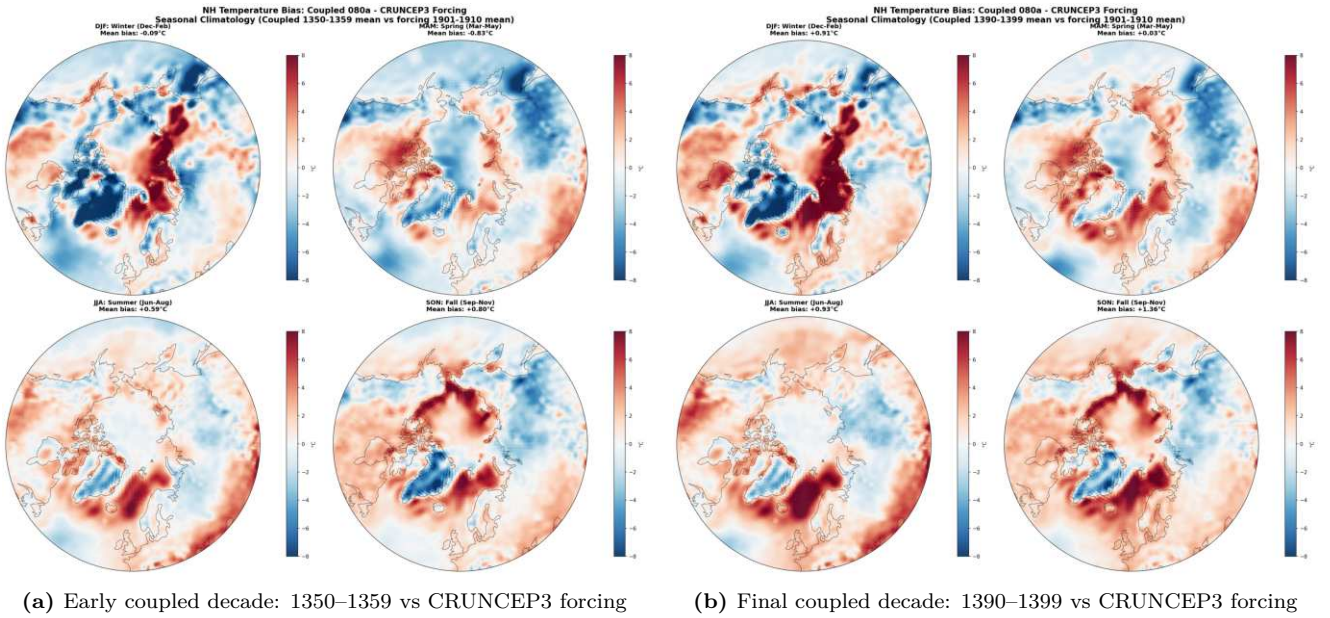


Figure 21: Seasonal NH T2m bias of the coupled CRUNCEP3-initialized branch against CRUNCEP3 forcing. The year-50 state is not a uniform NH cooling; instead it shows a land–ocean split, with boreal continental land still cold while oceanic sectors are relatively warm.

This follow-up changes the recommendation in one practical way: the next runs should **fix the physical coupled branch first, and only then add LPJ compensation**. In other words, the “warm-floor” concept of this report still holds, but the relevant floor is now more specific: **a warmer boreal land floor**, not merely a warmer NH mean. The cleanest next matrix is therefore:

1. **06T-only** physical branch: `oasis3mct.time_step=3600, use_momix=True, momix_kv=0.01, kpp_av0=0.003, kpp_kv0=0.003, kpp_kvbckg=5e-06`.
2. **06V-only** physical branch: 06T plus ENTSTPC3=1.
3. Best of 06T/06V plus **07A** (BNS `greff_min=0.035`).
4. Best of 06T/06V plus **07A+07C** (C3G `pstemp_low=12`).

The year-50 evidence also explains why **LPJ-only tuning should not be the first move**. The all-tuned coupled NH bias panels (seasonal and annual) show that the coupled atmosphere response already differs strongly among branches even before the LPJ compensation is evaluated. If the physical branch leaves East Siberia cold, then LPJ-only changes risk compensating the wrong climate state rather than fixing it. Conversely, if a 06T/06V-type branch can hold the global net-radiation target while reducing the East-Siberian land cold pool, then the LPJ levers can be used for their intended role — retaining boreal trees — rather than masking a regional coupled-climate defect.

Net Radiation Components: Baseline, 06A..06V, 07A..07C, HR goal, LR_offlineSpinup (last 30 years)

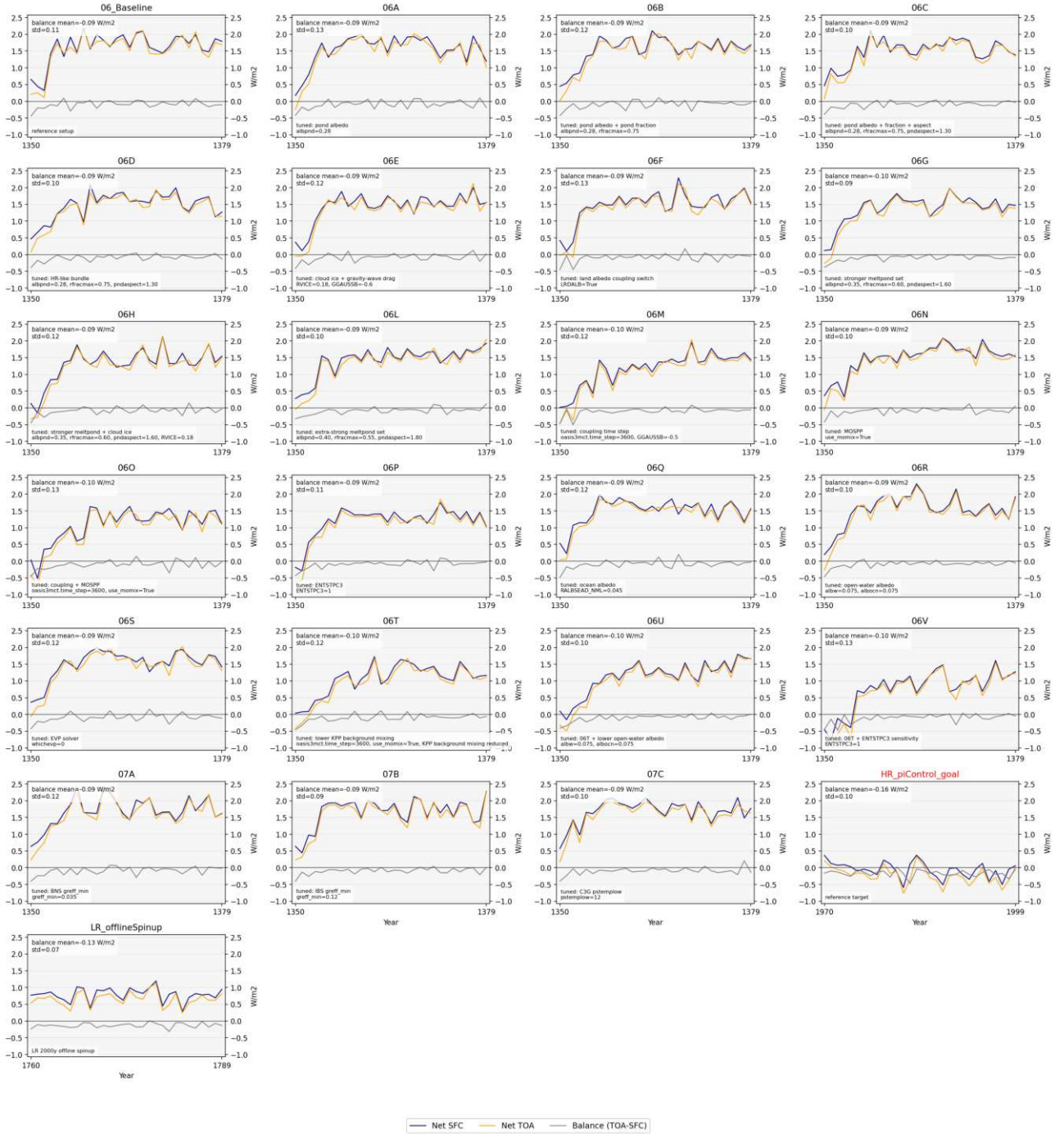


Figure 22: Global net-radiation history used to define the next physical branch. The preferred candidates remain the late 06-series branches that move closest to the target while staying within a plausible tuning envelope; the follow-up now adds the requirement that they must also improve boreal land climate, not only the global radiation budget.

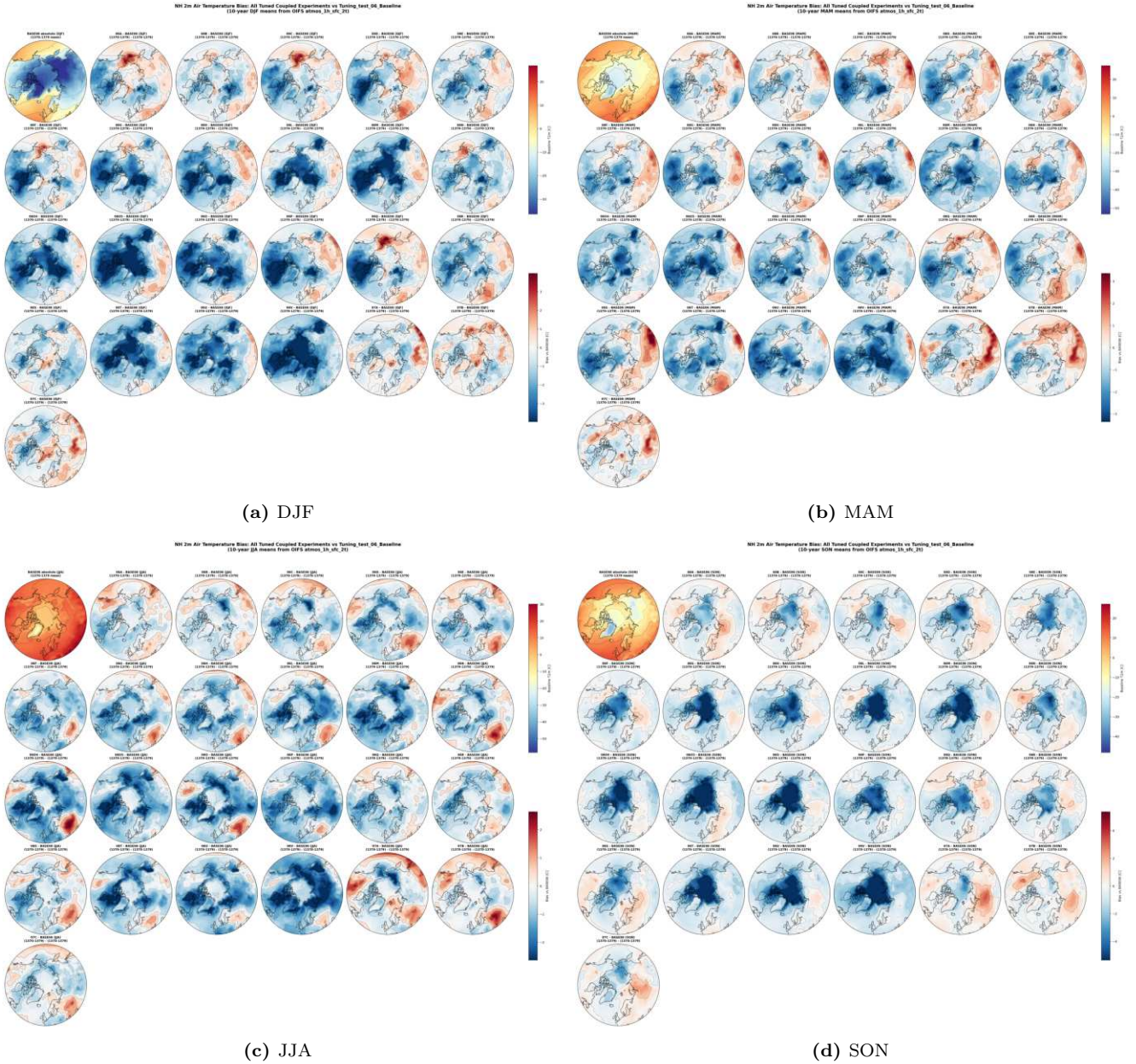


Figure 23: All tuned coupled experiments relative to 06_Baseline, resolved by season (10-year means). These panels help separate branches that reduce the global radiation problem by globally cooling everything from branches that have a more usable boreal-land signal.

10 Work plan: a systematic path to a warm-floor, surface-balanced PI control

Two revised targets. (i) net *surface* energy balance $|F_{\text{sfc}}| \lesssim 0.3 \text{ W/m}^2$ — *not* TOA. Surface and TOA net flux track to within $\approx 0.1 \text{ W/m}^2$ in every run (§4.1); they are one problem, so a surface-zero target is equivalent to $\text{TOA} \approx -0.1 \text{ W/m}^2$, marginally stricter than $\text{TOA} = 0$. (ii) global 2m-T bias vs ERA5 $\approx -1.3^\circ\text{C}$ — the physically-expected PI-minus-present-day offset, *not* zero: a run that reaches -1.3°C is *correct*, not cold-biased.

Ordering principle. The imbalance can only be cut by cooling, and every cooling lever over-cools if applied to today's cold-biased base. So we *first* raise the warm floor with two structural fixes that are *orthogonal to the TOA balance* (they change the land surface and BL mixing, not the radiation budget — §6.3 showed the no-LPJG floor sits at *near-unchanged* imbalance, $+0.2 \text{ W/m}^2$ within drift), *then* spend a *moderate* cooling lever to reach surface balance, now affordable because the floor is $\sim 3\text{--}4\text{K}$ warmer where it was worst. Phases 1A/1B are independent (parallel); Phase 2 depends on both; Phase 3 gates.

10.1 Phase 1A — Restore boreal tree cover (forcing first; LPJG residual second)

Ordering updated by Part IV. This phase was originally written as an LPJG-competition tuning step. The controlled transfer test now requires the opposite order: correct/regenerate the atmospheric forcing, re-spin with one LPJG binary, and tune competition only against the remaining vegetation residual (§14.3, §16). **Lit-**

erature & observational basis — what realistic trees require (so we don’t green the tundra). The observed arctic/boreal treeline sits near the July 10 °C isotherm (Köppen), refined by Körner & Paulsen (2004) to a growing-season mean of $\sim 6\text{--}7$ °C over ~ 90 days, with the northern boreal limit at $\lesssim 30$ days > 10 °C. **Crucially, LPJG’s current gates are already the standard, treeline-calibrated values** (Smith et al. 2001; Sitch et al. 2003 — byte-identical to the canonical `global.ins`): BNE/BINE `twmin_est`= 5 (already *below* the 10 °C isotherm) and `gdd5min_est`= 500 sit essentially *at* the observed treeline — so loosening them risks trees north of the real treeline. Two further realism facts constrain the fix: (i) NE Siberia’s forest is $\sim 80\%$ *larch* (deciduous BNS), because evergreen conifers cannot survive its extreme winters (Kharuk et al.; Herzsuh 2020, *Glob. Ecol. Biogeogr.* 29, 198–206, the deciduous-vs-evergreen distribution paper) — LPJG already encodes this (BNE/BINE `tcmin_surv`= -31 excludes them; BNS has *no* cold gate and `gdd5min_est`= 350), so the realistic tree there is larch, and it is *already permitted*. (ii) Closed taiga tree cover is $\sim 0.4\text{--}0.7$ (MODIS VCF), falling below 0.2 at the taiga–tundra ecotone — the spatial shape `cvh` must reproduce.

Larch caveat. The realistic NE-Siberian tree is *deciduous* larch, whose leafless winter canopy masks little snow — so the spring *albedo* mechanism is genuinely weaker over the larch zone (though it retains full force over the evergreen west Eurasia / N. America). This only re-weights the two mechanisms, it does not soften the conclusion: the *roughness* mechanism survives (even leafless, larch is $\gg$ rougher than grass/tundra), and the seasonality fingerprint (§6.4) shows the LPJG cold effect is co-equal in the *sunless* Nov–Jan months — i.e. the dark-season roughness/inversion part is what carries the larch zone.

The coupled winter cold is a self-inflicted trap, not the trigger — for the evergreens. **This paragraph applies to BNE/BINE only:** as §31.1 shows, larch (BNS) has no cold gate at all, clears its GDD gate on 84 % of NE-Siberian cells, and is lost to *growing-season competition with grass* ($5.8\times$ its LAI) rather than to any gate. Since NE Siberia is $\sim 80\%$ larch, the mechanism below does not explain the region that matters most. At the cells where prescribed veg has forest but LPJG does not (1785 boreal-land cells), the growing-season gates are already cleared in every run ($\text{JJA} \geq 12^\circ\text{C}$, $\text{GDD5} > 500$); only the winter `tcmin_est`= -30 evergreen gate binds. But against the *forested* WITHOUT-LPJG climate it excludes evergreens on just 23% of cells ($\approx$ ERA5’s 22%), vs 43% in the degraded WITH-LPJG run — the missing forest’s own -3.3°C winter cooling doubles the exclusion. So it is a **bistable trap** seeded by the under-forested spin-up, not a climate trigger.

The feedback-free tests now constrain the order. The full CRUNCEP–AMIP forcing change reproduces most of the regional loss with unchanged parameters (§14.3); the four-arm experiment shows that CERES net-radiation replacement removes only a small part of the NH `treeFrac` contrast (§7). Trees do establish and fire is not involved, but those facts do not isolate a competition error. Boreal `greff_min`, C3-grass competitiveness and cold-tree `pstemp_low` remain candidate levers only for a residual measured after a forcing-consistent re-spin. Phase-1B winter warming (§10.3) remains an independent test on the coldest evergreen-gated cells.

Why not just relax the gates? The summer gates are not binding (above), so loosening them cannot help; and the *one* binding gate — the winter `tcmin_est` — correctly excludes evergreens from the extreme-cold larch zone, so relaxing it would plant *unrealistic* evergreens where even ERA5 says larch. The fix is therefore the winter *climate*, not the limits. (Keep `iftreefracca` 1, `global.ins`:150: `cvh` is the summed tree *crown area*, so it tracks establishment success directly.)

Screen & deploy. First run the CRUNCEP and corrected-AMIP arms with the *same* LPJG binary and unchanged parameters. Only then screen any competition knobs on the cheap standalone spin-up, targeting the MODIS-VCF pattern ($\sim 0.4\text{--}0.7$ closed taiga, < 0.2 at the ecotone, larch winning the cold NE) and a ~~closed canopy~~ (`FORESTFPC` > 0)³ — tree NPP $>$ grass NPP — *not* merely a higher global mean. On a passing screen, re-do the coupled LPJG spin-up, regenerate the restart, and carry it into Phase 2.

10.2 Execution: the clean AMIP–noLPJG forcing generator

The decisive test (§6.9) needs a *feedback-free* climate to drive it — reusing coupled-LPJG output would carry the very cold bias we are testing. We therefore built a dedicated **OpenIFS–AMIP** generator at TCO95 (`oifsamip-cy48`, git-aligned to `awiesm3-develop`): AMIP prescribes observed SST/sea-ice, so the atmosphere has no coupled ocean/LPJG feedback. Building and running it required fixing three esm-tools bugs (a FESOM-less crash, missing XIOS/coupling compile flags, a missing WAM date-preprocessing recipe — all committed &

³**Void diagnostic.** `FORESTFPC` sums only PFTs with `landcover==FOREST` (`commonoutput.cpp`:1143), i.e. the *managed-forest* land-use class; natural trees are counted in `TREEFPC` via the `NATURAL` branch. In this PI configuration nothing is allocated to `FOREST` (`Natural_sum`= 0.684, `Total`= 0.665), so `FORESTFPC` $\equiv 0$ by construction while `TREEFPC`= 0.323. It is a land-use bookkeeping artifact and says nothing about canopy closure.

pushed; detail in App. B). A 10-yr $\sim$ PI decade (1870–79 observed SST, GHG fixed 1850) produced the daily forcing. *One methodological catch worth recording:* the first re-spin came out barren everywhere because the XIOS field-def de-accumulated fluxes over the 6 h `freq_op` cadence instead of the true `NFRHIS` $\times$ `TSTEP`= 1 h period, making `rsns/rlns/pr` exactly $6\times$ too low (net SW 28 vs 166 W/m²) — fixed by a validated $\times 6$ patch and, at root, by templating the divisor from esm-tools. **Expected payoff** of the whole Phase 1A (§6.6): ≈ -3 K off the Siberian DJF bias, ≈ -0.2 K global RMSD, at near-unchanged imbalance.

10.3 Phase 1B — Enhance stable-boundary-layer mixing (OpenIFS)

Change. The residual ≈ 1.4 K surface-trapped inversion (§6.7) is relaxed by strengthening stable-BL turbulent mixing in the EDMF K-diffusion scheme via the “long-tail” (LTG) stability functions — lengthening the tail keeps mixing non-zero at large Richardson number and weakens the over-strong surface inversion. There is no namelist for the stable branch, so this is ~ 2 single-line OpenIFS source edits (minimum stable mixing length `ZLMIN` and the LTG coefficient `ZCB`) plus recompile; the exact file:line targets and secondary companions are in App. B. *Bear the documented side-effects in mind* (§6.7): long-tail mixing beyond the observational range can deepen cyclones and degrade low cloud, so verify the storm-track/low-cloud climatology, and pre-test cloud phase / `LW`↓ (Pithan et al. 2014) since the residual may be partly a cloud-microphysics rather than turbulence error.

Test. Change *one* knob first (`ZLMIN`) to keep attribution clean; scan 2–3 values with short (1–5 yr) coupled sensitivity runs of the no-LPJG reference. **Gate:** the DJFM low-level inversion bias (`djf_second_driver.py`) moves from $+1.4$ K toward ERA5’s $+4.2$ K level and the residual boreal cold shrinks, *without* eroding legitimate inversions or warming the tropics (judge globally via `part8`/CMPI pattern). **Risk:** over-mixing warms too broadly / degrades low cloud — mitigate with small steps and global CMPI-pattern judging.

10.4 Phase 2 — Moderate imbalance tuning on the warm floor

With the floor raised (1A+1B) and global skill already better than Baseline (RMSD 1.514, CMPI 0.738; §6.3), spend a *moderate* cooling lever to bring the *surface* budget to zero. Build the combined config in layers, cleanest-first:

layer	lever(s)	Δ TOA [W/m ²]	raw bias [K]	rationale
base	melt-pond (06A–06C; opt. 06G)	-0.08 to -0.26	-1.0 to -1.3	collateral-free
localised	06P (<code>ENTSTPC3</code>)	-0.44 (-29%)	-1.36	best efficiency ~ 1.05 W/m ² /K
moderate	06O (<code>1hcp1+mospp</code>)	-0.55 (-36%)	-1.77	improves ocean/ice; <i>not</i> 06V
(spare)	<code>+kpplow = 06T</code>	-0.58 (-38%)	-1.79	if more cut needed

Why this stops short of 06V. 06V (`entstpc3+kpplow` all stacked) gives the biggest cut (-57%) but overshoots to -2.1 K and degrades `tas/zg/rlut` in *every* region including the tropics. The base+06P+06O stack cuts ~ 1.0 – 1.1 W/m² of surface imbalance (reference surface $+1.52 \rightarrow \sim 0.4$ – 0.5 ; iterate the coupling strength or add `kpplow` to close the last ~ 0.4). **Cold-bias accounting:** the ~ -1.7 K raw cooling of the coupling lever, applied on a floor lifted ~ 3 – 4 K where the bias was worst (and ~ 0.2 – 0.4 K globally), lands the global mean near the target -1.3 °C rather than the -1.8 to -2.1 °C the same lever produces today.

10.5 Phase 3 — Verification protocol and decision gates

Run the combined config 30 yr (evaluate the last decade). Judge with the `reval` toolchain *plus* the DJF driver diagnostics:

target	metric (tool)	accept
surface balance	<code>part2</code> “Surface total”	$ F_{\text{sfc}} \lesssim 0.3$ W/m ² (TOA ≈ -0.1)
cold bias	<code>part8</code> 2 m bias vs ERA5	-1.3 ± 0.3 °C; RMSD $\leq$ ref. 1.51
multivariate skill	<code>part4</code> CMPI	$\leq$ Baseline 0.745; <i>pattern</i> , not just global
no new regionals	CMPI by region (esp. tropics)	no family-typical global cooling leak
driver check	<code>djf_second_driver.py</code> / <code>...roughness...</code>	inversion bias $\rightarrow$ ERA5; cvh restored

Iteration & critical path. Each axis adjusts independently (the floor fixes are orthogonal to the imbalance lever): too cold $\rightarrow$ back off coupling; imbalance too high $\rightarrow$ add `kpplow`; inversion too strong $\rightarrow$ raise `ZLMIN`. Phases 1A and 1B run in parallel now (both cheap/offline); 1A’s coupled re-spin is the long pole, and the Phase-2 run starts once the re-spun restart and the chosen stable-BL build are both in hand.

The submission copy of all scripts, data, figures and this report lives in `/work/ab0995/a270270/analysis/investigation_awiesm3_high_lat_cold_bias_global_toa_positive/`: `data/MASTER_metrics.csv`, `data/cmpi_by_variable.csv`, `data/cmpi_by_region.csv`, `configs/`, `plots/`, `run_figures/<run>/`, `notebooks/`. Tool: `release_evaluation_tool2` (reval env); per-run diags via `part2_rad_balance.py` / `part8_t2m_vs_era5.py` / `part4_cmpi.py`.

Part IV

The atmosphere alone — attribution, and where the remaining error lives

Parts I–III argued from coupled runs, in which atmosphere, ocean and vegetation errors are inseparable. This part removes that ambiguity with a **feedback-free OpenIFS–AMIP reference**: prescribed observed SST/sea-ice, prescribed vegetation, same OIFS branch as the coupled campaign. Two results follow. (i) With SST prescribed, the model still runs $+0.67 \text{ W/m}^2$ out of TOA balance — that fraction of the imbalance cannot be ocean heat uptake and is a genuine atmospheric radiative error, which places a floor beneath what any ocean-side lever can achieve. (ii) The boreal summer cold bias survives in full without ocean or LPJ-GUESS, so it is the atmosphere’s — and it is a surface shortwave deficit, not the boundary-layer problem of §6.7. Together these redirect the work plan: the next tuning should happen on AMIP, where both remaining errors live and where an iteration costs $\sim 7\times$ less than a coupled run.

11 The feedback-free AMIP reference (2026–07–27/28)

amip_pi_base, /work/bb1469/a270092/runtime/oifsamip-cy48/: OpenIFS TCO95L91, GHG/ozone/ aerosol fixed at 1850 (NCMIPFIXYR=1850), driven by the earliest observed AMIP SST decade 1870–1879 — a self-consistent $\sim$ PI atmosphere directly comparable to the PI-control coupled runs. 10yr complete in 2:44 on 14 nodes.

Two configuration decisions make it a valid counterpart to the coupled model, and both were wrong in the earlier _lpjgforcing run that Part II used (§10.2):

- **OIFS branch.** The oifsamip tree was on local_combined_fixes@1303a90; the coupled campaign runs esm-tools version 48r1v5 = movcav-landice+co2-concdriven. The branches differ in *atmospheric physics*, not merely coupling plumbing (suecozv, updecozv, suecrad, suphec, srfi_mod, surfece, voskin_mod). Rebuilt at f3ccacb — the last non-debug commit, and the same state as our coupled tree.
- **Sea albedo.** The old runscript carried TMPRALBSEAD= 0.075. The coupled runs never set it, so they use the source default RALBSEAD= 0.06 (susrad_mod.F90:107); OASIS exchanges only the *ice* albedo (sia_feom→A_Ice_albedo) so OpenIFS computes open-water albedo itself in coupled mode too. A 0.015 albedo offset over ocean is worth order 1 W/m^2 of reflected SW — comparable to the quantity being attributed. Removed.

11.1 Which budget the target applies to: snow enthalpy

part2_rad_balance.py:255–268 computes

$$F_{\text{sfc}} = \text{ssr} + \text{str} + \text{sshf} + \text{slhf} - \text{sf} \cdot \rho L_f, \quad F_{\text{TOA}} = \text{tsr} + \text{ttr}.$$

The snowfall-enthalpy term enters the *surface* budget only, and correctly so: it is an internal atmosphere↔surface transfer and cancels at the top of the atmosphere. Every net-TOA number quoted in this report is $(\text{tsr} + \text{ttr})/\Delta t_{\text{acc}}$, i.e. exactly part2’s TOA total.

	TOA	SFC w/o snow	snow enthalpy	SFC (part2)	TOA–SFC
AMIP 1872–79	0.667	1.441	0.925	0.516	+0.152
09C 1370–79	0.918	1.897	0.944	0.953	–0.035
080a 1370–79	1.682	2.588	0.823	1.765	–0.083

The term is $0.82\text{--}0.94 \text{ W/m}^2$ — *larger than the entire remaining gap to target*. Without it the surface budget misses TOA by $\approx 1 \text{ W/m}^2$; with it, 080a closes at -0.083 , reproducing the ≈ -0.09 quoted in §4.1. **Convention warning:** the work plan’s acceptance target is on the *surface* (§10), where the same runs read $+0.52$ (AMIP), $+0.95$ (09C), $+1.77$ (080a). State which convention is in use whenever these numbers are compared. *Open item:* AMIP’s closure ($+0.152$) is $2\text{--}4\times$ worse than the coupled runs’. Small against the $+0.67$ signal, so the attribution is unaffected, but unexplained — suspect the prescribed SST/sea-ice surface tile. Chase it before quoting the AMIP floor to better than $\sim 0.15 \text{ W/m}^2$.

11.2 Result I — roughly 45 % of the imbalance is atmospheric

AMIP net TOA = $+0.67 \text{ W/m}^2$ (1872–79, after discarding two soil spin-up years; per-year 0.13–0.92; $\text{tsr} = 240.3 \text{ W/m}^2$ confirms the 3600s accumulation divisor). With SST prescribed this *cannot* be ocean heat uptake.

Against a coupled baseline of $+1.44$ (080a), that is $\approx 45\%$ of the coupled imbalance. The best branch in the entire campaign (09C, $+0.92$) now sits only 0.25 W/m^2 above this floor: of the 1.08 W/m^2 still separating it from the HR goal, ≈ 0.83 is atmospheric and only ≈ 0.25 is reachable by ocean-side tuning. **This is the campaign’s most consequential number** — it says the lever family of Part I is nearly exhausted.

Caveat. AMIP runs over *observed* SST/ice while the coupled model runs over its own biased state, so $+0.67$ is “what this atmosphere does over a correct ocean”, not a strict lower bound on the coupled value. But for a $\sim \text{PI}$ configuration, where the real world was near equilibrium, the expected AMIP TOA is ≈ 0 — so most of the $+0.67$ is model error.

11.2.1 Where does it live? Nowhere in particular — the global budget is already right

Decomposing the $+0.67$ against CERES-EBAF (`amip_toa_decomp.py`) was expected to localise it. It does the opposite:

AMIP global	model	CERES	diff	band	net TOA	contribution
absorbed SW	240.13	240.5	-0.37	90S–60S	-90.5	-6.06
OLR	239.46	240.2	-0.74	60S–30S	-22.1	-4.04
planetary albedo	0.2947	0.293	$+0.0017$	tropics 30S–30N	$+42.6$	$+21.30$
SW CRE	-44.40	-45.4	$+1.00$	30N–60N	-21.8	-3.98
LW CRE	$+24.88$	$+25.8$	-0.92	60N–90N	-97.7	-6.55
net CRE	-19.52	-19.6	$+0.08$	sum		$+0.67$

The cloud radiative effect matches observations to 0.08 W/m^2 and the planetary albedo to 0.002. The AMIP and 09C zonal profiles lie almost on top of one another: the coupled model’s radiation *is* the atmosphere’s.

~~The band contributions reproduce the textbook meridional structure with no band carrying an anomalous imbalance, so the Hyder et al. atmospheric-SW hypothesis does not appear as a gross cloud error and there is no region to fix.~~ **That conclusion was an artifact of the comparison.** The bands above were compared to *each other*, and only the *global* CRE to CERES. Differencing each band *against CERES* (§11.2.2) shows the global agreement is a **cancellation of large regional errors**, and that the Southern Ocean carries most of the imbalance.

Where the atmosphere-only TOA imbalance lives — zonal means

AMIP 1872–79 (prescribed SST) vs coupled 09C 1370–79. Positive net TOA = energy into the system.

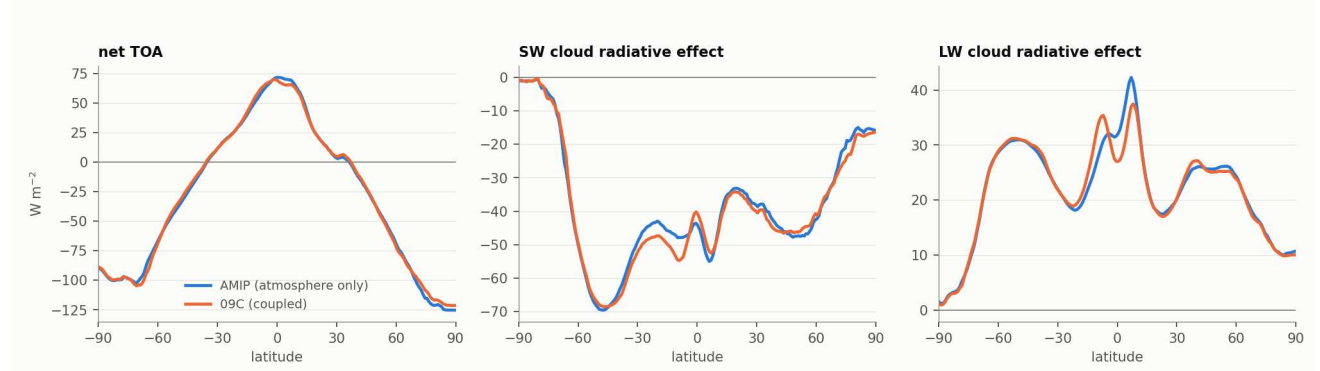


Figure 24: Zonal-mean net TOA and cloud radiative effect, AMIP (atmosphere only) vs coupled 09C. The curves are nearly indistinguishable, and the imbalance is the small residual of a large tropical surplus against extratropical deficits — not a localised error.

11.2.2 Differenced against CERES per band: the Southern Ocean carries it

band	model	CERES	diff	area	contribution
90S–65S	−97.0	−104.0	+7.03	4.8 %	+0.34
SO 65S–45S	−49.5	−55.3	+5.81	9.8 %	+0.57
45S–30S	−5.6	−1.4	−4.20	10.4 %	−0.44
tropics 30S–30N	+42.6	+45.1	−2.51	50.0 %	−1.26
30N–45N	−4.8	−5.7	+0.89	10.4 %	+0.09
45N–65N	−50.1	−54.1	+4.02	9.8 %	+0.40
65N–90N	−106.3	−106.1	−0.15	4.8 %	−0.01

The global budget is right by cancellation, not by being right. The Southern Ocean (65–45°S) runs +5.8 W/m² too much absorbed energy and contributes **+0.57 W/m²** to the global mean — on its own, comparable to the entire +0.67 imbalance. The cause is a cloud *deficit*: cloud area 83.1 % against CERES’s 89.7 % (−6.6 pp) with SW CRE −60.3 against −68.1 (+7.8 W/m² too little reflection). **This is precisely the Hyder et al. (2018) signature**, and it vindicates the compensating-error worry of §5: the ocean-mixing levers of Part I have been removing a Southern-Ocean warm bias that is substantially *atmospheric* in origin.

Consequence for the plan. The +0.67 is not irreducible after all — but neither is it fixable by a global knob, since the global mean is already correct. It is fixable *regionally*, and the two largest regional errors have *opposite* sign at TOA, which is what makes the round-10 design work (§16). Re-framing the campaign arithmetic:

	net TOA	= atmosphere +	coupled-state
080a (untuned coupled)	1.44	0.67 +	0.77
09C (06V + newSeaIce)	0.92	0.67 +	0.25

The ocean-side tuning has already delivered about two-thirds of what was available to it. Whether the HR piControl goal of −0.16 is even the right target for this configuration should be revisited, given that ≈ 0.67 of any coupled value is an atmosphere matching observations. *Revisited and answered on 2026-08-07: it is not the right target, and the goal is now 0 (§37).*

11.3 Result II — the boreal *summer* cold bias is atmospheric

Monthly 2 m-T bias against CRUNCEP3 (1901–10), land only, cos-lat weighted:

series	Boreal JJA/ann	Siberia JJA/ann	E. Siberia JJA/ann
AMIP (atmosphere only)	−1.64 / −1.05	−2.16 / −0.89	−2.24 / −0.49
080a (baseline, old sea ice)	−0.36 / −0.90	−1.41 / −1.36	−1.80 / −1.48
09A (baseline + newSeaIce)	−1.24 / −2.06	−2.19 / −3.00	−2.47 / −2.89
09B (06T + newSeaIce)	−1.56 / −2.33	−2.50 / −3.11	−2.72 / −3.04
09C (06V + newSeaIce)	−1.93 / −2.64	−2.54 / −3.08	−2.69 / −3.05

The atmosphere alone reproduces essentially all of 09A’s Siberian JJA bias (−2.16 vs −2.19) and most of 09B/09C’s. **Removing the ocean and LPJ-GUESS does not remove the summer cold bias.** The cold-season half behaves oppositely — AMIP’s Siberian *annual* bias is only −0.89 against −3.0 coupled — so September–April remains the coupled/vegetation problem of Part II, and JJA is now a separate, atmospheric one.

11.3.1 Correction (2026–07–29), **itself superseded 2026–07–30**: the reference period matters, but far less than estimated

Read §17.2 first. The indirect estimate in this subsection put the Siberian JJA period offset at ≈ 1.12 K and concluded the genuine boreal bias was ≈ -1.0 K. The `amip_presentday` run has since *measured* that offset: **+0.42 K**. The conclusions below are therefore **withdrawn** — the boreal bias is ≈ -2.0 K, close to the original figure. The reasoning is retained because the method error is instructive: it applied an *observed* epoch change to a model that does not reproduce that change. The number above is scored against **ERA5 1990–2014**, while these runs use **1870s observed SST with 1850 GHG**. Roughly 130 years of greenhouse warming therefore sits inside a quantity the campaign has been treating as model error — and tuning against. Measured from the DKRZ ERA5 pool (`scripts/analysis/era5_period_offset.sh`), Siberian JJA over the same box used for evaluation:

ERA5 period	offset vs 1990–2014
1979–1989	–0.670 K
1940–1969	–0.790 K

The interannual sd is 0.578 K, giving SE = 0.209 K on the 1979–89 vs 1990–2014 difference ($t \approx 3.2$): the offset is real. ERA5 stops at 1940, so chaining through HadCRUT5’s global series (1870s $\rightarrow$ 1990–2014 = +0.756 K) scaled by ERA5’s own Siberian-JJA-to-global amplification over the overlapping window ($\times 1.48$) implies a Siberian JJA period offset of ≈ 1.12 K, leaving a **genuine model cold bias of ≈ -1.0 K rather than -2.2 K**.

This changes the target, not the diagnosis. The attribution above is unaffected — it is a model-versus-model comparison (AMIP against coupled 09A) in which any reference offset cancels, so “the JJA bias is atmospheric” stands. What changes is *how much* of it we should try to remove: closing the full 2.2 K would manufacture a ≈ 1 K *warm* bias in the coupled pre-industrial state.

Caveats. The ERA5 box is unmasked whereas the model figures are land-masked; land warms faster than ocean, so the true land-only offset is if anything larger and this correction is conservative. The $\times 1.48$ amplification is extrapolated back from a single window, and HadCRUT5’s 1870s global mean rests on sparse coverage. **The same flaw applies to the CERES-based energy targets** (§13.1, §11.2.2): CERES EBAF here is the 07/2005–06/2015 climatology scored against an 1870s model, so the +8.08 W/m² SO SW CRE gap and the +0.53 W/m² imbalance carry an analogous, *as yet unquantified*, period offset. The decisive fix for both is a single present-day AMIP leg (1989–2015, transient GHG) scored directly against ERA5 1990–2014 and CERES 2005–2015; the AMIP SST forcing already covers 187001–201512.

11.3.2 Attribution (2026–07–29): the boreal forest is a *temperature* problem

The forcing-transfer test (§14.3) showed that swapping CRUNCEP $\rightarrow$ AMIP forcing alone costs –54 % of Siberian TREEFPC, but could not say *which* variable carried it: radiation and temperature both feed NPP, and therefore both feed the **greff_min** growth-efficiency mortality that §6.9 identifies as the actual killer (§6.9). A colleague’s spin-ups separate them by holding CRUNCEP temperature fixed and swapping *only* the radiation to CERES (LPJG–SPINUP_2000Y_TC095_CORE3_CRUNCEP and CERES). **AGDD5 is identical between those two arms** (775.8 Siberia, 748.0 E. Siberia, 1157.5 NH) — the control confirming that only radiation moved.

region	radiation only (CRUNCEP $\rightarrow$ CERES)	full swap (CRUNCEP $\rightarrow$ AMIP)	radiation share
NH 45N+	–1.7 %	–25.1 %	7 %
Siberia	–8.9 %	–54.3 %	16 %
E. Siberia	–10.5 %	–66.0 %	16 %

The radiation arm moves –21 W/m² while the full swap moves ≈ -32 , so scaling linearly puts radiation at ≤ 25 %. The daily-variability variant is indistinguishable (–7.8 vs –8.9 %), so it is not a sub-daily distribution effect either. **Temperature carries ≈ 75 –85 % of the boreal tree collapse.**

Consequence: prong A’s radiation work will not restore the forest. Closing the *entire* Siberian SW deficit recovers at most a quarter of the tree loss. The forest depends on the ≈ 1.0 K temperature bias established in §11.3.1, not on the Southern-Ocean/energy target. These are two separate jobs and should stop being described as one.

The period mismatch also sits on the LPJG side. CRUNCEP3 spans **1901–2015** — twentieth-century climate — yet was used to spin up the *pre-industrial* vegetation state that the coupled 1850 model restarts from. Part of the 131-GDD5 Siberian gap is therefore a real PI-versus-20thC climate difference rather than model error: 131 GDD5 over a ≈ 120 -day season is ≈ 1.1 K, of which the 1870s $\rightarrow$ 1901–2015 offset plausibly accounts for 0.3–0.45 K, leaving 0.6–0.8 K genuine — consistent with the ≈ 1.0 K derived against ERA5 1990–2014, since CRUNCEP’s period is cooler than ERA5’s. *Caveat:* the AMIP arm used the superseded forcing build, ≈ 0.9 K warmer over Siberian JJA, so the temperature share is if anything understated.

Does this shift the residual onto LPJ-GUESS’s CRUNCEP calibration? Partly, but the corroboration attempted on the *temperature* axis fails: NCEP2 Siberian JJA is –0.118 K *colder* than ERA5 over 1990–2014 (paired by year, $t = -2.6$) — real but negligible, and the wrong sign for a story in which LPJ-GUESS grew its trees under an over-warm forcing. That is consistent with CRUNCEP3’s construction: temperature derives from CRU, bias-corrected against stations, while *radiation* derives from NCEP, which is precisely where the +21 W/m² excess against CERES was measured (§13.1). The calibration mismatch sits on the radiation axis. The direct evidence remains the forcing-transfer test (§14.3), which does not depend on which variable carries it.

Round 09 against the atmosphere-only floor — boreal 2 m-T bias by month

model — CRUNCEP3 (1901–1910); land only, cos-lat weighted. Coupled: model yrs 1370–79. AMIP: 1872–79, prescribed SST + prescribed vegetation, same OIFS branch as round 09. Shaded band = JJA.

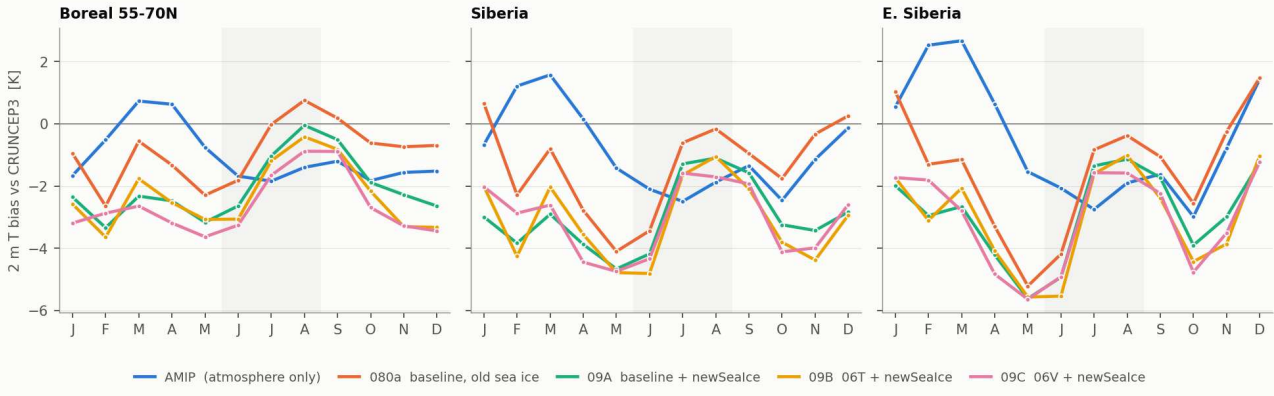


Figure 25: Boreal 2 m-T bias by month against CRUNCEP3, land-masked. The atmosphere-only AMIP curve (blue) sits at or below the coupled runs through JJA (shaded) but far above them in winter: **the summer bias is atmospheric, the cold-season bias is not.** *Caveat:* the “Boreal 55–70°N” panel is contaminated by the Greenland ice sheet (JJA snow depth 320 mm w.e. in that box vs 5.5 mm for Siberia); the Siberia and E. Siberia panels are clean and carry the argument.

Superseded numbers. AMIP Siberia JJA — 1.23, E. Siberia — 1.59, i.e. indistinguishable from the coupled baseline, so coupling adds nothing in summer. Those came from the earlier `amip_pi_forcing` run, built on the wrong OIFS branch and with `TMPrALBSEAD` = 0.075. The corrected build is ≈ 0.9 K colder in Siberian JJA; the conclusion strengthens (the atmosphere is now *colder* in summer than the coupled baseline) but the numbers changed.

12 Round 09 — new sea-ice thermodynamics (2026–07–27/28)

FESOM 2.7 `awiesm3-implicit-ice-surftemp` @db7dd9af makes the ice surface-temperature solve implicit (dQ/dT -linearised about the last transmitted `ist`) and hands ownership of the ice surface energy balance to OpenIFS, which now runs a slab-conduction ice tile using FESOM-supplied thickness; FESOM ice growth becomes $Q_{\text{atmice}} = -a2ihf$. The pairing is verifiable at runtime in the OASIS `namcouple`: the 09 runs carry a sixth `o2a` field, `sit_feom`→`A_Ice_thickness`, absent from every 08 run. Run identities and the 09B parameter block are catalogued in the 2026–07–28 update at the head of this report; the quantitative comparison is below. 09C was built and run on our own tree (`Tuning_test_09C_06V_CRUNCEPinit_newSeaIce`) to close the missing 06V corner.

run	net TOA [W/m^2]	global T2m [$^{\circ}\text{C}$]	Siberia JJA [K]	Siberia ann [K]
080a baseline, old sea ice	1.442	13.606	−1.41	−1.36
09A baseline + newSeaIce	1.286	13.148	−2.19	−3.00
09B 06T + newSeaIce	1.171	12.794	−2.50	−3.11
08B 06V, old sea ice	0.935	12.347	−3.11	−4.09
09C 06V + newSeaIce	0.919	12.492	−2.54	−3.08

On the clean old-vs-new 06V pair (08B → 09C) the radiation barely moves ($0.935 \rightarrow 0.919$) but the boreal **warms by +1.0 K in the Siberian annual mean**, +0.57 K in JJA, +0.15 K globally. **09C dominates 08B on both objectives simultaneously** — the first time in the campaign that has happened, and the “free warming” sought in §9. But the effect is *not* an offset: on the baseline (080a → 09A) the same code change goes the other way, cooling 0.46 K globally and 1.6 K in the Siberian annual mean. The sign depends on the ocean-mixing branch; no mechanism is established here, and pinning it down needs the surface energy budget over ice.

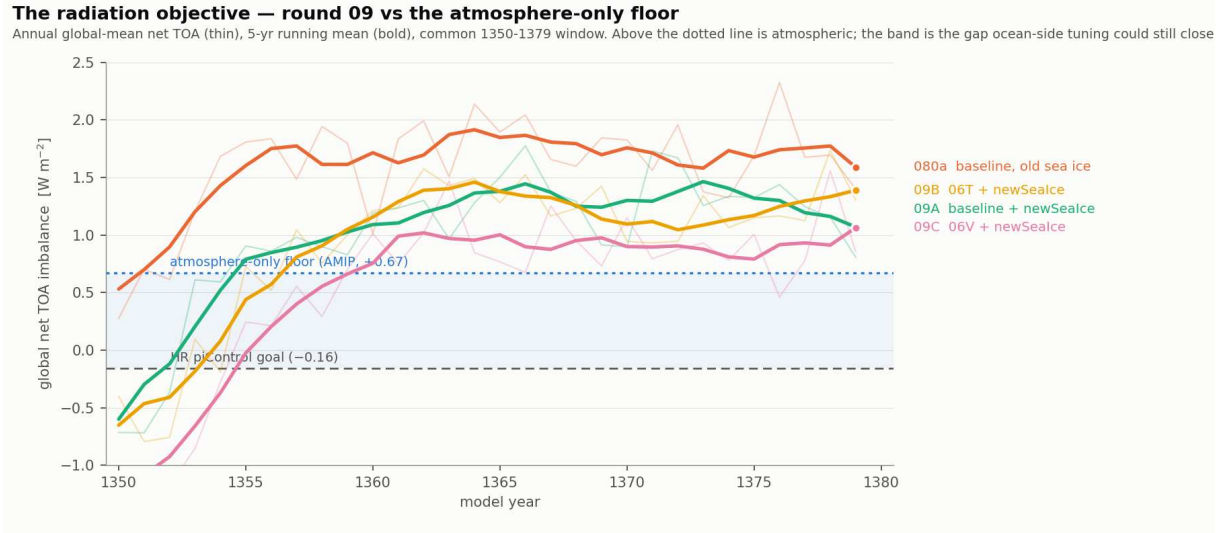


Figure 26: Annual global-mean net TOA (thin) with 5-yr running mean (bold), common 1350–1379 window. The dotted line is the **atmosphere-only floor** measured by AMIP ($+0.67 \text{ W/m}^2$); the shaded band is the only part of the remaining gap that ocean-side tuning can still close. Everything above the dotted line is atmospheric.

Bookkeeping caveat. `sacct` shows a `Tuning_test_08A_06T_1hCPL_MOSPP_KPLOW_CRUNCEPinit` completing all three chunks on 24–25 July, but no 08A directory exists on disk. 08A was the 06T-only physical control — Run 1 of the 2026-07-24 design note’s own sequence — so within round 08 every surviving 06T-family entry (08C, 08E, 08G) carries an LPJ lever and the intended 06T-vs-06V comparison cannot be made cleanly. Round 09 supersedes this, but conclusions drawn from round 08 alone inherit the hole.

13 Why the summer bias exists — a shortwave deficit (2026–07–28)

Siberia JJA surface energy budget, land-masked, cos-lat weighted (AMIP 1872–79 vs CRUNCEP3 1901–10, which carries `rsns/rlns` and so permits a direct radiative comparison):

	AMIP	CRUNCEP3	Δ
2 m temperature	9.74 °C	11.92 °C	−2.18 K
SW net at surface	155.1	188.3	−33.2 W/m ²
LW net at surface	−49.2	−64.5	+15.2 W/m ²
net radiation			−18 W/m ²

Supporting AMIP diagnostics: SW↓ 188.6, surface albedo 0.178 (`fa1` 0.172), LW↓ 315.6, SH/LH −24.6/ −56.8 (**Bowen** 0.43), cloud `tcc` 0.778 / `lcc` 0.505 / `mcc` 0.386, `skt−2t` +0.43 K, `swv11` 0.343, snow depth 5.5 mm w.e.

The LW term is *compensation* (a colder, cloudier surface loses less longwave), not cause. It is *not* albedo: a realistic boreal-summer 0.13–0.15 instead of the model’s 0.178 buys only 5–9 W/m². So SW↓ itself is too low. This is a different mechanism from the winter stable-BL inversion of §6.7, and it is not addressed by any lever proposed so far.

13.1 Re-measured against CERES: cloud, not clear-sky — and three times smaller

CRUNCEP3 is a reanalysis-derived forcing product, so the deficit was re-measured against **CERES EBAF Ed4.1** (satellite), which supplies all-sky *and* clear-sky net SW at the surface and so permits the decomposition $\Delta(\text{SW}_{\text{net}}) = \Delta(\text{SW}_{\text{net,clear}}) + \Delta(\text{SW CRE}_{\text{sfc}})$ with no assumption about surface albedo or the direct/diffuse split.

JJA	Siberia land			NH 55–75°N ocean		
	model	CERES	diff	model	CERES	diff
SW net at surface, all-sky	155.1	166.3	−11.2	142.4	155.4	−13.0
SW net at surface, clear-sky	246.6	241.4	+5.3	241.9	252.6	−10.7
SW CRE at surface	−91.6	−75.1	−16.5	−99.5	−97.3	−2.2
cloud area [%]	77.8	69.6	+8.2	84.6	83.0	+1.6

Three results, all of which revise §13:

(i) **It is unambiguously a cloud error.** The decomposition closes exactly ($-11.2 = +5.3 - 16.5$), and clear-sky surface SW is if anything slightly *too bright*. Aerosol, water vapour and surface albedo are therefore all exonerated — and since PI aerosol is lower than present-day, the PI-vs-CERES period mismatch works *against* this finding, making it conservative.

(ii) **The deficit is 11 W/m^2 , not 33.** ~~Surface net SW over Siberian land in JJA is 33 W/m^2 too low.~~ That figure was measured against CRUNCEP3, which is itself $\approx 21 \text{ W/m}^2$ brighter than satellite — **CRUNCEP3 was the outlier, not the model.** An 11 W/m^2 summer deficit remains the right order to explain a $\approx 2 \text{ K}$ land cold bias, so the mechanism survives; only its magnitude changes.

Verified independently, because the original comparison had assembled the two sides on different grids with different land masks — a way to manufacture exactly this kind of difference. Redone with all three sources on the CERES 1° grid under a single land mask (`verify_ceres_cruncep.py`): CERES 166.7, AMIP 155.8 (-10.8), CRUNCEP3 187.6 ($+21.0$). Raising the land-mask threshold $0.5 \rightarrow 0.8$ changes these by $< 1 \text{ W/m}^2$, and CERES's own JJA-mean file gives 168.4 against 166.7 from the monthly climatology. The grid inconsistency was therefore not the cause.

Why CRUNCEP3 is bright splits cleanly by season: in JJA it is the downward flux (234 W/m^2 in July against CERES's 209, the NCEP high-latitude brightness bias), whereas in spring it is the forcing file's fixed albedo (§14.1). CRUNCEP3 blends CRU *station* temperature with NCEP *reanalysis* radiation, which is the natural reading of why its T2m is usable while its SW is not.

Verification: AMIP and CRUNCEP3 against CERES EBAF on a common grid

Surface net shortwave, land only, all three regridded to CERES $1^\circ \times 1^\circ$ with a single land mask. Dashed box = Siberia ($55^\circ\text{--}75^\circ\text{N}$, $60^\circ\text{--}180^\circ\text{E}$).

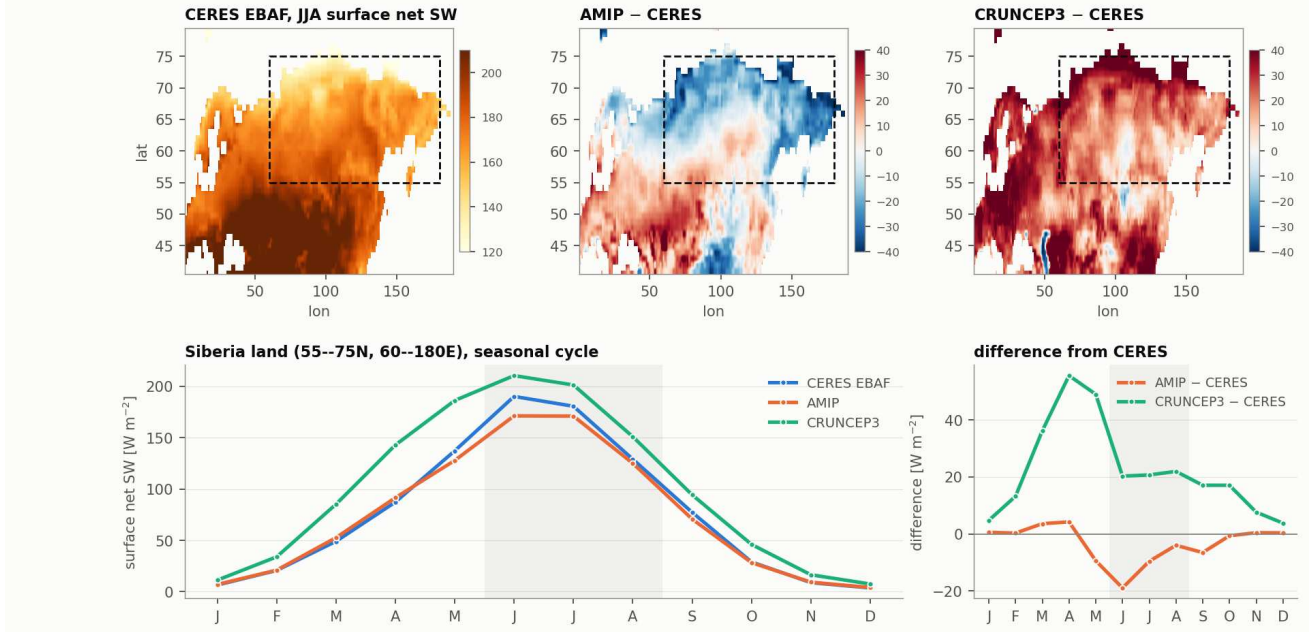


Figure 27: Surface net shortwave against CERES EBAF, all sources on a common 1° grid with one land mask. AMIP (orange) is $\approx 11 \text{ W/m}^2$ *dark* in JJA; the CRUNCEP forcing (green) is $\approx 21 \text{ W/m}^2$ *bright* in JJA and far more so in spring, where its fixed albedo does not mask snow. The two forcings differ from each other by $\approx 32 \text{ W/m}^2$ in the growing season — the transfer step of §14.

(iii) **It is specific to boreal land.** Over ocean in the same latitude band the cloud error nearly vanishes (-2.2 W/m^2 CRE, $+1.6 \text{ pp}$ area) while over Siberian land it is -16.5 and $+8.2$. A *global* cloud knob would therefore move the ocean, where nothing is wrong, and spend TOA budget for no benefit. The error is over land, in summer — pointing at surface-driven convection and boundary-layer cloud rather than the large-scale cloud scheme.

Amount or opacity? Both, roughly half each. At the observed CRE per unit cloud area ($-75.1/0.696 \approx -108 \text{ W/m}^2$), the $+8.2 \text{ pp}$ excess accounts for ≈ -8.8 of the -16.5 W/m^2 CRE error; the residual ≈ -7.7 is opacity or vertical placement. A cloud-*amount* lever alone therefore reaches at most half of it.

13.2 Does the cloud fix conflict with the energy target? Less than first thought

~~Burning off that cloud stops reflecting it at TOA, costing $\approx +0.18 \text{ W/m}^2$ on the annual global TOA — a 27% worsening — so a pure cloud fix cannot meet a fixed TOA constraint, and only a radiatively neutral surface flux lever can.~~ Two things falsify that framing:

- **The target is the *surface*, not TOA** (§11.1), and the drift it controls is driven by the flux into the *ocean*. The cloud error is over *land* (§13.1), where annual-mean energy storage is ≈ 0 — land returns what it receives. A boreal-land cloud correction therefore barely touches the quantity that governs spin-up drift.
- **The error is land-specific**, so the correction need not be a global knob at all (§13.1) — which was the whole reason a TOA cost was incurred in the first estimate.

The two objectives are consequently far more separable than the original framing implied, and the cloud error can be attacked at its diagnosed cause rather than through a proxy.

13.3 The stomatal-resistance sensitivity test (experiment A)

A second, radiatively neutral lever was nonetheless identified and is being tested, because the measured Bowen ratio over Siberian JJA land is $\text{SH/LH} = -24.6 / -56.8 = \mathbf{0.43}$, low against the general observed picture for boreal conifer forest. Minimum canopy resistance for boreal needleleaf (`susveg_mod.F90:266–267`) is 250 s/m, with 500 retained just above as a comment.

Provenance — and a correction. ~~500 is the IFS default and has been halved in this build.~~ Git refutes this: both the active 250 and the commented 500 are present in commit `cf82033`, “initial commit of 48r1 released version”, and no AWI commit has ever touched `RVRSMIN`. **250 is what ECMWF ships in 48r1**; 500 is the older HTESSEL-era value they themselves revised. Setting 500 moves *away* from the released configuration, so this is a **sensitivity test, not a default restoration**.

Status: `amip_expA_rvrsmmin500`, 6 yr, paired against `amip_pi_base` on 1872–1875. Verified compiled in by inspecting the rebuilt object file (the value appears twice for types 3–4; 250 survives only for Desert and Mixed Forest). Note this tests *partitioning*, not the diagnosed cause: if the cloud error is real then a resistance change compensates for it, and with 11 W/m^2 less available energy there is also less to repartition. *Caveat:* “Bowen 0.43 is too low” is inferred from the literature’s general picture, not measured — neither CRUNCEP3 nor CERES supplies SH/LH here, and FLUXNET boreal sites would settle it.

A connection worth testing in the same run. Excess summer evapotranspiration over boreal land would *both* cool the surface directly *and* moisten the boundary layer, feeding the excess low cloud (model `lcc= 50.5%` over Siberian land in JJA). If cutting transpiration also reduces low cloud, the partitioning error and the cloud error are the same problem, and experiment A tests both at once.

14 Forcing transfer: the spin-up was grown under brighter, warmer conditions than the model delivers

The vegetation state every coupled run in rounds 08 and 09 restarts from is the standalone offline spin-up `LPJG-SPINUP_2000Y_TC095_CORE3_CRUNCEPcalibrated_v3` (`/work/bb1469/a270270/runtime/lpjg-spinup/`), driven by `CRUNCEP_noLPJG_1d_1901-1910_TC095_calibrated_v3.nc` (verified at `lpj_guess_levante_spinup_..._2000y.yaml:38`). **That forcing and the atmosphere the coupled model provides differ substantially, and in the same direction for both light and temperature.**

Siberia land, JJA	surface net SW [W/m^2]	vs CERES
CERES EBAF (observed)	166.7	—
CRUNCEP forcing (what LPJG grew under)	187.6	+21 (+13 %)
AMIP (what our atmosphere delivers)	155.8	–11 (–7 %)
gap between the two forcings		32 W/m^2 ($\approx 20\%$)
plus 2 m temperature		2.2 K (11.9 vs 9.7 °C)

The errors point *opposite* ways and therefore compound: CRUNCEP is too bright (the NCEP high-latitude brightness bias — its downward SW is 234 W/m^2 in July against CERES’s 209, $\approx 12\%$ high), while our atmosphere is too dark by the cloud error of §13.1. Vegetation equilibrated to the first and then handed the second experiences a $\approx 20\%$ drop in growing-season radiation and a 2.2 K cooling — a step change in exactly the direction that pushes

a boreal system from trees toward grasses. All numbers here are computed on a common CERES 1° grid with a single land mask (`verify_ceres_cruncep.py`); mask threshold 0.5 → 0.8 moves them by < 1 W/m².

14.1 Ruled out by measurement

Several components were suspected along the way and cleared. Recording them so the same ground is not re-covered:

suspect	the observation that cleared it
LPJ-GUESS core radiation handling	PAR is inflated because no albedo is applied. Reading <code>rsns</code> as <code>NETSWRAD_TS</code> and applying a second correction would double-count. Model behavior is correct (see <code>driver.cpp:1435–1449</code>); a second correction would double-count. Model behavior is correct.
EC-Earth spin-up interface	Reads the right variable and sets the right type (<code>framework.cpp:2501–2507</code>). No further action is needed.
The forcing script's fixed albedo	<code>rsns=FSDS×0.861</code> is a fixed 0.139 albedo, so spring PAR is inflated ×1.66 and the model is biologically inert: measured over 687 Siberian cells at equilibrium, April carries 0.02 GPP and April LAI is 0.02 (no canopy), while JJA carries 80 %. Weighted by what is actually assimilated, the PAR excess is ≈ 13 %, not 66 %. True that the albedo is fixed (ratio 0.861 in every cell, April and July alike). But in April, the model's 0.139 against CERES's 0.142 — right to 0.003. The large spring error (0.139 vs 0.142) is biologically inert : measured over 687 Siberian cells at equilibrium, April carries 0.02 GPP and April LAI is 0.02 (no canopy), while JJA carries 80 % . Weighted by what is actually assimilated, the PAR excess is ≈ 13 %, not 66 %.
...and it is an unusual, campaign-specific defect	Standalone LPJ-GUESS assumes a constant albedo natively: <code>BETA= 0.17</code> (<code>driver.cpp:1435–1449</code>) in both the sunshine and <code>SWRAD</code> paths. The script's 0.139 is a slightly darker version of standard practice, and switching to its <code>bulk_albedo</code> option would give 0.195 in JJA, which is <i>worse</i> than the fixed value. Do not “fix” the script; it is not the problem.

14.2 What survives, and how to settle it

What remains is not a defect in any component but a **transfer problem**: a ≈ 20 % growing-season radiation step and a 2.2 K temperature step between the forcing the vegetation equilibrated to and the forcing it is then asked to live in. **Whether LPJG's *parameters* were tuned against CRUNCEP — the “calibrated” in the directory name — is not established**; the 07-series did change `greff_min` and `pstemp_low`, but no evidence was found of what was calibrated against what, and the folder name is not proof.

It still weakens the “competition, not climate” verdict of §6.9, though less than first argued. That test moved a standalone re-spin from the CRUNCEP forcing to the OIFS–AMIP forcing and read the resulting under-vegetation as an LPJG-internal competition failure. But that move alone removes ≈ 20 % of the growing-season light and 2.2 K, so under-vegetation is partly what the forcing change predicts on its own. The verdict should be read as provisional. (The forcing used there was also the *superseded* AMIP build, §11.3.)

14.3 The transfer test, measured: forcing alone halves boreal tree cover

The test needs no knowledge of what was tuned — take the *same* LPJG parameters and spin to equilibrium under each forcing; the difference *is* the transfer penalty. **Both arms already existed** and required no new computation:

CRUNCEP arm	lpjg-spinup/LPJG-SPINUP_2000Y_TCO95_CORE3_CRUNCEPcalibrated_v3
AMIP arm	lpjg-spinup-develop/LR_2000y_PIforcing_v3

Controlled: `global.ins` **byte-identical** between the two trees, firemodel "GLOBFIRM" in both `.ins` files, `nyear_spinup= 2000 / freenyears= 100`, fixed 1850 CO₂/Ndep/LU, both reaching year 3900, and **10 043 gridcells in common** (99.8 % of the AMIP gridlist). The material difference is the forcing file.

region		CRUNCEP-forced	AMIP-forced	diff	%
Siberia	TREEFPC	0.357	0.163	−0.194	− 54
	GRASSFPC	0.288	0.330	+0.042	+15
	AGDD5	776	645	−131	−17
E. Siberia	TREEFPC	0.359	0.122	−0.236	− 66
	GRASSFPC	0.316	0.398	+0.082	+26
	AGDD5	748	568	−180	−24
NH 45°N+	TREEFPC	0.383	0.287	−0.096	−25

Changing nothing but the forcing halves the Siberian tree fraction and cuts East Siberia’s by two-thirds, with grass taking up the space. For scale, the coupled 080a run lost TREEFPC 0.310 $\rightarrow$ 0.121 over Siberia in 50 years (§9.1); the forcing swap alone reproduces essentially that entire loss at equilibrium, with no parameter changed. AGDD5 falls 17–24 % in the same direction, so this is not a light effect alone — the 2.2K also does real work through the growing-degree-day gates.

Forcing transfer alone halves boreal tree cover — identical LPJ-GUESS, two forcings

Zonal means at equilibrium (yr 3900), 10043 common gridcells, byte-identical global.ins. Shaded band = the Siberian box.

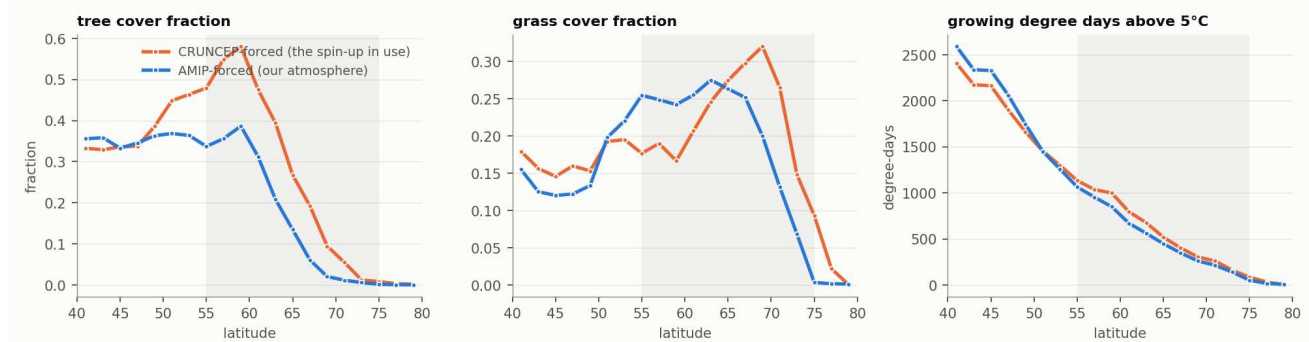


Figure 28: Zonal means at equilibrium (yr 3900), 10 043 common gridcells, byte-identical global.ins. South of $\approx 52^\circ\text{N}$ the two arms are indistinguishable; northward they diverge sharply and the AMIP-forced treeline retreats several degrees south. The deficit the campaign has been attributing to competition is reproduced by the forcing change alone.

Three caveats, all material.

1. The arms were built from *different binaries* (a270092 vs a270270) of the same LPJG version with identical global.ins. A compiled difference cannot be excluded; re-running both arms from one build (≈ 3 h) would close it, and should be done before this is published.
2. The AMIP forcing is the *superseded* build (§11.3), ≈ 0.9 K warmer over Siberian JJA land than the corrected one — so this is a **lower bound** on the penalty against today’s atmosphere.
3. ~~FORESTFPC=0 in both arms, so the absence of closed canopy survives as a separate unexplained problem.~~ FORESTFPC is a *void diagnostic* in this configuration — it counts only the managed-forest land-use class, which is empty here, so it is identically zero regardless of what the vegetation does. There is no separate closed-canopy failure to explain.

15 Revised plan: tune the atmosphere on AMIP

Both errors that remain — ≈ 0.83 W/m² of TOA imbalance (§11.2) and ≈ 2.2 K of Siberian summer cold (§11.3) — are atmospheric, and both are now *measured* in a configuration where neither the ocean nor LPJ-GUESS can confound the answer. They should therefore be tuned on AMIP, not in coupled mode. This does not retire Phases 1A/1B of §10 — the LPJG competition fix and the stable-BL mixing fix still target the *cold-season* bias — but it inserts a cheaper, better-posed stage ahead of Phase 2.

Two objectives, now known to be largely separable (§13.2):

1. **Surface energy balance** $\rightarrow 0$ to cut spin-up drift. This is an *ocean* surface-flux problem; the relevant number is F_{sfc} over ocean, which has *not* yet been computed — all figures quoted so far (+0.52 AMIP, +0.95 09C, +1.77 080a) are global. **Open item.**
2. **Boreal summer warmth** to restore GDD5 and let trees compete. This is a *land* problem, caused by an 11 W/m² cloud-driven SW deficit that is absent over ocean (§13.1).

	change	where	status / expectation
A	RVRSMIN(3,4): 250 $\rightarrow$ 500	susveg_mod.F90:266–267	<i>running</i> ; radiatively neutral; tests partitioning, not the diagnosed cause
B	RVLAI(3,4): 5.0 $\rightarrow$ 4.0	susveg_mod.F90:133–134	same sign, smaller
C	RCLDIFF $\uparrow$, global	NAMCLDP	+surface SW, but +TOA
C'	a <i>land</i> -acting cloud lever	convection / BL cloud	the error is land-only, so a global knob would move the ocean where no
D	A + C'		joint target

Gates: Siberian JJA bias toward 0 *and* SW CRE toward CERES; ocean F_{sfc} not degraded; no new error outside the boreal (judge on the global **part8**/CMPI pattern, §4.3). Since the cloud error splits roughly half amount, half opacity (§13.1), a lever acting only on cloud amount can close at most half of it, and the diagnostic to watch is SW CRE against CERES rather than temperature alone.

Why AMIP is the right testbed. 2.3 node-hours per model year against 15.8 coupled — a $\approx 7\times$ saving — and 10 yr completes in one job. Measured interannual scatter of the boreal JJA mean is $\sigma = 0.25$ K (Boreal), 0.59 K (Siberia), 0.72 K (E. Siberia), so with 8 usable years the standard error is 0.09–0.26 K against biases of 1–2 K: sampling is not the binding constraint. *Land spin-up is* — with **lresume: false** the deep soil (**st14**, **swv14**) carries initialisation shock, so discard the first 1–2 years. For paired tuning experiments 5 yr (1 discarded + 4 used) suffices, since identical prescribed SST makes most interannual variance cancel in the difference.

16 Round 10 design: a two-pronged, targeted attack (2026-07-28)

Original design as proposed. Operator notes to be appended in §16.4.

The structural fact the design rests on. The two largest regional radiation errors have *opposite* sign at TOA (§11.2.2, §13.1), so fixing them together is self-balancing, whereas fixing either alone is not:

	error	fixing it	Δ TOA
A1 Southern Ocean	too <i>little</i> cloud (−6.6 pp, SW CRE +7.8)	adds cloud, reflects more	−0.5 to −0.6
A2 boreal land	too <i>much</i> cloud (+8.2 pp, surface SW −11 in JJA)	removes cloud, absorbs more	+0.1

16.1 Prong A — AMIP (atmosphere), A1 before A2

A1 first. It moves the global target in the right direction ($+0.67 \rightarrow \approx +0.1$) *and* attacks the error the campaign has suspected since §5. A2 then costs back only ≈ 0.1 of the ≈ 0.55 that A1 buys, so the boreal fix becomes affordable — the “warm floor” logic of Part III, relocated from the vegetation to the atmosphere.

Critical: in AMIP the Southern-Ocean target is *radiation only*, never temperature. SST and sea ice are prescribed, so the SO surface temperature cannot respond to a radiation error — the excess absorbed shortwave is simply exported through the fixed boundary. **A1 must therefore be judged on SW CRE and cloud area against CERES, and it is meaningless to look for an SO temperature response in AMIP.** This is also why the SO *warm* bias appears only in AWI-CM3/AWI-ESM3: it is a *coupled* phenomenon, in which the same $+5.8 \text{ W/m}^2$ of excess absorbed energy has somewhere to go and warms the ocean. The causal chain is then explicit and closes the compensating-error argument of §5:

AMIP SO cloud deficit (−6.6 pp) $\rightarrow$ $+5.8 \text{ W/m}^2$ excess absorbed $\rightarrow$ (*coupled only*) SO warm SST bias $\rightarrow$ suppressed by
the 06T/06V ocean-mixing levers

i.e. Part I was damping Weddell/Ross convection to compensate an *atmospheric* cloud error. It follows that the only test of whether A1 fixed the real problem is a *coupled* run — AMIP can confirm the radiation is right, but not that the SO warm bias is gone.

On levers. OpenIFS has no land-only cloud knob, and that is informative rather than obstructive: the land-specificity of A2 (-16.5 W/m^2 CRE error over Siberian land against -2.2 over ocean at the same latitude) is itself evidence that the cause is **land-surface moisture supply**, not the cloud scheme — which makes the RVRSMIN sensitivity test (§13.3) potentially the actual fix rather than a proxy for one. A1, being over ocean, is a genuine cloud-scheme / microphysics target.

Guardrails on every A-run — all four, every time:

1. global net TOA *and* surface flux (state which convention, §11.1);
2. SO 45–65°S SW CRE and cloud area against CERES — *radiation only*, since prescribed SST makes an SO temperature response impossible in AMIP by construction (above);
3. **the tropics** — 50 % of the globe and the largest single contribution (-1.26), and the term the 06O/06T/06V levers degraded (§4.3);
4. CMPI by *pattern*, not the global number.

16.2 Prong B — LPJ-GUESS, and it must go second

Do not tune competition parameters yet. The tree deficit is largely forcing-driven (-54% Siberian TREEFPC from the forcing swap alone, §14.3), so tuning LPJG while its forcing is 11 W/m^2 dim and 2.2 K cold would bake the atmospheric error into vegetation parameters. That is the compensating-error trap, and it is what the 07-series was already doing — which is the natural reading of why it achieved nothing (§12).

1. Complete Prong A.
2. **Regenerate the LPJG forcing** from the corrected atmosphere. Enabling work required: `amip_pi_base` uses the default `file_def` and emits the evaluation fields but *not* the daily LPJG forcing set, so a merged `file_def` (or a second short run) is needed.
3. **Re-spin and re-derive the residual.** Whatever tree deficit survives against a corrected forcing is the real LPJG target. Only then are `greff_min` / `pstemp_low` legitimate levers.

Gates, replacing the void one. `FORESTFFPC>0` is unachievable by construction (§14.3) and must not be used. Use instead: `TREEFFPC` against the CRUNCEP-arm equilibrium (0.357 Siberia, 0.359 E. Siberia); `AGDD5` recovery toward 776/748; and `cvh` toward the MODIS 0.4–0.7 closed-taiga range.

16.3 Ordering

A1 → A2 → regenerate forcing → re-spin LPJG → B. Prong B cannot be evaluated honestly until Prong A has fixed what feeds it.

16.4 Operator notes

(reserved — to be filled in)

17 Round 10 results (2026–07–30): 46-year runs rebuild the ranking

All 19 prong-A runs were extended from 6 to 46 years (evaluated 1872–1915, 44yr) after a statistical-power check showed the original 4-year window could not resolve the boreal signal at all.

17.1 The 4-year window was resolving nothing in the boreal

A run × year ANOVA (`scripts/analysis/noise_floor.py`) decomposes each per-year value into a tuning signal, the SST-forced excursion common to every run, and internal noise. At 4 years the 95 % detection threshold on Siberian JJA T2m was ± 0.89 K and **1 of 18 runs cleared it** — A1a, the overshoot already rejected. Siberian JJA surface SW: 1 of 18. Siberian JJA cloud area: **0 of 18**. The energy target was by contrast well resolved (± 0.23 W/m² on global TOA, 9 of 18 significant), so the campaign’s Southern-Ocean and global conclusions were safe while *every boreal ranking was noise*.

At 44 years the threshold falls to ± 0.240 K, and the ranking inverts:

run	4-yr Δ	44-yr Δ	t	verdict
B5 RCAPDCYCL 2 → 0	+0.001	+0.407	+3.32	significant — best
B2 RCLDIFF_CONVI= 25	+0.290	+0.344	+2.80	significant
B3 RCLDIFF= 1.5E-5	+0.366	+0.306	+2.50	significant
A1a ov1= 0.10	−1.149	−0.749	−6.11	significant, wrong way
B8 RVLAMSK= 5	+0.502	−0.038	−0.31	noise
AB, B7, C2, E1, A2, A1c, B6	—	$\leq +0.12$	—	noise

Two reversals matter. **B8, promoted to “best lever of the campaign” on its +0.502 K, is noise**; it had already been retracted on 07–29 when the noise floor was first measured, and the 46-year runs confirm that retraction. **B5, dismissed as “inert” at +0.001 K, is the strongest lever in the campaign.** The caution recorded at the time — *unresolved, not proven inert* — is what kept it in play.

Method note for future rounds: B8 was promoted and then built upon twice (ABB8, E1) before its uncertainty was ever estimated. Three of that round’s four runs were spent following a result inside the noise. The noise floor cost one post-processing script and no node-hours.

17.2 Period offset, measured rather than estimated

`amip_presentday` is the same model and configuration run over 1989–2015, so the epoch offset is measurable directly instead of chained through observations (§11.3.1).

arm	Sib JJA T2m	vs ERA5 90–14	SO SW CRE	vs CERES	net TOA
control (1872–1915)	9.73	−2.45	−60.29	+7.85	0.64
presentday (1990–2014)	10.14	−2.04	−60.36	+7.78	2.20
measured offset		+0.42		−0.07	+1.56

Both 07–29 corrections were too large. The boreal offset is +0.42 K, not +1.12, so the period-clean bias is **−2.04** K; the SO SW CRE offset is −0.07, not −1.43, so that gap is **+7.78** — essentially the original +8.08, and the “ ≈ 6 W/m²” claim is withdrawn.

The real finding is underneath the error. The model warms Siberian JJA by only +0.42 K between the epochs where observations imply $\approx +1.1$ K, so **the model under-warms the historical period by ≈ 0.7 K** — a *trend* error distinct from the mean-state bias. Taking observations as correct for the epoch change puts the 1870s bias at ≈ -1.3 K; taking the model’s own offset puts it at ≈ -2.0 K. The honest target range is **−1.3 to −2.0 K**.

Caveat: **presentday** global net TOA is +2.20 against CERES’s +0.97. Aerosols were verified correct (MACv2-SP transient, years 1989→2002 logged), so this is a genuine radiative bias rather than a missing-forcing artefact, consistent with the SO cloud deficit being identical across epochs.

17.3 The runs were never 1850 GHG — a dead namelist block

NCMIPFIXYR: 1850 never took effect in any oifsamip run. All 24 runscripts hand-wrote a &NAMECECMIP6 block; OpenIFS 48r1 renamed that namelist to &NAMECECMIP (ece_cmip.F90:68 POSNAMs 'NAMECECMIP' only, and NAMECECMIP6/SSPNA appear *nowhere* in the source). The legacy block survives in esm-tools’ fort.4 template, so f90nm1 patched it without complaint and the generated namelist looked correct — while the model read only NAMECECMIP and left NCMIPFIXYR at its default −1. Since LFIXYEAR = NCMIPFIXYR > 0, forcing was **transient**. From the model’s own output:

```
UPDRGAS: greenhouse gas concentrations for YEAR/MONTH 1870/01
CO2 = 286.94 -> 287.12 -> 287.31 -> 287.49 ... (rising through the run)
SP_SETUP: IYR = 1870, 1871, 1872, ... (MACv2-SP aerosols advancing)
```

So every run is a transient CMIP7 *historical* run at its actual calendar year, Krakatoa (1883: −0.98 K in Siberian JJA) included, not a fixed-1850 PI run.

The tuning results are unaffected: all runs share that identical forcing trajectory, the perturbations live in NAMCLDP/NAMVDF/source which *are* read, and the ANOVA absorbs the shared trend and the volcanoes in $g[\text{year}]$. **The absolute targets are affected**, since they are all read off the control: CO₂ drifts 287 → 301 ppm ($\approx +0.15$ W/m² mid-run vs 1850, partly offset by aerosols) and the volcanoes depress the 44-yr mean by ≈ 0.05 K. Small — but the global TOA target is only +0.53 W/m², so ≈ 0.1 W/m² of spurious forcing is not negligible *there*. **amip_presentday** is unaffected: **historical** is *correct* for 1990–2014.

The fix is the setup’s own switch, not a hand-patch: oifs.scenario: "piControl" routes through configs/components/oifs which writes NCMIPFIXYR into the live block and sets the CMIP7 stratospheric-aerosol switches in NAERAD. The dead block has been removed from all 24 runscripts and **amip_picontrol** is running as a true PI reference; **amip_pi_base** is deliberately untouched because all 19 lever deltas reference it. **The CMIP7 production spin-up is not affected** — its NCMIPFIXYR= 1850 is in the live block and CO₂ is pinned at 284.30 ppm at calendar year 2018. *Residual code gap:* nothing pins MACv2-SP tropospheric aerosols (updtim.F90:812–818 takes the calendar year unconditionally).

17.4 Can the levers be combined? Not as they stand

Targets from the 44-year control, and the significant per-lever responses:

	SO SW CRE	Sib JJA T2m	global TOA	tropics TOA
required Δ	−7.85	+1.3 to +2.0	−0.64	+2.50
A1a ovl= 0.10	−6.46	−0.749	−1.90	−1.07
A1b ovl= 0.35	−2.17	−0.135	−0.58	−0.29
B5	−0.23	+0.407	−1.00	−1.94
B2	+0.11	+0.344	+0.39	+0.65
B3	+0.67	+0.306	+1.34	+2.02

Note the tropics are 2.5 W/m² *too low*, not too high, so B5’s −1.94 is a real cost and only B3 helps there. Added linearly:

stack	SO	Sib	global TOA	tropics
A1b+B5	−2.40	+0.27	−0.94 ✗	40.4 ✗
A1b+B2+B5	−2.30	+0.62*	−0.55 ✗	41.0 ✗
A1b+B2+B3+B5	−1.63 ✗	+0.92*	+0.79 ✗	43.0
A1a+B2+B3+B5	−5.91	+0.31 ✗	−0.53	42.3 ✗

*optimistic: AB measured −0.053 K where additivity predicted +0.21, so A1b *cancels* B2’s boreal gain.

Two structural walls, not bad luck.

- **The Southern Ocean is reachable only through A1-strength mixed-phase overlap, which cools Siberia.** That is the same process on both ends: more supercooled liquid brightens SO cloud *and* boreal mid-level cloud. A1a buys 82% of the SO target and spends -0.75 K of Siberia doing it.
- **Siberia’s levers are too small and do not add.** The three significant ones sum to $+1.06$ K against a $+1.3$ – 2.0 K need, before the non-additivity that AB demonstrated.

The LPJ-GUESS fallback is closed: §11.3.2 showed the forest is 75–85% temperature-driven, so prong B cannot absorb a boreal residual the atmosphere does not deliver. The boreal problem must be solved in OpenIFS.

18 Round 11 design (2026–07–30): attack the mechanisms, not the parameters

Round 10 closed on two structural walls (§17.4). Round 11 therefore targets the underlying mechanisms. Two proposals were written and then **withdrawn for being tuning-to-fit**, and recording why is part of the result:

- ~~Land/sea split on RCL_OVERLAP_LIQUICE.~~ **Anti-physical.** That parameter is the sub-grid *overlap fraction of supercooled liquid with ice* (range $[0, 1]$, default 0.65, EC-Earth4 uses 0.1), entering $ZDEPOS = \max(ZOVERLAP_LIQUICE \cdot ZA \cdot (ZINew - ZICE0), 0)$. The code’s own comment states the intended selectivity — “Reduce in shallow convection because assume SLW in active updraught is less overlapped with ice in less active part” — i.e. convective segregation *lowers* overlap. But ocean mixed-phase cloud is predominantly stratiform (well mixed, so *high* overlap) and land cloud more convective (segregated, *low* overlap), the **opposite** of what the tuning wants.
- ~~RCAPDCYCL mode 3 = disable the CAPE diurnal-cycle correction over land.~~ **A regression.** That correction is Bechtold et al. (2014), which exists to fix the documented error of land convection triggering too early in the day.

18.1 D1 — RCAPDCYCL mode 4: reformulate the land closure

B5 was misdiagnosed. RCAPDCYCL is a *mode selector*, not a magnitude (`cumastrn.F90:773,777`): 1.0 = land only, scaled on ZKHVFL (surface kinematic virtual heat flux, `:464`); 2.0 = everywhere, land via ZCAPPBL · ZTAU/ZTAURES and ocean via a wind-based ZTAUPBL; 0.0 = off entirely. So B5’s $2.0 \rightarrow 0.0$ **deleted the correction globally**: its $+0.407$ K boreal gain came from the land branch, its -1.94 W/m² tropical cost from losing the ocean branch. The two are separable.

The physical argument is in the scaling. Mode 1 is proportional to the *actual surface buoyancy flux* and vanishes where that forcing is weak; mode 2’s land branch uses boundary-layer CAPE instead. Boreal summer has weak insolation, a shallow boundary layer and small surface fluxes, so the buoyancy-flux form self-limits while the BL-CAPE form need not — a real reason to suspect **mode 2’s land branch is mis-scaled for weak-forcing high-latitude convection**, and it predicts the observed sign. **Mode 4** (added this build) is mode-1 land physics with mode-2’s ocean branch verbatim; both are existing code. *Prediction*: recovers part of $+0.407$ K with little of the tropical cost. If the boreal gain vanishes too, the effect was never about the land closure formulation and the B5 result needs re-reading.

18.2 D2 — ice nuclei, not overlap

The Southern-Ocean supercooled-liquid deficit is an **ice-nucleation** problem, and the scheme has no aerosol dependence whatsoever:

$$ZICENUCLEI = 1000 \cdot \exp(12.96 \cdot ZSUPERSATICE - 0.639) \quad (\text{cloudsc.F90 : 2516, : 2618})$$

That is **Meyers et al. (1992)** — a function of ice supersaturation *only*, documented to overestimate INP in clean environments, which is precisely the remote Southern Ocean; DeMott et al. (2010/2015) make INP a function of aerosol number instead. Reducing INP over ocean therefore has a real mechanism (INP sources are continental: mineral dust, biological) with the correct sign in both regions at once, and strong leverage since $ZCVDs \propto ZICENUCLEI^{0.666}$ — a factor 0.1 cuts deposition to ≈ 0.22 .

Three better routes are closed. (i) PNICE is already a `cloudsc` argument (`:265`) and already used for autoconversion (`:2855`), but it is filled only when `NAERCLD > 0` (`callpar.F90:1155`, else zeroed at `:1166`) and `NAERCLD` defaults to 0; that configuration is several times slower and **would not finish CMIP7 in time**. (ii) `onlinedust_v4_rmp.nc` is staged and *opened* unconditionally, but its arrays (`POTSRC`, `SOILTYPE`, `CULT`) are

allocated only by `TM5M7_SRC_DUST_INIT` via `tm5m7_init.F90` — **only under m7** — so its contents are never populated here (`suecrad`’s `YDUSTCLIM` is a different object, dust *optical properties* for the radiation climatology). (iii) MACv2-SP plumes are anthropogenic sulphate/BC/OC: poor INP, sulphate coating *deactivates* mineral dust INP, and in a piControl anthropogenic aerosol is near zero, giving no land/ocean contrast.

18.2.1 Two physical caveats that shape the test

The tropics will pay, and “ocean” is the wrong discriminator. The scaling acts only inside the mixed-phase window ($RTHOMO \leq T < RTT - 5$, i.e. 235.15–268.15 K, with liquid present), which in the tropics sits at ≈ 500 –300 hPa and *is* populated — deep convective updraughts carry supercooled liquid well above the freezing level, and anvil detrainment feeds it. Reducing INP there retains more supercooled liquid, brightening tropical mid-level cloud and pushing tropical TOA *further below* CERES. Expect a cost of order A1a’s -1.07 W/m^2 or worse. More fundamentally, tropical and subtropical oceans are *not* dust-remote (Saharan dust over the tropical Atlantic, Asian dust over the North Pacific); the Southern Ocean is uniquely far from continental sources. The property that distinguishes it is **remoteness**, which a local land-fraction mask cannot encode.

Sea salt sets an INP floor. Sea salt itself is a poor INP (highly soluble, inefficient in the immersion mode at these temperatures), but sea spray carries **marine biogenic organics** — sea-surface microlayer material, phytoplankton exudates — which are a modest INP source (Wilson et al. 2015; DeMott et al. 2016). So the ocean INP factor **must not approach zero**: ≈ 0.1 –0.3 is defensible, not 0.01. Sea salt’s larger role is as CCN, which is moot here since PCCN is zeroed with `NAERCLD = 0` and droplet number is the constant `RCL_KK_CLOUD_NUM_SEA`. A tempering caveat: marine biogenic INP peaks with the austral-summer bloom (DJF), exactly when SO insolation and our SW bias peak, so the SO is *least* INP-starved in the season we care most about.

18.3 The three runs

All on the same forcing as the 19 round-10 levers (`oifs.scenario` defaults to `historical`), so deltas are directly comparable to B5’s; 46 yr, evaluated 1872–1915.

run	change	purpose
D1	<code>RCAPDCYCL = 4</code> (source + namelist)	separate B5’s boreal gain from its tropical cost
D2a	<code>ZICENUCLEI $\times f(\text{PLSM})$</code> , no remoteness	does the INP route have leverage at all?
D2b	tile-based two-term INP + remoteness gate	the physically honest version

D2a is run *knowing* it should damage the tropics: the question it answers is whether INP has Southern-Ocean leverage, which decides whether D2b is worth building. D2b computes the INP field in `callpar` from `SURFL%ZFRIT` (tile fractions, already used at `callpar.F90:2006`) and `PAUX%PGEMU` (sine of latitude, available at `:1801`), passing one new array to `cloudsc`:

$$\text{INP} \propto a \cdot [\text{bare soil (tile 8)} \times \text{wind}] + b \cdot [\text{low veg (4)} + \text{high veg (6)}]$$

Tile convention from `srfi_mod.F90:52`. The two terms matter: a *dust-only* proxy would give the **wrong sign** over boreal forest, which is a weak dust source — lowering its INP would brighten boreal cloud and cool it further. Vegetated land is instead a strong **biological** INP source in summer (pollen, bacteria, fungal spores, litter), peaking in exactly our season, while snow tiles (5/7) collapse that term in winter and leave the cold-season bias untouched.

All three are source edits on the single `oifsamip-cy48` tree and therefore **serialise**: edit, `esm_master recomp-oifsamip-cy48`, verify the binary md5 changed, submit, wait for staging, repeat.

19 Round 11 results: D2b adopted for the Southern Ocean, D1 falsified

	SO SW CRE	SO SW RMSE	Sib JJA T2m	global TOA	tropics
control	−60.29	6.88	9.73	0.64	42.61
piCTRL 1850	−60.65	6.83	9.74	0.79	42.69
D1 <code>RCAPDCYCL = 4</code>	−60.38	6.51	9.56	0.61	42.53
D2a INP, no gate	− 65.13	5.07	9.56	−0.28	42.23
D2b INP + p700	−62.93	4.96	9.51	0.36	42.70

D2b is adopted as the Southern-Ocean lever. It gives the best SO surface-SW RMSE of any run ($6.88 \rightarrow 4.96$, −28 %) at *zero* tropical cost (+0.09), beating both A1b (5.54) and the ungated D2a (5.07) — the latter despite

a larger mean-CRE change, because D2b improves the spatial *pattern*, which is what the priority metric exists to measure. The ice-nucleation hypothesis (§18.2) is therefore supported, and the pressure gate did what it was designed to do: D2a’s tropical cost (-0.38) vanishes ($+0.09$) while 55 % of the SO gain is retained.

D1’s prediction is excluded, but as a NULL, not a sign reversal. It was predicted to recover part of B5’s $+0.407\text{ K}$ by switching the land CAPE closure to mode-1’s buoyancy-flux scaling; it delivered -0.170 K — which does *not* clear the $\pm 0.242\text{ K}$ detection floor (§17.1), so the predicted recovery is excluded but the sign is unresolved. Quoting it as “the wrong sign” was the very error §17.1 exists to prevent. So B5’s boreal gain requires *removing* the land correction, not rescaling it — the effect is not about which quantity the closure scales on. Stating the prediction in advance is what made the refutation unambiguous.

amp_picontrol settles two things. The energy target is $+0.79\text{ W/m}^2$, not $+0.64$ — slightly *larger*, so A1b does not overshoot. And Siberian JJA T2m differs by only $+0.008\text{ K}$, so the dead-namelist forcing bug (§17.3) never affected the boreal target and every round-10 boreal conclusion stands.

20 Round 12: boreal surface exchange, and the best lever of the campaign

Round 11 left the boreal unsolved. Round 12 changed target from convection and cloud to **boreal surface exchange**, anchored on one observation: the measured Siberian JJA Bowen ratio is **0.43** against $\approx 0.5\text{--}1.5$ observed for boreal conifer, so the model puts too much surface energy into latent heat. Four levers, each raising Bowen by an independent route, all indexed on vegetation types 3/4 so that types 5/6/18 — and hence the tropics — are untouched *by construction*. That is exactly the property B5 lacks.

run	change	ΔT2m	t	verdict
F4	RVRSMIN(3,4) 250 $\rightarrow$ 1000	+0.749	+6.06	significant, best ever
F5	all four combined	+0.746	+6.03	significant, $\equiv$ F4
B5	(previous best)	+0.407	+3.29	significant
F2	RVLA1(3,4) 5 $\rightarrow$ 3	+0.321	+2.59	significant
F1	RVZOH(3,4)=z0m/10	+0.185	+1.49	marginal
F3	RVC0V(3,4) 0.9 $\rightarrow$ 0.7	≈ 0	—	noise

F4 costs the tropics nothing (42.68 vs control 42.61) where B5 wrecked them (40.67), and it gives the best global T2m RMSE of any run (1.54). Bowen moves $0.44 \rightarrow 0.60$, into the observed range *without* overshooting — the risk named in advance did not materialise.

F5 equals F4 to 0.003 K. Against a naive sum of $\approx 1.25\text{ K}$, adding F1, F2 and F3 on top of F4 contributes nothing. The advance prediction — $F5 \ll \sum$, closer to the largest single lever — held, and more strongly than expected: near-total saturation of a single shared latent-heat pathway with a floor at zero. *Use F4 alone.*

A prediction that failed. F1 was called strongest a priori because $kB^{-1} = 0$ is the largest departure from observation (every closed-forest type sets $z_{0h} = z_{0m}$, i.e. $kB^{-1} = 0$, where observations give $kB^{-1} \approx 2$). It is marginal and barely moved Bowen ($0.44 \rightarrow 0.43$). “Largest departure from observation” is not a reliable guide to leverage.

B5 carries a hidden -0.72 K DJF cooling (F4: $+0.06$), invisible for the whole campaign because only JJA was ever evaluated. Given that the coupled model’s original complaint is a *cold-season* bias, that is disqualifying for B5, and DJF now belongs in the standard guardrails.

20.1 Is F4 fixing a cause or a symptom?

Two tests, both partly exonerating. *Seasonal selectivity:* DJF $+0.061$, MAM $+0.047$, JJA **$+0.749$** , SON $+0.110$ — exactly the signature a transpiration mechanism must have, since stomata are shut and the canopy snow-covered in winter. *Vertical targeting:* F4 warms 1000/925/850 hPa by $+0.83/+0.84/+0.83$, closing 56–83 % of the *Siberia-specific* layer bias (§21.1), and only $+0.53$ aloft.

What remains true is that RVRSMIN 250 $\rightarrow$ 1000 is an unanchored $4\times$ excursion from what ECMWF ships, and that the vertical profiles show it removes moisture that is already deficient. **G1 = F4 + D2b** tested this: the first pairing whose components are disjoint in both process and geography, hence expected not to conflict — but AB and ABB8 both got superposition wrong in sign, so it was measured, not assumed. It superposes (§20.2).

20.2 G1 = F4 + D2b: the best configuration of the campaign

G1 is the first combination to improve *both* targets at once, and the first whose superposition is additive rather than wrong in sign.

metric	control	G1	Δ	target
SO SW RMSE (priority)	6.877	4.809	−2.067	—
SO TOA SW CRE [W m^{-2}]	−60.29	−63.13	−2.84	−68.14
SO cloud area [%]	83.07	83.59	+0.52	89.72
Siberia JJA T2m [$^{\circ}\text{C}$]	9.73	10.25	+0.521 ($t=4.22$)	≈ 12.2
Siberia sfc net SW [W m^{-2}]	153.78	159.31	+5.54	166.26
global net TOA [W m^{-2}]	+0.64	+0.45	−0.195	~ 0
tropics net TOA [W m^{-2}]	42.61	42.71	+0.096	45.11
global T2m RMSE [K]	1.579	1.553	−0.026	—
Nordic Seas SW RMSE	9.058	9.358	+0.300	—

Superposition, measured: Siberia JJA T2m F4 +0.749 + D2b −0.222 = +0.527 predicted vs **+0.521** actual; SO SW RMSE −2.131 vs −2.067; SO CRE −2.719 vs −2.840; global net TOA −0.188 vs −0.195. Additive to within noise on every metric. The plausible reason it holds here and failed for AB/ABB8: F4 acts on a land surface flux in boreal summer, D2b on ice nucleation over ocean below 700 hPa, whereas AB and ABB8 combined levers modifying the *same* cloud scheme in the *same* regime.

The one cost is the Nordic Seas, +0.300 SW RMSE against a ± 0.052 threshold, and worse than additive (+0.184 expected). No other lever in the campaign has ever moved Nordic RMSE; this one moves it the wrong way. Small box, not the deep-water priority, but it must not be allowed to grow.

G1 closes 26 % of the boreal bias and 36 % of the SO CRE gap. The SO *cloud-area* deficit of ~ 6 pp remains untouched by every lever tried.

20.3 Seasonal audit: the noise floor is not the same in every season

The campaign evaluated JJA only for eleven rounds, while the coupled model’s original complaint is a *cold-season* bias. `scripts/analysis/seasonal_by_run.py` applies the run $\times$ year ANOVA of §17.1 separately to each season. The detection threshold varies by a factor of 2.4 across the year:

season	control	sd(ϵ)	95 % threshold
DJF	−29.35 $^{\circ}\text{C}$	1.407 K	± 0.588 K
MAM	−9.28 $^{\circ}\text{C}$	0.923 K	± 0.386 K
JJA	+9.73 $^{\circ}\text{C}$	0.580 K	± 0.242 K
SON	−10.41 $^{\circ}\text{C}$	1.031 K	± 0.431 K

Applying the JJA threshold to a winter delta overstates significance by 2.4 $\times$, and a first pass through these results did exactly that, briefly recording D2b (−0.455) and B3 (−0.486) as winter-damaged. Both are *within* the DJF noise floor. Corrected:

- **B5 capdcyc10 is the only lever with genuine winter damage** (DJF −0.720, clears ± 0.588) and the only one that significantly warms JJA while significantly cooling DJF. Its rejection stands, now properly thresholded.
- **D2b’s real seasonal cost is spring, not winter:** MAM −0.467 clears ± 0.386 ; its DJF −0.455 does not clear ± 0.588 .
- **F4 is seasonally clean** — DJF +0.061, MAM +0.047, SON +0.110, all within noise, the entire signal in JJA. A further argument for F4 over the cloud levers.
- **G1 is seasonally clean:** DJF +0.070, MAM −0.296, SON +0.280, none significant; D2b’s MAM cooling is diluted below threshold.
- A1a warms DJF/SON and cools MAM/JJA by large amounts — a global cloud change, not a boreal lever. B6, B7 and F3 cool MAM significantly; F5 inherits F3’s MAM −0.507.

21 Vertical structure vs ERA5: where the biases live

21.1 The Siberia-specific cold bias is confined below 850 hPa

Model minus ERA5 on the model’s 19 pressure levels, `amip_presentday` vs ERA5 1990–2014 so the comparison is period-clean. ERA5 comes from the DKRZ pool; all 19 model levels exist exactly in ERA5’s 37, so no interpolation is needed.

hPa	Siberia	Global	Siberia-specific
1000	−1.90	−0.73	− 1.17
925	−2.17	−0.65	− 1.52
850	−1.99	−0.99	− 1.00
700	−1.22	−1.15	−0.07
500	−1.67	−1.49	−0.18
300	−2.32	−2.22	−0.10

This separates two problems that had been conflated. Above 700 hPa Siberia merely shares a **global tropospheric cold bias of 0.7–2.9 K** that no boreal lever will touch; the boreal-specific excess lives entirely in the bottom three levels, which is the signature a surface-exchange problem should have and which validates the F-series framing.

The excess boreal cloud is thermally driven, not moisture-driven. Siberian RH is +5.6 to +8.2% too high while q is *too low* (−0.2 to −0.3 g/kg): a 2 K cold bias cuts q_{sat} by $\approx 14\%$ while q falls only $\approx 6\%$. So a large F-response must not be read as confirming the cause.

Independent support for D2: the Southern Ocean has RH +5.8 to +6.6% too high through the mixed-phase layer while reflecting *too little* SW — humidity present, liquid absent, exactly the supercooled-liquid deficit the INP hypothesis predicts.

Broken diagnostic, recorded: the model’s pressure-level relative humidity r is **identically zero in every run checked** (0 nonzero of 17510400 values) while q is fine. The XIOS r output yields silent garbage and must not be used; RH above is recomputed from t and q .

21.2 Extending the column downward: the soil is the least-biased part

§21.1 stops at 1000 hPa. Continuing into the skin and the four soil layers (`scripts/analysis/vertical_bias_column.py`), JJA, model `amip_presentday` minus ERA5. Low-lying land (< 500 m) is shown alongside because the all-land figure is inflated by orographic mismatch (§21.3):

JJA	global land	land < 500 m	Siberian box
700 hPa	−0.83	−0.89	−1.22
850 hPa	−0.79	−0.92	−1.99
925 hPa	−0.82	−1.02	−2.17
2 m	−1.34	− 1.00	−2.04
skin	−0.83	− 0.53	−1.66
soil L1 (7 cm)	−0.40	− 0.36	−1.73
soil L2 (28 cm)	−0.42	−0.38	−1.90
soil L3 (1 m)	−0.78	−0.72	−2.14
soil L4 (2.9 m)	−1.09	−0.89	−1.02

Over ordinary low-lying land the **soil and skin are the least-biased parts of the entire column**, and 2 m sits about 0.5 K *below* the skin under it. The ground is not what holds the near-surface cold; if anything it is doing better than the air above it. Siberia is the exception that §21.1 already identified — there the soil is cold too (−1.7 to −2.1 K), which is the surface-confined excess the F-series was built to attack.

Caveats. The 1000 hPa row is unreliable over land, where much of that level is below ground and both datasets extrapolate; 925/850 are the lowest honest levels. ERA5’s soil and skin are HTESSEL output, so those rows are model-versus-sibling rather than observation — the 2 m and pressure-level rows are the observation-constrained ones.

21.3 The near-surface cold is not boreal: every surface type is cold

The campaign has only ever scored one box, so it had never checked whether its levers address a local fault or a global one. Resolving the 2 m bias by dominant surface type (`scripts/analysis/bias_by_tile.py`) gives an unambiguous answer: **all 20 surface types are cold, in every season. Not one is warm.**

dominant type	area	DJF	MAM	JJA	SON
bare soil	25.3 %	−1.86	−2.12	−2.16	−2.03
Evgr Broadleaf (tropical)	8.7 %	−1.78	−1.79	−1.46	−1.42
Tundra	5.2 %	−2.51	−0.18	−1.43	−1.84
Evgr Needleleaf	4.7 %	−2.00	−1.84	−1.16	−1.05
Dec Needleleaf	3.0 %	−1.22	−1.55	−1.87	−1.69
ALL LAND	100 %	−1.73	−1.60	−1.35	−1.39

Two controls bound how much of this is real. The land figure is *elevation-dependent* — -0.70 K below 200 m rising to -2.43 K above 2000 m, the expected signature of the model’s smoothed orography against ERA5’s 25 km — so the all-land number overstates it and the low-lying value is the honest one. And over **prescribed-SST ocean** 2 m is -0.72 K: about 0.7 K of near-surface cold exists where the land surface is absent entirely.

Consequence for the campaign. Siberia’s JJA -2.58 K decomposes roughly as ~ 0.7 K universal, ~ 0.3 K orographic, and 1.3–1.6 K boreal-specific. G4 has taken ~ 0.95 K of that boreal budget — which is a far better explanation of the F-series saturation ($F5 = F4$ to within noise, §20) than any parameter limit. *The boreal budget is nearly spent, and the remainder is not reachable by a boreal-indexed lever.*

One benign consequence. Because every type is cold, a lever indexed by vegetation type does not damage the regions it leaks into: the non-boreal areas of types 4 and 9 are cold too (JJA -0.68 and -1.06). That is what licenses the round-17 restriction to those two types (§28).

22 Surface albedo: half the Siberian SW deficit is not cloud

Every boreal lever so far has attacked cloud. Decomposed against CERES on the identical box and mask, the 12.5 W/m² Siberian JJA surface-SW deficit is only half cloud: **7.0 W/m²** from too little SW reaching the surface, and **5.5 W/m²** from **surface albedo too high** — 0.1753 model versus 0.1461 CERES, entirely independent of cloud. F4 does not touch it (0.1749).

22.1 Not a global albedo error, and not the forest

Temperate and tropical land is essentially perfect (Europe, Amazon, Great Plains, steppe, Australia all within ± 0.004), so the albedo *scheme* is sound and a global fix would be wrong. *This held up, with one amendment.* §29.1 later measured land albedo globally and by type with snow masked per cell per month: every forest type is inside $+0.006$ and tropical broadleaf is at -0.0003 , confirming the scheme — but a residual $+0.0080$ does exist on crops, semidesert and bare soil, so “a global fix would be wrong” is right about the *vegetation table* and too strong about the *soil background*. The error is concentrated at high latitude: Siberian tundra $+0.094$ (-16.0 W/m²), Canada boreal $+0.032$, Siberia boreal $+0.029$, Sahara $+0.020$ — but **Fennoscandia, the same needleleaf type, is correct at $+0.002$** . Splitting Siberia by latitude gives $+0.011$ in the forest core and $+0.087$ at 70–75N.

22.2 How the albedo is built, and two hypotheses that died

All configurations — AMIP, coupled round-09, *and the CMIP7 production piControl* — run `LRDALB = .false.` with `NALBEDOScheme = 3`, which **discards the MODIS albedo present in the input file** and constructs albedo each step (`surfbc_ctl_mod.F90:512`) as $\alpha = \text{RVVEGALB}(t_{vl})c_{vl} + \text{RVVEGALB}(t_{vh})c_{vh} + \alpha_{soil}(1 - c_{vl} - c_{vh})$.

~~Hypothesis 1: the soil-albedo field dominates tundra.~~ The bare fraction is only 0.01–0.03, so the soil term is negligible. ~~Hypothesis 2: the soil fields are corrupt.~~ GRIB 117–120 do carry an inconsistent header (N=128 with a 256-row `p1` but only 40320 values, the O96 count), which **breaks cdo and grib_get_data** on those fields and should be reported upstream — but the values are O96-ordered and correct: zero over 99.2% of ocean cells, $\text{corr}(\alpha_{soil}, \text{lsm}) = +0.894$. Not our bias.

22.3 The real signature is snow-melt timing

The blend at 70–75N yields 0.158 while the model runs at 0.303; the extra ≈ 0.145 is snow. Monthly, the model is **too dark in April** (-0.06 to -0.07), **near-perfect in July/August** when snow-free ($+0.004$), and **too bright in June** ($+0.086$ to $+0.116$) and **September** ($+0.08$ to $+0.10$). So RVVEGALB is essentially right and the *snow season is too long at both ends*.

Cause and effect. Through February–April the model is too cold *and* too dark simultaneously; if bright snow caused the cold those signs would agree, so the winter cold is **not** albedo-driven. The albedo bias turns positive only in June, after the surface crosses freezing. But the June snow is a genuine defect in its own right: **June is**

+6.2°C with 213 W/m² of insolation and 0.020 m w.e. still lying, and the implied snow-cover fraction is ≈ 17 % where CERES implies ≈ 2 %. That points at the **snow-cover-fraction formulation**, not snow albedo (too *low* in April) and not melt energy (ample).

22.4 Growing season: June is albedo, July is cloud

Siberia 55–75N, W/m ²	Jun	Jul	Aug	JJA
cloud	6.9	9.9	4.4	7.0
albedo / snow	13.4	1.7	1.3	5.5
total	20.2	11.6	5.7	12.5

Almost cleanly separated in time. The residual snow term is 5.5 W/m² on the JJA mean, essentially all in June, worth ≈ 0.2–0.7 K depending on which sensitivity applies — potentially another F4-sized gain, and *orthogonal* to F4, which fixes SW-down while this fixes the absorbed fraction.

But albedo does not explain the land cold bias. Insolation-weighted over land 60S–75N excluding Greenland, the mean albedo-induced SW loss is −3.00 W/m² against a mean T2m bias of −1.35 K, with spatial correlation $r = +0.145$ and slope +0.033 K per W/m². Decisively, **cells where the model is too dark are still −1.16 K cold**. A ≈ 1.2 K land-wide cold bias therefore has no albedo signature at all, consistent with the global tropospheric cold bias of §21.1. Fixing albedo is worth doing; it is not the root cause.

23 Round 13: snow cover fraction, the June lever

23.1 Design and prediction

The albedo decomposition (§22) splits the 12.5 W m^{−2} Siberian JJA surface-SW deficit into ~7.0 cloud and ~5.5 **surface albedo**, with the albedo half concentrated almost entirely in **June** (§22.4; bias +0.086, July and August near-perfect). That is a snow-*melt-timing* signature, not a snow-*brightness* one. Cause and effect were established before touching anything: winter cold is *not* albedo-driven, since February–April is simultaneously cold and dark (§22.3), which rules out the albedo → cold direction for those months.

Snow cover fraction (`surfbcrctl_mod.F90:317–322`) is

$$ZCVS = \min(1, d_{\text{snow}}[\text{cm}] \cdot \text{RQSNCR}), \quad \text{RQSNCR} = 1/10 \text{ under LESN09=T.}$$

A grid box is therefore fully snow-covered at **10 cm**, linearly below that, with no dependence on vegetation height or orographic roughness. For boreal forest this is wrong in an obvious direction: 10 cm of snow does not hide a spruce canopy, and the observed masking depth over forest is several times larger. The H-series sets $\text{RQSNCR} = 1/30$.

Why the ice sheets are safe: Antarctica and Greenland carry snow far deeper than 30 cm, so $\min(1, \cdot)$ saturates at 1 for any value in this range and they are untouched. Only shallow, marginal, melting snow responds — which is the target.

H1 applies the change alone against control (clean attribution); **H2** applies it on top of G1’s F4 source change and D2b namelist (the deployable stack). H2 – G1 measured against H1 – control gives the interaction directly.

Falsifiable prediction, recorded before the runs finished: ~~H1 recovers a good part of June’s 13.4 W m^{−2} albedo term, worth ~ +0.2 to +0.7 K on Siberian JJA T2m, and does little in July and August.~~ **Falsified** (§23.2): JJA came out at +0.020 K, $t = 0.17$. The failure mode was not the anticipated one — it did not warm July as much as June, it failed to warm *any* summer month.

23.2 Results: prediction falsified, lever rejected

Predicted +0.2 to +0.7 K on Siberian JJA T2m; measured +0.020 K, $t = 0.17$. And the lever inflicts the largest winter cold bias in the campaign, −1.233 K DJF against a ±0.588 threshold, beating the −0.720 for which B5 was rejected.

	DJF	MAM	JJA	SON
H1 snowcr30	− 1.233*	−0.473*	+0.020	+0.428
H2 = G1 + snow	−0.973*	−0.342	+0.662*	+0.863*
(G1, reference)	+0.070	−0.296	+0.521*	+0.280

Why it failed: the mechanism was pointed at the wrong month. The lever did exactly what it was built to do to the albedo — but not where the reasoning assumed. H1 all-sky surface albedo change ($\times 100$, * = clears that month’s threshold):

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
H1	−0.51*	−0.21	−0.42*	−0.84*	−1.43*	−0.95*	−0.10*	−0.32*	−3.08*	−8.16*	−4.66*	−1.60*

The response concentrates in **September–November**, the *snow-onset* season, and is small in June. The design error is plain in hindsight: $ZCVS = \min(1, d_{cm} \cdot RQSNCR)$ can only respond where snow is **shallow**, and snow is shallow while it is *accumulating in autumn*, not while it is melting in June — by June the pack either still saturates $\min(1, \cdot)$ under both settings or is already gone. **Shallow snow is an autumn phenomenon, not a melt phenomenon.** Consistently, June/July/August T2m move $-0.076 / -0.067 / +0.205$ (none significant) while **October T2m is $+1.135$ K.**

The winter cooling is not radiative. February and March are dark yet cool by -1.65 and -1.19 K. Reducing the snow-covered fraction moves area from the exposed and sheltered **snow tiles (5, 7)** onto the vegetation tiles, and the snow tile is what buffers the surface against radiative cooling in polar night. The lever trades a small autumn warming for a large winter cooling through **tile fractions, not albedo.**

Verdict: rejected, and RQSNCR reverted. Three things survive:

1. The June surface-albedo bias is **not** dominated by the snow-cover-fraction formulation. Its cause — snow-albedo decay, vegetation masking of snow albedo, or the snow-free surface albedo itself — remains unidentified.
2. RQSNCR has large leverage on *autumn* albedo (-8 points in October). If an autumn or snow-onset bias is ever diagnosed, this is the knob.
3. H2 posts the campaign’s best SO SW RMSE (4.787), best subpolar N Atl (4.713) and best global T2m RMSE (1.530) — but that is D2b’s doing, and the winter damage disqualifies the combination. **G1 remains the configuration to carry forward.**

24 Round 14 (2026–08–03): the June albedo bias is a SPRING MELT bias

Round 13 established what the June albedo bias is *not* (§23.2): not the snow-cover formulation. This section identifies what it is, entirely from output already on disk plus ERA5 — no new integration was needed to get here.

24.1 Against ERA5: too much snow, and snow that is too DARK

scripts/analysis/albedo_decompose.py, model amip_presentday vs ERA5 1990–2014, both land-masked to the Siberian box, June:

June	model	ERA5	Δ
surface albedo fa1	0.2147	0.1730	+0.042
snow cover fraction	0.380	0.261	+0.119
snow albedo asn	0.658	0.759	−0.102

Both **f_snow** columns are the *same* HTESSEL formula $\min(1, d_{cm}/10)$ applied to each dataset’s snow mass, so the gap is **snow AMOUNT, not the cover formula** — which is precisely why round 13, which changed only the formula, could not move it. The model carries roughly twice ERA5’s June snow mass.

Corroborated independently by CERES (§25.1), which puts the June excess slightly higher at $+0.046$; and the cover excess is larger still against satellite, $+0.176$ rather than $+0.119$ (§25.2).

And the model’s snow is too dark, not too bright (-0.10 in June, -0.11 in May). Making snow albedo more realistic would make June *worse*. Any lever that brightens snow is pushing the wrong way; the excess reflection comes from snow *extent*, not snow *brightness*.

Two things this analysis does not claim. The naive two-term split $\alpha_{box} = f \alpha_{snow} + (1 - f) \alpha_{free}$ does **not** close: backing out α_{free} gives negative values over the box, because HTESSEL does not apply **asn** to the whole snow fraction — tile 7 is snow *sheltered* under high vegetation and carries a canopy-shaded albedo far below **asn**. That the residual is large and negative in exactly the melt months is direct evidence the canopy-masking term already

does most of the work there, so no snow-free albedo is reported. Separately, ERA5’s albedo scheme is a relative of HTESSEL, so **asn** agreement is not independent evidence; only ERA5’s snow *extent* is observationally constrained, via assimilated IMS snow cover.

24.2 Accumulation is right; the melt is a month late and 30 % too slow

Snow water equivalent over the box, model vs ERA5 [mm], with the month-to-month increment:

	model	ERA5	Δ	$\delta(\text{model})$	$\delta(\text{ERA5})$
Jan	89.8	82.3	+7.5		
Feb	106.5	97.6	+9.0	+16.7	+15.3
Mar	122.1	114.5	+7.7	+15.6	+16.9
Apr	131.6	112.8	+18.7	+9.4	−1.6
May	98.1	65.3	+32.8	−33.5	−47.5
Jun	14.1	9.6	+4.5	−84.0	−55.7
Nov	49.2	41.5	+7.7	+25.9	+25.4
Dec	70.7	63.8	+6.9	+21.5	+22.3

Accumulation is essentially correct. Through the accumulation season the model is only 7–9 % high and the monthly increments track ERA5 almost exactly (+16.7 vs +15.3, +15.6 vs +16.9, +25.9 vs +25.4, +21.5 vs +22.3). The divergence is entirely in spring:

- **April:** the model is still *gaining* (+9.4 mm) while ERA5 has already peaked and begun to melt (−1.6). **The snowpack peaks a month late.**
- **May:** the model melts 33.5 mm against ERA5’s 47.5 — **30 % too slow** — leaving it with 50 % more snow by the end of May.

Both figures come from ERA5. The satellite record confirms the timing directly and makes it worse: melt-out is 13 days late, and June cover is nearly double observed (§25.2).

24.3 May is albedo, not cloud — and it is a runaway

Model `amip_presentday` vs CERES on the same box:

	SW _{net} mod	SW _{net} CERES	Δ	SW _↓ mod	SW _↓ CERES	Δ
Apr	87.8	86.6	+1.2	176.9	187.7	−10.8
May	122.5	136.4	−13.9	220.7	224.2	−3.5
Jun	167.6	189.7	−22.2	215.7	228.2	−12.5
Jul	165.9	180.4	−14.5	193.8	208.7	−14.9

In May the downwelling flux is nearly right (−3.5) while the net is −13.9: the May deficit is *almost pure surface albedo*, i.e. the excess snow reflecting. June is roughly half cloud (−12.5 of downwelling) and half albedo. July is essentially all cloud (−14.9 down, −14.5 net), consistent with §22.4.

This closes a **self-reinforcing loop**: late melt → more snow in May → higher albedo → -13.9 W m^{-2} less absorbed → less melt energy → snow survives into June. It also explains why the model melts slower *despite* having darker snow than ERA5: the brightness term is working against the bias, and the extent term is winning.

24.4 The gap F4 leaves: tundra is the largest cover type in the box

Area-weighted vegetation cover over land in the evaluation box, from the model’s own `tvh/tv1/cvh/cv1` output:

	type	cover	
low	9 Tundra	25.6 %	F4 does not touch this
high	4 Deciduous Needleleaf	19.2 %	
high	18 Mixed Forest	5.2 %	
low	13 Bogs/Marshes	3.9 %	
high	5 Deciduous Broadleaf	3.6 %	F4
high	19 Interrupted Forest	3.5 %	
high	3 Evergreen Needleleaf	3.4 %	
high	16 Evergreen Shrubs	3.0 %	

Tundra is the single largest cover type in the box, and the campaign's best lever misses it entirely. F4 reaches about 24 % of the box (types 3, 4); tundra is another 26 %.

Its stomatal resistance is a table-internal outlier. $\text{RVRSMIN}(9) = 80 \text{ s m}^{-1}$ is the *lowest value of any vegetated type* in HTESSSEL:

crops 100 short grass 100 tall grass 100 semidesert 150 decid. broadleaf 175
interrupted forest 175 evergreen shrubs 225 decid. shrubs 225 evergr. broadleaf 240
bogs/marshes 240 desert 250 needleleaf 250 mixed forest 250 TUNDRA 80

The model therefore has arctic tundra transpiring *more freely than tropical grassland*. Tundra's closest physiognomic analogues in the same table — evergreen and deciduous shrubs — are both 225, and bogs/marshes, which co-occur with tundra in this very box, are 240. This is a **table-consistency** argument, checkable against the table itself and independent of our bias; it is not a value chosen because it helps. Contrast the standing criticism of F4, whose $250 \rightarrow 1000$ remains an unanchored $4\times$ excursion.

24.5 Round 14 runs, and their predictions

All three are **namelist-only** (§24.7) — one shared binary, no rebuild, no serialisation against the model tree. That is what the NAMSURFTUNE work bought.

- **G2** — $\text{RVRSMIN}(3,4) = 500$. Half-strength F4.
- **G3** — $\text{RVRSMIN}(3,4) = 2000$. Double-strength F4.

G2/G3 bracket G1 and map the response curve, answering the standing criticism that 1000 was never chosen from a measurement. *Prediction:* G2 keeps $> 60\%$ of G1's gain ($\geq +0.31 \text{ K}$) ~~and G3 adds little ($\leq +0.15 \text{ K}$ beyond G1)~~. First half held (64 %); second half **failed** — G3 added $+0.355$ (§24.6). *The response is a ramp, which was flagged in advance as an argument against the approach, not for a larger number.*

- **G4** — G1 plus $\text{RVRSMIN}(9) 80 \rightarrow 225$ (tundra), keeping 1000 for types 3 and 4. *Prediction:* $+0.2$ to $+0.5 \text{ K}$ beyond G1's $+0.521$ (held: $+0.431$), ~~accompanied by a measurable drop in May/June snow water equivalent~~ — **falsified**, §24.6. As stated in advance, **T2m rose with no change in May–June snow mass, so the melt mechanism is unsupported** and the gain is the plain sensible-heat route. Guardrails: tundra is confined to high latitudes so tropical net TOA should stay near 42.6; check DJF against its own $\pm 0.588 \text{ K}$ threshold and the Nordic Seas SW RMSE that G1 already degraded.

24.6 Round 14 results: G4 adopted, mechanism falsified, RVRSMIN capped

G4 is the best configuration of the campaign and closes 48 % of the boreal bias, against G1's 26 % — and its stated mechanism is falsified. Both matter.

metric	control	G1	G2 (500)	G3 (2000)	G4
Siberia JJA T2m [°C]	9.73	+0.521	+0.336	+0.876	+0.952 ($t=7.68$)
Siberia sfc net SW	153.78	+5.54	+3.21	+7.27	+8.10
SO SW RMSE	6.877	4.809	4.816	4.903	4.800
subpolar N Atl SW RMSE	5.007	4.872	4.866	4.728	4.738
Nordic Seas SW RMSE	9.058	+0.300	+0.098	+0.304	+0.089
tropics net TOA	42.61	+0.096	+0.041	+0.082	+0.124
global T2m RMSE [K]	1.579	1.553	1.558	1.546	1.543
DJF	—	+0.070	−0.325	−0.798*	−0.385
MAM / SON	—	−0.296/+0.280	−0.462*/−0.030	−0.217/−0.012	−0.360/+0.185

G4 is seasonally clean on every season's own threshold, and it **largely repairs G1's Nordic Seas damage** ($+0.300 \rightarrow +0.089$) — which nothing predicted and nothing yet explains. An unexplained improvement is as much a loose end as an unexplained degradation.

The melt mechanism is falsified

The prediction on record was a measurable drop in May–June snow water equivalent. Siberian land box SWE [mm], with run×year ANOVA thresholds:

	Mar (± 3.53)	Apr (± 3.85)	May (± 5.25)	Jun (± 2.72)
control	121.98	131.07	97.12	15.39
G1	+0.88	+0.37	+5.58*	+5.68*
G4	−1.44	−2.39	+2.75	+1.60

G4’s snow mass did not move. T2m rose +0.952K with May and June SWE unchanged within noise, so the gain is the **plain sensible-heat route**, exactly as for F4 — *not* the melt-timing route that motivated the run. The tundra lever stands on its independent table-consistency justification (§24.4) and it works; the mechanism claimed for it does not.

And F4 makes the snow bias worse. G1 *significantly increases* May–June SWE (+5.58, +5.68, both clearing threshold). The campaign’s best boreal lever improves temperature while **aggravating** the very snow bias it was meant to address. Leading suspect: cutting evapotranspiration also cuts **sublimation**, a genuine snowpack sink. Untested.

RVRSMIN does not saturate — cap it at 1000

500/1000/2000 give +0.336/ + 0.521/ + 0.876: increments per doubling of +0.185 then +**0.355**, i.e. *accelerating*. **There is no knee.** 1000 is defensible only as *the largest value before the winter damage starts* — G3 at 2000 costs DJF −0.798, clearing ±0.588 and joining B5 and H2 in the “warms JJA, cools DJF” list. F4’s magnitude is a fitted ramp, not a corrected bias, and §24.5 flagged this outcome in advance as an argument against pushing it further.

Two mechanisms for the June albedo bias are now dead: the snow *cover formulation* (round 13) and the snow *melt rate* (round 14). Untried candidates: the snow-albedo decay timescale; the canopy-masking tile-7 formulation, which the round-13 residual showed already does most of the work in the melt months; and the RVVEGALB vegetation albedo table, which the campaign has never touched.

24.7 Tuning without rebuilding: NAMSURFTUNE

The ECMWF `surf` library deliberately ships no namelist, so every HTESSEL lever in this campaign was a source edit plus a rebuild. That serialises experiments against the single model tree, and it puts AWI tuning inside upstream ECMWF files where it collides with EC-Earth’s tuning of the same tables at every merge.

`surf/module/surfece.F90` is already an AWI module *inside* `surf` with its own namelist, so the overrides live there: `ECE_TUNE_RVRSMIN`, `_RVLAI`, `_RVCOV`, `_RVZOH`, `_RVLAMSK` (all 0:20, by vegetation type) and scalar `ECE_TUNE_RQSNCR` — covering the whole F-series, the H-series and the skin-conductivity levers. They are applied by `SURFECE_APPLY_TUNING` from `susurf.F90` *after* `SUSURF_CTL` has filled the tables; nothing in the setup path derives from these entries, so a late override is equivalent to editing the table in place. Every element defaults to a sentinel meaning “keep the as-released value”, so a run setting none of them is bit-identical to untouched code, and every applied override is logged to `NULOUT`.

They occupy their own `&NAMSURFTUNE` group, *not* `NAMECECFG`: `NAMECECFG` is read twice from the same `fort.4`, by `ECE_CONFIG` in `arpifs/ecearth.F90` and by `SURFECE_CONFIG` in `surf`, and a Fortran namelist read aborts with *invalid reference to variable* on any name the reading module does not declare — so adding the tune entries there killed the `arpifs` read at `su0yoma.F90:152`. Found by a one-day test run, which is the argument for always doing one. Exactly **one line** changes in an ECMWF file. G1 is now

```
NAMCLDP: RCL_INPSEA=0.2, RCL_INPPMIN=70000.0  NAMSURFTUNE: ECE_TUNE_RVRSMIN(3)=ECE_TUNE_RVRSMIN(4)=1000.0
```

25 Observational grounding: what ERA5 can and cannot settle

Rounds 13 and 14 leaned on ERA5 for the snow diagnosis. That is a methodological weakness worth stating plainly: **ERA5’s land surface is the same HTESSEL family as ours**, so comparing our snow and flux fields to ERA5’s is partly comparing a scheme against a sibling of itself. This section replaces ERA5 with genuine observations wherever one exists, and marks the places where none does.

25.1 Surface albedo: CERES corroborates ERA5, and the model bias is real

CERES supplies `sfc_sw_up` and `sfc_sw_down` separately, so their ratio is a surface albedo derived from a *radiative retrieval* rather than a land model. Siberian land box, 1990–2014, against model `amip_presentday` and ERA5 `fal`:

	CERES	ERA5	model	ERA5–CERES	model–CERES	SW _↓
Mar	0.5540	0.5325	0.5055	–0.021	– 0.049	109
Apr	0.5386	0.5045	0.4990	–0.034	– 0.040	188
May	0.3918	0.3722	0.4285	–0.020	+0.037	224
Jun	0.1685	0.1730	0.2147	+0.005	+0.046	228
Jul	0.1357	0.1362	0.1487	+0.001	+0.013	209
Oct–Feb	—	—	—	+0.10 ... +0.13	—	6–42

ERA5 and CERES agree to ± 0.02 in every month with enough sun to retrieve (Mar–Sep), which is what licenses ERA5’s albedo for this purpose. The June excess is confirmed *independently* at +0.046 — slightly larger than the +0.042 measured against ERA5 (§24.1). The two diverge badly Oct–Feb, but at 6–42 W m^{–2} downwelling no albedo retrieval is trustworthy at these latitudes, so that is expected rather than disqualifying.

25.2 Snow cover and melt date: satellite, and ERA5 was flattering us

The Rutgers NH 24 km Weekly Snow Cover Extent CDR (NSIDC G10035, 1980–2024; NOAA visible-band snow charts to 1999, NOAA/NIC IMS thereafter) gives snow *extent* and *melt timing* from observation. `scripts/analysis/snowcove`

snow cover fraction	Rutgers (sat.)	ERA5	model	G4
Apr	0.950	0.991	0.964	0.961
May	0.647	0.796	0.890	0.893
Jun	0.203	0.261	0.380	0.447
Sep	0.075	0.178	0.257	0.249
Oct	0.612	0.755	0.825	0.834
melt-out (50 % crossing)	11 May	~17 May	24 May	28 May

Three consequences.

(i) **ERA5 was flattering the model.** ERA5 itself runs +0.058 high in June and +0.149 in May against satellite, so the model excess quoted in §24.1 as +0.119 is really **+0.176** — June snow cover is close to *double* the observed value.

(ii) **The model melts about 15 days late** (control; `amip_presentday` is 13), now measured against an observation rather than inferred from a sibling model. *Note the two runs differ* — June cover 0.413 / melt-out 26 May for the control, 0.380 / 24 May for `amip_presentday` — and earlier drafts of this report quoted them interchangeably. This is round 14’s “peaks a month late” (§24.2) confirmed independently and given a number.

(iii) **G4 makes the snow field worse.** June cover 0.380 → 0.447; melt-out 24 May → 28 May. The campaign’s best *temperature* configuration pushes the spring snow cycle *further* from satellite, consistent with G1/F4 significantly increasing May–June SWE (§24.6). **This matters for the coupled goal:** a later melt is a later growing-season onset for LPJ-GUESS, which is a plausible route to losing boreal forest even while AMIP T2m improves. It should be weighed before G4 is committed to the PI spin-up.

Comparability caveat: Rutgers SCE is a **binary** classification per 24 km cell, whereas HTESSEL’s ZCVS is a *sub-grid* fraction within a ~100 km cell. The two agree on “how much of the region is snow” but differ in how a partially covered cell is counted, which matters most mid-melt. The **crossing date** is the robust comparison; absolute mid-melt fractions are not over-read here.

25.3 Where no observation exists: the turbulent fluxes

A claim is withdrawn here. The surface energy budget (`scripts/analysis/snow_energy.py`) appeared to show the model venting 3–5 W m^{–2} more upward sensible heat than ERA5 in February–April, which would have made turbulent exchange over snow the round-15 target. ~~The model loses its March–April melt energy to the atmosphere as sensible heat.~~ **That comparison has no observational authority:** ERA5’s `sshf/slhf` are pure HTESSEL output, unconstrained by any assimilated observation, so it compares our scheme to a sibling of itself. It survives only as a hypothesis. The same caveat applies to the cold-content test on `tsn` (model within a few tenths of a K of ERA5), which is likewise model-versus-model — though as a *null* result it is the less dangerous of the two.

No satellite product measures surface turbulent fluxes. Only eddy-covariance towers can, and the Siberian candidates (Spasskaya Pad, ZOTTO) are point sites, not a gridded field. Any round-15 lever aimed at turbulent exchange over snow would therefore be tested against its *consequences* — melt date and snow cover, both now observationally anchored — rather than against the flux itself.

25.4 Summary: what each reference can support

quantity	reference	status
surface albedo	CERES up/down	observational ; confirms model +0.046 June
snow cover extent, melt date	Rutgers/IMS	observational ; model 13 days late
TOA and surface SW fluxes	CERES	observational
snow water equivalent	ERA5	partly constrained (SYNOP, IMS); forest bias
snowfall, snowmelt	ERA5	model output, weakly constrained
sshf, slhf, str, tsn	ERA5	unconstrained — HTESSEL output

ERA5 assimilates IMS snow cover, so ERA5 agreeing with Rutgers on extent is *expected* and is not independent corroboration — but it is what justifies having treated extent as ERA5’s trustworthy field throughout.

26 Route B resolved: HTESSEL has no sub-grid snow depletion

Rounds 13 and 14 both failed on Route B, and the observational work (§25) sharpened the target without naming the cause. It is now named, and it also explains *why both rounds failed in the way they did*.

26.1 The arithmetic that settles it

Snow cover fraction is set from the **cell-mean** snow depth alone (`surfbc_ctl_mod.F90:317–322`):

$$\text{ZCVS} = \min(1, \bar{d}_{\text{cm}} \cdot \text{RQSNCR}), \quad \text{RQSNCR} = 1/10.$$

So cover can only fall below 1 once the *cell mean* drops under 10 cm. The model’s actual depth distribution over the box:

	mean d	p_{10}	p_{50}	p_{90}	frac < 10 cm	ZCVS
Mar	51.1	31.4	48.7	75.6	0.037	0.964
Apr	48.9	24.8	47.2	75.8	0.044	0.964
May	28.8	4.5	27.1	56.5	0.181	0.890
Jun	4.8	0.0	2.7	13.5	0.842	0.380

The median May cell still holds 27 cm, so the formula returns essentially full cover — while the satellite record shows 0.647. Real 100 km regions at 27 cm mean depth are patchy: ridges blow bare, hollows hold metre drifts, canopies intercept. The model has no representation of that variance.

26.2 Why both previous attempts had to fail

This is the part that makes the diagnosis convincing rather than merely plausible.

- **Round 13 was doomed by arithmetic.** Changing RQSNCR $1/10 \rightarrow 1/30$ moves the threshold from 10 to 30 cm — which does nothing at all where depths are 27–51 cm. It could only bite where depths sit *near* the threshold, i.e. 13–27 cm, which is October and November. That is precisely where the response appeared (–8.2 albedo points in October, –0.95 in June, §23.2). The lever was not weak; it was arithmetically incapable of acting in spring.
- **The commented-out `tanh` and `sqrt` variants** at `surfbc_ctl_mod.F90:325` share the defect — still functions of the cell-mean depth. $\tanh(27/10) = 0.99$.
- **Snow brightness is the wrong direction** (model snow is darker than both ERA5 and CERES) and **energy supply is not short** in Mar–Apr (§25.1).

26.3 The fix, validated offline before implementation

A depth-only function cannot work; the depletion must respond to something that tracks the *state* of the pack. Niu & Yang (2007), as used in Noah-MP and CLM:

$$\text{SCF} = \tanh\left(\frac{d}{2.5 z_{0g} (\rho_{\text{bulk}}/\rho_{\text{new}})^m}\right)$$

The density ratio supplies exactly the missing behaviour: aged, melting snow is denser, so it patches out at a given depth. It needs **no new prognostic** — depth and density are already carried.

Tested on existing output *before* writing any code, Siberian box, `amip_presentday` vs Rutgers 1990–2014:

	satellite	as-released	$z_0 = 0.01$	$z_0 = 0.02$
Jan–Mar	0.999	0.964	0.964	0.952
Apr	0.950	0.964	0.948	0.893
May	0.647	0.890	0.772	0.588
Jun	0.203	0.380	0.321	0.180

$z_0 = 0.02$ matches the melt season closely at a small cost in midwinter cover; $z_0 = 0.01$ is conservative, leaving winter untouched and closing about half the May–June gap.

Provenance. The problem and the two standard remedies are well established in the land-surface literature: Liston (2004) sets out why a cell-mean depth cannot represent depletion; Niu & Yang (2007) give the density-dependent tanh form used here, calibrated against North American basin observations and adopted in Noah-MP and CLM; Swenson & Lawrence (2012) give the main alternative, separate accumulation and depletion curves driven by a snow-depth standard deviation, which is more faithful but needs an extra prognostic; Roesch et al. (2001) take the third route of adding sub-grid *orographic* variance in a GCM. Douville et al. (1995) is the classic depth-threshold form that HTESSEL’s RQSNCR formulation resembles, and Dutra et al. (2010) documents the LESN09 scheme this campaign has been modifying. That CMIP-class models share the bias found here is documented by Thackeray et al. (2015), and its consequence for the snow-albedo feedback spread by Qu & Hall (2014) and Lorant et al. (2014). Betts & Ball (1997) remains the reference for boreal canopy masking, which is why the tile-7 residual behaves as it does (§24.1).

Implemented as ECE_SNOW_SCF in &NAMSURFTUNE (§24.7), with ECE_SNOW_SCF_ZO, _M, _RHONEW. Default is 0 = the as-released formula, so the change is inert unless switched on, consistent with every other entry in that namelist. Verified by a 1-day run: the switch logs to NULOUT, no namelist errors, the model steps normally.

26.4 What this still does not tell us

- **The temperature leverage remains unproven (§58.4).** G4 gained a full kelvin with snow mass unchanged and cover *worse*. Fixing the cover may buy much less T2m than its -5.6 W m^{-2} suggests. Where it should matter is *coupled*, through LPJ-GUESS growing-season onset.
- **The March–April mass deficit is a separate, weaker claim.** It rests on ERA5 SWE (forest-biased) and may be a second effect or partly an artifact (§24.2).
- **Comparability.** Rutgers is a *binary* 24 km classification, so its 0.647 is “65 % of cells classed snow”, not a true areal fraction. The melt-out *date* is the robust comparison; mid-melt absolutes are softer, and the z_0 choice should not be over-fitted to them.

27 Round 15 (2026–08–04): the snow fix works, buys no summer, and cools winter

Three runs, all namelist-only on one binary: **I1** = G4 + depletion, **I2** = depletion alone (isolating the snow route from the vegetation route), and **I3** = G4 + the SDOR-scaled variant. Sections §27.1–§27.5 were written on I1 and I2 alone; **I3 is evaluated in §27.6**, and it changes the round’s verdict — the snow route *does* buy summer temperature, but only in the scale-aware formulation, and it presents a bill in the Arctic energy budget.

27.1 The scheme does exactly what it was built to do

Siberian snow cover	satellite	control	G4	I1	I2
Apr	0.950	0.962	0.961	0.931	0.931
May	0.647	0.891	0.893	0.703	0.721
Jun	0.203	0.413	0.447	0.160	0.175
melt-out	11 May	26 May	28 May	12 May	13 May

Melt-out went from **15 days late to 1 day late**, against the satellite record.

27.2 And it delivers no summer temperature

	surface SW gain	JJA T2m	K per W m^{-2}
G4 (vegetation route)	+8.10	+0.952	0.12
I2 (snow route alone)	+2.91	+0.094 (noise)	0.03

This settles §58.4: the link is weak. Route B was a genuine, satellite-measured radiative bias with almost no temperature leverage — per watt it is **four times less efficient** than the vegetation route. The reading is clean: energy added over a *melting* snowpack goes into melt, not sensible heat, which is precisely what it did. Fixing it remains right — the snow field is now correct, which is what matters for LPJ-GUESS growing-season onset — but Route B is not where the remaining boreal kelvin lives.

27.3 The winter response is a compensating error being removed

Scored against ERA5 rather than against the control (period-clean, `amip_presentday`):

bias vs ERA5 [K]	DJF	MAM	JJA	SON
control	−1.95	−1.97	−2.58	−2.69
G4	−2.33	−2.33	−1.63	−2.51
I1	−4.71	−2.28	−1.33	−3.46
I2	−4.63	−2.36	−2.49	−3.77

The control is too cold in every season, so there was never a warm bias to protect — scoring against observations rather than against the control is what makes that visible, and eleven rounds had not done it.

What this does NOT establish. ~~The excess snow cover had been propping DJF up by ~ 2.7 K, masking a ~ 4.7 K bias that was there all along.~~ That reading was asserted here and is **not supported**: winter cover is *unchanged* between the two schemes (0.963 vs 0.964, both saturating at 43 cm, §27.4), so excess *winter* cover cannot have been doing the propping. What is measured is only that **I1 cools DJF by -2.76 K by an unidentified route**. Whether that unmasks a pre-existing bias or creates a new one is **untested**.

The point is sharpened by a precedent this report got the other way round: H1 produced the same signature (-1.233 K DJF by cutting snow-tile fraction, §23.2) and was *rejected* for it. Identical evidence cannot be disqualifying in one run and revelatory in another. Until the route is identified, the honest position is that the snow fix and the winter response are two findings, not one story — which is also why round 16 tests the winter on its own terms rather than assuming it is recoverable.

27.4 Ruled out by measurement: it is not a radiation-supply problem

Downward longwave over the box, model vs CERES [W m^{-2}]: Oct **−12.9**, Nov **−11.9**, Dec -3.1 , Jan $+0.1$, Feb $+5.6$ — **DJF mean $+0.9$, essentially exact**. That kills the Pithan et al. (2014) supercooled-liquid route for midwinter. The Oct/Nov deficit is real and is a *separate* autumn problem. The $+30$ pp polar-night cloud excess against CERES is retrieval failure over snow at night, not evidence.

Still unresolved: winter cover is **unchanged** (0.963 vs 0.964 — both formulas saturate at 43 cm) and winter SWE is slightly *higher*, so the DJF response is not a direct cover or mass effect. `str` and `sshf` shifts are the surface being colder, i.e. responses. ~~`stl1` and `skt` are unusable (~ 0.07 K — the same dead-field trap as pressure-level `r`).~~ **That was wrong**: both fields are sound (210–319 K). The 0.07 was 273/3600 — an analysis-side bug that divided an *instantaneous* temperature by the accumulation period. Fixed in `boxcache.py`; unlike pressure-level `r`, which really is identically zero, these fields were never broken. Using them (§27.5) shows I1 and I2 cool `stl4` by ~ 2.1 K, so the unexplained winter signal reaches the *deep soil* — which matters because LPJ-GUESS is fed `ST1L–ST4L`. H1 cooled DJF by -1.233 through a different route to the same tile fractions, so this is the second time cutting snow-tile fraction has cooled winter hard.

27.5 Does the warming reach the skin and the soil?

A lever that raises 2 m air temperature by mixing heat *away* from the surface would improve the metric we score while degrading the surface itself. That is not a cosmetic concern: **LPJ-GUESS is fed `ST1L–ST4L_LPJ` (soil temperature, layers 1–4) and `SM1L–SM4L_LPJ` alongside `T2M_LPJG` (`lpj_guess/framework/OasisCoupler.cpp:17–95`)**, so a warmer-air/colder-ground configuration would hand the vegetation model an incoherent state. Skin temperature itself is not coupled.

Siberian land box, JJA, deltas against control [K]:

	T2m	skt	stl1	stl4	$\Delta(\text{skt}-\text{T2m})$
<i>control (absolute)</i>	9.73	10.18	9.31	−1.31	0.45
G4	+0.952	+0.990	+0.883	+0.157	+0.037
I1	+1.250	+1.318	+1.160	−2.092	+0.068
I2	+0.094	+0.100	+0.046	−2.190	+0.006

G4’s warming is genuine, not a redistribution: skin and layer-1 soil warm *as much as* the air and the skin-to-2 m gradient barely moves (+0.037 K). What LPJ-GUESS would receive from G4 is coherent.

But I1 and I2 cool the deep soil by ~2.1 K, I2 doing so with essentially no surface change at all. The unexplained winter response of the snow scheme therefore reaches **st14**, which *is* a coupled field. That is a second, independent reason to keep the snow-depletion scheme out of a coupled run until round 16 explains the winter behaviour.

This subsection exists because the diagnostic was wrongly believed impossible: the soil and skin fields had been written off as dead (§27.4) on the basis of an analysis-side scaling bug.

27.6 I3 (2026–08–04): the strongest summer lever, and the Arctic bill

I3 = G4 + mode 2, the SDOR-scaled depletion, calibrated to match I1 *over the Siberian box* so that any remaining difference would be spatial structure (§26.3). It does not behave like I1, and the differences are the point.

What it wins.

	control	G4	I1	I3	CERES
Siberian JJA T2m [°C]	9.728	+0.952	+1.250	+1.322 ($t=10.61$)	—
Siberian surface net SW [W m^{-2}]	153.78	+8.10	+12.77	166.58	166.26
global T2m RMSE vs ERA5 [K]	1.579	−0.036	+0.048	−0.023	—
global net TOA [W m^{-2}]	0.643	−0.128	−0.021	+0.056	—
SO SW RMSE	6.877	−2.076	−1.906	−2.065	—

+1.322 K is the **largest JJA gain of 36 evaluated tuning runs**, and the Siberian surface SW deficit that §22 opened this whole thread with **closes to within 0.3 W m^{-2}** of CERES. Global TOA does not move, and where I1 *degrades* global T2m RMSE (+0.048), I3 improves it. Per season it dominates I1 outright — more JJA, 0.32 K less DJF damage, and I1’s significant autumn penalty (−0.766*) collapses to a null:

Δ vs control [K]	DJF	MAM	JJA	SON
threshold (own season)	± 0.601	± 0.394	± 0.244	± 0.441
G4	−0.385	−0.360	+0.952*	+0.185
I1	−2.759*	−0.306	+1.250*	−0.766*
I3	−2.439*	−0.214	+1.322*	−0.105

What only the full guardrail set revealed. The Siberian-box scripts alone would have promoted I3. They were run first, and the verdict “best lever of the campaign” was drafted on them — prematurely, and caught only because the guardrails were demanded before writing. Four guardrails move together, all one way:

guardrail	control	I1	I3	CERES
net TOA 60–90N [W m^{-2}]	−97.74	+0.87	−96.02 (+1.72)	−97.98
net TOA NH [W m^{-2}]	0.280	+0.245	0.952 (+0.672)	0.08
NH–SH albedo	−0.003	−0.002	−0.007 (−0.004)	—
Nordic Seas SW RMSE	9.058	+0.196	9.557 (+0.499)	—

This is the mechanism *working as designed*: removing high-latitude snow cover admits more shortwave at high latitude. The Arctic did not have that energy spare. The control sat almost exactly on the observation at 60–90N (−97.74 vs −97.98); I3 leaves it 1.96 away. The NH net TOA change (+0.672) and the NH–SH albedo change (−0.004) are each the **largest of all 37 runs**, and Nordic Seas is the **only significant SW RMSE degradation in the campaign** — `rmse_significance.py` flags exactly two runs there and I3 is one (+0.07 against a ± 0.051 threshold).

Caveat against these four, stated so it is not over-read: these are PI runs scored against present-day CERES, and this report’s own rule (§58.7) is that the PI target is net TOA ≈ 0 , **not** the CERES column. That weakens the “away from CERES” reading of the two regional TOA rows. It does *not* weaken the NH–SH albedo or Nordic Seas results, which are model-versus-model and model-versus-RMSE.

Why this matters more for the coupled run than for AMIP. All four are Arctic energy-*gain* guardrails, and AMIP structurally hides their consequence: prescribed SST and prescribed sea ice absorb 1.7 W m^{-2} without responding. In the coupled PI spin-up — the configuration I3 is a candidate for — they would not. This is the same structural blindness as §58.5, in the opposite direction.

27.7 The winter penalty is not radiative

§27.4 left the DJF route unidentified. It can now be excluded as radiative. Siberian land, DJF, all four configurations:

Siberian DJF	snow albedo	SWE [m]	absorbed SW [W m^{-2}]
control	0.6789	0.6933	8.16
G4	0.6760	0.6901	8.18
I1	0.6800	0.6926	8.16
I2	0.6796	0.6958	8.16
I3	0.6797	0.6944	8.22

Albedo, snowpack and absorbed shortwave are **unchanged to three decimals**, and there are only $\sim 8 \text{ W m}^{-2}$ of insolation in the box in midwinter regardless. **A -2.4 K penalty cannot be constructed from that budget.** This generalises the finding that killed the compensating-error story (§27.3, winter cover 0.963 vs 0.964) from cover to the whole radiative chain, and it holds for I2, which carries the penalty with no vegetation change at all. The route must be **thermal** — **the snow/exposed-tile coupling** — which is precisely what round 16 (§28) tests, so the diagnosis and the running experiment agree.

27.8 Direct observations at last: RIHMI-WDC station soil temperature (2026–08–05)

Every soil claim in this report had rested on ERA5, whose land surface *is* HTESSEL — the same snow scheme, the same soil thermal conductivity — so it is a sibling, not a witness. **RIHMI-WDC (Sherstyukov v3)** ends that: daily soil temperature at Russian meteorological stations, 1963–2024, twelve depths from 2 to 320 cm, measured **under natural cover** so the snowpack is intact. 43 of the 110 stations lie in the campaign box. Model and ERA5 are sampled *at the station points*, not box-averaged, which is the one representativeness control available without land-cover weighting.

What the archive cannot do: depths 2/5/10/15 cm carry *no* winter data — the Russian network withdraws the shallow Savinov thermometers for the cold season. So **st11** (0–7 cm), where the damage is largest, **has no observational counterpart**; **st12** is the shallowest verifiable layer and bounds it. Only `tsoil_qc == 0` is used.

	obs depth	OBS	model	bias	OBS offset	model offset
st12	20 cm	−6.07	−5.03	+1.04	+17.78	+20.11
st13	40/80 cm	−3.58	−3.19	+0.39	+20.28	+21.95
st14	120–240 cm	+0.34	−0.23	−0.57	+24.18	+24.85

Model is `amip_presentday` (1990–2014), period-clean. Offset = soil minus 2 m air, each dataset using its *own* air, which is what makes it insensitive to the 2.8 K air-temperature change the I-series also carries.

Two results, both DJF-only. In *winter* the control soil is correct to $\sim 1 \text{ K}$ at every verifiable depth — there was no pre-existing *winter* soil bias to remove — and the model is mildly *over*-insulated, by +0.7 to +2.3 K of offset.

~~The control soil is correct to $\sim 1 \text{ K}$ at every verifiable depth.~~ **That was measured in DJF only and does not hold in summer** (§27.9): the same control is -3.4 to -5.4 K too cold in JJA, a bias *larger* than the 2.58 K Siberian air bias this campaign has been chasing. The winter statement stands; the unqualified one does not.

~~The winter soil-air offset under a deep Siberian snowpack is +10 to +20 K.~~ That was quoted from memory of the permafrost literature and is too low: the station record gives +17.8 to +24.2 K. The correction strengthens rather than weakens the conclusion below.

Scored on offset (the tuning runs are 1872–1915, so absolutes are not period-comparable):

	st11	st12	st13	st14
control	19.87	20.39	22.24	25.13
G4	20.27	20.81	22.72	25.82
I1	1.53	2.61	6.36	15.25
I3	1.41	2.54	6.46	15.63
OBSERVED	—	+17.78	+20.28	+24.18

I1 is **-15.2 K wrong at 20 cm**, -13.9 at 40–80 cm, -8.9 at 1.2–2.4 m. In raw terms its soil sits at -25.8°C where stations measure -6.1°C . The snowpack has stopped insulating at every depth.

27.9 The summer soil bias, and why it dwarfs the air bias

The DJF validation above was extended to JJA at the same 43 stations, and it changes the picture completely.

amip_presentday	OBS	DJF model	bias	OBS	JJA model	bias
boreal st12 20 cm	−6.07	−5.03	+1.04	13.68	10.33	− 3.35
boreal st13 40/80 cm	−3.60	−3.19	+0.42	10.17	5.84	− 4.33
larch st12 20 cm	−11.05	−7.76	+3.29	12.52	8.33	− 4.19
larch st13 40/80 cm	−7.80	−5.54	+2.26	7.64	2.29	− 5.35

The soil is 3–5 K too cold in summer while being slightly too warm in winter, and the summer soil bias is *larger than the 2.58 K Siberian air bias* the campaign has spent 45 runs on. It was never looked at because no soil diagnostic was scored until §27.8.

It is far worse coupled. LPJ-GUESS’s soiltemp25cm is the OIFS soil temperature received through OASIS, so it measures the same quantity:

larch zone, ≈ 25 cm	DJF	JJA
RIHMI obs (20/40 cm bracket)	−11.4 / −8.3	+ 12.5 / + 10.0
06 Baseline	−15.96	+1.80
080a CRUNCEP	−16.35	+1.38
09A newSeaIce	−18.09	+0.72
10A (G4)	−21.04	− 0.37
10B (G4 + snow depletion)	−20.85	+ 2.31

Against an observed $\approx +11.8^\circ\text{C}$ at 25 cm, every coupled run sits at $+0.7$ to $+2.3^\circ\text{C}$: a **~ 10 K summer soil cold bias**. **Larch cannot build cover in soil near freezing all summer**, which makes this a more plausible driver of the forest collapse than either the evergreen gate or grass competition (§31) — those describe *which* PFT wins, this describes why none of them grow.

Representativeness caveat, and it is a real one. RIHMI plots are open, bare or short-grass sites in full sun; a forested cell’s summer soil is legitimately cooler under canopy shade. Part of AMIP’s -4 K is therefore representativeness and cannot be separated without land-cover weighting. What is *not* representativeness is the ~ 6 K gap between AMIP and coupled at the same stations with the same comparison — that is real degradation contributed by the coupled vegetation feedback.

27.10 The snow scheme, reconsidered on coupled evidence (2026–08–05)

~~ECE_SNOW_SCF is switched off and stays off; no production run should set it.~~ **Retracted the same day.** That verdict was reached on AMIP alone, and **AMIP was the wrong testbed for it**: it has no vegetation response, which is precisely the thing the scheme was helping. On the coupled 10A/10B pair — identical binaries, identical settings (ECE_SNOW_SCF=1, ZO=0.016), differing *only* in the scheme:

larch zone	BNS LAI	C3G LAI	grass : larch	TREEFPC
09C reference	0.096	0.400	4.2×	0.048
10A (G4)	0.019	0.388	20.4×	0.019
10B (G4 + scheme)	0.134	0.830	6.2×	0.065

10B carries $7\times$ the larch and $3.4\times$ the tree cover of 10A, and G4 alone (10A) is *worse than the 09C reference* — it actively degrades the forest. NH total LAI bias against the Lund CRUNCEP reference halves, -0.62 (-28.1%) $\rightarrow$ -0.28 (-12.9%).

And the winter soil damage does not appear coupled. 10A and 10B are -21.04 and -20.85 in DJF — identical — while JJA goes $-0.37 \rightarrow +2.31$. The ~ 5 K winter cooling relative to the 06/070/080a baselines came in with G4 + CERES-init, *not* with the scheme. And 10A’s summer soil is the outlier against every earlier baseline ($+1.4$ to $+3.0$); **the scheme restores the baseline range rather than departing from it**, recovering 2.7 K of the ~ 12 K deficit.

Standing position: the scheme is not rejected. The AMIP soil error is real, but it is not disqualifying for coupling, where the scheme is the difference between a badly degraded and a substantially recovered boreal forest. **RETRACTED the same day — see §32.** The claim that the winter damage does not appear coupled was an *artefact of where I sampled*: I measured only the larch zone, where 10A is already saturated near -21°C , so both runs looked alike. Across the full station network 10B is -16.2 K in DJF and -18.3 K in January. The AMIP testbed had predicted exactly this.

Why. Verified against *direct observations* — RIHMI-WDC station soil temperature, 43 stations in the box, under natural cover (§27.8) — the scheme destroys the winter snow insulation. Observed thermal offset (soil minus 2 m air) is +17.8 K at 20 cm, +20.3 at 40–80 cm, +24.2 at 1.2–2.4 m. The control reproduces those to within 0.7–2.3 K. I1 gives +2.6, +6.4, +15.3: an error of –15 K at 20 cm. The control soil itself is correct to ~1 K at every depth the Russian network reports in winter, so this is a **new error, not the removal of an old one**. Since ST1L–ST4L feed LPJ-GUESS, that alone disqualifies it for coupling.

What we still do not know, and it matters. The mechanism is *unidentified*. The monthly decomposition rules out the obvious candidates: Siberian snow cover changes by –0.02 in October and SWE not at all, so it is neither a cover nor a mass effect. The soil *leads* the air (October: soil –4.19 K, air –0.02), so it is a surface-coupling change rather than an atmospheric one. Winter snow density falls 20–27 kg m^{–3}, which with unchanged SWE means a *deeper, less dense* pack — that insulates *better*, so density is a consequence of a cold pack, not its cause. ~~The tanh formula bites hardest in October when snow is shallow, cutting cover and exposing the soil.~~ Measured: cover is essentially unchanged. That explanation was written into `soil_temp_vs_obs.py` before it was checked and is wrong.

Reactivation therefore needs three things, in order:

1. **Diagnose ZCVS.** It is not in the output stream, which is why the mechanism is still open — every field we do write exonerates itself. Nothing should be rebuilt before the snow-tile fraction can actually be seen.
2. **Accumulation/ablation hysteresis.** Our implementation applies one depletion curve year-round. Both Niu & Yang (2007) and Swenson & Lawrence (2012) use *different* curves for accumulation and ablation, and that omission is the most likely structural defect. It would also match the measured timing: the benefit is May–June, the damage begins in October, so the two are cleanly separable in season.
3. **Re-verify on the SOIL OFFSET, not on snow cover.** The scheme was accepted in round 15 because it fixed melt-out to within a day of satellite. That was true and it was not sufficient. §27.8 is the test any successor must pass.

And weigh the prize before spending the effort. I2 — the scheme alone, no vegetation levers — bought +0.094 K JJA, inside the noise floor. The value of the snow fix is not temperature; it is that the snow field becomes correct, which matters for LPJ-GUESS growing-season onset. That is a real but narrow benefit, and it does not justify a source change plus a run family while a 20 K soil error remains unexplained.

28 Round 16 (2026–08–04): the snow-tile skin conductivity

`vlamsk_mod.F90` hardcoded the snow-tile skin conductivities, so they were untunable. λ_{sk} sets how tightly the skin couples to the medium below it; a small value lets it radiatively decouple and crash in polar night. Physically $\lambda_{sk} \approx k_{\text{snow}}/d_{\text{skin}}$ with k_{snow} 0.1–0.4 W m^{–1} K^{–1} and d_{skin} 0.01–0.05 m, so **roughly 2–40**. As released:

constant	value	tile
ZSNOW	7	5, exposed snow — the weakly coupled one, and the tile covering tundra (25.6 % of the box)
ZSNOW_GLACIER	8	5, where SWE > 9000 mm
ZSNOWHVEG	20	7, snow under high vegetation
ZLARGESN	50	5, when melting

The as-released 7 sits at the decoupled end of the physical range, on precisely the tile whose *fraction* round 15 corrected. Now exposed as `ECE_LAMSK_SN/_SNGL/_SNHV/_SNMLT`, all defaulting to the shipped values.

run	setting	prediction
J1	I1 + <code>ECE_LAMSK_SN</code> : 15	DJF +1 to +3 K, JJA barely moves
J2	I1 + <code>ECE_LAMSK_SN</code> : 25	same, stronger

Falsifier on record: if DJF does not respond, skin conductivity is not the route, and the winter bias lies in boundary-layer mixing instead (Holtzlag et al. 2013; Sandu et al. 2013). Also watched: melt-out and May/June cover must *stay* fixed — drift back toward the control would mean the two changes interact and the pair is not separable.

28.1 Round 16 result (2026–08–05): falsified, and the falsifier was on record

Δ vs control [K]	DJF	JJA	SON	global T2m RMSE
I1 (the base)	−2.759*	+1.250*	−0.766*	1.627
J1 (λ_{sk} 15)	− 2.834*	+1.313*	−1.132*	1.581
J2 (λ_{sk} 25)	− 3.226*	+1.246*	−0.879*	1.597
<i>control</i>	—	—	—	1.579

The prediction was **DJF +1 to +3 K**. Measured: DJF got **worse**, and *monotonically with the knob* — −2.834 at $\lambda_{sk}=15$, −3.226 at 25, against I1’s −2.759. Both runs carry a global T2m RMSE *worse than the control*.

The pre-registered falsifier fires: skin conductivity is not the winter route. By §27.7 the route is non-radiative, and λ_{sk} was the leading non-radiative candidate; with it excluded, **boundary-layer mixing** (Holtslag et al. 2013; Sandu et al. 2013) is what remains. The round is doubly moot in practice, since its base I1 is now deactivated (§27.10) — but the falsification stands on its own and the diagnosis it leaves behind is the useful part.

Both J runs were built on I1, before §27.6 showed I3 to be the better base in every season. That was the correct decision with the information available and the round is still diagnostic — it tests whether λ_{sk} is the winter route at all, and §27.7 shows I1 and I3 share the same non-radiative penalty, so an answer on I1 transfers. But if λ_{sk} does prove to be the route, **the production pair must be re-run on I3**, and the Arctic guardrails of §27.6 re-checked on the combination rather than assumed to superpose — superposition has been wrong in sign twice in this campaign (§58.7).

29 Land albedo (2026–08–04): half of it is the snow tile

With the boreal budget nearly spent (§21.3) the target moved to the ~ 0.7 K *universal* land bias, and §21.1 localised its energy to a clear-sky shortwave deficit — cloud is the wrong sign. Decomposing that deficit assumption-free (each dataset divided by its *own* downward flux) gives an all-sky land surface albedo +0.0154 too high against CERES.

29.1 Snow is a tile, not a vegetation type

The first attribution split that +0.0154 by *dominant vegetation type*, built from tvh/cvh/tvl/cv1. That map has no snow in it: in HTESSEL snow is a separate tile (5 on low vegetation, 7 under high) that **overlies** every vegetation type. So each type was part-time snow, and the ranking the split produced was largely a map of *snow duration* — evergreen needleleaf is snow-covered 53.9% of the year, tundra 63.2%, “bare soil” 49.2% (a bucket holding the Sahara and high-Arctic bare ground together). An annual-mean $T_{2m} > 5^\circ\text{C}$ filter does not fix this: a cell averaging 6°C still has months of snow.

Masking snow **per cell per month** (SWE < 1 mm, sunlit months only) splits the error almost exactly in half:

Δ albedo vs CERES	all sunlit months	snow-free only
ALL LAND	+0.0154	+0.0080
$\Rightarrow$ <i>snow contributes +0.0074, the snow-free surface +0.0080</i>		

and the per-type ranking *inverts*:

type	snow-covered	all months	snow-free
Evgr Needleleaf	53.9%	+0.0250	+0.0032
Bogs/Marshes	37.1%	+0.0143	+0.0039
Evgr Shrubs	41.2%	+0.0150	+0.0058
Evgr Broadleaf (tropical)	6.9%	+0.0017	−0.0003
Crops	25.9%	+0.0215	+0.0204
Semidesert	22.5%	+0.0342	+0.0169
bare soil	49.2%	+0.0240	+0.0145
Irrig Crops	24.6%	+0.0253	+0.0138
Tundra	63.2%	+0.0064	+0.0105

Every forest type falls inside +0.006 once snow is removed, tropical broadleaf to −0.0003. **This vindicates §22.1 and §22.3:** RVVEGALB’s high-vegetation entries are sound and the snow season really is the story. What is *new* is the residual +0.0080, which sits on the sparse and cultivated surfaces where the soil background shows through — and on tundra, which gets **worse** under masking (+0.0064 $\rightarrow$ +0.0105) because snow was hiding a genuine snow-free summer error. Nothing in 41 runs has touched that half.

29.2 The snow half already has a lever

Land albedo over snow-covered sunlit months, model-internal:

run	snow-month albedo	vs G4	all-land
G4	0.5185	—	0.2586
I1 = G4 + mode 1	0.5011	−0.0174	−0.0024
I2 = mode 1 alone	0.4996	−0.0189	−0.0018
I3 = G4 + mode 2	0.4689	−0.0496	−0.0052

I3 removes ~70 % of the snow half, I1 ~32 % — and I3 was calibrated to match I1 *in the Siberian box*, so globally it is twice as strong as intended. That is new information about mode 2 and it is the same over-reach that §27.6 bills at 60–90N: the SDOR scaling does most of its work outside the box it was tuned in.

29.3 Why the two halves cannot simply be stacked

The obvious next move — keep I3 and add a vegetation-albedo correction on top — is **not additive in the direction that matters**. A darker land surface at high latitude adds absorbed shortwave to precisely the region I3 has already pushed 1.72 W m^{-2} past CERES. The two changes compete for one energy budget. Consequences for the plan:

- **The mid-latitude and subtropical entries are the safe ones.** Crops (+0.0204), semidesert (+0.0169), bare soil (+0.0145) and irrigated crops (+0.0138) lie outside the Arctic, so correcting them attacks the $\sim 0.7 \text{ K}$ *universal* bias without adding to the 60–90N surplus.
- **Tundra (+0.0105) is the dangerous one.** It is the single entry that *would* stack with I3, being boreal and in-season — which makes it both the most attractive for the summer target and the one most likely to break the Arctic guardrails.
- **The snow-free half is not reachable by RVVEGALB alone** where bare fraction dominates; there the background soil albedo climatology carries it, and that is an ancillary field, not a namelist parameter (§22.2).

Assets: land_albedo_snow_split.py, chained into evaluate.sh -obs.

30 Round 17 (2026–08–05): K1 adopted, and land albedo closed out

Two families on the G4 base, run concurrently and deliberately orthogonal: **L** puts a moss/organic layer on the boreal types, **K** corrects the snow-free land albedo diagnosed in §29.

30.1 L: the moss proxy is falsified, monotonically the wrong way

λ_{sk} $10 \rightarrow 2.9$ (L1) and $10 \rightarrow 1.7$ (L2) on types 4 (larch) and 9 (tundra) only — the two most boreal-confined types, 62 % and 70 % of their area poleward of 50N. Gaillard (2025) attributes most of a $>2 \text{ K}$ high-latitude summer gain to a surface organic layer, which HTESSEL lacks entirely.

	JJA	vs G4	verdict (threshold ± 0.244)
G4	+0.952*	—	the base
L1 (λ_{sk} 2.9)	+0.784*	−0.168	inside $\rightarrow$ null
L2 (λ_{sk} 1.7)	+0.638*	−0.314	outside $\rightarrow$ significant degradation

More moss is worse, the opposite of the prediction, and the monotonic ordering makes a noise explanation implausible. **Caveat that limits the claim:** λ_{sk} is only a *proxy* for an organic layer — it changes the skin-to-substrate coupling but not the heat capacity, not the moisture retention, and not the summer evaporative regime. This falsifies *the proxy*, not necessarily the Gaillard mechanism. A real organic layer would need a soil-column change, which is a much larger piece of work.

30.2 K: adopted

K1 = G4 + RVVEGALB on crops ($\times 0.887$), irrigated crops ($\times 0.910$) and semidesert ($\times 0.906$) + ECE_TUNE_SOILALB 0.95. K2 is the soil background *alone*, isolating the $\sim 25 \%$ of land that RVVEGALB cannot reach. **Tundra was deliberately excluded** so K stayed orthogonal to L and did not add absorbed shortwave to the Arctic (§29.3).

	Siberian JJA	DJF	net TOA 60–90N	global T2m RMSE
control	—	—	–97.737	1.579
G4	+0.952*	–0.385	–97.603	1.543
I3	+1.322*	–2.439*	– 96.017	1.556
K1	+1.036*	–0.496	–97.463	1.524
K2	+0.913*	–0.144	–97.525	1.518
<i>CERES</i>	—	—	–97.98	—

Both are clean — every season inside its own threshold — and they are the **two best global T2m RMSE of the campaign**. The tundra exclusion did its job: 60–90N stays near the control, against I3’s +1.72 W m^{–2} excursion. In the Siberian box neither K is significant against G4 (+0.084, –0.039, both inside ±0.244), *which was predicted in advance* — K’s mass is crops, semidesert and the subtropical desert belt.

30.3 What K bought, measured on the metric it was designed for

global LAND, vs G4	land T2m	land SW _{net}	land albedo
K1	+0.089 K	+0.77 W m ^{–2}	–0.0059
K2	+0.076 K	+0.48	–0.0038

The lever did what it was built to do. The snow-free residual was +0.0080; K1 removed –0.0059 of it, **74 %**. The measured sensitivity is 0.089/0.77 = **0.116 K** per W m^{–2} against the 0.12 assumed — essentially exact. The geography is right too: land equatorward of 60N warms +0.109 K while land poleward of 60N *cools* 0.055 K, exactly what excluding tundra was meant to achieve.

~~Predicted ~1.1 W m^{–2} and 0.1–0.2 K of land warming.~~ Measured 0.77 W m^{–2} and 0.089 K — the SW gain was 70 % of the prediction and the temperature at or below its low end. The sensitivity assumption was right; the forcing estimate was optimistic.

THE NEGATIVE RESULT IS WORTH MORE THAN THE 0.089 K. Three quarters of a *directly measured* land albedo bias has now been removed, and it recovered 0.089 K of the ~0.7 K universal bias. Extrapolating, a *perfect* albedo fix buys ~0.12 K — about **one sixth** of the target. **Land surface albedo is therefore closed out as the explanation for the universal cold bias**, which is a real constraint on where the remaining ~0.6 K lives.

~~K3 should push the soil-albedo scale further, since K2 alone gives the best global RMSE.~~ **Retracted before it was run.** The remaining 26 % of the residual is worth roughly 0.03 K, and the 0.95 scale was calibrated to the *measured* –0.0163 bias — pushing past it means tuning *against* the observation, which is the same overfitting-to-present-day trap that ruled out editing the soil ancillary (§29.3). **K1 is adopted and albedo tuning stops here.**

31 The boreal forest fails by TWO different routes (2026–08–05)

The report has carried a single story for the missing forest: the winter evergreen cold gate (§6.6). Evaluating the Tuning_test_07*/08* LPJ-GUESS families (lai.out, fpc.out, est_limits.out) shows that story is **correct but only half the picture**, and that it does not apply at all to the region that matters most — NE Siberia, which is ~80 % larch.

31.1 BNE is gate-excluded; BNS is permitted and outcompeted

Larch zone, 60–72°N × 100–160°E, 2520 cells, last decade:

	080a (CRUNCEP base)	09A (newSeaIce)
mTmin20	–38.94 °C	–41.89 °C
BNE gate t _{cmin} _est= –30	0.7 % clear	0.0 % clear
BNS cold gate	none, 100 %	none, 100 %
agdd5 vs BNS gdd5min_est= 350	83.6 % clear	79.0 %
BNE LAI	0.000	0.000
BNS LAI	0.129	0.103
C3G LAI	0.751	0.510
grass:larch	5.8×	5.0×
cells with C3G > BNS	65 %	60 %

The two PFTs fail for completely different reasons.

- **BNE/BINE: a climate gate.** 0.7% of larch-zone cells clear `tmin_est`= −30, and their LAI is *exactly* 0.000. The winter-cold story of §6.6 is right, and this is where it applies — the evergreen zone of west Eurasia and North America, *not* NE Siberia.
- **BNS (larch): no gate at all, and it loses.** It has no cold limit and clears its `gdd5min_est`= 350 on 84% of cells, so it is **climatically permitted** across the zone — yet grass carries 5–6× its LAI, grass exceeds larch on 65% of cells, and larch is absent outright from half the cells where it is allowed. **Larch is not excluded; it is outcompeted in the growing season.**

This is a correction to this report, not a new hypothesis. §6.6 already states that BNS has no cold gate and is “*already permitted*” — and then never explains why it is absent anyway. A reader was left applying the winter-gate mechanism to a zone where it explains nothing, with only the roughness leg (§6.5) for the 80%-larch region.

31.2 Reconciling this with the falsified “competition, not climate”

§60 carries “*the boreal failure is an LPJG-internal competition error*” as dead, killed by the forcing-transfer test reproducing most of the loss with byte-identical parameters. **That falsification stands and is not weakened here**, because it and the result above are different claims:

claim	status	
competition is an LPJG <i>parameter error</i>	dead	forcing reproduces the loss unchanged
competition is the <i>pathway</i> by which climate removes larch	supported	permitted on 84%, loses 5.8:1

Forcing is the *driver*; grass-versus-larch is the *mechanism*. Both hold at once. As written, the report recorded only the falsification, so a reader would reasonably conclude that growing-season competition had been ruled out — it had not been tested.

No contradiction with §6.4 either, though the two look opposed. That section measures vegetation→climate (the LPJG cold effect on T2m, which vanishes in July–August). This measures climate→vegetation (how the growing season sets larch cover). Opposite directions of the same coupling; both can be true.

Weak evidence, flagged as such. Weakening grass (C3G `pstemp_low`= 12, run 08G) moves BNS 0.129 → 0.156 and TREEFPC 0.060 → 0.072 while C3G falls 0.751 → 0.583 — directionally exactly what competition predicts. But the 08 runs sit on *different atmospheric bases* (06T vs 06V), so this is not a clean attribution and should not be quoted as one. The gate/permission evidence above is the load-bearing part; the knob evidence is suggestive only. On the old forcing, **07A (BNS `greff_min` relaxed) did not recover BNS at all** (0.024 → 0.021), which is why a forcing-consistent re-spin remains the precondition for any competition tuning (§14.3).

32 The snow scheme, settled on coupled soil (2026–08–05)

A colleague’s figure — OpenIFS `stl2` against RIHMI-WDC at 105 stations, coupled gen-10 runs — overturned §27.10. Reproduced independently to within rounding:

stl2 – RIHMI 20 cm [K]	DJF	MAM	JJA	SON	annual
AMIP K1 (adopted, tuned)	+0.49	−1.60	−1.44	−1.30	−0.96
COUPLED 10A (G4)	−4.33	− 6.57	−3.62	−3.61	−4.53
COUPLED 10B (+scheme)	− 16.19	−6.04	− 2.44	−5.95	−7.66

January reaches −18.3 K. The scheme buys 1.2 K of summer and pays 12 K of winter. **It is rejected for coupling.**

Why the earlier verdict was wrong, recorded because the error is repeatable. I sampled the larch zone alone (60–72°N, 100–160°E), where 10A already sits near −21 °C in DJF — *saturated*. Both runs looked identical there. The damage is only visible across the wider, more southerly station network where 10A still retains its snow insulation. **I measured in the one place the signal was flat.** Worse, the AMIP run had already predicted it: the I1 delta is DJF −13.57 K, composing to ≈ −13.3 against the −16.2 measured. When the coupled LAI figure appeared I treated it as overturning the AMIP evidence instead of asking why the two disagreed.

32.1 Reconstructing ZCVS, and what it excludes

ZCVS is not an output field, and neither are the tile fractions ($ZCVS = FRTI(5) + FRTI(7)$ exactly, from `surfbc_ctl_mod.F90:421,4` since the three cover fractions sum to one). But it is a pure function of `sd` and `rsn`, both of which *are* written, so it can be reconstructed offline from each run’s own snowpack:

Siberian land	Sep	Oct	Nov	Dec	Jan	May
Δz_{cvs}	+0.026	−0.075	−0.008	−0.001	−0.000	−0.188
$\Delta soil$ [K]	+0.0	−4.2	−16.2	−22.2	−23.8	+2.4

Midwinter cover is identical (−0.0004 in January): at 0.43 m depth the tanh is saturated. So the −19 K January soil **cannot be a cover effect**. The cover difference *peaks in October exactly as the soil begins to diverge*, then vanishes while the soil runs on to −24 K — autumn seeding plus thermal memory, with nothing in midwinter needing to differ at all.

A mechanism proposed and falsified within the hour. ~~The snow fraction is applied twice on the conduction path: ZHFLUX is already area-weighted while ZSNCONDH is multiplied by PFRSN again, so the pack drains through a conductance scaled by area a second time.~~ **Wrong.** ZHFLUX is W per m² of *grid box* (which is why the /ZFRSN at `srfsn_driver_mod.F90:271` is commented out), ZDSN uses *grid-mean* SWE, and ZSNCONDH is therefore also per m² of grid box. Flux, conductance and heat capacity are all in the same units. There is no second division for the factor to pair with.

Supporting evidence for the surviving reading: the snowpack cools only 3.9 K and the air 2.6 K while the soil cools 23.8 K. In January the control holds a +21.7 K gradient between snow surface and soil; I1 holds +1.8 K. The insulation has gone, with SWE identical and density *lower* — deeper, less conductive snow. No physical field changed in a way that could do it, which is what pointed at autumn seeding rather than a winter process.

32.2 Rounds 19: what is running, and what would falsify it

Four AMIP runs on the K1 base at **full campaign length** (1870–1916, evaluate 1872–1915), with daily `sd/rsn/tsn/asn/st11/st12` added through a per-run `add_config_sources` override — so all four difference directly *and* are scorable against the standard thresholds. An earlier 10 yr version was cancelled: at that length they could not be judged against the 44-yr detection floors, which would have made them incomparable with the other 43 runs.

run	role	z_0	ρ_{new}	m	SCF Oct	SCF May
N1	reference, scheme off	—	—	—	—	—
N2	current parameters	0.016	100	1.6	0.894	0.817
O1	re-parameterised	0.018	170	4.0	0.995	0.495
O2	re-parameterised, milder	0.018	170	3.0	0.995	0.764

The O series rests on one piece of arithmetic. With $SCF = \tanh(d/L)$ and $L = 2.5z_0(\rho/\rho_{new})^m$,

$$\frac{x_{Oct}}{x_{May}} = \frac{d_{Oct}}{d_{May}} \left(\frac{\rho_{May}}{\rho_{Oct}} \right)^m = 0.469 \times 1.853^m.$$

Depth works *against* us — October’s pack is half as deep. Density works for us, and $m > 1.23$ is needed merely for October to hold more cover than May. $m = 1.6$ **clears that bar but gives a ratio of only 1.23**, hence a separation of 0.077: the −0.075 October deficit that seeds the winter collapse buys a spring depletion only 0.08 deeper. $m = 4$ gives 5.5. Raising m alone makes October *worse* (0.383), because $\rho_{Oct} = 170$ already exceeds $\rho_{new} = 100$ — so ρ_{new} must be recentred on autumn density first.

~~Density cannot substitute for melt state; the two regimes are inseparable by density at their respective depths.~~ **Wrong, corrected the same day:** that was argued from the current parameters rather than from what the formula can do.

Why this replaced the hysteresis it was meant to be. A ripe/not-ripe state test is *not* hysteresis — it is reversible, so cover would flip back every night and every cold snap, injecting a spurious diurnal oscillation into the tile fractions. Real melt-out is irreversible within a season. Swenson & Lawrence (2012) avoid this with a prognostic seasonal SWE_{max} , which is a new surface prognostic. **Density sidesteps the problem entirely:** it is a genuine physical memory variable, ripening monotonically and not walking backwards. And it is namelist-only.

Pre-registered falsifiers. (i) Winter soil must return to the N1 reference; if O1 still shows the −16 K DJF collapse the density route is dead and only the SWE_{max} rewrite remains. (ii) May/June cover must still deplete, or the summer gain is lost and O1 is all cost. *And this is curve-fitting until the daily data supports it:* $m = 4$ is far from Niu & Yang’s calibrated 1.6 and was derived from two box-mean monthly numbers, while a steep exponent amplifies cell-to-cell spread.

33 Round 19 results and Round 20 (2026–08–06): the tanh is falsified on observations, and the curve is rebuilt

The O series failed its own pre-registered falsifier, and so did every mechanism proposed to explain the failure. Three hypotheses died in sequence before the real one was found, and *two of the deaths were caused by my own analysis errors* — both recorded in §33.4 because both are easy to repeat.

33.1 The measurement: cover is identical, the soil is 23 K apart

All four runs at 46 yr on the K1 base, Siberian box, differenced against N1:

	$\Delta\text{cover DJF}$	$\Delta\text{SWE DJF}$	$\Delta\text{depth DJF}$	$\Delta\text{soil DJF}$	Jan f_{full}
N1 reference	—	—	—	—	0.960
N2 current $m=1.6$	−0.002	+0.65	+0.064	−22.77	0.773
O1 $m=4$	+0.001	−0.60	+0.047	−18.72	0.947
O2 $m=3$	+0.001	+0.12	+0.043	−16.93	0.958

DJF snow cover is identical to within 0.002 across all four runs while the soil differs by 23 K. October soil damage is $-4.94/ -5.02/ -4.95$ K — identical to 0.08 K — while October cover spans 0.10. Mean cover is not the winter channel, and no re-parameterisation of a cover curve could ever have been.

33.2 The mechanism: same mean, opposite state

The exposed area is $1 - \overline{\text{cover}}$ by definition, so it is *identical* between the schemes. Where it sits is not, and the soil response is nonlinear in it.

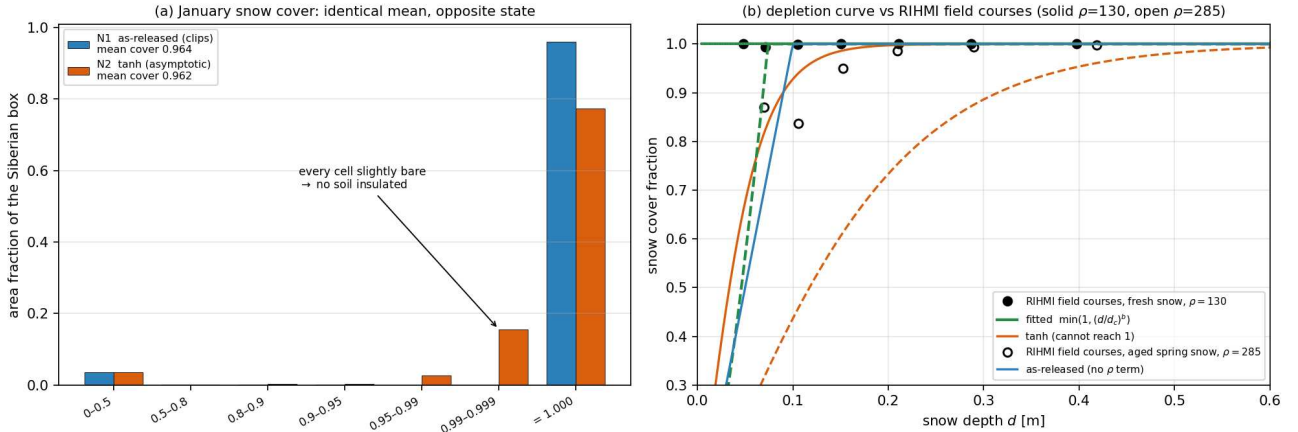


Figure 29: (a) January snow cover across the Siberian box. N1 and N2 have the same box-mean cover to within 0.002 and the same total exposed area, but N1 (clipping ramp) puts 93.5% of the box at cover *exactly* 1.0 while N2 (**tanh**) dumps a quarter of it into 0.99–0.999. A sliver of bare ground in *every* cell instead of a few genuinely bare cells: each sliver couples that cell’s soil to the air through exposed tiles at $\lambda_{sk} = 10 \text{ W m}^{-2} \text{ K}^{-1}$, about $16\times$ stiffer per unit area than the whole snow column (0.61), so no soil is left insulated. **(b)** The depletion curve against 36 492 RIHMI snow-course surveys. **tanh** has range $(0, 1)$ *open at 1* and so cannot represent complete cover at any parameters; the as-released ramp reaches 1 but ignores density and so has no spring depletion; the fitted form does both.

The signature in the model is unmistakable: **N1 holds a 22 K gradient from soil (265.0 K) to snowpack (242.7 K); N2 holds 2 K (240.4/238.6).** A 2 K gradient under half a metre of snow cannot be produced by conduction — N2’s soil sits at air temperature under a deep pack. `RTHRFTI = EPSILON (su0phy.F90:1308)`, so no tile is ever culled and every sliver is fully active.

f_{full} , the area fraction at cover ≥ 0.999 , is the only quantity found that predicts the damage: per-cell correlation with Δsoil is $+0.75/ +0.52/ +0.52$ against $+0.32/ +0.19/ +0.10$ for mean cover.

33.3 RIHMI-WDC settles it: the as-released cover was right

Snow-course transects (`snmar`, 1–2 km, the direct observational analogue of ZCVS): **14 331 of 14 369 Siberian DJF surveys report cover of exactly 10/10** — 99.74%, mean 0.9995. Complete cover is real, not an artefact

of clipping. Soil temperature (`tpg`, 43 stations, 174 676 DJF observations at 0.2 m against `st12`):

	f_{full} DJF	soil at 0.2 m	bias
observed	0.997	−6.1 °C	—
N1, scheme off	0.990	−5.3	+0.8
N2, tanh	0.678	−26.3	−20.2
O1	0.932	−22.2	−16.1
O2	0.965	−20.8	−14.7

Field January f (complete cover): **observed** 0.996, **tanh** 0.290. Period caveat: observations are 1963–2024 against a PI model, so the PI-equivalent observation is nearer $−9$ °C, making N1 $\sim +3$ K warm — the scheme is still $−13$ to $−18$ K wrong. **This retracts §27.3**: for soil temperature the as-released cover was correct and the scheme replaced a good field with a broken one.

33.4 Two analysis errors, and four dead mechanisms

~~The soil collapse is seeded by an October cover deficit and carried by thermal memory.~~ Killed by its own pre-registered test: O1/O2 produced *positive* autumn cover anomalies and still lost 18–20 K.

~~A melt-state (ripe/not-ripe) gate can separate spring from autumn.~~ May ripe-day fraction is 0.04 in every run; the gate would fire in June, after melt-out.

~~The residual exposed fraction at saturation explains it — tanh never reaches 1 while the ramp clips to exactly 1.~~ Right about the codomain, wrong about the box: N1’s own January cover is 0.963, not 1.0, so both schemes leave the *same* exposed area. The distribution, not the total, is what differs.

~~Drying the soil before freeze-up removes the latent-heat buffer.~~ The freeze-up buffer differs by only $−0.51/−0.72/−0.68$ K against 17–25 K to explain, and is not ordered with the damage — O1 has the largest water deficit and not the largest cooling.

~~More cover means a colder soil, because $ZSNCONDH \propto PFRSN$.~~ $dG/df = 0.61−10 \approx −9.4$: the exposed route swamps the snow route, so more cover means a *warmer* soil. The blanket intuition holds; the exposed fraction is what matters.

Both of my errors were the same error: averaging before applying a nonlinear function. `snow_daily_diag.py` averaged depth and density over 40 years and *then* applied the **tanh**; its Δcover numbers are void and the conclusions drawn from them were withdrawn. The first monthly fit check evaluated $\text{SCF}(\bar{d}, \bar{\rho})$ against $\overline{\text{cover}}$, which made a good fit look like it saturated in May. $\text{SCF}(\bar{x}) \neq \overline{\text{SCF}(x)}$ — **apply the formula per sample, always.**

33.5 Round 20: `ECE_SNOW_SCF` = 3, fitted to observations

A source change this time (`surfbc_ctl_mod.F90` branch, `surfece.F90` parameters); modes 1/2 retained so N/O stay reproducible.

$$\text{SCF} = (1 - c_{vh}) \min(1, (d/d_{cl})^{b_l}) + c_{vh} \min(1, (d/d_{ch})^{b_h}), \quad d_c = \min(D_{max}, S \cdot D \cdot (\rho/\rho_{ref})^M)$$

$\min(1, \cdot)$ **reaches 1 exactly** — that is the entire fix; everything else is calibration. It nests the as-released ramp at $D = 0.1$, $M = 0$, $b = 1$. Split by vegetation **type and fraction only** (via PCVH), never gridpoint or hemisphere, so it stays portable to other climate states and follows the vegetation if LPJ-GUESS moves the treeline. **No new prognostic and no new input**: PSNM1M, PRSNM1M and PCVH are already arguments of `SURFBC_CTL`, and **nothing knows the date** — the seasonal cycle is an output of the snow physics, not an input to the formula.

Parameters are fitted to 36 492 snow-course surveys, each measuring depth, density and covered fraction on the same transect, split by course type (1 = *pole*/field $\rightarrow$ low vegetation, 2 = *les*/forest $\rightarrow$ high). Constrained fit: hold Oct–Feb saturated, then minimise Mar–May error.

	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
field observed	0.998	0.999	0.999	0.999	0.999	0.994	0.974	0.914
field fit	0.999	1.000	0.998	1.000	1.000	0.996	0.970	0.915
forest observed	0.999	1.000	1.000	1.000	1.000	1.000	0.987	0.931
forest fit	0.999	1.000	1.000	1.000	1.000	1.000	0.987	0.931

Mar–May RMSE 0.0025/0.0001 against 0.0229/0.0156 (as-released) and 0.1207/0.0964 (**tanh**). **The frontier is flat** — the same parameters win at every winter floor from 0.000 to 0.998, so the winter/spring trade-off that dominated rounds 15–19 was an artefact of the **tanh** being unable to do either, not a real tension. Kept: $D_{cl} = 0.014$, $D_{ch} = 0.026$ m at $\rho_{ref} = 200$, $M = 4.70$, $b_l = 1.46$, $b_h = 0.40$, $D_{max} = 0.30$.

The physical reading is that d_c runs from ~ 2 mm at $\rho = 130$ to 8–14 cm at $\rho = 285$: fresh low-density snow drapes over everything, while old snow is dense *because* it has been wind-redistributed and melt-metamorphosed, and that same history is what leaves bare patches. **Density is the proxy for age and melt state** — the hysteresis the campaign spent two rounds trying to build, delivered by a state variable.

33.6 The runs, and what would falsify them

P1 (`amip_P1_scffit`, $S = 1$) takes the observations literally. P2 (`amip_P2_scffit_x3`, $S = 3$) scales d_c for 100 km sub-grid variance. Both on the K1 base at 46 yr with the daily snow/soil output retained so f_{full} can be measured directly.

S is the one uncalibrated knob and is labelled as such. A snow course is 1–2 km; a TCO95 box is ~ 100 km, and sub-grid variance grows with scale, so course-fitted d_c is a *lower bound* and $S > 1$ is the physically expected direction. At the Siberian box-mean May state ($d = 0.31$ m, $\rho = 313$) $S = 1$ gives $d_c = 0.118$ m, i.e. $d/d_c = 2.6$ and **no depletion at all** — P1 is expected to behave much like as-released in spring. Rutgers/IMS 24 km cover aggregated to the model grid is what should set S , and **that calibration is not done**; P2 is an exploratory bracket, not a candidate for adoption.

Pre-registered falsifiers. (i) The DJF soil must return to the N1/K1 reference; if it does not, the distribution diagnosis is wrong and this whole section fails. (ii) January f_{full} must return to ~ 0.96 from N2's 0.773. (iii) If the scaled variant recovers the JJA gain *but* reopens the winter soil bias, then spring depletion and winter insulation are genuinely coupled at grid scale and the flat frontier found on station data does not survive.

33.7 P1 crashed: the fit extrapolated where it had no data (2026–08–06)

P1 aborted at 1888–03–22 with ABOR1: `Very snow cold temperature (srfsn_webal_mod.F90:451, the PTSN < 100 K sanity guard):`

```
Tsn 91.93613   SWE-1 5.6058891E-02   SWE 0.0   Snow frac 1.000000   heat 10.88281
```

A pack of 0.056 kg m^{-2} — half a millimetre — was given *full* snow cover. The snow tile then absorbed the entire grid-box flux into essentially zero heat capacity and the snow temperature collapsed. The cause is that the fitted power law was extrapolated below any depth the snow courses ever sampled: there is no survey under ~ 5 cm, and at $\rho = 100$ the formula gives $d_c = 0.014(0.5)^{4.7} = 5.4 \times 10^{-4}$ m, so a 0.56 mm dusting reaches $d/d_c = 1.04$ and saturates. The governing ratio is SCF/SWE: as-released gives 0.100 at that state, mode 3 unfloored gives 17.86 — **180 $\times$ worse**. I had written down that the low-depth end was unconstrained and then extrapolated into it with no guard.

The fix is `ECE_SNOW_SCF_SWEMIN`, which floors d_c at SWEMIN/ρ : a minimum snow *mass* before cover may saturate, which is the right currency because the abort is a heat-capacity failure, not a depth one. $\text{SWEMIN} = 3 \text{ kg m}^{-2}$ is the *smallest* value that beats the as-released margin ($\text{SCF/SWE} = 0.053$ against 0.100) while barely touching the calibration — October field cover $0.999 \rightarrow 0.996$ against an observed 0.998, and Dec–May unchanged to three decimals. $\text{SWEMIN} = 5$, which I had picked by guess first, would have cost 0.983 in October for no extra safety that matters.

P2 was cancelled un-crashed and is also withdrawn: $\text{SCALE} = 3$ only makes d_c 1.6 mm instead of 0.5 mm, so it carries the same defect and its output is **TAINTED** wherever thin snow occurs, which includes every autumn onset and spring melt-out. **P3** ($\text{SCALE} = 1$) and **P4** ($\text{SCALE} = 3$), both with $\text{SWEMIN} = 3$, supersede them.

~~P1 was stopped by an unexplained scancel from the account UID rather than by a model failure.~~ Wrong, and the reasoning behind it was worse than the conclusion: the model aborted normally on `PROC=48` and `esm_tools` tore the job down afterwards, exactly as designed. I searched `NODE.001_01` (rank 1 only) for `fortrl/nan/ abort` and for the `esm_tools` trigger strings, none of which match `ABOR1` or `MPL_ABORT`, concluded “no model error”, and then constructed an elaborate infrastructure story on top of that gap. **Method note: an OpenIFS abort is ABOR1/MPL_ABORT and can land on any rank — grep the whole compute log for those tokens before concluding a job died externally.**

One tension, found while making Figure 29 and not resolved. The constrained fit optimises the *monthly climatology*, but the model applies the curve *pointwise*. Scored on the pointwise (d, ρ) bin surface instead, the adopted field parameters give RMSE 0.0624 against 0.0384 for a fit made directly to that surface ($b \approx 0.1$) — though both beat as-released (0.0771). For forest the adopted parameters are better on *both* metrics (0.0420 vs 0.0483; as-released 0.0644). So the field curve is a compromise: optimal on the seasonal cycle, second-best on the surface, and an improvement on the incumbent either way. Whether the climatology or the surface is the right objective is not obvious — the model integrates the curve over a cell distribution, which is closer to the climatology aggregation — and it should be settled alongside the Rutgers calibration of S .

Residual hysteresis is real but small. At matched (d, ρ) the ablation season is lower in *24 of 24* non-zero bins — systematic, $p \approx 10^{-4}$ — but the weighted mean is only -0.005 (field) and -0.001 (forest). Capturing it would mean plumbing PWSNM1M through the caller chain for ~ 0.005 of cover; not done. The matched test can only compare bins populated in both seasons, so spring’s extreme states ($\rho > 300$) are structurally untestable this way.

34 Round 20 results (2026–08–07): P3 adopted, and the fix costs the autumn

Both pre-registered falsifiers passed, and by more than they required.

	DJF soil vs N1	Jan f_{full}	soil→snow gradient, Jan
N1, scheme off	—	0.961	22.4 K
N2, tanh	−22.71 K	0.755	1.8 K (impossible)
P3	+0.30 K	0.965	22.9 K
P4	+0.16 K	0.965	21.8 K

AMIP OpenIFS/HTESSSEL Subsurface Soil Temperature versus RIHMI-WDC

OpenIFS st12: 7–28 cm layer (centre 17.5 cm) · RIHMI-WDC: observed 20 cm · equal-weight mean across 105 stations

Observations: QC=0, 1991–2020 · AMIP models: final 10 years (1908–1917, pre-industrial) · climatological, not date-matched

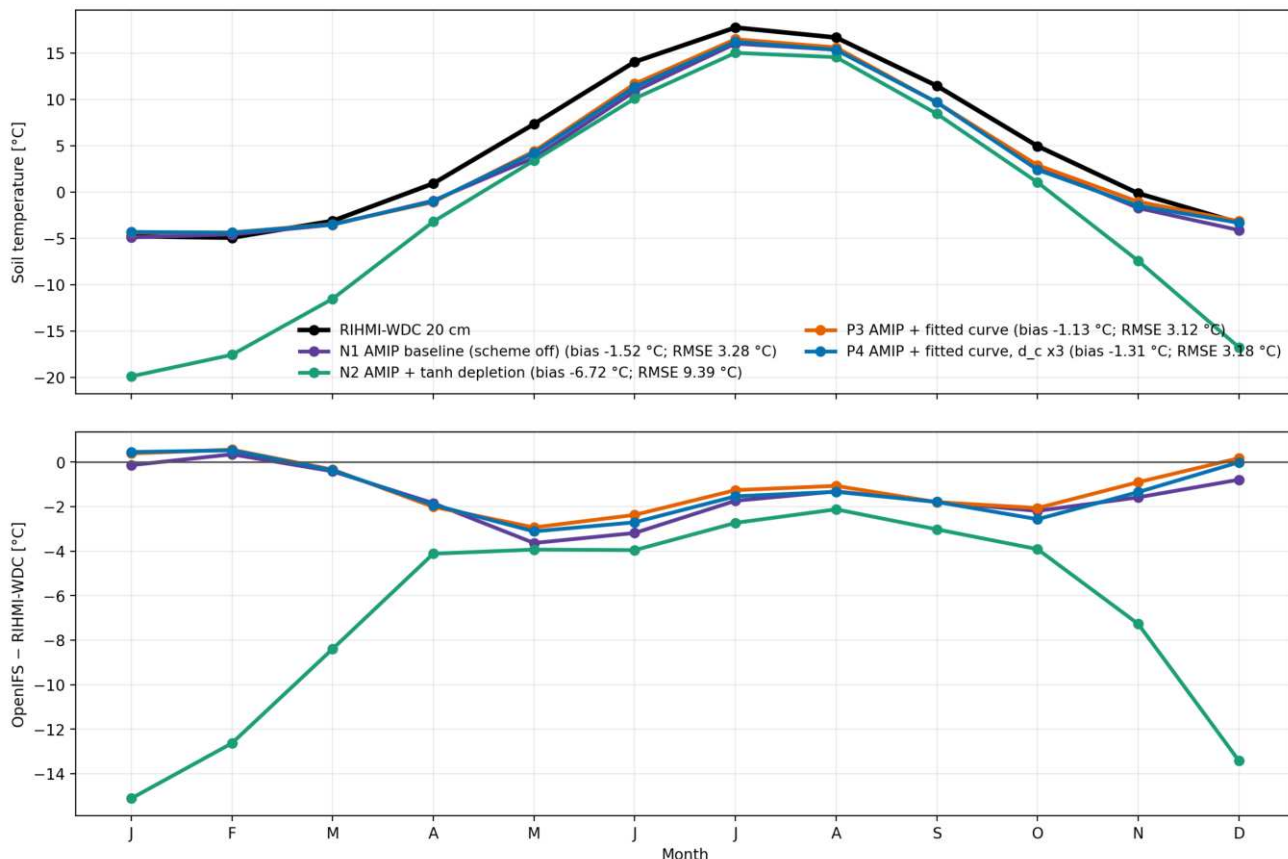


Figure 30: AMIP soil temperature against 105 RIHMI-WDC stations, built with the same methodology as the coupled 09C/10A/10B figure so the two are directly comparable (st12 7–28 cm vs observed 20 cm, QC=0, 1991–2020, final model decade). The **tanh** run (green) peels away from October and reaches -15°C in January — the same catastrophe 10B shows coupled. **P3 and P4 lie on top of the scheme-off baseline all winter, and slightly above it.** P3 beats that baseline on every metric: bias -1.13 vs -1.52 , RMSE 3.12 vs 3.28 , JJA -1.56 vs -2.08 . The residual bias is now entirely a growing-season deficit that no snow scheme can touch. Absolute offsets are not comparable with the coupled figure (prescribed vs interactive SST); the spread between lines is.

Seasonal ANOVA against the control, each season on its own threshold: P3 -0.584 / $+1.341^*$ and P4 -0.132 / $+1.092^*$ (DJF / JJA). **Neither appears on evaluate.sh’s “warms JJA significantly but cools DJF**

significantly” list, which contains I1, I3, J1, J2, N2, O1 and O2. The winter penalty that has followed this scheme since round 15 is gone.

Marginal over the K1 base, which is the honest accounting: P3 +0.305 JJA / -0.088 DJF; P4 +0.056 / +0.364. Most of the +1.3 is K1’s. P3 beats P4 despite SCALE = 1 being the conservative choice, which *falsifies my own prediction* that P3 would show no spring depletion — cutting cover *slows* box-mean melt, because the snow tile receives energy in proportion to its area.

~~SCALE is the parameter that controls the spring depletion, and the uncalibrated one that matters.~~ Varying it 1 → 4 moves the offline Sep–May RMSE by 0.002. P4 was a wasted run; SWEMIN is the whole lever.

34.1 Rutgers: the fix made the autumn 2.4× worse

Rutgers 24 km, Siberian box, area-weighted, model minus observed:

	Sep	Oct	Jan	Apr	May	Jun
observed (absolute)	0.067	0.599	0.999	0.956	0.655	0.202
N1 as-released	+0.107	+0.103	-0.035	-0.012	+0.118	-0.011
P3	+0.259	+0.249	-0.034	-0.019	+0.074	-0.035

The fitted d_c for fresh low-density snow is ~ 2 mm, so a dusting saturates. That is *correct* at 1–2 km, which is what the snow courses measure, and wrong at grid scale, where September Siberia is 7% covered rather than 33%. The uniform -0.035 in midwinter is the model’s permanent lake/water bare fraction, present in every run including scheme-off — not the scheme.

An offline SWEMIN sweep scored over the whole Sep–May season gives an *interior* optimum at 30 (RMSE 0.045 against P3’s 0.139); past it the September gain is paid for by November and December going deficient. **P5** (SWEMIN = 15) and **P6** (= 30) bracket the albedo→melt→SWE feedback the offline sweep holds fixed, so those numbers are lower bounds on the sensitivity.

34.2 Round 21: resolution awareness, and a cap that disabled the floor

Three source changes (562df81), one of them a real bug.

DCMAX was capping the mass floor. The order was $d_c = \min(D_{max}, \max(\text{fitted}, \text{floor}))$, so the 0.30 m cap killed the floor above SWEMIN = $D_{max}\rho - 30 \text{ kg m}^{-2}$ at $\rho = 100$. Every grid coarser than TCO95 therefore got no floor beyond the reference value, and it is *also* part of why the sweep flattened past 30 and reversed past 40, which I had attributed entirely to November. Now $d_c = \max(\min(D_{max}, \text{fitted}), \text{floor})$: the cap guards the steep $M = 4.7$ extrapolation of the *fit*, the floor is a separate physical statement about how much box-mean mass fills a box of this size. The two were nested and should not have been.

A resolution preset. ECE_SNOW_SCF_DXKM rescales SWEMIN from its TCO95 reference, $\text{SWEMIN}_{\text{eff}} = \text{clamp}(\text{SWEMIN} \cdot (\Delta x/100)^{0.624}, 3, 120)$, off by default so every earlier run reproduces. The exponent $\ln 10/\ln 40$ comes from the only two *measured* anchors: 3.0 at 2.5 km (the snow courses are 1–2 km transects, i.e. the high-resolution limit observed directly) and 30 at 100 km (Rutgers). Presets: TL21 94, TL63 48, TL255 26, TCO95 30, TCO399 13, TCO1279 6, TCO4000 3.

Below 2 km the scheme disables itself, falling back to the as-released ramp with a loud log line. At that scale the model resolves individual patches, so a subgrid curve double-counts them, and 2 km is the finest scale any calibration exists for.

~~The scale dependence can be measured as a power law by coarse-graining Rutgers and the model together.~~ It cannot, and the attempt is kept because the negative result is the useful part. The ladder confirms the *sign* — SWEMIN grows with box size, 41.5 at 66 km against ~ 114 at ≥ 133 km — but the values **plateau** beyond 133 km rather than following a power law; the per-cell fit implies ~ 80 at 100 km where the seasonal fit gives 30, a 3× disagreement that is the same climatology-versus-pointwise tension as the (d, ρ) surface fit; its own extrapolation to 2.5 km gives 9.4 against the course anchor of 3; and absolute skill is poor at every scale (RMSE 0.20–0.25 on a 0–1 field), which indicts the functional form rather than the parameter. **The preset is an interpolation between two trusted anchors, not a measured law, and a new resolution needs validation rather than a DXKM value.**

35 Round 18 results (2026–08–07): the aerosol term is six times smaller than inferred

M1 removes the anthropogenic MACv2-SP plumes entirely; M2 gives the same CAMS mass its 3D vertical distribution. Both sit on the present-day base, so the comparison is against `amip_presentday` over the same years.

Units first, because they invalidated the first pass. IFS TOA fluxes are *accumulated* J m^{-2} over the 3600s step, not W m^{-2} ; the raw numbers come out $\sim 3600\times$ too large. Dividing by the timestep reproduces the campaign’s own recorded decomposition almost exactly — global +2.68 clear-sky and -1.18 cloud against the recorded +2.68 / -1.17 — which is the check that the pipeline is right.

band	M1 clear-sky	t	M1 cloud	t
60–90N	+0.381	+3.8*	−1.014	−7.7*
30–60N	+0.699	+9.4*	−1.397	−13.3*
tropics	−0.037	−2.0	−0.439	−5.9*
SO 45–65S	−0.116	−18.9*	+0.022	+0.1
60–90S	+0.010	+0.3	+0.079	+1.0
GLOBAL	+0.126	+11.1*	−0.542	−8.9*

Paired annual differences over 25 yr; the runs share prescribed SSTs and years, so the pairing removes the forcing variability. * clears two-sided 95 % ($|t| > 2.06$).

~~Aerosol contributes $\sim 0.7 \text{ W m}^{-2}$, 13 % of the Southern Ocean’s +5.10 clear-sky excess, inferred from a +0.033 AOD excess against MISR.~~ **Measured, it is -0.116 W m^{-2} ($t = -18.9$) — six times smaller.** Removing *all* anthropogenic aerosol moves SO clear-sky reflection by a tenth of a W m^{-2} . Aerosol is $\sim 2\%$ of the SO excess, not 13 %, and **the unexplained residual grows from ~ 1.3 to $\sim 1.9 \text{ W m}^{-2}$.**

The two reflection columns are not separable. M1’s effect is overwhelmingly on *cloud* — -1.40 against $+0.70$ at 30–60N, -0.54 against $+0.13$ globally — through `LMACV2SP_CCNF`. Any future aerosol lever must be scored on both columns; a clear-sky-only argument is invalid. M2 moves the tropical clear-sky column by $+0.553$ ($t = +79$) at *fixed column mass*, so where the aerosol sits is a first-order term of its own.

Open and untraced: the M1 clear-sky deltas are mostly *positive* — removing aerosol *increased* clear-sky reflection ($+0.126$ global, $t = +11.1$). That is backwards for a direct effect and is significant, so it is not noise; most plausibly a surface-albedo or water-vapour adjustment to the changed surface SW, but it has not been chased down.

36 Round 21 results (2026–08–07): P5 adopted, and the summer ceiling

run	DJF	MAM	JJA	SON
K1 (base)	−0.496	−0.265	+1.036*	+0.238
N2 tanh	−2.783*	−0.599*	+1.350*	−0.888*
P3 SWEMIN = 3	−0.584	−0.333	+1.341*	+0.217
P5 SWEMIN = 15	−0.060	−0.140	+1.149*	+0.004
P6 SWEMIN = 30	−0.850*	−0.288	+1.096*	+0.487*

P5 is the cleanest run in the campaign — every season inside its own threshold bar a significant summer gain. Against Rutgers it cuts the autumn excess P3 introduced by $\approx 40\%$ (Sep $+0.259 \rightarrow +0.151$, Oct $+0.249 \rightarrow +0.148$) while leaving Nov–Jan intact, and its DJF soil is $+0.22 \text{ K}$ against the scheme-off reference.

~~Raising SWEMIN cannot reopen the winter, because at midwinter $\text{SWE} \approx 93$ the floor is 0.15 m against a 0.43 m pack — safe by construction.~~ True in *January*, and I never checked **November**, where the pack is still shallow. P6’s November f_{full} falls to 0.752 against N1’s 0.884 and its DJF soil to -1.00 K . The floor binds in early winter; that is the interior optimum the offline sweep predicted, arriving by the mechanism I had dismissed.

36.1 The JJA ceiling: the lever was nearly exhausted before round 15

Marginal over the K1 base, against the ± 0.242 threshold: P3 $+0.305$ (marginal), P5 $+0.113$ (noise), P6 $+0.060$ (noise). Only P3 buys a measurable summer gain, and P3 is the version with the worst autumn cover. The reason is structural. May cover against Rutgers’ 0.655 :

	May cover	vs observed	
N1 as-released	0.773	+0.118	(the entire available error)
N2 tanh	0.615	−0.040	(overshot <i>past</i> observations)
P3 / P5	0.729 / 0.732	+0.074 / +0.077	

The whole spring cover error is 0.118. P5 removes 0.044; a perfect correction removes 0.118, worth roughly +0.2 K of JJA over K1 on a crude scaling of N2’s response. **And N2’s larger gain was never legitimate** — it took 0.158 out of a 0.118 error, overshooting into a −0.040 deficit, so that extra warmth came from having too little May snow. It is the same over-depletion that destroyed the winter soil, and any comparison against N2’s +1.350 is a comparison against an error.

Honest accounting for the whole depletion line: worth $\approx 0.1\text{--}0.3$ K of Siberian JJA, not the ≈ 1 K the early rounds implied. What it delivered instead is a scheme that is no longer *wrong*. **The missing summer is elsewhere**: every AMIP run including scheme-off tracks RIHMI soil through Dec–Mar and falls 2–3.5 °C short from April to September (Fig. 30), a deficit common to all of them, and that is the global tropospheric cold bias of §21.1 — still untouched after 50 runs, and now the sharpest remaining target. **Further SWEMIN optimisation is not worth runs**: P3 and P5 differ by 0.19 K in JJA, at or below noise.

37 The coupled radiation objective, and why the goal moves to zero (2026–08–07)

The 10 series branches from the same point as round 09 and runs twenty years longer, so it extends the campaign’s central radiation figure rather than replacing it. 10A is 09C plus G4; 10B adds the **tanh** depletion since falsified (§34).

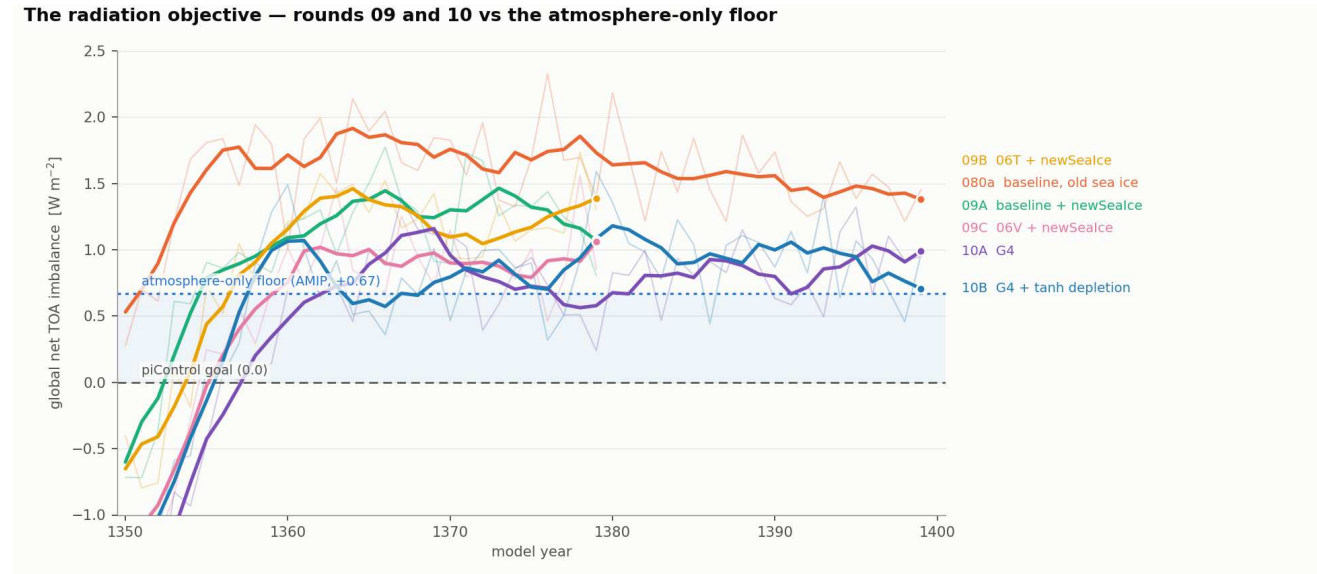


Figure 31: Global-mean net TOA by model year, annual (thin) and 5-yr running mean (bold). **G4 is worth $\approx 0.25 \text{ W m}^{-2}$ at top of atmosphere**: 10A settles at +0.854 against 09C’s ≈ 1.06 , the best of the previous round, and both 10-series runs spend 1362–1380 at or below the +0.67 atmosphere-only floor. **10A and 10B are indistinguishable** (+0.854 vs +0.899, curves interleaving throughout), so the depletion scheme that destroyed the winter soil by 20K cost essentially nothing at TOA — consistent with its damage being a distributional surface effect invisible to box-mean energy diagnostics. The dashed goal is now 0, not the inherited −0.16; the shaded band is the part of the gap ocean-side tuning could still close, and everything above the dotted line is atmospheric. 11C is absent: it aborted in its first leg on the crash of §33.7.

37.1 Why the goal is 0 and not −0.16

−0.16 was never derived. It is whatever the HR piControl settled at, adopted as a matching target. A target displaced from zero is only defensible if the atmosphere fails to conserve energy: with a spurious atmospheric source S , equilibrium gives $\text{SFC} = \text{TOA} + S$, so zero *ocean* drift would require $\text{TOA} = -S$. Whether the two conditions differ is measurable, not arguable.

Measured on 10A over 1390–1399, global, cos-latitude weighted, using the release tool’s own formula (**ssr** + **str** + **sshf** + **slhf** − **sf** · 3.3355 × 10⁸, each divided by the accumulation period):

term	W m^{-2}
ssr + str + sshf + slhf	+1.843
snow enthalpy (sf = 1.042×10^{-5} m/step)	−0.966
net surface	+0.877
net TOA (tsr +238.749, ttr −237.895)	+0.854
residual	−0.023

The atmosphere conserves to 0.02 W m^{-2} . There is no leak for an offset target to absorb, so “the ocean is not drifting” and “the system is not drifting” are the same condition, and both put the goal at **0**.

Snow enthalpy is doing nearly all of the reconciliation — 0.97 W m^{-2} of a 0.99 discrepancy. Omitted, the surface reads +1.84 against a TOA of +0.85, and that artefact is exactly the kind of apparent imbalance an offset target gets invented to absorb. Anyone comparing a raw surface flux against TOA without it will see a spurious $\approx 1 \text{ W m}^{-2}$ gap.

Re-framing the campaign arithmetic against the corrected goal:

	net TOA	= atmosphere +	coupled-state	gap to 0
080a (untuned coupled)	1.38	0.67 +	0.71	1.38
09C (06V + newSeaIce)	1.06	0.67 +	0.39	1.06
10A (+G4)	0.85	0.67 +	0.18	0.85
10B (+ tanh)	0.90	0.67 +	0.23	0.90

The conclusion is unchanged in kind and smaller in size: the ocean-side coupled-state term is now only 0.18 W m^{-2} in 10A, so **almost the entire remaining imbalance is the atmosphere** — the same +0.67 the AMIP testbed measures directly. Chasing −0.16 would have meant over-tuning by that margin against a target with no physical basis in this configuration.

~~The HR piControl goal of −0.16 is the target for the coupled spin-up.~~ Inherited, not derived; the budget closes to 0.02 W m^{-2} , so the goal is 0.

38 Round 22 (2026–08–09): the energy target re-based, and three targets relocated

Four diagnostics, no new model time except the coupled runs a colleague had already finished. Three of them move a target, and two of them falsify something this report was recommending.

38.1 Coupled round 11: the first runs carrying G4 and D2b

The report has said the coupled branch carries only G4 out of the whole AMIP campaign. **That is out of date.** 11A, 11B and 11D all carry `ECE_TUNE_RVRSMIN 1000/1000/225` and `RCL_INPSEA= 0.2` with `RCL_INPPMIN= 70000` — G4 plus D2b — on top of a new `useIFSsoiltemp` coupling. 110Baseline is 09C plus that coupling change alone and is therefore the correct control; comparing against 10A would confound the snow scheme with the soil-temperature coupling.

Δ vs 110Baseline	net TOA	glob T2m	Sib JJA	Sib DJF	soil DJF	SO cloud
11A +G4+D2b	−0.233	−0.150	−0.16	+0.13	−0.41	+0.84
11B + tanh snow	−0.060	−0.019	+1.81	−1.62	−19.40	+0.76
11D +fitted, <code>SWEMIN= 30</code>	+0.023	−0.031	+0.16	+0.05	−0.45	+0.41

The tanh falsification reproduces coupled on a clean pair: 11B minus 11A is −18.99 K of Siberian DJF soil, against the −13 to −16 K the AMIP run predicted. This is a better-controlled version of the 10B result, because 11A and 11B differ *only* in the snow scheme.

11D is not catastrophic, but it has not tested what it looks like it tested. 11D runs the fitted scheme at `SWEMIN = 30` — that is P6, the setting AMIP *rejected* (DJF −0.850*, November f_{full} 0.752, soil −1.00 K), not the adopted P5 at 15. Against 11A it gives soil −0.04 K and DJF T2m −0.08 K. Before reading that as vindication, the detection threshold was measured from the interannual scatter of these very runs:

	interannual sd	95 % threshold, 10-yr pair
Siberian DJF soil (<code>st12</code>)	0.80 K	$\pm 0.75 \text{ K}$
Siberian DJF T2m	1.33 K	$\pm 1.25 \text{ K}$

So the soil half of the P6 falsifier (-1.00 K) is marginally testable and comes out null; **the DJF T2m half (-0.850 K) is not testable at all on ten coupled years**, and a null there was the expected outcome whether the scheme is safe or not. What 11D *does* establish is that this diagnostic has ample power for damage of the kind that mattered — it sees 11B’s -19 K without difficulty. **The outstanding coupled run is 11D at SWEMIN = 15 plus K1**, which would bring the coupled configuration level with the atmosphere-only one for the first time.

38.2 The energy target, period-clean

Every band target in this report was measured on `amip_pi_base` against CERES 07/2005–06/2015 — an 1870s atmosphere against a present-day satellite record. `amip_presentday` overlaps CERES directly and its forcing is *correct* for 1990–2014 (unlike the PI arm, whose NCMIPFIXYR never took effect, §17.3), so the epoch offset is measurable per band rather than caveated. Band means over full latitude circles are exact on each source’s own grid, so nothing is remapped. The CERES read is validated against the published EBAF globals first: ASR 241.18 (published ~ 240.5), OLR 240.21 (~ 240.2), net $+0.970$.

band	PI vs CERES	PD vs CERES	offset	verdict
90S–65S	+7.24	+8.11	−0.87	robust to the epoch
SO 65S–45S	+6.32	+7.46	−1.15	robust, and larger
45S–30S	−4.46	−2.49	−1.97	partly epoch
tropics 30S–30N	−2.51	−0.67	−1.83	epoch-dominated
30N–45N	+0.81	+2.08	−1.27	partly epoch
45N–65N	+4.40	+5.35	−0.96	robust to the epoch
65N–90N	+0.39	+1.53	−1.14	partly epoch
GLOBAL	−0.33	+1.23	−1.56	

Two targets move. **The tropics are epoch-dominated** — period-clean the band is only -0.67 W m^{-2} from CERES, not -2.51 , so most of what §58.8 has carried as a tropical deficit was never model error. **The Southern Ocean survives the correction and grows**, $+6.32 \rightarrow +7.46$, and its contribution to the global mean rises to $+0.735\text{ W m}^{-2}$. It is now unambiguously the largest single regional term.

A small discrepancy, recorded. §11.2.2 quotes $+5.81$ for this band against a CERES value of -55.3 ; reading the gridded climatology directly gives -55.81 and hence $+6.32$ on the same PI arm. The new number comes with a validated global check and is preferred, but the difference is not chased further — it does not change any conclusion.

The OLR test, which does not close. A cold troposphere under-emits, so the measured OLR deficit and the measured cold bias should be quantitatively consistent. Period-clean, $\Delta\text{OLR} = -2.529$ and $\Delta\text{ASR} = -1.299\text{ W m}^{-2}$, which is where the $+1.229$ net comes from. But the effective emitting temperature from CERES OLR is 255.1 K , giving a Planck slope of $3.77\text{ W m}^{-2}\text{ K}^{-1}$, and the model is -1.62 K through 700–300 hPa: a purely radiative response would be -6.10 W m^{-2} . **Measured minus predicted is $+3.57\text{ W m}^{-2}$** , and §38.4 identifies what is offsetting it.

38.3 The Southern Ocean deficit is cloud AMOUNT — and no lever moves it

The band error has been tracked by CRE, which cannot distinguish too little cloud from cloud that is not bright enough. Splitting it at the *observed* CRE per unit area (`amip_presentday` vs CERES, period-clean):

SO 65–45S	model	CERES	diff
cloud area [%]	83.06	89.49	−6.43
SW CRE [W m^{-2}]	−60.39	−67.84	+7.45
CRE per unit area	−72.71	−75.80	+3.10
<i>decomposition of the +7.45:</i>			
AMOUNT (missing area $\times$ observed CRE/area)		+4.87	(65.5 %)
OPACITY (residual)		+2.57	(34.5 %)

Amount-dominated, and zonally uniform — Atlantic -5.40 , Indian -6.82 , Australian -6.55 , Pacific -6.81 pp, a sector spread of 1.42 pp, which points at microphysics rather than circulation or the ice edge. It is worst in austral spring and summer (SON -7.03 , DJF -6.94 pp), when there is shortwave to reflect: DJF alone carries $+11.67\text{ W m}^{-2}$ of net TOA error in the band. That seasonality is what the supercooled-liquid hypothesis predicts.

Then the part that falsifies the plan. Scored on cloud *area* for the first time, against the $+6.43$ pp that needs closing:

lever (44-yr AMIP, Δ vs control)	Δ area [pp]	Δ SW CRE	
D2a inpsea.2	+0.63	−4.80	9.7 % of the gap
D2b inp+p700	+0.43	−2.61	6.7 %
A1a ovl=0.10	+0.62	−6.44	9.7 %
G4 tundra225	+0.48	−2.73	a LAND lever
K1 landalb	+0.45	−2.62	a LAND lever
B7 rvice.22	−0.60	+0.66	wrong sign

G4 and K1 are boreal-land levers that cannot physically change Southern Ocean cloud area, and they move it by +0.48 and +0.45 pp, so ≈ 0.5 pp is the noise level of this diagnostic and every INP run sits inside it. **Wrong, and corrected the same day.** G4 and K1 are not independent of D2b — they *contain* it ($G4 = F4 + D2b + \text{tundra}$, $K1 = G4 + \text{land albedo}$), so their +0.48/+0.45 is D2b’s +0.43 inherited, not a null. Differenced against D2b the pure land contribution is +0.050 and +0.022 pp, which is the clean control and is genuinely negligible.

The real threshold, measured. Interannual scatter of SO cloud area in the 44-year control is 0.232 pp, giving a 95 % detection threshold of ± 0.098 pp on a paired difference — five times smaller than the figure asserted above. **Every lever in the table is therefore statistically significant.** The conclusion survives, but it rests on MAGNITUDE rather than on significance:

lever	Δ area	% of the 6.43 gap	Δ CRE	of which opacity
D2a inpsea 0.2	+0.626	10 %	−4.80	90 %
A1a ovlliqice 0.10	+0.623	10 %	−6.44	93 %
D2b inp+p700	+0.432	7 %	−2.61	87 %
A1b ovlliqice 0.35	+0.331	5 %	−2.17	88 %
<i>G4 minus D2b</i>	+0.050	—	−0.12	<i>the land control</i>

The best lever in the campaign closes one tenth of the area gap and takes ≈ 90 % of its CRE from opacity. Two independent mixed-phase routes agree — ice nuclei (D2a/D2b) and the liquid/ice deposition overlap (A1a/A1b, the latter pushed to EC-Earth4’s 0.1) — so this is a property of the mixed-phase branch, not of one parameter. **The amount deficit, two thirds of the error, is 90 % intact.**

~~The Southern Ocean cloud-area deficit should be attacked by pushing the INP branch harder, since D2b is the only lever that has moved the SO.~~ Written into the plan on 2026–08–09 and falsified the same day. The INP branch acts on opacity, which is 34.5 % of the error; its area response closes only one tenth of the gap, and the same is true of the deposition-overlap route even at EC-Earth4’s setting. A cloud-*amount* knob is required, and it is a different knob.

One more pointer. CERES puts the SO effective cloud pressure at 656 hPa, i.e. just *above* the 700 hPa ceiling of D2b’s pressure gate, with optical depth 7.41. The gate was designed to protect the tropics and it does, but it may also be excluding the layer where the observed cloud actually sits.

38.4 The cold bias is extratropical and UTLS — and the atmosphere is dry as well as cold

§21.1 established that above 700 hPa Siberia merely shares a global cold bias. Where that bias lives has never been asked. amip_presentday vs ERA5 1990–2014, annual, all levels exact in both so nothing is interpolated:

hPa	90S–60S	60S–30S	tropics	30N–60N	60N–90N	GLOBAL
850	+0.02	−1.17	−1.10	−0.90	−0.87	−0.99
700	−0.53	−1.40	−1.23	−1.00	−0.86	−1.15
500	−1.81	−1.75	−1.31	−1.56	−1.57	−1.49
300	− 3.65	−2.48	− 1.72	−2.34	− 3.43	−2.22
200	− 4.20	−3.28	− 2.09	−3.53	− 4.45	−2.87
100	+0.25	−1.52	−2.07	−1.30	+0.26	−1.51

The tropics are the *least* biased band, −1.72 at 300 hPa against an extratropical mean of −2.97. That excludes the hypothesis a reader would reach for first: a cold tropical upper troposphere over prescribed, correct SST would have meant a convective heating-profile error propagating globally. It is not that. The bias peaks at 200 hPa over *both* poles (−4.20 and −4.45) and **reverses sign at 100 hPa** (+0.25, +0.26) — a tropopause-region signature, not a tropospheric one. The seasonal tables behave the same way, so it is not a summer or winter effect.

And the atmosphere is dry, increasingly so with height:

hPa	ΔT [K]	q model	q ERA5	$\Delta q/q$ [%]
850	−0.99	5.7843	5.9887	−3.4
700	−1.15	2.9203	3.1596	−7.6
500	−1.49	1.0082	1.1018	−8.5
300	−2.22	0.1454	0.1695	−14.2
200	−2.87	0.0208	0.0260	−20.2
100	−1.51	0.0014	0.0026	−46.9

This is the missing 3.57 W m^{-2} of §38.2. The two errors oppose each other in OLR: the cold depresses emission, while the dryness lowers the effective emission level into warmer air and raises it. Over 700–300 hPa the model is -1.62 K and -10.1% relative humidity content — both large, both real, and their sum is small.

The consequence is a constraint on all future radiation tuning. Fixing either one alone moves net TOA by several W m^{-2} in the *wrong* direction. The global budget that §11.2.1 found “right by cancellation” is cancelling in the vertical as well as in the horizontal, and any lever scored on a global mean is being scored on top of that. *Caveat:* ERA5 upper-tropospheric humidity is itself weakly constrained by observation, so the 100 hPa row is indicative only; the 300–200 hPa rows carry the argument.

38.5 What round 22 changes about the plan

- **The Southern Ocean needs a cloud-amount lever**, not more ice nucleation. Two thirds of the error is missing area; the INP family moves area at the noise level. Whatever is chosen must be scored on area, and the D2b gate ceiling should be revisited against the observed 656 hPa cloud pressure.
- **The tropical TOA deficit is largely not model error** and should come off the open list at its old size.
- **The cold-bias investigation starts at the extratropical tropopause**, and it must treat cold-and-dry as one problem. A lever that warms without moistening will make the energy target worse.
- **The next coupled run is 11D at SWEMIN = 15 plus K1.** Ten years cannot resolve a 1 K DJF signal, so if the P6-versus-P5 question is to be settled coupled at all, it needs either a longer run or the AMIP answer taken on trust — and the AMIP answer says 15.

39 Round 23 (2026–08–09): the reference check, and a round withdrawn

Round 23 was designed to close the Southern Ocean cloud-amount deficit. Three of its four runs were withdrawn before completion — one as a duplicate, two on a design error — and the round’s lasting product is a reference check that decides which observation the campaign is allowed to tune against.

39.1 Prior work this campaign had not incorporated

Two AWI-ESM project issues (#169, #170, 2025–03 to 2026–04) cover the same Southern Ocean warm bias from the coupled side and contain results this report was working without.

- **The origin is cy48’s moist-physics upgrade.** Cycle 48r1 raised high cloud cover everywhere relative to 43r3, strongest in the SH high latitudes, and lowered SO low cloud (#170). The exact diff is the ECMWF Newsletter 164 moist-physics patch, which was evaluated mainly at $\sim 5 \text{ km}$ and mainly in low latitudes.
- **The lead/lag test, which orders the causal chain.** Scored on *year one* rather than the 50-year mean, the hcc offset is already present while the T2m bias and the SO lcc dip are *not*. So hcc leads and lcc lags: **fix hcc $\rightarrow$ reduces t2m $\rightarrow$ fixes lcc**. In the coupled model the SO low-cloud deficit is partly a *consequence* of the warm bias.
- **RVICE is already the hcc lever.** The v3.4 tuning raised it from the 0.13 source default to 0.16–0.19 (TUNE1–TUNE5, with NDECOLAD 2 $\rightarrow$ 0 and RALBSEAD 0.06 $\rightarrow$ 0.05) precisely to suppress SO high cloud. Every run in this campaign carries RVICE= 0.16 for that reason.
- **The compensating-bias trap is documented.** “All further tests regarding reducing HCC resulted in Arctic and Tropics becoming too cold”, and RALBSEAD was found “not a strong enough counter tuning method for the combined RVICE + ENTSTPC3”.

~~Lowering RVICE below 0.16 would add tropical cirrus and warm the tropics, providing the counter-lever the Southern Ocean levers need. Mechanically correct — B7 at 0.22 gives $\Delta_{\text{high}} -2.31$ pp and $\Delta_{\text{LW CRE}} -1.35$ — but it would undo the v3.4 SO tuning, which raised RVICE for exactly this reason. RVICE is global and the two regions need opposite signs, so it cannot serve both.~~

39.2 Four-way attribution: what kind of error each band has

Net TOA cannot distinguish a surface-albedo error from a cloud-reflection one. Splitting each band against CERES into clear/cloud $\times$ SW/LW, period-clean, positive = the model gains energy:

band	net	SW clr	SW cld	LW clr	LW cld	contribution
90S–65S	+8.11	+1.53	+3.98	+4.35	–1.75	+0.390
SO 65S–45S	+7.46	–3.79	+7.45	+4.21	–0.40	+0.735
45S–30S	–2.49	–2.54	–2.82	+4.66	–1.79	–0.257
tropics	–0.67	–1.39	–0.40	+3.28	–2.16	–0.336
30N–45N	+2.08	–4.90	+4.67	+4.36	–2.05	+0.215
45N–65N	+5.35	–2.67	+2.45	+5.11	+0.45	+0.527
65N–90N	+1.53	–3.79	–1.45	+6.30	+0.46	+0.073

The two problems are different cloud types. The Southern Ocean loses its energy through **SW cloud** (+7.45): the low cloud does not reflect enough. The tropics lose theirs through **LW cloud** (–2.16): the high cloud does not trap enough. With SST prescribed and correct, a tropical LW-cloud deficit is an anvil/cirrus problem — too low, too thin, or radiatively too weak — not a surface or a convective-heating one.

That is why no single microphysics knob has ever satisfied both, and it is also the shape of a solution: a lever that makes ice cloud *fewer but optically stronger* would move the Southern Ocean and the tropics in opposite, both-correct directions. RVICE changes amount alone, which is why it trades one for the other.

Scanning all 15 NAMCLDP-family levers for tropical response, only three warm the tropics at all: B3 clddiff= 1.5e-5 (+2.02, via SW not LW), B2 clddiff_convi (+0.65) and K1 (+0.37, clear-sky, its subtropical crop/semidesert albedo). **Nothing in 50 runs warms the tropics through the longwave**, which is the term that is actually deficient.

39.3 Which reference is right about Southern Ocean cloud?

The round-22 amount deficit rests on one source. Bringing ERA5 in appeared to contradict it outright, so before spending runs the references were tested against each other.

SO 65–45S	model	ERA5	CERES	MODIS
cloud area [%]	83.1	81.6	89.5	89.3
SW CRE [W m^{-2}]	–60.39	–61.52	–67.84	—
vs CERES	+7.45	+6.31	—	—

The test is what the cloud DOES, not how much is counted. Cloud area is not one quantity: CERES cldarea is a MODIS *mask* fraction that flags optically thin cloud in full, while ERA5 and model tcc are radiative covers built by overlapping layer fractions. Comparing them directly is a category error. Shortwave CRE is a difference of two broadband fluxes and *is* identically defined across all three.

If ERA5’s lower area were a definition artefact, ERA5 would reflect like CERES. **It does not: ERA5 has low area and low reflection, in proportion** — 6.31 W m^{-2} short where the model is 7.45 short. ERA5 is reproducing the model’s bias because ERA5 *is* IFS. **So CERES is the arbiter and the Southern Ocean amount deficit is real.**

~~The Southern Ocean cloud-area deficit may be an artefact of comparing a satellite mask against a model tcc, since ERA5 puts SO cloud below the model’s.~~ Raised as a concern and refuted the same day by the CRE test above. **ERA5 cannot be used as a cloud reference for an IFS-derived model at all** — its global cloud area is 62.8 against CERES’s 67.4, and in the tropics it reads 55.3 where the model’s 61.6 matches CERES’s 61.7 almost exactly. On cloud, ERA5 is the outlier, not the model.

A limit on CERES, in the other direction. At 90S–65S CERES reads 67.5 and MODIS 59.8 — a 7.7 pp disagreement between two products of the same instrument. Passive cloud masking over bright ice is unreliable. ~~The Arctic has +11.49 pp too much cloud (§38.2 band table).~~ Not supportable: that number is a cloud mask over sea ice and snow, exactly where the two retrievals disagree most. **CERES cloud area is trustworthy over dark ocean and should not be used over ice.**

39.4 The runs, and why three were withdrawn

Four levers were designed on the P5 base (which already carries G4, K1 and D2b):

run	lever	outcome
S1	RCLDIFF 6e-6 → 3e-6	cancelled — would cool the tropics
S2	RCL_OVERLAPLIQICE 0.65 → 0.35	withdrawn — this is A1b, already run
S3	RCLCRIT_SEA 2.5e-4 → 6e-4	cancelled — same reason as S1
S4	RCL_INPPMIN 70000 → 50000	running

~~RCL_OVERLAPLIQICE 0.65 → 0.35 is an untested mixed-phase lever.~~ It is **A1b**, run for 44 years in round 3; A1a is the same knob at EC-Earth4’s 0.1. The A-series label “ovl” denotes the liquid/ice *deposition* overlap, not the radiative cloud overlap, and it was read as the latter. The duplicate is not a total loss: A1a/A1b give a *second, independent* mixed-phase route, and it behaves exactly like the INP route (+0.62 pp of area for −6.44 of CRE, i.e. 93 % opacity), which is far stronger evidence that opacity-not-amount is a property of the whole mixed-phase branch than one parameter could be.

S1 and S3 were cancelled for the tropics, not for the Southern Ocean. Both add low cloud, and B3 — the same parameter as S1 moved the other way — *warms* the tropics by +2.02 W m^{−2}. S1 would therefore cool a band that is period-cleanly only −0.67 from CERES, overshooting it roughly threefold. The deficit they were built to close is real; a *global* low-cloud knob is the wrong instrument for it. Their runscripts are kept under runscripts/oifsamip/withdrawn/.

RCLCRIT_SEA and RCLCRIT_LAND were exposed to NAMCLDP for S3 (sucldp.F90: moved from the ASSOCIATE block onto the POINTER mechanism every other NAMCLDP variable uses, with the association placed before the defaults are assigned so a silent namelist still reproduces). The exposure is kept — it works, and the lever may be wanted later with better targeting.

39.5 What round 23 changes about the plan

- **The Southern Ocean deficit is confirmed and stays the largest single term**, but it needs a *regime-selective* reflection lever. Global low-cloud knobs overshoot the tropics; the D2b pressure gate is the template for what selective looks like. *Answered in §40*: marine CCN is that lever, it is selective by construction, and it reaches the opacity third only.
- **The tropics need longwave cloud *effectiveness*, not more cloud.** The deficit is −2.16 W m^{−2} of LW CRE, and no lever in 50 runs has moved the tropics through the longwave at all.
- **Reference hygiene is now a stated rule.** ERA5 is not an independent cloud reference for this model; CERES cloud area is valid over dark ocean and not over ice.
- **Check the prior issues before designing a round.** Both errors this round — the A1b duplicate and the RVICE proposal — were already answered in #169/#170.

40 Round 24 (2026–08–09): the marine CCN path, traced end to end

No runs. The question was whether cloud condensation nuclei offer the *regionally selective* Southern Ocean lever round 23 concluded was needed. The answer is a qualified yes, by a route nobody had looked at, and the trace corrected one of my own claims on the way.

40.1 The path, and where it is decided

Marine CCN reaches cloud albedo through four hops, each gated by a switch whose *runtime* value had to be read from `fort.4` rather than a source default:

where	what	runtime state
surfrad_ctl_mod.F90:772	Genthon (1992) sea salt from wind	LCCNO=. TRUE .
liquid_effective_radius.F90:153	CCN → N_d → r_{eff}	NRADLP= 2
cloudsc.F90:1023	MACv2-SP CDNC factor	LMACV2SP_CCNF=. true .
cloudsc.F90:2050	CCN → autoconversion (lifetime)	LAERLIQAUTOLSP=. FALSE .

With `LCCNO` true the code takes the `PCCN_SEA` branch, so **RCCNSEA is inert** — its own source comment (“now irrelevant”) is correct, but only because of a switch three files away. The live formula is

$$Q = e^{0.16u+1.44}, \quad \text{CCN} = 10^{1.2+0.5 \log_{10} Q}, \quad N_d = -1.15 \times 10^{-3} \text{CCN}^2 + 0.963 \text{CCN} + 5.30.$$

~~Marine CCN is pinned at a uniform 125 cm^{-3} because `NAERCLD=0` switches the aerosol–cloud path off. Read from a *default assignment* in `sucldp.F90` and reported as the runtime state, while `fort.4` carries `LMACV2SP_CCNF=.true.`, PCCN had already been printed as a 2-D field, and §35 of this report already documented that CCN path. Three independent disproofs, two of them on screen at the time.~~

40.2 The correction that matters: N_d is right

Evaluating Genthon on *monthly-mean* winds gives a Southern Ocean wind of 6.54 ms^{-1} and $N_d = 55$, which against an observed $60\text{--}120 \text{ cm}^{-3}$ looks like a clear deficit. **That was wrong.** Monthly *vector* averaging cancels the transient eddies that are the entire Southern Ocean wind field, and the parameterisation acts on instantaneous speed. On 6-hourly output:

SO 65–45S	wind [m s^{-1}]	CCN	N_d
monthly-mean wind (wrong)	6.54	55.6	55.3
6-hourly wind	10.28	77.6	73.5

~~Southern Ocean droplet number is too low, so the cloud is too dark, and raising marine CCN is the opacity lever. $N_d \approx 73 \text{ cm}^{-3}$ sits squarely inside the observed $60\text{--}120$. **The mean is correct.** This is the same error class as $\text{SCF}(\bar{d}) \neq \overline{\text{SCF}(d)}$ from round 19 — a nonlinear operator applied to a mean — committed hours after writing it into a skill as a rule.~~

What survives is the phase. The model’s seasonal cycle is inverted:

SO 65–45S	DJF	MAM	JJA	SON
N_d [cm^{-3}]	68.5	74.9	76.1	74.5
SW CRE error [W m^{-2}]	+10.86	+3.71	+2.09	+13.11

Model N_d is *lowest* in DJF because it is wind-driven and Southern Ocean winds peak in winter. Observations put the maximum in DJF, driven by biogenic sulphate. So the model has its fewest droplets exactly when the sun is up and the measured error is largest. Crudely — $\tau \propto N_d^{1/3}$, two-stream albedo $A \approx \tau/(\tau + 7.7)$ on CERES’s $\tau = 7.41$ — lifting DJF N_d from 68.5 to an observed ~ 95 is worth about -6 W m^{-2} of DJF SO CRE, roughly half that month’s error.

40.3 The amount two-thirds is unreachable from here

N_d acts through r_{eff} , i.e. **opacity**, which round 23 measured as 34.5% of the SO error. The cloud-*lifetime* route that would reach the amount two-thirds runs through `LAERLIQAUTOLSP/LAERLIQCOLL`, and both are `.FALSE`.

Enabling them would divide by zero. Traced: `callpar.F90:1165` sets `AUXL%ZCCN=0` when `NAERCLD ≤ 0`; `cloud_layer.F90:191` passes that array to `CLOUDSC` as `PCCN`; `cloudsc.F90:2050` then evaluates $(\text{RCCN}/\text{PCCN})^{0.333}$. Using the lifetime effect requires `NAERCLD > 0` and prognostic aerosol, which also changes the direct effect — so it stops being one lever. **Recorded so nobody enables the switch.**

40.4 The missing term, and that the data already exists

Genthon is sea-salt-from-wind with *no biogenic term*, and MACv2-SP — the only other live CCN path — is anthropogenic plumes only. Natural marine biogenic sulphate is absent from this model by construction, which is exactly why the seasonal phase is wrong.

The input already ships with OpenIFS and nothing reads it:

<code>climate.v020/95_4/month_dms</code>	<code>emdms, paramId 210043</code>
<code>units</code>	$\text{kg m}^{-2} \text{ s}^{-1}$ (DMS surface <i>emission</i>)
<code>grid</code>	N96 reduced Gaussian, 40 320 points — TCO95’s own grid
<code>time</code>	12 monthly fields
<code>land</code>	–99 sentinel

It exists at every resolution from TL21 to TCO1279. It is not ingested: ICMCL carries only `al`, `aluvp/d`, `alnip/d`, `lai_lv/hv`, and OpenIFS references DMS only in the TM5/M7 chemistry paths, which need `NACTAERO` $\neq 0$.

Route, via ocp-tool rather than mkclimr.sh. ocp-tool already owns ICMCL (`config.py:180–186`, `paleo_input.py:278–320`, `run_ocp_tool.py:189` step 14b) and, decisively, already applies a changed land–sea mask to it. `emdms` is an *ocean* field with the opposite land convention to albedo and LAI, so under a moved coastline new ocean points need infill and new land points need `–99`. `mkclimr.sh` cannot do that; ocp-tool can, which is what makes the field paleo-safe.

1. ocp-tool: append `emdms` to ICMCL in deterministic order — `ece_updcliclie_climr.F90:233` aborts on “Inconsistent order of fields”.
2. ocp-tool: extend step 14b for the inverted land convention.
3. Reader: one new `paramId` case (210043) beside the existing 174 / 15–18 / 66–67 branches. `max_fields=10` **against 7 in use**, so there is headroom for an eighth field without touching the limit.
4. Physics: an additive biogenic term on `PCCNO` in the `LCCNO` branch.

The interface is a DMS surface flux in $\text{kg m}^{-2} \text{s}^{-1}$, which is exactly what a REcoM DMS module would produce. So a prescribed climatology now and a prognostic source later are a drop-in swap at the ocp-tool boundary, with no change to items 3–4. That is a better seam than a chlorophyll relation, which would need its own empirical fit on both sides.

Unpriced, all cheap: the provenance of ECMWF’s `emdms` (Kettle & Andreae 2000 or Lana et al. 2011 — matters for citation); the flux→CCN relation, which is the one genuinely scientific choice and needs a published basis, not a fitted curve; and whether `–99` is the sentinel everywhere or only where it was sampled.

Why this is not scheduled first: the SO band is 9.8% of the globe, so closing the opacity third perfectly is 2.57 W m^{-2} in-band but only $\approx 0.25 \text{ W m}^{-2}$ global, against a +1.23 error. 11E is a namelist edit on a lever already validated over 44 AMIP years. Do the cheap decisive run first.

~~Prognostic marine biogenic CCN needs the `–cc` setup with REcoM ocean biogeochemistry.~~ REcoM carries no DMS tracer; the only DMS in the tree is Icepack’s sea-ice biogeochemistry (`dmspp`, `dmspd`, `dms`), which is DMS *within* sea ice and is in a component this configuration does not run (`whichevp=1`, FESOM’s native EVP). And there is no coupling field or atmospheric sulphur chemistry to receive it. The prescribed `emdms` route needs none of that.

40.5 T1 read test: passed, and the emdms data characterised

The 1-year read test completed with no abort and the reader’s own diagnostics confirm every pass criterion: Found 12 dates, Found 8 fields, `emdms` discovered as the eighth GRIB message matching `NCLIGC(8)`, and CLIMR slot 8 filled. Slot 8 takes **twelve distinct values through the model year**, following a clean seasonal march on rank 1’s northern high-latitude strip: 1.13 in January, peaking at 3.04 in June, minimum 0.30 in October, returning to 1.13. **The DMS ingest works.**

The data, characterised at full precision (source file and built ICMCL are identical, so the ocp-tool step is also verified):

land sentinel (<code>–99</code>)	29.9 % of points
ocean values	min 0.043, median 2.05, max 53.25
exact zeros	240 of 339 360 (0.1 %)
SO 65–45S, September	0.52
SO 65–45S, December	5.06
austral seasonal ratio	9.8×
tropics, all months	2.0–3.1 (flat)

The phase is exactly the missing term. Model marine CCN is wind-driven and so minimises in DJF ($N_d = 68.5$) and peaks in JJA (76.1), anti-phased with observation. `emdms` peaks in DJF at 9.8× its September value — largest precisely when and where the SO cloud error is worst (DJF $–6.94$, SON $–7.03$ pp).

~~`emdms` is a DMS surface *emission flux* in $\text{kg m}^{-2} \text{s}^{-1}$, which is exactly what a REcoM DMS module would produce, so a prescribed climatology now and a prognostic source later are a drop-in swap.~~ The GRIB carries that unit label, but values of $\sim 2 \text{ kg m}^{-2} \text{s}^{-1}$ of DMS are physically absurd. A global mean of 2.36 with maxima of 53 matches **DMS seawater concentration in nmol L^{-1}** — Lana et al. (2011) put the global surface mean at ≈ 2.3 . **The unit label is wrong.** That inserts a step the round-24 design did not price: a sea–air flux parameterisation (wind-dependent piston velocity, Nightingale or Wanninkhof) is needed *before* any flux→CCN relation. The interface

argument survives — REcoM would also produce a concentration — but the flux calculation belongs on the atmosphere side.

Two diagnostic traps recorded, both mine. The reader printed paramIds with an `i5` format, so the 6-digit 210043 came out as `*****`; widened to `i8`. And the field was twice declared broken from too small a sample — once from an `awk` column error that reported the ocean maximum as 0, once from reading four *consecutive* timesteps, which all fall in the same month, and concluding the field never varied. Both were wrong; the sampling was.

40.6 T2: the sea–air flux, validated in the model

§40.5 established that `emdms` is a *concentration*, so a flux has to be formed before the field can drive anything — and the flux is what must be validated first. Implemented as a diagnostic in `surfrad_layer.F90`, where the DMS field, the model’s own instantaneous wind and the surface temperature are all already in scope:

$$\begin{aligned} \text{Sc} &= 2674.0 - 147.12 T + 3.726 T^2 - 0.038 T^3 && \text{Saltzman et al. (1993)} \\ k_{600} &= 0.222 u^2 + 0.333 u \text{ [cm hr}^{-1}\text{]} && \text{Nightingale et al. (2000)} \\ k &= k_{600} \sqrt{600/\text{Sc}} / 360000 \text{ [m s}^{-1}\text{]} \\ F &= k C_w \times 10^{-6} \text{ [mol m}^{-2} \text{s}^{-1}\text{]} \end{aligned}$$

with $1 \text{ nmol L}^{-1} = 10^{-6} \text{ mol m}^{-3}$. The wind is the model’s instantaneous lowest-level speed, *not* a monthly mean — k is quadratic in u , and §40.2 showed monthly vector averaging halves the Southern Ocean wind.

The pass criterion was set before running. Hand-evaluating the chain at $C_w = 2 \text{ nmol L}^{-1}$, $u = 7 \text{ m s}^{-1}$, $T = 15^\circ\text{C}$ gives $k_{600} = 13.2$, $\text{Sc} = 1177$, $k = 2.6 \times 10^{-5} \text{ m s}^{-1}$ and $F = 5.2 \times 10^{-11} \text{ mol m}^{-2} \text{s}^{-1}$; scaled by an ocean area of $3.6 \times 10^{14} \text{ m}^2$, $3.156 \times 10^7 \text{ s}$ and 32.06 g mol^{-1} that is $\approx 19 \text{ Tg S yr}^{-1}$, against a published 18–34 (Lana et al. 2011 give 28.1). So the model should print a mean flux of order 10^{-11} – $10^{-10} \text{ mol m}^{-2} \text{s}^{-1}$.

The run completed a full year, 8760 steps, no abort, 70 088 diagnostics (8 blocks $\times$ 8760).

flux, minimum	3.78×10^{-13}
flux, median	2.33×10^{-11}
flux, mean	3.52×10^{-11}
flux, maximum	3.31×10^{-10}
predicted window	10^{-11}–10^{-10}
indicative global budget	$12.8 \text{ Tg S yr}^{-1}$
published	18–34 (Lana et al. 2011: 28.1)

The unit chain is correct. The mean flux sits inside the predicted window, and it varies while C_w is uniform on a block — so the variation is coming from wind and temperature through k , which is the intended behaviour.

Three limits, stated so the number is not over-read. The diagnostic covers rank 1’s blocks, a northern high-latitude strip, not the globe. Land points enter as zero because the flux clamps C_w at 0 to reject the -99 sentinel, which drags the sampled mean down. And the 12.8 is that non-global mean scaled as though it applied over all ocean. Landing within a factor of ≈ 2 of 28.1 on so crude an extrapolation is a *pass* for a unit test — the purpose was to catch an error of orders of magnitude, not to measure the budget. A real budget needs a global reduction the diagnostic does not do.

The run changes no physics. The flux is computed and printed, deliberately not wired into PCCNO, because the flux→CCN step is a separate scientific choice that needs a published relation rather than a fitted constant.

40.7 T3: DMS-Rev3 ingested through ocp-tool

The shipped `month_dms` is Lana et al. (2011). DMS-Rev3 (Hulswar et al. 2022) rests on 865 109 observations against L11’s 47 313, and the paper states the largest revisions are in the polar oceans — the only region this campaign is looking at. Measured on the model’s own grid after regridding, SO 65–45S:

SO 65–45S	L11	DMS-Rev3	change
annual	1.93	2.24	+16 %
DJF	4.23	4.36	+3 %
SON	1.34	2.02	+51 %
November	2.18	4.24	+95 %
global maximum	53.25	26.56	–50 %

Rev3 does not raise the summer peak, it broadens it — the bloom starts a month earlier — and it halves the maximum, which is the paper’s “reduced patchiness”. The model’s SO shortwave CRE error is worst in SON (−7.03 pp) and DJF (−6.94), so the upgrade lands in the season the old climatology was weakest.

Units derived, not read. `DMS_REV3.nc` carries no attributes at all. Integrating the shipped `DMSflux` globally gives $28.19 \text{ Tg S yr}^{-1}$ (`DMSfluxSI` 28.36) against a published 27.1–28.1, which fixes the flux as $\text{mol m}^{-2} \text{ d}^{-1}$ and the concentration as nmol L^{-1} ; every other assumption is out by orders of magnitude. Only the *concentration* is ingested — the shipped flux uses Rev3’s own wind climatology, whereas the model must form the flux online from its own wind, which is quadratic in speed. Their flux is retained as a validation target.

Built through ocp-tool (`dms_rev3_to_grib.py`), which is what makes it paleo-safe. Nearest *valid* neighbour in 3-D — 1° source against $\approx 0.9^\circ$ target is matched resolution, so nothing is invented, no land NaN bleeds into coastal ocean, and the dateline and poles cannot yield a spurious neighbour. Values are overwritten into the shipped messages, so grid, `dataDate`, `paramId` and all headers stay byte-identical to what the reader accepts. The land sentinel is taken from the template, preserving the model’s own mask; a Rev3 hole at a template ocean point would read as land, so L11 is retained there and counted — for TCO95 that count is **zero**, with 12 040 land points held at −99. `icmcl_dms.py` then interleaves it: $84 \rightarrow 96$ messages, 12 dates $\times$ 8 fields.

T3, one year, verified. Found 12 dates, Found 8 fields, no ordering abort, full year, no abort. Against T2 on the identical wind field:

	C_w [nmol L ^{−1}]	flux mean	indicative global
L11	1.825	3.515×10^{-11}	$12.8 \text{ Tg S yr}^{-1}$
DMS-Rev3	2.044	4.218×10^{-11}	15.4
ratio	1.120	1.200	

The flux rises by 20 % where the concentration rises by only 12 %. That is not an inconsistency — the flux is $\bar{k} C_w$, not $\bar{k} \bar{C}_w$, and k varies spatially, so the ratio of means need only agree if the spatial pattern is unchanged. It means **Rev3 places more DMS where the wind is stronger**, a positive C_w – k covariance relative to L11, and the flux is what feeds CCN.

Caveat on the $12.8 \rightarrow 15.4$. That is rank 1’s blocks — a northern high-latitude strip with land entering as zero — not a global budget. Rev3 and L11 have near-identical *global* fluxes (28.19 against a published 28.1), so the +20 % is regional sampling, consistent with the polar oceans being where Rev3 revised most, and is not a global statement.

40.8 Screening at one year: measured, with the threshold that makes it safe

Project issue #170 found the cy48 high-cloud offset already present in year 1 while the T2m bias and the SO low-cloud dip were not, and concluded that one-year runs would suffice to test tuning parameters. That claim is about `hcc`; the campaign’s Southern Ocean metric is SW CRE, a different quantity with different noise, so the transfer was measured rather than assumed.

lever	44-yr Δ	year-1 Δ	ratio	screens at 1 yr?
A1a ovlliqice 0.10	−6.44	− 6.56	1.02	yes
D2a inpsea 0.2	−4.80	− 5.23	1.09	yes
D2b inp+p700	−2.61	− 2.01	0.77	yes
A1b ovlliqice 0.35	−2.17	−1.74	0.80	no
B3 clddiff 1.5e−5	+0.67	−0.17	− 0.25	no — <i>wrong sign</i>
B7 rvice 0.22	+0.66	+1.22	1.85	no

Control interannual scatter in SO SW CRE is 0.712 W m^{-2} , so a *pair* of single years resolves ± 1.97 against ± 0.30 for a 44-year pair. **The claim holds, with a threshold: a lever screens at one year if its effect exceeds $\approx 2 \text{ W m}^{-2}$, and is unreliable below it** — B3 comes out with the *opposite sign* at one year.

Why this is worth having. Closing the SO amount deficit requires $\approx 4.9 \text{ W m}^{-2}$, so any lever large enough to matter would show at one year. A 1-year AMIP leg is ≈ 11 minutes against 9.5 hours for the 44-year standard, so candidate triage is $\approx 50\times$ cheaper and only survivors need full length. Round 23 spent four 48-year submissions on levers that could have been screened in 45 minutes together.

What it does NOT license. Temperature. #170’s own lead/lag result is that T2m and the SO low-cloud dip lag the cloud change, and the campaign’s seasonal thresholds (DJF $\pm 0.588 \text{ K}$ on 44 years) are far beyond a 1-year window. **Screen the radiative response at one year; never adopt on one.**

41 Round 25 (2026–08–09): the tropopause is not displaced, and cold-and-dry share one entry point

No runs. Round 22 found the cold bias extratropical, peaking at 200 hPa over both poles and reversing sign at 100 hPa, and read that shape as a displaced tropopause. It is not, and what is there instead is better news.

41.1 The tropopause height is right

WMO lapse-rate tropopause, model and ERA5 on identical levels, annual, period-clean:

band	model [hPa]	ERA5 [hPa]	diff
90S–60S	250	250	0
60S–30S	200	200	0
tropics	100	100	0
30N–60N	200	200	0
60N–90N	250	250	0

~~The sign reversal at 100 hPa is a displaced tropopause: the model is still on a tropospheric lapse rate at 200 hPa where the real atmosphere has turned isothermal.~~ **Falsified.** The tropopause sits on the same level in every band. The vertical resolution here is one level, so a displacement smaller than ~ 50 hPa would not be seen — but a displacement large enough to produce a 4 K bias would be.

What the lapse rates show instead is a *sharper* transition, not a moved one: the model cools faster than ERA5 through 300–250 hPa ($+1.76$ and $+1.78$ K km $^{-1}$ at the two poles, against $+0.21$ in the tropics) and then slower than ERA5 above 200 hPa (-1.49 , -1.85). Same tropopause, too much curvature at it.

41.2 The cold point, and one caveat on reading it

band	T model	T ERA5	ΔT	p model	p ERA5
90S–60S	208.77	212.03	-3.26	250	70
60S–30S	211.80	213.32	-1.52	100	100
tropics	193.44	195.51	-2.07	100	100
30N–60N	211.54	212.84	-1.30	100	100
60N–90N	213.21	217.55	-4.33	250	30

The two polar rows are not like-for-like and must not be quoted as 4 K cold points. Model and ERA5 locate their profile minimum at different levels there (250 against 70 and 30 hPa), because at high latitudes the upper stratosphere is near-isothermal and the minimum is ambiguous. What those rows actually say is a structural difference: the model’s polar stratosphere is too *cold* at 250 hPa and too *warm* above 100 (§38.4 gives $+0.25$ and $+0.26$ there). The three rows where both datasets pick the same level are clean, and the tropical one carries the next result.

41.3 Cold and dry are one problem with one entry point

Stratospheric water vapour is set by the temperature of the tropical cold point, which freeze-dries air entering the stratosphere; the Brewer–Dobson circulation then carries that air poleward. So a cold-biased tropical cold point dries the stratosphere *everywhere*, including over the poles. Testing it quantitatively with Clausius–Clapeyron over ice (Murphy & Koop 2005) on the saturation mixing ratio at each dataset’s own cold point:

tropical cold point, model	193.44 K at 100 hPa
tropical cold point, ERA5	195.51 K at 100 hPa
saturation mixing ratio	3.575 vs 4.998 (ppmv-mass)
predicted entry-air drying	-28.5%
measured drying at 100 hPa	-46.9%
share explained	61 %

A 2 K cold bias at the tropical cold point accounts for three fifths of a 47 % stratospheric dry bias. That reframes §38.4, which recorded cold and dry as a compensating *pair* — two independent errors whose opposing OLR effects happen to cancel. They are not independent. They share an entry point, and the polar UTLS dryness is the tropical cold point seen downstream.

Why that is much better news. Two independent cancelling errors would have to be fixed together or not at all, and fixing either alone moves net TOA several W m^{-2} the wrong way (§38.4). One error with one entry point means **a lever that warms the tropical cold point should move both**, in the right direction, at once. The remaining 39 % of the drying is a second source — transport, or the model’s own UTLS microphysics — and is not yet identified.

Caveat carried forward: ERA5’s upper-level humidity is weakly constrained by observation, so the -46.9% denominator is the softest number here. The -2.07 K cold point and the -28.5% it implies rest only on temperature, which ERA5 constrains well.

42 Round 26 (2026–08–09→10): marine biogenic CCN wired and priced, and the tropical longwave becomes the gate

Round 24 traced the marine CCN path end to end and found the DMS field read but *unused*. This round wires it into the radiation, brackets the one free coefficient across its published range, and reaches a verdict that changes the shape of the remaining campaign: the Southern Ocean lever works, and the tropical longwave deficit is what stops it being adopted. The two are no longer separate items on the open list.

42.1 The wiring, and why it is default-off

In `radpar.F90`, immediately after `SURFRAD` fills `PCCNO` from Genthon (1992) sea salt, ocean points gain a DMS term: Saltzman et al. (1993) for the Schmidt number, Nightingale et al. (2000) for the transfer velocity $k_{600} = 0.222u^2 + 0.333u$, flux $F = kC_w$, and

$$\text{PCCNO} += S \cdot F_S, \quad S \equiv \text{ECE_DMS_CCN_SENS}, \quad F_S \text{ in } \text{mg S m}^{-2} \text{d}^{-1}.$$

`PCCNO` then flows to `radiation_scheme`, through the Martin et al. (1994) quadratic to droplet number and effective radius — the Twomey effect. The input is DMS-Rev3 (Hulswar et al. 2022), ingested through `ocp-tool` and verified in T3 (§40.7).

S is a `NAMECECFG` namelist knob defaulting to 0.0, and the `ICMCL` field is registered only when `ECE_CLIMR_DMS` is set, so `NUMFLDS` is unchanged and **every existing configuration reproduces bit-for-bit with the switch off**. That is what allows one library to serve both arms of every pair below.

Two near-misses worth recording. The first attempt wired CCN into `surfrad_layer`, which would have been a **no-op**: the radiation’s CCN comes from `radpar`, where `PCCNO` is `INTENT(OUT)` and any earlier value is overwritten. The second was a build failure, **error #6404**, diagnosed twice as a stale `.mod` and once as a CMake dependency-scanner gap — *all three wrong*. A full clean `esm_master recomp` reproduced it. `nm` showed the symbol present in the object file: the OpenIFS modules declare `PRIVATE` by default and export an explicit list, so a new module variable is invisible at every USE site until it is added to `PUBLIC`. **Check accessibility before blaming the build system, and diagnose with `nm` rather than by theory.**

42.2 U1, the kill test at the top of the published range

Woodhouse et al. (2010) measure $\text{dCCN}/\text{d}F_{\text{DMS}}$ anywhere from -43 to $+166\text{ cm}^{-3}$ per $\text{mg S m}^{-2} \text{d}^{-1}$ depending on season and hemisphere. That range is too wide to pick a number from, so the coefficient was made a namelist knob and the range *sampled*, starting at the most optimistic end. Control is T3: identical DMS-Rev3 input, identical winds, $S = 0$.

$S = 166, 1\text{ yr}$	DJF	MAM	JJA	SON
SO 65–45S $\Delta\text{SW CRE}$	−7.25	+0.03	−0.63	−1.14
tropics $\Delta\text{SW CRE}$	−1.56	−2.17	−2.44	−2.39

Annual: SO -2.25 , tropics -2.14 , global -1.72 W m^{-2} . December alone is -8.54 and January -7.34 .

The lever is real and it is seasonally selective in exactly the right way. The SO DJF deficit is $+10.86\text{ W m}^{-2}$ and this closes 67 % of it, in the season the deficit is worst, at a DJF tropical cost of only $-1.56 - 4.6:1$ in our favour. **The selectivity is the bloom’s, not a tuned parameter**; no hemispheric constant was invented to produce it. That also confirms round 24’s phase argument quantitatively: the model’s SO droplet number was right in the mean but inverted in phase, and this is the correction.

The annual means look like a wash only because the SO gain is concentrated in one season while the tropical cost is flat year-round.

42.3 U2 and U3: the bracket, and what one year can actually resolve

Two more arms at $S = 43$ (the published floor) and $S = 90$ (the midpoint), to read an operating point off a curve instead of guessing a fourth value. Detection thresholds were computed first, per band, from the control’s own interannual scatter as $1.96 \sigma \sqrt{2}$ for a pair of single years — the construction that gives the already-known ± 1.97 for SO SW CRE:

metric	σ (interannual)	1-yr pair, 95 %
SO SW CRE, annual	0.712	± 1.97
SO SW CRE, DJF	1.379	± 3.82
tropics SW CRE, annual	0.357	± 0.99
tropics SW CRE, DJF	0.441	± 1.22
global SW CRE, annual	0.402	± 1.11

S	SO annual	SO DJF	tropics annual	resolved?
43	−0.32	−1.78	−0.76	nothing
90	−0.38	−1.22	−1.11	tropics only, marginally
166	−2.25	−7.25	−2.14	all three

The bracket produced exactly one usable point. SO DJF is not even monotonic in S (−1.78, −1.22, −7.25), which is the signature of the two sub-floor arms being internal variability rather than signal.

~~The tropical response is monotonic and clean at 0.0124 W m^{-2} per unit S , fitted through three arms.~~ **Withdrawn the same day.** Two of those three points sit below the ± 0.99 tropical threshold, so the fit was reading a line through noise. The analysis script now refuses to interpolate the SO points and guards the tropical extrapolation explicitly — `np.interp` would have *clamped* to the sampled range and returned a confident wrong answer. The same paragraph also treated the $\pm 0.5 \text{ W m}^{-2}$ tropical guardrail as though it were a detection threshold. It is not: it is a *tolerance*, set because the tropics are only -0.67 W m^{-2} from CERES period-clean.

42.4 The verdict, and why scaling cannot rescue it

From the single resolved arm the tropical cost is $2.14/166 = 0.0129 \text{ W m}^{-2}$ per unit S , so the ± 0.5 tolerance binds at $S \approx 39$ — **below Woodhouse’s published floor of 43**. Twomey scales with $\Delta \ln N_d$, so the response is *sub-linear* and the per-unit cost at low S is if anything larger; the bound holds a fortiori. Third branch of the pre-registered verdict logic: **no coefficient in the published range works as a tuning knob**.

Why the tropics respond more per unit ΔCCN , and why that is not a bug. Twomey goes as $\Delta \ln N_d$, and tropical background CCN ($\sim 52 \text{ cm}^{-3}$) is *lower* than the Southern Ocean’s (~ 78), so the same absolute increment is a larger fractional change there. Physically the model has *no* marine biogenic CCN anywhere — Genthon is sea salt from wind, MACv2-SP is anthropogenic plumes — so adding it in the tropics is *more* correct, not less. What fails is a tuning tolerance, not the physics. And because both ends scale together, no choice of S separates them.

42.5 The tropical longwave deficit is now the gate

This reframes the open list. The two remaining regional failures act through *different* channels (§58.6): the Southern Ocean through SW cloud (+7.45), the tropics through LW cloud (−2.16). The tropical cost of the DMS lever is a *shortwave* cost; the tropical bias is a *longwave* one. Those are independent terms, so:

- **The tropical LW deficit is no longer a parallel item — it gates the Southern Ocean fix.** Close it and the SW cost of DMS CCN becomes affordable, and the best SO lever this campaign has found goes in unchanged.
- **The pair is complementary in the global budget.** DMS CCN *removes* 1.72 W m^{-2} globally, which helps the +0.52 imbalance; closing the tropical LW deficit puts roughly +1 back once the tropics are area-weighted. Together they fix both regional biases and partly cancel globally — a better position than either alone, since the DMS lever’s problem was never its SO effect but its global reach.
- **The requirement on the LW lever is sharper than “acts through the longwave”.** RVICE is the counter-example: an ice fall speed applies wherever there is cloud ice, so it *trades* SO high cloud against tropical anvils, and it was raised $0.13 \rightarrow 0.16$ precisely to suppress SO hcc (project issues #169/#170). A usable lever must be keyed on **deep convection**, which barely exists over the Southern Ocean, so that selectivity comes from where convection happens rather than from a latitude band — the same structural trick that made the DMS lever seasonally selective.

The cheap check comes first: ~ 52 arms have been scored on SO SW and global RMSE and *none* on the tropical LW term. `tropical_lw_scan.py` is written to answer that from data already on disk before any new run is spent; it has not yet been run.

Not Blomquist. The standing todo to switch the transfer velocity from the Nightingale quadratic to Blomquist et al. (2017) should be dropped for this purpose: the DMS-specific scheme is *less* steep in wind, which shifts flux away from the windy Southern Ocean toward the calm tropics. It would make selectivity worse, not better.

42.6 Two coupled runs, on a campaign-branch coupled build

The coupled tree was on `movcav-landice+co2-concdriiven`, which carries neither the snow-depletion knobs nor the CCN wiring. The campaign branch is a *strict descendant* of that branch’s tip (`f3ccacb` is an ancestor), so it contains every coupled commit, and every campaign feature is namelist-gated off by default — switching the tree in place removes nothing and changes no existing configuration’s behaviour. One clean OpenIFS rebuild then serves both runs. Six files completing the DMS ICMCL chain, plus the `RCLCRIT_SEA/RCLCRIT_LAND` exposure, had been left uncommitted in the AMIP tree; U1–U3 ran against them, so the library had them and only the commit was missing.

run	change vs. its control	question
11E	11D + <code>SWEMIN 30 → 15 + K1</code>	is the coupled configuration level with AMIP?
11F	11E + <code>ECE_DMS_CCN_SENS= 166</code>	does coupling change the Southern Ocean answer?

Both are 50 years, both submitted 2026–08–10. **11F is deliberately not a tuning candidate** — AMIP has already said no published *S* clears the tolerance. It runs at the only setting AMIP could resolve, to answer what AMIP structurally cannot: with SST and sea ice free to respond, does brightening Southern Ocean cloud in bloom season move the coupled SO state and the TOA imbalance by more than the tropical cost? Fixed SSTs suppress exactly the feedback that makes an SO cloud change matter. Fifty years also prices the tropical response with feedbacks and proper statistics instead of from one AMIP year.

Status update, 2026–08–10. Both started the same day (11E job 26837423, 11F 26837527) and are through their first ten-year legs.

11F is reclassified as a MECHANISM PROBE, not a production candidate, and should not later be read as one. Two things changed within hours of submitting it. First, the bracket verdict (§42.4) means *S* = 166 is a configuration we will not adopt under any circumstances. Second, S4 (`RCL_INPPMIN 70000 → 50000`) turned out to deliver -1.98 W m^{-2} of Southern Ocean SW CRE — nearly the same forcing as 11F’s -2.25 — at a tropical cost of 0.10 against DMS’s 2.14, i.e. **twenty times the selectivity, namelist-only, and already run**. So the specific lever inside 11F has been superseded before its first leg finished.

What 11F still buys is the *generic* question, and it is worth 50 years for it: what does $\approx 2 \text{ W m}^{-2}$ of Southern Ocean SW brightening do once SST and sea ice can respond? That answer transfers to the whole S4/DMS family, and AMIP cannot produce it at any run length — prescribed SST absorbs the perturbation without responding. It is being kept for that reason alone. If compute becomes tight, **11F is the run to cut and 11E is not**.

43 Round 27 (2026–08–10): the tuned stack nobody ran, and a lever that fixes the tropics by breaking the Southern Ocean

The largest single-day gain of the campaign, and the largest single-day correction. A jointly-tuned eight-parameter cloud stack for *this resolution* was found sitting unused in the runscript directory; one of its parameters closes the tropical longwave deficit outright; and scoring it properly showed the same parameter warms the Southern Ocean, putting the campaign’s two remaining regional targets in direct conflict.

43.1 The archive has no tropical longwave lever

Round 26 made the tropical LW deficit the gate on the Southern Ocean fix, so the free check came first: score all 50 arms on a term none had ever been scored on.

The term is extraordinarily stiff — interannual sd 0.101 W m^{-2} , so the 44-year pair threshold is ± 0.042 and the -2.16 deficit is **51 times** the detection limit. Yet the largest positive response in the whole archive is **A1a at $+0.373$** , 17% of the deficit, and it costs SO SW -6.44 and global -1.90 . Everything else is $\leq 6\%$. **No clean tropical-longwave lever exists in anything tried.**

The useful result was B1. It *lowered* DETRPEN $0.75 \rightarrow 0.45\text{E-}4$ and tropical LW fell 0.510 — second-largest in the archive — while the Southern Ocean did not move at all (-0.067 against its own ± 0.298). Against RVICE, which moves tropical and SO longwave $\approx 1:1$, that is **57:1 selectivity**. B1’s recorded verdict was “failed — more SW yet colder; *worst tropics*”; the tropical half of that failure *is* this term, so its reverse is a lever.

43.2 Two levers screened, one of them the wrong way round

1 yr vs control	trop LW ± 0.279	trop SW ± 0.990	trop net ± 1.656	SO LW ± 0.584	global net ± 1.333
V1 ENTRORG 1.25E-3	+0.288	+3.583	+4.536	-0.830	+2.734
V2 DETRPEN 1.11E-4	+0.583	-1.295	-0.398	+0.171	-0.428

V2 landed on prediction: B1’s slope forecast $+0.61$, measured $+0.583$, with the Southern Ocean untouched. **V1 failed, and it should not have been run in that direction.** Project issue #170, comment 7, already says “there seems some potential for *increasing* ENTRORG”. V1 lowered it on Semmler’s 43r3 value and cost $+2.73\text{ W m}^{-2}$ globally. Eleven minutes, not 9.5 hours — but the desk check existed and was not done.

43.3 A tuned stack for TCO95L91, written and never run

awiesm3-v3.4.2-levante-TCO95L91-DARS2_{1d,2y}.yaml carry eight jointly-tuned cloud parameters at *our exact resolution*:

	NAMCUMF default	DARS2		NAMCLDP default	DARS2
RPRCON	1.4E-3	1.48E-3	RVICE	0.16	0.18
ENTRORG	1.75E-3	2.07E-3	RLCRITSNOW	2.0E-5	1.46E-5
RMFDEPS	0.30	0.48	RSNOWLIN2	0.030	0.04
DETRPEN	0.75E-4	1.32E-4			
ENTRDD	3.0E-4	1.08E-4			

Three-figure precision — an optimiser’s output, consistent with the EPPES table in the logbook. **No fort.4 on disk sets any of them.**

Two of this campaign’s own proposals were superseded on the spot: V2 ran DETRPEN at $1.11\text{E-}4$ as “the largest defensible value” from the EPPES 1σ bound, and DARS2 uses $1.32\text{E-}4$, 19% beyond; and V1’s ENTRORG direction is the opposite of DARS2’s.

43.4 Why it was invisible: preflight looked in one place

~~RPRCON: never set in any run — genuinely untested. RSNOWLIN2: never set in any run — genuinely untested. Both false, on the same day.~~ preflight.py scanned one runtime tree and one runscript directory, so it could only ever see *this* campaign.

RPRCON was tuned at TCO319/43r3 in the DART line (#87): $1.4\text{E-}3 \rightarrow 0.7\text{E-}3$, adopted through DARTC/D/E/F/J/K, relaxed to $1.0\text{E-}3$ from DARTL; and launched once at TCO95L91-CORE2 in #95, where **the result was never posted** and it did not reach the AWI-CM3.1 final configuration. RSNOWLIN2 was set to 0.04 in DARS2 and to 0.025 in awicm3/MELDPOND.

preflight now globs every runtime tree and every runscript directory, separates “tried here” from “tried elsewhere at AWI”, and reports the *value* — because “mentions it” and “sets it” are different facts and only the second is prior art. Its remaining blind spot is recorded in its own docstring: **the GitHub issues**, where the *why* lives, and which issue search does not index because it ignores comment bodies.

~~The logbook’s EC-Earth3 ENTRORG row ($1.80\text{E-}4$) is a transcription error; 48r1 defaults to $1.75\text{E-}3$. Also wrong.~~ The EC-Earth3 files genuinely say $1.80\text{E-}4$ with their own reference column in E-4 as well. It is a real formulation difference between EC-Earth3’s IFS cycle and OpenIFS 43r3/48r1 — which matters far more than a typo would, because “correcting” the logbook would invite someone to set $1.8\text{E-}4$ in 48r1 and cut entrainment tenfold. **EC-Earth3’s ENTRORG is not transferable; its DETRPEN and RPRCON are.**

43.5 W: the stack taken apart, and the lever that works

Eight one-variable arms plus the combination, one year, against amip_pi_base.

	trop LW	trop SW	trop net	SO SW	SO LW	global net
W4 DETRPEN 1.32E-4	+1.025	-2.355	-0.959	-0.087	-0.429	-0.804
W8 RSNOWLIN2 0.04	+2.581	-2.204	+0.671	-0.884	+1.223	+0.321
W9 whole stack	+1.939	-6.494	-4.518	-0.983	+0.854	-2.659
interaction W9 - $\sum$	-0.458	-0.449	-1.194	-1.763	+2.415	-0.270

RSNOWLIN2 = 0.04 gives +2.581 W m⁻² of tropical LW CRE — 120 % of the deficit, nine times its threshold, and five times the largest response among the other fifty arms — with tropical and global net TOA both *below* their thresholds.

It works for a reason that was predicted before it was run. It is the *temperature exponent* of ice→snow autoconversion, ZZCO = PTSPHY · RSNOWLIN1 · e^{RSNOWLIN2(T-T₀)} (cloudsc.F90:2858). Raising it slows conversion 31 % at a 200 K cirrus temperature but only 11 % at 250 K, so it builds **cold thin cirrus rather than thickening anvils** — and thin cirrus is longwave-dominant while a thick anvil is bright in both. The contrast is measured: DETRPEN gives LW:SW of 0.44, RSNOWLIN2 1.17.

~~You cannot buy tropical longwave without paying tropical shortwave; every lever that raises tropical LW does it by adding condensate, which also raises SW reflection.~~ **Withdrawn the same day.** It was generalised from two SW-dominant arms. The archive already contained LW-dominant responses — D1 at 4.0:1, B6 at 1.4:1, B7 at 1.3:1 — and W8 settles it.

The full stack overshoots and is significantly non-additive: SO longwave interaction +2.415, with the sum of parts predicting -1.561 and the combination measuring +0.854. The arms are not a menu.

A near-miss worth recording. RK00PTAU, the Koop homogeneous-freezing timescale, was the first candidate for a temperature-selective cirrus lever: it is in NAMCLDP, never set, and physically exactly right. Its only use in cloudsc.F90 is **commented out**. Setting it is a no-op — caught by tracing the code path rather than by running it.

43.6 X and Y: on the real base, and the seasonal reason DMS survives

Ladder from T3 (adopted stack, DMS field read at $S = 0$), one variable per rung.

cumulative vs T3	trop LW	trop SW	trop net	SO SW	SO LW	global net
X1 +RSNOWLIN2	+2.740	-2.631	+0.311	-0.507	+2.626	+0.419
X2 +DMS $S=166$	+2.547	-4.062	-1.438	-2.765	+2.240	-1.259
X3 +INPPMIN 50k	+2.687	-4.299	-1.274	-5.051	+2.841	-1.434
X4 +INPPMIN, no DMS	+2.563	-2.450	+0.225	-2.595	+2.718	+0.067
Y1 +ov1 0.35	+2.677	-2.729	+0.088	-1.963	+2.827	-0.005
Y2 +ov1 0.10	+2.852	-3.742	-0.718	-5.029	+3.563	-1.184

RSNOWLIN2 reproduces on the real base (+2.740 vs +2.581 alone) and DMS reproduces its Southern Ocean effect exactly (-2.258 vs -2.250).

~~RCL_INPPMIN buys the same Southern Ocean shortwave as DMS for free (+0.164 tropical cost against DMS's -1.750), so the DMS rung is redundant and the marine CCN line can be retired.~~ **Withdrawn.** Their *annual* means agree by coincidence; their seasons do not:

SO 45-65S SW CRE	DJF	MAM	JJA	SON	ANN
INPPMIN 50k (S4-P5, 44 yr)	-3.72	-1.30	-0.51	-2.35	-1.97
DMS $S=166$ (U1-T3, 1 yr)	-7.25	+0.03	-0.63	-1.14	-2.25

DJF is austral *summer* and carries the shortwave: incoming solar is 460 W m⁻² against 87 in JJA, and the control's SO SW CRE is -115.5 against -18.7. The deficit is +10.86 there. DMS reaches 67 % of it, INPPMIN 34 %. **DMS is the only lever that reaches the season the error lives in.**

43.7 Scoring the Southern Ocean on NET — and the correction that follows

~~Southern Ocean levers scored on SW CRE.~~ Used throughout rounds 22-27 and **wrong**: the coupled complaint is a warm SST bias, which responds to Δ SW CRE + Δ LW CRE.

Re-ranked on net, the 44-year archive is clear and consistent — longwave cancels 25-29 % in *every* arm, so SO SW $\times$ 0.72 is a good rule:

44 yr	SO SW	SO LW	SO net
A1a ovl= 0.10	−6.438	+1.871	− 4.567
D2a inpsea 0.2	−4.802	+1.200	−3.602
S4 inppmin50k	−4.669	+1.195	−3.474

The new levers break the rule. True net TOA at one year: X3 −2.024; **DMS alone** −1.976 with only 17 % longwave cancellation; Y2 −1.300; X4 +**0.377**; Y1 +**0.764**; and **X1 +2.294, resolved.**

RSNOWLIN2 warms the Southern Ocean. Its +2.6 of SO longwave swamps its shortwave. On the bare base it was neutral (−0.100); on the adopted stack it flips hard positive — a sign reversal produced by the base, not a small non-additivity. Every configuration built on it inherits the penalty.

So the campaign’s two remaining regional targets are in direct tension through one parameter: RSNOWLIN2 closes the tropical longwave deficit partly by putting cirrus over the Southern Ocean, where the model already traps too much. And DMS — dismissed as redundant that morning — is on net forcing the cleanest new lever there is, and the only reason X3 points the right way.

43.8 S4, evaluated at last

S4 completed 48 years on 2026–08–09 and had *never been added to runs.py*, so `evaluate.sh` had never scored it. Its exact one-variable partner is P5 — the two `fort.4s` differ in one line.

S4 − P5, 44 yr	value	threshold
SO TOA SW CRE	−1.98	±0.298
global net TOA	−0.41	±0.201
tropics net TOA	−0.10	±0.250
Siberia JJA T2m	−0.120	±0.242
SO SW RMSE	+0.255	±0.019

Best mean, worse pattern: it deepens opacity where cloud already exists rather than adding cloud (area +0.105 pp). Siberia is untouched, and the code says it cannot be otherwise — `cloudsc.F90:2520` scales ice nuclei by `RCL_INPSEA + (1 − RCL_INPSEA) · PLSM`, which is exactly 1 over land.

43.9 The AMIP runs are not at 1850 forcing

~~All AMIP runs use 1850 GHG (NCMIPFIXYR=1850).~~ **False for every AMIP run in the campaign, including the control.** NCMIPFIXYR=1850 sits in `&NAMECECMIP6`, the block 48r1 renamed and no longer reads. The runtime value is −1 and the model reads *yearly* GHG, ozone and aerosol: the log shows 1869 in the first leg and 1911 in the last.

The base runscript already documents the trap and removed the dead block on 2026–07–30, but its replacement (`oifs.mip + oifs.scenario`) was never added, because that path aborts with `GRID_IN MISSING FIELD` on the `awi3 INIT`. **The coupled runs are correctly pinned** — 11E logs `Set JYEAR=... = 1850` — so this is AMIP-only.

Z1 verified the one-line fix: NCMIPFIXYR: 1850 in the *live* `&NAMECECMIP` block reaches the model, the run completes CNT0 with zero aborts, and it does *not* trip the missing-field abort.

Consequences: **every lever delta stands**, because both arms of every pair share the forcing. The *absolute* energy baseline does not — mean CO₂ over the evaluation window is ≈294 ppm against a PI 284.3, worth ≈+0.15 W m^{−2}, or about a quarter of the control’s +0.64. The fix is therefore deliberately *not* applied to the runs in flight: it would break comparability with P5 and all 50 prior arms to buy an offset that one dedicated control run can measure later.

The irony worth remembering: `amip_pi_base`’s `fort.4` *contains* “NCMIPFIXYR = 1850”. Anyone grepping it sees the right number. Only `NODE.001_01` disagrees.

43.10 Coupled: 11F retired, 11E resumed

11F was cancelled at 20 of 50 years. Its lever had been superseded within hours of starting (§42.6), and the generic question it was kept for — what ≈2 W m^{−2} of SO brightening does with free SST — did not justify the remaining 30 years once S4 and the net-forcing re-ranking landed.

11E was cancelled by accident in the same operation and **resumed cleanly** from year 1370 (`run_number=3`), with 20 years banked and restart state intact at the end of 1369. Nothing was lost but the killed leg.

43.11 Three runs in flight, and what would disqualify them

run	configuration	why
LX1	stack + RSNOWLIN2	the gating run; isolates the SO penalty and the Siberian cost
LX3	+ DMS $S=166$ + INPPMIN	only arm with resolved net SO cooling
LY2	+ ov10.10	best radiative package; A1a's value, rejected in round 3

Control is P5, verified **bit-for-bit identical to T3** at year 1870 on all three radiative metrics — ECE_CLIMR_DMS reads the DMS field but does nothing at $S = 0$. That check saved a 9.5-hour run.

LX4 was cancelled after 12 minutes as dominated: it warms the SO (+0.377), leaves energy untouched (+0.067), and its unique content already exists as S4 – P5.

What would disqualify them, and what one year could never see. Siberia. The JJA threshold is ± 0.242 K and DJF ± 0.588 , both 44-year numbers; every arm changes cloud, which is the boreal energy supply; and LY2 carries the value round 3 rejected *for boreal damage*. The land stack currently recovers 1.15 K of a ≈ 2 K period-clean bias, leaving Siberia **still ≈ 0.9 K cold** — so there is no headroom to spend.

43.12 The boreal forest: the coupled model destroys it, and warming brings it back

11E's Siberian result prompted the question the campaign actually exists to answer — not whether T2m improved, but whether the forest is there. It is not, and chasing that properly produced one retraction, two analysis bugs of my own, and a calibration the campaign has never had.

The claim, and the retraction. ~~The forest failure is not a growing-season problem: 11E clears the larch GDD gate on 91 % of cells and the trees still do not come, so no amount of additional summer warming will bring the forest back.~~ **Refuted by experiments already on disk.** Three 2000-year offline LPJ-GUESS spin-ups exist at TCO95 — AMIP forcing, and the same forcing warmed by +1 and +2 K:

offline 2000-yr equilibrium, Siberian box	BNS	TREE	GRASS	tree>0.5	BNS>0.1
CRUNCEP+CERES (observed forcing)	0.227	0.415	0.281	50.3 %	36.2 %
AMIP forcing	0.034	0.202	0.313	18.2 %	7.4 %
AMIP +1 K	0.058	0.261	0.332	24.5 %	12.5 %
AMIP +2 K	0.099	0.327	0.344	31.7 %	19.7 %

Warming restores the forest, monotonically and without saturating. BNS roughly triples from AMIP to AMIP+2 K and tree cover recovers 59 % of the gap to the observed-forcing equilibrium.

This is the campaign's founding premise, calibrated for the first time. The AMIP Siberian cold bias is ≈ 2 K period-clean (§38.2), and +2 K of offline warming buys back three fifths of the forest. Closing the cold bias should therefore recover most of it — previously an assumption, now a measured transfer function.

Why 11E looked like a counter-example, and was not. The coupled runs are **50-year transients**; these are **2000-year equilibria**. 11E is not showing an endpoint but the *decline* from its CRUNCEP restart toward whatever equilibrium its own climate supports, and forest composition takes centuries:

Siberian box, 473 cells	year	BNS	TREE	tree>0.5	grass>tree
offline CRUNCEP (the restart state)	3900	0.227	0.415	50.3 %	29.2 %
11E at start	1350	0.213	0.394	47.8 %	28.3 %
11E after 50 coupled years	1399	0.017	0.083	2.1 %	69.6 %
11D after 50 coupled years	1399	0.019	0.074	0.6 %	68.1 %

Year 1350 reproduces the offline restart almost exactly, as it must. By 1399 larch has fallen **92 %** and grass goes from beating trees on 28 % of cells to 70 %. **The coupled model inherits a real boreal forest and dismantles it in fifty years**, from a healthy state, while clearing the GDD gate on 91 % of cells — which is exactly what a transient toward a much poorer equilibrium looks like.

Consistent with the established mechanism, not a new one. §31.2 already separates the two claims: competition as an LPJG *parameter* error is dead (the forcing-transfer test reproduced the loss with byte-identical parameters), while competition as the *pathway* by which climate removes larch is supported — larch is climatically permitted on 84 % of cells and loses to grass 5.8:1. The offline warming series is the driver half of that statement, measured: **grass>tree** stays at 43–45 % across AMIP, +1 and +2 K against CRUNCEP's 29 %, so warming raises tree cover *without* dissolving the competition. Forcing is the driver; grass-versus-larch remains the mechanism.

Two of my own analysis errors, recorded because both produced plausible numbers. ~~11E-Siberian FPC over 2365 cells: TREE 0.109, GRASS 0.203.~~ The glob matched all five decade directories and took each one's

own final year, so it averaged 1359–1399, not 1399. The true final-year values are *worse*: TREE 0.083. ~~Offline CRUNCEP TREE 0.617, BNS 0.617, forest on 100% of cells.~~ That came from **tail-ing** a 13 GB `fpc.out` — but LPJ-GUESS writes *all 2000 years per gridcell* before moving to the next, so the tail sampled the last ~30 gridcells, of which three fell in the box. Both were caught only by checking cell counts against each other; neither looked wrong on its face. **For these files, stream with `awk` filtered on year and box — 13 GB in 20 seconds — and never trust a tail.**

Unread, and deliberately not speculated about. Three offline experiments target the competition directly: LPJG_COMP_EXP1_BNS_GREFF0035, EXP2_IBS_GREFF012, EXP3_C3G_PSTEMLOW12. They belong to the final land round.

44 Round 28 (2026–08–10→11): the L-series at full length, and the arm that was already on disk

The three 46-year runs landed, and with them the first honest scoring of the round-27 levers. Two of the three fail. The configuration that wins is S4 — one namelist number, finished on 2026–08–09, and unevaluated until now.

44.1 The three candidates, attributed

Each effect is the difference from the arm above it, 44 years, against `amip_pi_base`:

effect	SO SW	SO net	Siberia JJA	global net	global LW CRE
RSNOWLIN2 (LX1–P5)	–1.13	+0.61	+0.14	+0.42	+2.06
DMS + INPPMIN (LX3–LX1)	–4.53	–3.87	+0.04	–1.96	+0.21
ov1 0.10 (LY2–LX1)	–4.73	–3.34	–0.67	–1.66	+0.85

Three things settle. **RSNOWLIN2’s Southern Ocean penalty is real at 44 years** — it delivers the tropical longwave gain exactly as the one-year screen predicted (+2.06 global LW CRE) and makes SO net forcing and global energy worse. **A1a’s boreal damage reproduces on the new base**: LY2 costs –0.67 K of Siberian JJA, nearly 3× the ±0.246 threshold, and A1a alone on the bare base gives –0.749 — the same number. Round 3 was right. And **DMS+INPPMIN dominates ov1 0.10**: more SO cooling, no Siberian cost.

LX3 is the best boreal arm in the campaign on the seasonal table — JJA +1.331, the highest of all 54 arms, with every season neutral or positive, and it is not on the report’s standing “warms JJA but cools DJF” list of eleven levers.

44.2 Two fields, not one: what the maps resolved

LX3’s band mean improved while its “SW RMSE” worsened, which looked like a pattern degradation. It is not. **The two numbers are different fields**: the band mean quoted throughout this campaign is TOA SW CRE, while the RMSE is *surface* net SW over ocean (`eval_round10_A.py`:190).

~~LX3 overshoots reflection in some places and not others, so the pattern becomes patchy.~~ **Wrong.** On TOA CRE the pattern improves monotonically (RMSE 11.45 → 8.78 → 7.74). On the surface field LX3 overshoots into a *uniform* negative bias of –4.75, and since $\text{RMSE}^2 = \text{mean}^2 + \text{variance}$ that mean is most of the degradation. The field did not become noisier; it sailed past zero — a much better problem, because it is fixable by scaling rather than by finding a different shape.

The global frame then undercut the whole framing. Extending the maps from the Southern Ocean strip to 80S–30N:

	TOA SW CRE			surface net SW	
	global	45–65S	tropics	global	tropics
P5	+0.60	+4.98	–0.71	–0.60	–0.15
LX3	–3.17	–0.63	–4.76	–4.40	–4.30

The tropics move –4.05 while the Southern Ocean moves –5.61. The change is near-uniform blue in every basin. **LX3 is a global dimming knob, not a Southern Ocean lever**, and it survives on *net* only because RSNOWLIN2 adds +2.46 of global LW CRE in compensation. The seasonal selectivity DMS showed at one year does not become spatial selectivity at 44.

Two further things only the global frame shows. Global RMSE barely moves (TOA 11.48 → 12.40, surface 14.44 → 15.18) because the error field is dominated by the tropical Pacific dipole, the stratocumulus decks and the

land signals — none of which any lever touches, and against which **the Southern Ocean box is comparatively well-behaved**. And the surface field carries a *larger* error than TOA everywhere (14.4 against 11.5), which is an atmospheric shortwave-absorption signature and is global rather than a Southern Ocean artefact.

44.3 Scoring every arm on all three constraints at once

Southern Ocean *surface* bias against CERES, ocean points only, with the tropics and both Siberian seasons as constraints and global net TOA carried alongside:

	SO sfc	SO rmse	SO CRE	Δ trop	Δ JJA	Δ DJF	global TOA
control	+3.66	7.04	+7.69	—	—	—	+0.64
S4 inppmin50k	−0.96	5.31	+3.02	+0.26	+1.03	+0.05	+0.25
LX1	−0.52	5.15	+3.86	+0.75	+1.29	+0.54	+1.08
P5	+0.74	5.12	+4.98	+0.35	+1.15	−0.06	+0.66
LX3	−4.75	7.31	−0.63	−0.97	+1.33	+0.26	−0.88
LY2	−4.75	6.81	−0.87	−0.32	+0.61	+2.14	−0.58
A1a	−2.32	5.70	+1.25	−1.06	−0.75	+0.83	−1.25

S4 wins. LX1 is nominally closer to zero on the SO surface (−0.52 against −0.96) and pays 0.83 W m^{−2} of global imbalance for that last 0.4 — a bad trade. LX3 and LY2 reach the Southern Ocean by dimming the planet. A1a is the only arm that damages the tropics *and* Siberian summer, confirming its round-3 rejection on both counts. **S4 is the only arm that improves the Southern Ocean surface while landing global net TOA near the piControl target, at no cost in either Siberian season or the tropics — and it is one namelist number.**

~~LX1 fails the tropical guardrail at +0.75.~~ **My own sign error.** The test compared $|\Delta$ trop| against ± 0.5 regardless of sign, but the tropics sit *below* CERES (42.61 against 45.11) and need *more* absorbed energy, so +0.75 is an improvement. The test is now one-sided. LX1 reads CLEAN; it is still disqualified, on global energy.

44.4 S4 coupled: 11G

Every coupled arm that sets RCL_INPPMIN — 10A, 10B, 11A, 11B, 11C, 11D, 11E, 11F — uses 70000. **The 50000 value has only ever run in AMIP.** So 11G is 11E with that one line changed: a strict one-variable pair against a run already complete at 50 years, needing no new control. Submitted 2026-08-11 as job 26868368.

~~The AMIP number should be a *lower* bound. The coupled amplification measured in round 27 is 1.40×... A lever that cools the SO surface is exactly the kind that should amplify.~~ **Falsified by 11G itself (§44.5):** the amplification is a property of the *lever*, not of coupling, and S4 does not have it. The 1.40× came from sea-ice albedo — clear-sky, not CRE — with SST falling 0.598 K and ice growing 3.2 pp, and S4 moves neither enough to engage it. 11E’s remaining gap is SO CRE 5.65 short of CERES with net TOA +0.79.

11G’s leg history, and the decade that is not a clean pair (2026-08-11→12). 11G reached 40 years on 2026-08-12 and leg 5 (1390-99) is running. Getting there was not clean, and one decade of it cannot be used.

Legs 3-5 were killed by a defect of mine in an unrelated branch: `couple_put` called `put_2d(GUE_ZOHV)` unconditionally, so any binary carrying the Raupach roughness code aborts with “*oasis_put is called for a variable not in namcouple*” unless the `namcouple` declares the field. 11G was rewound to 1370 and restarted on the production binary. Every one of the five legs is confirmed to have loaded the same LPJ-GUESS executable, md5 8c5ab467, so there is no binary discontinuity.

The soil map is a different matter. Staged `slt_TC095.nc` per leg, counted by organic (type 6) cells:

years	11E type 6	11G type 6	
1350-59	547	547	matched
1360-69	547	547	matched
1370-79	547	233	NOT a one-variable pair
1380-89	547	547	matched
1390-99	547	547	matched (in flight)

The repaired CORE3 soil map slipped into 11G’s third leg alone and was reverted after. So for 1370-79 the arm differs from 11E in *two* things, not one: RCL_INPPMIN and 314 cells moved off soil code 6 — a code `soil.cpp` treats on its own branch. **Score 11G against 11E on 1350-69 plus 1380-99, 40 years, and exclude 1370-79.** The campaign minimum is 30 years, so this costs nothing except the temptation to use the most recent 30, which is exactly the window that contains the contaminated decade.

A stale log file can eat a completed leg. Leg 4 ran to completion — job COMPLETED in 4:27:46, all ten years present *and already remapped* in the run directory — and delivered nothing, because the file-movement step hit

```
ERROR: File movement conflict
log/oasis3mct/nout.000000_13800101-13891231 already exists.  Skipping movement
```

left behind by one of the killed attempts at the same leg. `esm_tools` aborts the *whole* movement on the first conflict, so 10 years of model output stayed in `work/` while `outdata` showed 30 years and the date file still read 1380. The run looked like it had failed and was one resubmission away from recomputing 4.5 hours that were already on disk. Moving the dead attempts’ leftovers aside and re-running `esm_runscripts -t tidy` recovered all ten years and advanced the chain. *A leg that looks unfinished should be checked in work/ before it is rerun: ifs.stat* and the presence of `atm_remapped_*` files settle it in seconds.

44.5 S4 coupled, scored: real, resolved, and three times smaller than AMIP (2026–08–12)

11G against 11E over the **30 matched years** 1350–69 plus 1380–89; 1370–79 excluded because 11G alone carries the repaired soil map there. Thresholds are $1.96 \text{ sd} \sqrt{2/n}$ from 11E’s own interannual scatter over the same years.

	11E	11G	Δ	threshold
net TOA [W m^{-2}]	+0.387	+0.205	−0.182	± 0.362
SO SW CRE [W m^{-2}]	−62.472	−63.946	− 1.474	± 0.479
SO net TOA [W m^{-2}]	−51.417	−52.499	− 1.082	± 0.778
SO clear-sky SW [W m^{-2}]	217.847	217.294	−0.553	± 0.902
SO SST [$^{\circ}\text{C}$]	3.561	3.297	− 0.264	± 0.098
SO sea ice [%]	9.162	9.758	+0.595	± 0.806
global T2m [$^{\circ}\text{C}$]	12.604	12.388	− 0.216	± 0.123
Siberia JJA T2m [$^{\circ}\text{C}$]	9.670	9.777	+0.107	± 0.397
Siberia DJF T2m [$^{\circ}\text{C}$]	−32.957	−33.322	−0.365	± 0.736
Siberia DJF soil [$^{\circ}\text{C}$]	−12.606	−13.015	−0.409	± 0.469

S4 transfers, and costs nothing. SO SW CRE, SO net TOA, SO SST and global T2m all resolve, all in the intended direction, and *no* Siberian term does — the AMIP promise of a free Southern Ocean gain holds coupled. The CERES gap narrows from +5.36 to +3.89, i.e. 27 % of what 11E had left.

But it is three times smaller than AMIP, not $1.40\times$ larger. Against S4’s own AMIP deltas (−4.704 SO CRE, −3.813 SO net TOA) the coupled ratios are **0.31** $\times$ and **0.28** $\times$. The prediction from round 27’s DMS pair was ≈ -6.5 ; the measurement is −1.474.

The discriminator is sea ice, and it is a property of the lever. The amplification mechanism is sea-ice albedo, which is clear-sky and invisible to CRE. DMS at $S = 166$ cooled the SO surface 0.598 K and grew 3.2 pp of ice; S4 manages 0.264 K and 0.595 pp, the latter not resolved, and its clear-sky SW response is not resolved either. The amplifier never engages. So “*treat the AMIP magnitude as a lower bound*” is not a rule about coupling — it is a rule about levers large enough to move ice, and S4 is not one.

What this does to the decision. S4 remains the best available arm: it is one namelist number, it improves the Southern Ocean, and it is free everywhere else. It is simply much smaller than advertised, and on its own it does not close the SO gap. The global energy target is untouched — −0.182 against a ± 0.362 threshold is not a result. Leg 5 will take this to 50 years and 1390–99 is matched, so the clean window becomes 40 years and every threshold above tightens by $\sim 15\%$.

44.6 Krakatoa in a pre-industrial run

The net-TOA history was regenerated with the 11 series and given the AMIP companion it had always implied (Fig. 32). The AMIP panel shows a deep *synchronised* dip in 1883–86 in every arm. That is **Krakatoa** — and a run at fixed 1850 forcing cannot contain a volcano. It is independent visual confirmation of §43.9, and stronger than the CO_2 argument: not a slow drift, but real volcanic events inside runs the logbook describes as pre-industrial.

The AMIP arms also no longer cluster near the “atmosphere-only floor” the coupled figure draws at +0.67. They straddle it: LX1 +1.08, P5 +0.66, control +0.64, S4 +0.25, LY2 −0.58, LX3 −0.88. The line is relabelled as what it is — the AMIP *control*, one arm among many.

The radiation objective — atmosphere-only (AMIP), the round-27 candidates

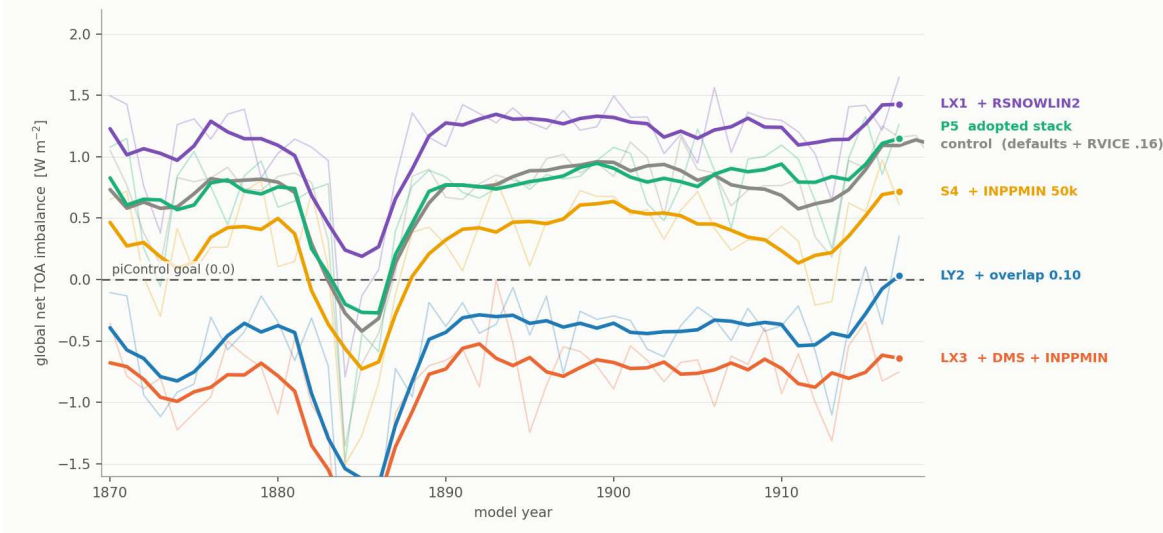


Figure 32: The AMIP radiation objective, round-27 candidates. The synchronised 1883–86 dip is Krakatoa: these “pre-industrial” runs read transient forcing, because NCMIPFIXYR sits in the dead &NAMECECMIP6 block and the runtime value is -1 (§43.9). S4 lands nearest the piControl goal; LX3 and LY2 overshoot past it; RSNOWLIN2 alone (LX1) moves it the wrong way.

44.7 Where the SO/tropics phase ends

Adopt S4 — RCL_INPPMIN 70000 $\rightarrow$ 50000 — pending 11G. It is the only arm that improves the Southern Ocean without spending the tropics, Siberia or the global budget.

Retire the round-26/27 lever line for tuning purposes. RSNOWLIN2 closes the tropical longwave deficit and is the largest such response ever measured here, but it warms the Southern Ocean and worsens global energy; DMS and ov1 0.10 reach the Southern Ocean only by dimming the planet. None is adoptable. RSNOWLIN2 remains interesting as *physics* — it is the one lever that moves the tropical longwave at all — and should be revisited if the atmospheric-absorption question below is resolved.

The open structural question. The surface shortwave error exceeds the TOA error everywhere (14.4 against 11.5 globally), and the TOA and surface constraints disagree about which arm is best: P5 has the surface nearly right and TOA 5 W m^{-2} short, LX3 the reverse. Closing a TOA gap of ≈ 5 costs the surface ≈ 5 , and it did, almost exactly. That is an *atmospheric absorption* signature, not a cloud-amount one, and if it is the underlying error then no cloud-reflectivity lever can satisfy both — the campaign has been optimising against whichever metric happened to be quoted. Measuring it is one pass over data already on disk.

45 Round 29 (2026–08–13→16): the land-surface repair priced, and roughness falsified structurally

Three arms differing by one thing each — **11G** (adopted stack, S4), **11I** (+ land-surface repair), **11J** (+ Raupach canopy roughness) — run at 50 years to price the two remaining land changes. 11I completed. 11J reached 30 years, lost its fourth leg to a filesystem error, and was restarted. The roughness lever is dead, and not for a statistical reason.

45.1 Getting the arms to start: a fork, and two coupling defects (2026–08–13→14)

Both arms build from `run/50yr-arms = 168a98b` (LPJ-GUESS gridlist from TC095-land rather than L096) plus a cherry-picked 0218bbb (cold-start a gridcell that carries no spin-up state). They *exclude* 46e7a84; see §45.2.

Two defects had to be cleared first, both in the roughness coupling and both fatal at about 80s:

- `put_2d(GUE_ZOHV)` was *ungated*. `ifraupachz0` gated the *computation* but not the OASIS put, so 11I — which does not request the field — sent it anyway: `oasis_put` is called for a variable not in `namcouple`, `mod_oasis_getput_interface.F90:685`. This is the same defect that cost 11G three legs on

2026–08–11; it had been fixed once and lost. Gating the put is what allows both arms to run from *one* binary (md5 ff9b5507), so 11J–11I carries no build difference at all.

- **11J’s coupling fields were declared in the wrong section.** A new field needs `lpj_guess.add_veg_atm_fields` on the sending side and `oifs.add_atm_veg_fields` on the receiving side; 11J had the latter under `oasis3mct` and lacked the former. `esm_runscripts` then exits with `Coupling var not found without creating the experiment directory`, so there is no run to inspect — only the driver log.

45.2 46e7a84: 107 new fail() calls, classified (2026–08–14→15)

Laszlo’s 46e7a84 (“Fix historical run crashes during land-use restart years”, 2026–08–11, 2065 changed lines in `landcover.cpp` alone) adds **107** hard `fail()` calls and took `landcover.cpp` from 9 to 74. Three fired on our restart. Rather than patch them one at a time, the twelve failing cells — *all* between 61 and 71°N, this campaign’s own region — were classified:

- **Nine are floating-point roundoff**, 3.3×10^{-9} to 4.2×10^{-7} , against `lc_bookkeeping_tol() = INPUT_RESOLUTION  $\times$   $10^{-9}$` . That tolerance is scaled off the precision of the land-use *input*, but is applied to a sum of restart-state floats whose error floor is `float epsilon`; 4.2×10^{-7} is $\approx 3.5 \times$ float32 eps. **No restart can pass it.** A later run with the check relaxed measured 14909 metadata repairs and 1099 renormalisations across a ≈ 10 200-cell gridlist — essentially the whole domain fails it, not twelve cells.
- **Three are real:** stand metadata against physical stand fraction, differing by 8.0×10^{-4} , 6.1×10^{-3} and 3.3×10^{-2} . These are ours, not his; the check found something wrong in our own spin-up state.
- **One is the peat conflict** (§45.3).

The function carrying the first two is named `synchronize_carried_landcover_state` and its next statement is `gcst.frac = st_sum[s]` — **it is already a repair routine that refuses to repair**. Both CMIP6 predecessors of this configuration repair rather than assert: EC-Earth3 `lpjg_v4p1` rescales land-cover fractions by their sum with a warning and pushes the residual into `NATURAL` (it tests closure at 10^{-14} , *tighter*, but fixes); JSBACH3 `scale_cover_fract` iterates `ntiles` times toward `sum=1` and calls `finish()` only if a fraction is still out of range afterwards.

45.3 Peat versus LUH3: two prescribed inputs pinned to different epochs (2026–08–16)

At 67.7917°N 19.2857°E the peat map prescribes 0.2264 and the spin-up carries a 0.1917 physical stand, while LUH3 leaves only **0.088604** `NATURAL`. Reproducing that number exactly required two details: our configuration is `run_barren 0`, so `NATURAL` also absorbs the LUH3 closure gap rather than being the raw `primf+primn+secdf+secdn` (0.043008); and cells with no usable LUH3 state fall back to `NATURAL=1` instead of taking the $\approx 2 \times 10^{20}$ fill value into the arithmetic.

The conflict is **not** static-peat-versus-dynamic-land-use. Both are prescribed and both are static here: `externalinput.cpp:779` pins the LUH3 year to `fixedLUafter` unconditionally, ignoring the model calendar. Read at the run’s own year the same cell gives `NATURAL = 0.617646` (1350), `0.796720` (1399) — **no conflict at all**. The entire problem is an artefact of the 1850 pin, which is not negotiable because 1850 forcing *is* the piControl convention.

Neither CMIP6 predecessor could hit this. EC-Earth3 reads `PEATLAND` from the land-use file itself (`externalinput.cpp:670`), so there is no second dataset to disagree with — its `file_peat` is a dead declaration, referenced nowhere — and its coupled configuration runs `run_landcover 0` regardless. JSBACH3 has no peatland type at all.

Two policies were coded behind `peat_lu_conflict_policy`, default 0 so an unmodified `.ins` is CMIP7-compliant:

- **0, LUH3 authoritative** — required for CMIP7, where the land-use dataset is a common forcing. Peat is capped and the surplus *area and its pools* are moved by `landcover_dynamics`. Under `fixed_LU` that machinery never runs (`landcover_init` returns straight after `landcover_reconcile_restart_state`, `landcover.cpp:156–163` and `lc_changed()` is false), so this policy *refuses with an explanatory error* rather than truncating. A first attempt that capped anyway rewrote the bookkeeping while the stand kept its area and was caught one call later by the very next check (`st=2 metadata=0.0886 physical=0.1917`) — Laszlo’s comment that capping “would discard state without a transfer” is not a preference but structurally true here.
- **1, physical peat authoritative** — the stand keeps its area and the non-peat categories are scaled into the remainder. Nothing is discarded, so no transfer is needed. Defensible only for fixed-forcing control runs: north of 60°N mean LUH3 rangeland at 1850 is 0.0040, while northern peat holds order 500–1000 PgC and sets the soil thermal regime this campaign’s bias lives in.

Only 73 of 10211 cells have a peat-map value above the available NATURAL, and only one aborted, because elsewhere the spin-up had already capped the physical stand lower.

45.4 11I: the repair is not free (2026–08–16)

11I–11G is not a one-variable pair, despite its runsript header saying “ONE input file”. Three things differ: the `_v2` soil map, the L096→TC095–land gridlist, and the cold-start binary path; the last two are inseparable, because the new gridlist is what adds the 307 cells the cold-start path lets run. Read it as *the land-surface repair bundle*.

Scored over the 40 matched years (1350–69 + 1380–99; 1370–79 excluded because the repaired soil map slipped into 11G’s leg 3 alone, 233 organic cells against 547):

metric	11G	11I	diff	paired <i>t</i>
net TOA [W m ^{−2}]	0.321	0.301	−0.020	−0.33
Siberia DJF soil [°C]	−13.079	−12.771	+ 0.308	2.49
Siberia JJA soil [°C]	8.160	7.939	−0.221	−1.08
Siberia JJA T2m [°C]	9.664	9.434	−0.230	−1.19
SO SW CRE [W m ^{−2}]	−63.964	−63.779	+0.185	1.71
global T2m [°C]	12.365	12.335	−0.030	−1.25

The statistic mattered. Every arm branches from the same 1350 state, so the difference series is *paired* and starts near zero. Scored against the control’s absolute interannual scatter ($1.96 \text{sd} \sqrt{2/n}$, threshold 0.345) the DJF soil warming reads as null; paired, it is $t = 2.49$ and **resolved**, and it is still growing (first half +0.212, second +0.404).

But against the target it is a trade, not a freebie. Siberian JJA T2m bias against ERA5:

$$11E - 1.013 \rightarrow 11G - 0.673 \rightarrow \mathbf{11I - 1.123} \rightarrow 11J - 0.844$$

11I hands back S4’s entire Siberian summer gain and lands *worse than the 11E baseline*. The increment (−0.230, $t = -1.19$) is not resolved, but the ranking is unambiguous and JJA Siberia is the bias the campaign exists for. Vegetation moves the other way: on matched decade and matched cells, TREEFPC +0.0094 and C3G +0.0276, recovering most of what S4 cost (0.0787 → 0.0649 → 0.0743).

45.5 Raupach: the lever cannot bootstrap the canopy it needs (2026–08–16)

11J–11I *is* a clean one-variable pair: same binary, one namelist flag. Over all 30 years, paired, nothing Siberian resolves — DJF soil, the *pre-registered* target, is +0.054 ($p = 0.670$); JJA soil +0.312 ($p = 0.167$); JJA T2m +0.161 ($p = 0.432$). Resolving the JJA soil point estimate at 95% would need $(1.96 \text{sd}/\text{effect})^2 = \mathbf{57}$ years, so **even a completed 50-year 11J is underpowered for its own point estimate**: with a 1.2 K interannual noise floor this design cannot detect anything below ≈ 0.35 K at 50 coupled years.

That is a statement about the diagnostic. The *mechanism* was then tested at the top of the causal chain, where a roughness change acts first and the noise is far lower:

Siberian land, 30 yr paired	11I	11J	diff	<i>p</i>
DJF $ \tau $ [N m ^{−2}]	0.0955	0.0916	−0.0039	0.113
DJF 10 m wind [m s ^{−1}]	2.1282	2.1911	+0.0629	0.253
DJF skt−2t [K]	−2.2719	−2.2761	−0.0042	0.787
JJA $ \tau $ [N m ^{−2}]	0.0753	0.0706	−0.0047	0.016
JJA 10 m wind [m s ^{−1}]	1.2821	1.3637	+0.0816	0.037
JJA skt−2t [K]	0.8456	0.9497	+ 0.1041	0.00003

JJA skt−2t survives Bonferroni over all 14 tests. But the pre-registered sign test — *noise cannot know to reverse between seasons* — fails: winter was predicted rougher (stress up, wind down) and instead has the *same* signs as summer. The winter lever does not act at all.

Why. Measured in 11J’s Siberian box (728 cells, 1370–79): tree height in January **2.89** m, density 0.161 indiv m^{−2} (of which BNS 0.112), implied stem diameter 0.0198 m, and total LAI collapsing Jan/Jul = 0.131. `add_woody` is called only for `lifeform == TREE`, so grass contributes nothing to λ by construction, and in winter everything rests on the stem term:

$$\lambda_{\text{stem}} = \text{dens} \cdot \text{diam} \cdot h = 0.161 \times 0.0198 \times 2.89 = \mathbf{0.0092}$$

Fed to the code's own constants ($C_S=0.003$, $C_R=0.3$, $C_{D1}=7.5$, $\kappa=0.41$), that gives $z_0 = \mathbf{0.0132\text{ m}}$ — the bare-soil value — against the IFS lookup's 0.058 m for the same box in DJF. **Raupach makes the winter surface 4.4× smoother, not 2.4× rougher**, which is exactly the sign the momentum budget measures.

“A leafless canopy is the roughest state” is true — for a canopy. A real boreal larch stand ($h = 18\text{ m}$, $\lambda_{\text{stem}} = 0.12$) gives $z_0 = 1.51\text{ m}$. At 2.9 m and 2 cm stems, λ is an order of magnitude below the peak of the Raupach z_0/h curve, on the rising branch where $z_0 \rightarrow 0$. The offline calculation that predicted DJF $0.058 \rightarrow 0.140\text{ m}$ assumed a forest; the coupled model grows scrub.

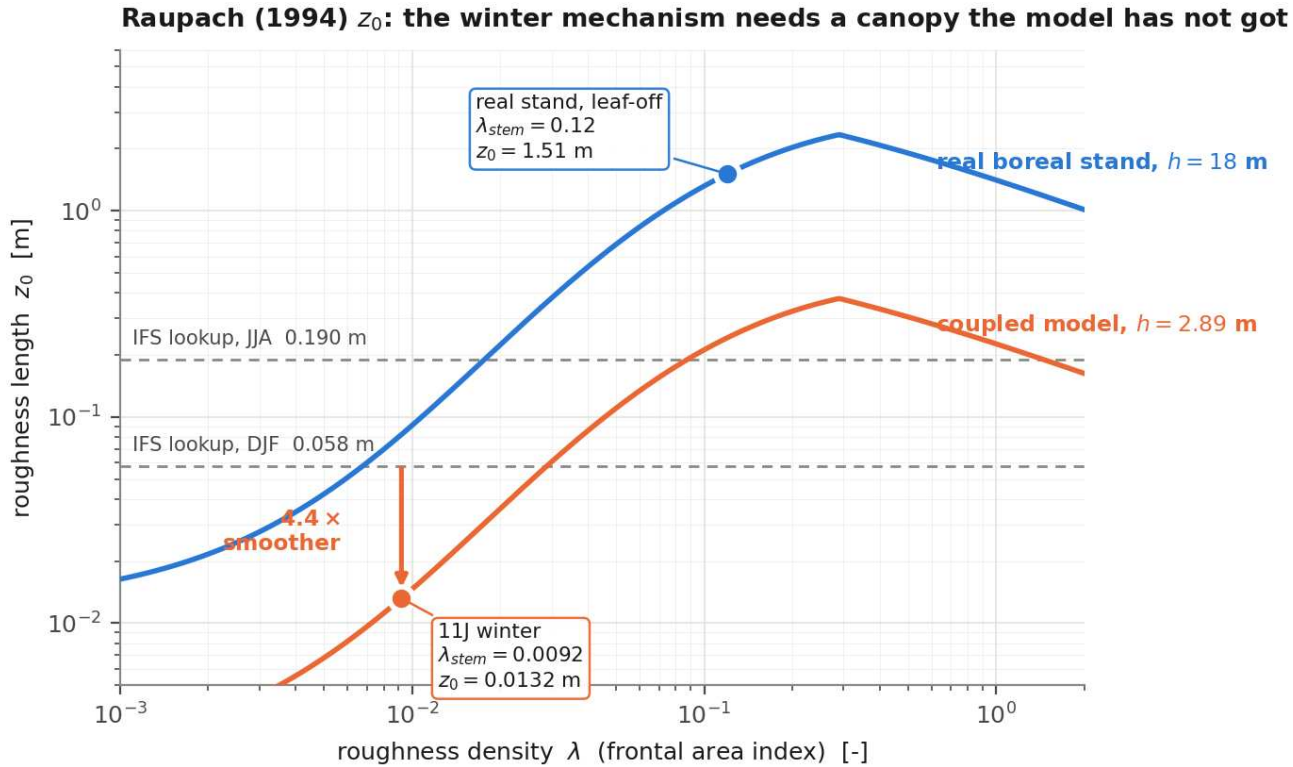
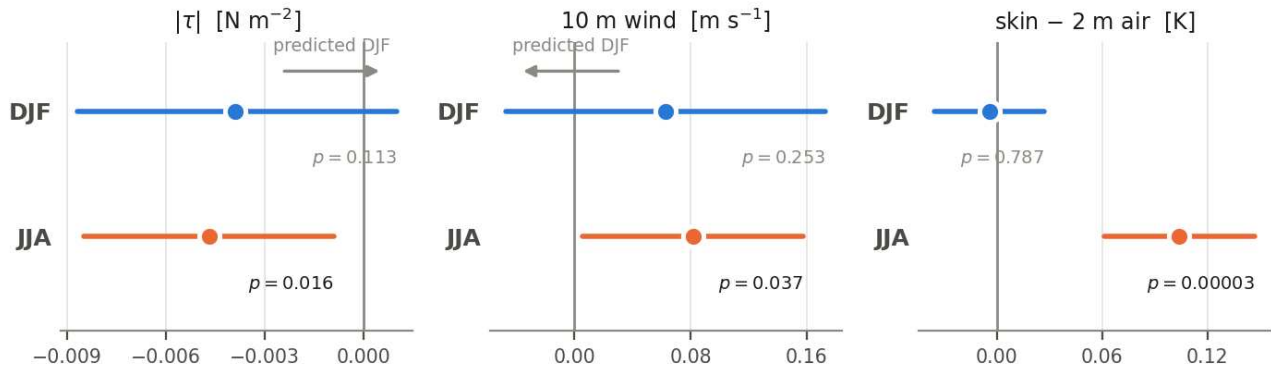


Figure 33: Why the winter lever cannot engage: the model has no canopy for it to act on. $z_0(\lambda)$ from the implementation's own constants ($C_S=0.003$, $C_R=0.3$, $C_{D1}=7.5$, $\kappa=0.41$), for the 2.89 m scrub the coupled model grows in Siberia (orange) and an 18 m boreal larch stand (blue). Measured in 11J's Siberian box (728 cells, 1370–79): density $0.161\text{ indiv m}^{-2}$, stem diameter 0.0198 m, so $\lambda_{\text{stem}} = 0.0092$ once the leaves are gone (Jan/Jul LAI = 0.131). That lands an order of magnitude below the peak, on the rising branch, at $z_0 = 0.0132\text{ m}$ — the bare-soil value — against the 0.058 m IFS lookup it replaces. **Switching Raupach on makes the winter surface 4.4× smoother, not 2.4× rougher.** A real stand at leaf-off sits at 1.51 m: “a leafless canopy is the roughest state” is true, for a canopy.

11J – 11I, Siberian land, 30 paired years: summer responds, winter does not



Points are paired mean differences, bars 95 % CI. Winter was predicted rougher ($|\tau|$ up, wind down) and instead has the same signs as summer. Only JJA skin-air survives Bonferroni over all 14 tests.

Figure 34: The momentum budget confirms it, where soil temperature could not. 11J–11I over 30 paired years, Siberian land, in native units (small multiples rather than a shared axis). Summer responds as a *smoothing*: $|\tau|$ -6.3% ($p = 0.016$), 10 m wind $+6.4\%$ ($p = 0.037$), and skin-to-air decoupling $+0.104\text{ K}$ ($p = 3 \times 10^{-5}$, the only result surviving Bonferroni over all 14 tests). Winter shows the *same* signs, where the pre-registered mechanism required the opposite (grey arrows), and skin-air is flat at -0.004 K . The sign test was built as a falsification check — noise cannot know to reverse between seasons — and it fails. This is why Siberian DJF soil is $+0.054\text{ K}$ ($p = 0.670$): not a signal buried in noise, but no winter roughness change to propagate.

The structural finding. Raupach was proposed to fix a cold bias severe enough to lose the boreal forest, and its winter mechanism *requires* that forest. The lever cannot bootstrap the canopy it needs in order to work. This is stronger than the statistical null: with 57 years we would merely be measuring the response to a roughness *decrease*. It is the same trap as \$43.12 — BNS is outcompeted by C3 grass rather than climate-gate limited — seen from the other side.

Not explained: summer also got smoother, whereas a rough summer λ suggests Raupach should exceed IFS's 0.190 m. That estimate used box-total LAI, which includes grass contributing nothing, so it is untrustworthy; tree-only LAI would settle it. It does not affect the winter conclusion, which is the hypothesis under test.

45.6 A leg destroyed at finalisation, again (2026–08–15)

11J's leg 4 ran to **step 87648 — the full ten years** — and then died in finalisation: **error in line 261 io_netcdf_file_module.F90 NetCDF: HDF error on FESOM rank 129**. The XIOS buffer flush is destroyed by the abort, so ten completed years became nothing, the date file walked to 1390, and the run stopped. Ruled out: not our code (no oasis_abort, no LPJ-GUESS fail()), not Raupach (the failure is in FESOM's writer), not quota (2.81P of 4.22P), not systematic (11I never saw it in five legs). A transient Lustre/HDF fault. Recovered by resetting the date file to 1380–01–01 run 4; all three restart sets were intact, and the leg-4 oifs restart directory was *empty*, confirming nothing was written.

LPJ-GUESS output survived: it writes .out files per rank continuously, outside XIOS, so the vegetation diagnostics for the lost decade are on disk even though every netCDF field is gone. That is worth knowing for any future leg loss.

46 Round 30 (2026–08–18→19): the Southern Ocean cloud round, coupled

Six weeks of Southern Ocean work reduced to one question — *is there a lever that brightens SO cloud without wrecking Siberia?* — and it now has an answer, with the candidates named and two of them measured coupled.

46.1 The reframing: 84 % of the coupled SO error is inherited from AMIP

The same diagnostic applied to AMIP and coupled, 30yr each, SO 45–65°S TOA SW CRE against CERES (scripts/analysis/amip_vs_coupled_so.py):

run	SW CRE	vs CERES	tcc
AMIP P5 control	−63.097	+ 4.739	0.8356
AMIP LY2 ov1 0.10	−68.983	−1.147	0.8474
AMIP LX3 DMS+INPPMIN	−68.737	−0.901	0.8435
CPL 11E	−62.186	+ 5.651	0.8365
CPL 11G +S4	−63.673	+ 4.163	0.8359

Cloud cover is 0.8356 in AMIP against 0.8365 coupled, and the CRE differs by 0.91 W m^{-2} : **the cloud field transfers**. 84% of the coupled error is already present in AMIP over a prescribed, correct ocean, and 11G coupled (+4.163) is *closer* to CERES than the AMIP control (+4.739). The Southern Ocean bias is mostly an unclosed AMIP problem, not a coupling problem — which is why the remaining effort belongs in AMIP, where an arm costs a fraction of a coupled one.

46.2 The overlap lever taken coupled: 11L and 11M, both rejected

Both are 11G plus one namelist number, 40 yr, scored paired over the 30 clean matched years 1350–69 + 1380–89. 11G’s 1370–79 carries the repaired soil map and has to be excluded; `coupled_dms_pair.py` takes an exclusion argument for exactly this.

	11L ov1 0.35	11M ov1 0.10	threshold
SO SW CRE	−1.565 *	−5.938 *	0.517
SO SW CRE DJF	−3.399 *	−11.433 *	1.167
vs CERES	+3.89 → + 2.33	+3.89 → − 2.05	
net TOA	−0.281	− 0.698 *	0.403
tropics SW CRE	−0.297 *	−1.370 *	0.209
global T2m	−0.242 *	−0.604 *	0.162
Siberia JJA T2m	− 1.000 *	− 2.321 *	0.406
Siberia JJA soil	−1.014 *	−2.377 *	0.402

It transfers — the first SO lever that does. AMIP predicted 11L at +1.992 against CERES; coupled lands at +2.33. The transfer ratios are $0.57\times$ and $1.01\times$ on annual SO CRE, against DMS’s $0.19\text{--}0.44\times$ and S4’s $0.28\text{--}0.31\times$. The discriminator, consistent with §58.5, is how much sea ice the lever grows.

Both are rejected anyway. 11M fails net TOA outright. 11L passes net TOA but costs 1.000 K of Siberian JJA — it spends the entire budget of target 2 to buy 40 % of target 1. The vegetation cost is proportionate: against 11G, 11L loses 39.5 % of boreal needleleaf and 11M loses 89.2 %, all but eliminating larch.

46.3 Ocean drift, and a correction that inverts its own conclusion

Whole-ocean potential temperature (`thetaoga`), steady rate after the first decade:

arm	K/century	per millennium
11E	+0.22	+2.2 K
11G	+ 0.19	+1.9 K
11L	+0.12	+1.2 K
11M	−0.02	~ 0

Not decelerating, so a multi-millennial spin-up on 11G is untenable.

Corrected the same day. An earlier reading of this table reported a “ -1.0 W m^{-2} structural leak, invariant across arms” between TOA and surface, and concluded the drift could not be fixed radiatively. **That -1.0 is snow enthalpy.** $sf \cdot L_f$ measures $0.962/0.976/1.018/1.073$ against residuals of $-1.006/-0.996/-1.022/-1.088$. The convention was already on record in §11.1: $\text{surface} = \text{ssr} + \text{str} + \text{sshf} + \text{slhf} - \text{sf}\rho L_f$. With the term included the column conserves to $0.02\text{--}0.04 \text{ W m}^{-2}$ and the ocean warming matches what TOA implies, so **tuning net TOA to zero does stop the drift** — 11M, at TOA -0.016 , drifts -0.02 K/century . The conclusion inverts.

A second error in the same reading was a window mismatch: 11G’s net TOA is $+0.205$ over 1350–69+1380–89 but $+0.732$ over 1380–99. Net TOA is itself drifting upward through these runs, which matters separately for spin-up planning.

47 The 11-series coupled campaign (2026–08–16→21): LX4 taken coupled

47.1 The roster

arm	content	role
11E	campaign base	the reference the others are read against
11G	11E + S4 (RCL_INPPMIN 50000)	the adopted base for rounds 29–30
11I	11G + v2soil (LPJG repair)	prices the land-surface repair
11J	11I + ifraupachz0	the roughness pair, one binary, run-time gate
11L	11G + ov1 0.35	SO overlap lever, rejected (§46.2)
11M	11G + ov1 0.10	SO overlap lever, rejected (§46.2)
11N	LX4 stack, 1850 forcing	the best AMIP tuning, taken coupled
11P	LX4 stack, constant 1990 forcing	the same stack, period-matched to ERA5

47.2 Is the 1990 forcing actually 1990? Everything except LPJ-GUESS

11P exists to remove a confound that had been quietly distorting every comparison against ERA5: a PI run read against a present-day reanalysis is cold *by construction*, so part of the reported land bias was never a model error at all. NCMIPFIXYR pins GHG, ozone, composition, stratospheric aerosol, SO₄, solar TSI *and* MACv2SP. It does **not** pin LPJ-GUESS, which has its own `fixed_C02`, `fixed_ndep` and `fixed_LU`.

Two traps were hit while verifying this and are worth recording:

- The `&NAMECECMIP6` block accepts NCMIPFIXYR and is **never read**. Liveness must be checked as `Set JYEAR` in `NODE.001_01`, not by finding the name in `fort.4`.
- **A correct run was cancelled on a misreading.** The log line `SP_SETUP: IYR` reports the MACv2SP *file* that was opened, not the year the profile is evaluated at. The evaluation year is `YEAR_FR`, which honours NCMIPFIXYR in both source trees (`radiation_scheme.F90:387`). 11P was already right; the patch written to “fix” it was reverted and the arm resubmitted unchanged.

47.3 What the stack bought: the land bias by band and season

Land T2m bias against ERA5, 1380–89, from `scripts/analysis/land_bias_season_band.py`:

band	11G					11N (1850)					11P (1990)				
	DJF	MAM	JJA	SON	ANN	DJF	MAM	JJA	SON	ANN	DJF	MAM	JJA	SON	ANN
60–90N	−6.35	−4.60	−1.96	−3.65	−4.14	−4.49	−2.99	−1.52	−2.26	−2.82	−3.63	−1.64	−0.58	−1.29	−2.11
30–60N	−3.83	−3.56	−0.80	−1.84	−2.51	−3.18	−3.37	−0.17	−0.93	−1.91	−2.90	−2.49	+0.72	−0.22	−2.11
0–30N	−2.96	−2.21	−2.46	−2.33	−2.49	−2.59	−1.98	−1.99	−1.88	−2.11	−1.90	−0.98	−1.16	−1.19	−2.11
30S–0	−2.75	−2.83	−2.21	−1.68	−2.37	−2.30	−2.20	−1.61	−1.00	−1.78	−1.76	−1.76	−0.90	−0.63	−2.11
60–30S	−0.89	−1.46	−2.29	−1.52	−1.54	−0.32	−0.90	−1.73	−1.00	−0.99	+0.07	−0.66	−1.13	−0.86	−2.11
ALL LAND	−3.47	−3.27	−2.14	−2.29	−2.79	−2.76	−2.66	−1.48	−1.54	−2.11	−2.23	−1.85	−0.73	−0.83	−2.11
ocean	−1.86	−1.64	−1.53	−1.49	−1.63	−1.44	−1.24	−1.15	−1.10	−1.23	−0.86	−0.71	−0.64	−0.56	−2.11

Read the two steps differently. 11G → 11N is a genuine tuning gain at a fixed forcing basis: all-land annual −2.79 → −2.11, and 60–90N DJF −6.35 → −4.49, the largest single winter improvement in the campaign. 11N → 11P is *not* a tuning gain — it is a change of comparison basis, from a 1850-forced run read against a present-day reanalysis to a period-matched one. Only 11P should be quoted as a bias against ERA5.

And the shape of what remains is the finding. In 11P the land–ocean gap is −1.37 K in DJF against −0.09 K in JJA. Siberian JJA is essentially spot on (10.649 against ERA5’s 10.666). The campaign was built around a boreal *summer* cold bias; what is left is a *winter* one, and §48 resolves where in the column it sits.

48 Round 31 (2026–08–21): the winter land bias is above the screen

Three surface-side levers are scored null in this round, and the reason turns out to be the same for all three.

48.1 Round 16 scored at last: skin conductivity is not the route

Round 16 (ECE_LAMSK_SN, the exposed-snow skin conductivity $Z_{\text{SNOW}} 7 \rightarrow 15 \rightarrow 25$) ran 48 years on 2026-08-04 and then sat unscored for 17 days. Scored paired against I1 over 1872-1915:

arm	60-90N DJF	threshold	60-90N ANN	land-ocean gap DJF
J1 LAMSK_SN 15	+0.125	0.41	+0.062	+0.009
J2 LAMSK_SN 25	-0.039	0.45	-0.015	-0.073

The round predicted +1 to +3 K. It measured nothing, at $3.6\times$ the default, with power to detect 0.4 K.

This is **not** a seventh silent no-op, and that was checked before the null was believed: the symbol lives in `libsurf.SP.so`, *not* `libarpifs.SP.so` — grepping only `arpifs` returns zero and looks exactly like the W1 failure — `fort.4` carries 15/25, and the arms differ from I1 by up to 21 K locally in year 1 across 53% of cells. The lever is live and its mean response is genuinely zero. Round 16’s own pre-registered falsifier therefore fires: the winter bias lies in boundary-layer mixing instead.

48.2 Raupach roughness, measured: null, as §45.5 predicted

11J = 11I + `ifraupachz0`, one binary with a run-time gate, both branched from 1350. Verified before scoring: 11J’s `guess.ins` carries `ifraupachz0 1` and `fort.4 ECE_CPL_LPJG_Z0 = .true.`; 11I has neither.

window	60-90N DJF	30-60N DJF	land-ocean gap DJF
1380-1389	+0.302 (0.89)	-0.234 (0.41)	-0.017 (0.20)
1360-1379	-0.172 (0.58)	-0.161 (0.43)	-0.003 (0.19)

The sign flips between windows and the DJF gap is zero in both. This is the T2m confirmation of the mechanism argument already made in §45.5: Raupach derives z_0 from FPC and *canopy height*, so wherever LPJ-GUESS has lost the forest it too returns a low z_0 . It changes the functional form, not the missing canopy.

48.3 The decomposition: the skin is already right

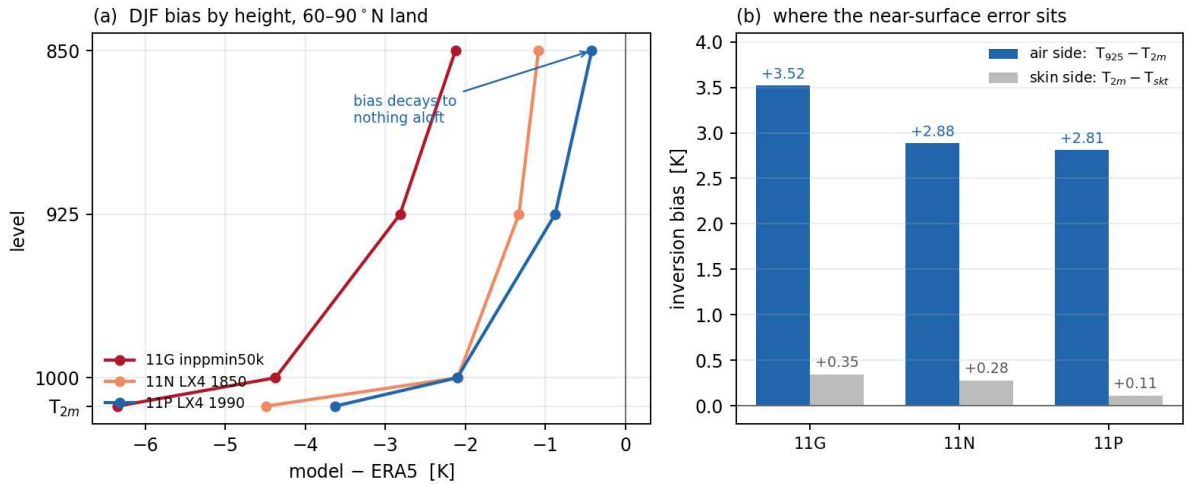


Figure 35: The DJF land cold bias is surface-confined, but the surface is not what is broken. **(a)** Model-ERA5 DJF bias over 60-90°N land against height. In 11P the bias runs -3.63 K at the screen to -0.42 K at 850 hPa, so it decays to nothing aloft and a *redistribution* lever is the right class. **(b)** Splitting the near-surface error: the air-side inversion bias ($T_{925} - T_{2m}$) is +2.81 K in 11P while the skin-side one ($T_{2m} - T_{skt}$) is +0.11 K. The skin is already right, so the entire excess inversion sits in the layer between the screen and 925 hPa. That is why skin conductivity (§48.1), Raupach z_0 (§48.2) and F1 RVZ0H all scored null: each adjusts a part of the column carrying no error. Note also that the campaign’s tuning improved both terms, 11G→11N→11P.

Measured against ERA5 monthly pressure-level T (`/pool/data/ERA5/E5/p1/an/1M/130`, DJF 1990-2014, remapped to the model grid), with land cells excluded *per level* where the level lies below ground. That mask is not cosmetic: without it the “bias” at 1000 hPa reads +28 K, which is mountains being extrapolated rather than error.

11P, DJF	T _{2m}	T ₁₀₀₀	T ₉₂₅	T ₈₅₀	lowinv bias	sfcinv bias
60–90N	−3.63	−2.10	−0.88	−0.42	+ 2.81	+ 0.11
30–60N	−2.90	−1.86	−1.31	−1.31	+1.07	−0.27
ALL LAND	−2.23	−1.43	−1.25	−1.22	+0.86	−0.36

Both inversion columns are differences of differences, so any systematic model–ERA5 offset cancels out of them. Splitting the all-land number: of the −2.23 K DJF bias, roughly −1.0 K is excess inversion and roughly −1.2 K is a **deep tropospheric cold bias** still present at 850 hPa, which no mixing lever can remove and which tracks the ocean bias (−0.69 K) and the TOA budget. **Only about half of the remaining land bias is addressable by a surface or boundary-layer lever at all**, and that half is concentrated at high latitude.

48.4 The lever: &NAMVDFS, and why it had to be created

The 2 m–925 hPa layer is mixed by the stable branch of VDFEXCU (Sandu & Beljaars 2013, cited in the routine’s own header at `vdfexcu.F90:81`), and every constant governing it was a hardcoded literal. `scripts/patches/expose_sbl_mixing.py` moves five into YOEVDFS, set in SUVDFS from a new namelist:

name	default	what it does
RSBLB	5.0	coefficient b of the LTG stable stability functions
RSBLD	1.0	coefficient d of the same
RSBLLMIN	30.0	asymptotic mixing length above the stable BL, and floor within it [m]
RSBLLMAX	300.0	cap on the stable mixing length [m]
RSBLPBLF	0.1	fraction of stable BL depth taken as the mixing length

Defaults equal the removed literals, so a `fort.4` without the group is bit-identical to the as-released model. ZCB, ZCD and ZLMIN were checked to appear only inside IF ($ZRI > 0$) before patching, so the knobs cannot reach unstable or convective mixing.

With $F_H = 1/(1 + 2b Ri\sqrt{1 + d Ri})$, **lowering b lengthens the stable tail** and the gain grows with Ri , concentrating the change in exactly the deep-inversion columns that carry the error:

Ri	$b = 5$	$b = 3$	$b = 2$
0.5	0.140	0.214 ($\times 1.5$)	0.290 ($\times 2.1$)
2.0	0.028	0.046 ($\times 1.6$)	0.067 ($\times 2.4$)

Which knob bites where. Inside the stable BL the mixing length is `clamp(0.1h, 30, 300)`, so the $0.1h$ term only takes over once $h > 300$ m and the cap needs $h > 3000$ m. An Arctic winter SBL is typically 50–300 m deep, so in the regime being targeted λ is pinned at the 30 m floor and RSBLPBLF never fires. The operative pair is therefore RSBLB (always evaluated) and RSBLLMIN; RSBLLMAX is effectively inert.

POSNAM is deliberately not used: it aborts when a group is absent, and no existing `fort.4` carries &NAMVDFS. A plain READ with IOSTAT keeps the defaults when the group is missing (< 0) and **aborts** when it is malformed (> 0), rather than swallowing a typo into an arm identical to its control. SUVDFS echoes the values to NULOUT unconditionally so NODE.001_01 proves what ran.

48.5 The SB arms, and the premise check they rest on

run	setting	prediction
SB0	LX4, no &NAMVDFS	rebuilt control (see below)
SB1	LX4 + RSBLB: 3.0	60–90N DJF warms; JJA and tropics barely move
SB2	LX4 + RSBLB: 2.0	the same, roughly twice the size

SB0 exists because the AMIP library was last built 2026–08–09 and commit 75a4e45 has since completed the DMS ICMCL ingest chain — not an inert namelist exposure but a change to which surface field is ingested and under which GRIB code. LX4 reads DMS, so SB1 minus the old LX4 output would mix the RSBLB effect with an ingest change. The Raupach pair was deliberately run from one binary for this reason; the same standard applies here, and since a rebuild cannot be a run-time gate, the control is re-run instead.

Does the AMIP base share the defect? It has to, or the screen is meaningless, and the bias being screened was measured on *coupled* 11P. Checked before the compute was spent (`scripts/analysis/amip_vs_coupled_inversion.py`), low-level inversion bias, DJF:

band	AMIP LX4	coupled 11P
60–90N	+1.54	+2.81
30–60N	+0.37	+1.07
ALL LAND	+0.43	+0.86

Same sign and same vertical shape, so the screen tests the right defect, and +1.54 K against a 44-year paired threshold of ~ 0.4 K resolves a lever that removes a quarter of the bias. But T_{925} is *identical* in the two (-0.88), so the whole AMIP–coupled difference sits at the screen: about 1.3 K of the coupled excess at 60–90N has no AMIP counterpart. AMIP prescribes sea ice, which is the obvious home for it. **A clean AMIP result bounds the direct land effect and cannot close the coupled gap**, and the sea-ice guardrail cannot be tested in AMIP even in principle.

Scored in §49. The falsifier below did not fire: the inversion responded, and coupled the screen followed.

Falsifier on record. If 60–90N DJF does not respond, the excess inversion is not set by the stable stability function, and what remains is the mixing length (RSBLLMIN) or a radiative rather than turbulent cause. **Guardrails:** 60–90N DJF must not overshoot warm, the inversion bias must not cross zero, global net TOA must stay within ± 0.3 at 44 years, Siberian JJA must not go backwards, and the Southern Ocean must be re-checked — $Ri > 0$ does not know about land, and the stable regime is rare over open ocean but common over ice.

48.6 Infrastructure fixed on the way

- `esm_master recomp-oifs-*` died in the config parser with `KeyError: 'Key general.with_co2_oce_coupling was not defined'`. The `feat/awiesm3-v3.4-co2` merge added a `choose_general.` block to the `component` yaml, and a standalone component build has no setup to define the key. Fixed with `esm_parser`'s documented `"*"` wildcard, which drops the block when the key is undefined; every setup defines it, so the wildcard never fires for them.
- The build then failed three times with `ifort: error #10103: can't fork process`. Not a code or HPC change: `ulimit -u` counts *threads* on Linux, and the IDE/agent session alone held ~ 1310 of the 2048 (`devin` 399, `language_server` 300, `github-mcp` 218, `claude` 206), leaving ~ 700 for a build that spawns `make -j32` with `fcm make -j 256` inside it. The first failure also orphaned ~ 250 `fcm` workers, which then broke the retries. Resolved by building on a compute node (`scripts/build/build_oifs_compute.sh`); the as-released `FCM_PARALLEL` is left untouched. Completed in 8:16 with zero fork failures.

49 Round 32 (2026–08–21→23): the stable-BL lever, scored

Round 31 ended with a diagnosis and an untested knob. This round scores it: four AMIP arms, one falsification, a coupled pair, and the first coupled state in this campaign that improves the winter land bias without paying for it somewhere else.

49.1 The AMIP screen, and the shortwave bill

SB0/SB1/SB2 ran 46 years each on one binary, scored paired over 1872–1915. SB0 exists because the AMIP library predated commit [75a4e45](#), which changed the DMS ICMCL ingest — LX4's own output was therefore not a valid control (§48.5).

DJF, 44 yr	SB1 $b=3$	SB2 $b=2$
inversion 60–90N	-0.424^*	-0.713^*
inversion all land	-0.148^*	-0.282^*
T2m 60–90N	+0.029	+0.285
T2m 30–60N	+0.060	+0.492 [*]
T2m all land	+0.077	+0.233 [*]
T925 60–90N	-0.396^*	-0.428^*
Siberia JJA	-0.002	+0.000
global net TOA	+0.090 [*]	+0.175 [*]

Three things to read here. The inversion responds strongly and **dose-consistently**. The Arctic *screen* does not — +0.285 against a 0.345 threshold — and the inversion weakens roughly 40 % by warming the screen and 60 % by cooling the air above it. And **Siberian JJA is exactly null**, which makes this the first lever in the campaign to move the winter bias without spending boreal summer.

The TOA bill is **shortwave, and cloud-mediated**: of SB2's +0.175, SW carries +0.319 against LW -0.144, and SW CRE carries +0.284 of the SW with total and low cloud both falling (-0.007, -0.005). Enhanced stable mixing dries and erodes low cloud. Regionally the tropics carry all of it (SW CRE +0.515, net +0.235) because *subtropical stratocumulus decks are stable boundary layers*; the Southern Ocean mildly improves (-0.182); and the Arctic's +0.620 is almost all LW, a symptom of the cooling aloft reducing outgoing longwave rather than a gain.

49.2 SB3: the mixing-length route, falsified

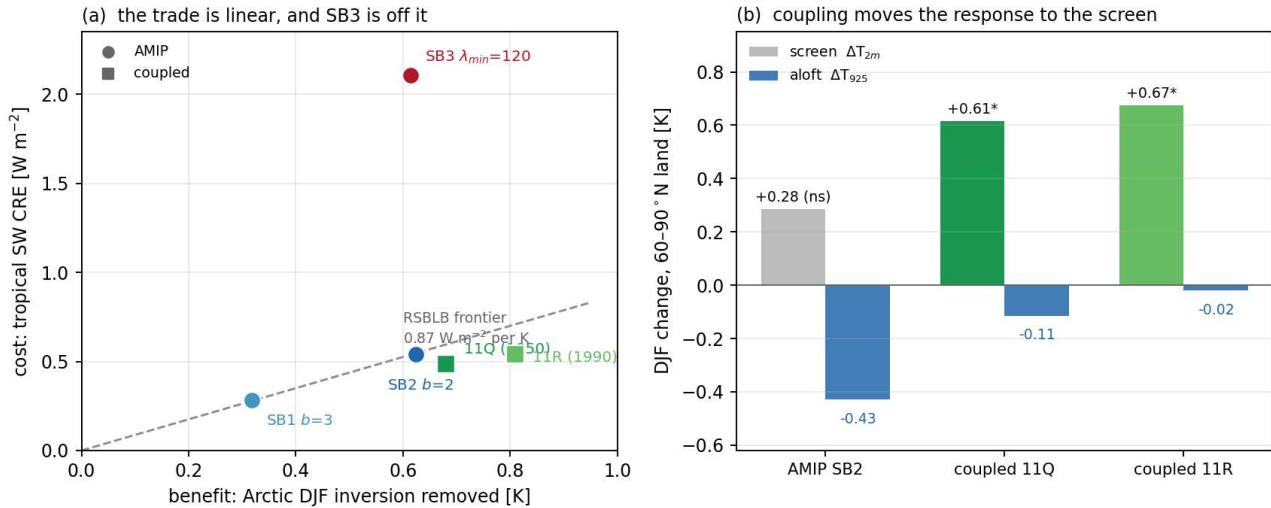


Figure 36: (a) What each arm pays (tropical SW CRE) against what it buys (Arctic DJF inversion removed). SB1 and SB2 lie on one line through the origin at 0.87 W m⁻² per K, so the trade is *linear in dose* and cannot be dodged by tuning gentler — halving RSBLB halves both terms. SB3, the mixing-length route, sits far off that line at 3.43. The coupled arms land on the same frontier. (b) Why the coupled pair was necessary: in AMIP the inversion weakens mostly by cooling the air aloft and the screen response is not significant, whereas coupled — with sea ice free to respond — it is essentially all screen.

RSBLLMIN 30 → 120 m was a regime-discriminator hypothesis: inside the stable BL the length is clamp(0.1h, 30, 300), so the 30 m floor binds only where the stable layer is shallower than 300 m, which a shallow Arctic winter SBL is and a deep subtropical Sc boundary layer is not. Crucially this is a *regime* discriminator, not a geographic one; no latitude-dependent parameter is invented anywhere.

The runscript recorded, before the run, the reason it might invert: RSBLLMIN also sets the length *above* the stable BL, because the ELSE branch of `vdffexc` assigns `ZKLENT = ZLMIN` in free-shear layers. Over a Sc deck that is enhanced entrainment across the capping inversion, a known way to destroy stratocumulus. That is what happened.

1874–1901, 28 yr	SB1 <i>b</i> =3	SB2 <i>b</i> =2	SB3 λ_{min} =120
Arctic DJF inversion	-0.318 *	-0.624 *	-0.614 *
tropics SW CRE	+0.281 *	+0.541 *	+2.106 *
global net TOA	+0.094 *	+0.187 *	+1.673 *
SO SW CRE	-0.170	-0.182	+1.395 *
cost per K of inversion	0.88	0.87	3.43

SB3 buys SB2's Arctic benefit at four times the tropical cost and five times outside the ±0.3 TOA guardrail, and it damages the Southern Ocean where SB2 mildly helped. **The exchange rate is the durable result**: constant across a factor-two RSBLB dose change, so the tropical shortwave cost is intrinsic to enhancing stable mixing in this scheme rather than an artefact of which knob is turned. RSBLB is the efficient frontier.

49.3 Coupled: 11Q and 11R

11Q = 11N + RSBLB 2.0 (1850) and 11R = 11P + RSBLB 2.0 (1990), 40 yr, paired against their controls. No re-run controls were needed here, unlike AMIP: the coupled source has no commits since the 08–12 build the controls used, all four pre-existing working-tree modifications predate it, and the patch is bit-identical with &NAMVDFS absent.

1360–1389	11Q – 11N	11R – 11P	AMIP SB2
DJF inversion 60–90N	–0.680 *	–0.809 *	–0.713 *
DJF T2m 60–90N	+0.614 *	+0.674 *	+0.285
DJF T2m all land	+0.566 *	+0.535 *	+0.233 *
DJF T925 60–90N	–0.114	–0.020	–0.428 *
ANN T2m all land	+0.516 *	+0.494 *	—
global net TOA	–0.001	–0.027	+0.175 *
tropics SW CRE	+0.486 *	+0.540 *	+0.541 *
tropics net TOA	–0.059	–0.048	+0.247 *
SO SW CRE	–0.367 *	–0.713 *	–0.182
Siberia JJA	+0.767 *	+0.045	+0.000

All three questions AMIP could not settle come out favourably.

The Arctic screen follows, 2.2–2.4× the AMIP response and significant in both arms. That is the sea-ice amplification the pair was run to test: AMIP prescribes ice and suppresses it, which is also why 45 % of the coupled Arctic inversion bias has no AMIP counterpart (§48.5).

The cooling aloft disappears. T925 and T850 are null in both arms. Coupled, the inversion weakens by warming the screen rather than by chilling the layer above — the mechanism, not merely the metric.

The TOA cost vanishes. Tropical SW CRE is unchanged at +0.5, so the cloud is still lost, but tropical *net* is null and global net TOA is zero to 0.03 W m^{–2}: the ocean warms 0.13–0.15 K and the longwave response cancels the shortwave gain. The cost was paid as warming rather than as imbalance, and in a model this cold that is where it was wanted.

49.4 cmpitool: the best-scoring run in the campaign

run	CMPI
11E base	0.8237
11G	0.8446
11N LX4 1850	0.8198
11P LX4 1990	0.7709
11R +RSBLB2 1990	0.7577

Against 11P, the largest gains are **tas** –0.109 and **zg** –0.088 — *precisely the two fields where 11G regressed against the base* (+0.160, +0.149, §44.4). Sea ice also improves (–0.024, with Arctic SON 0.891 → 0.764).

A prediction of this report, falsified by its own test. §49.1 argued the tropical low-cloud loss would land on **nino34**, where 11G had already regressed. It does not: **nino34** is the *most improved* region at –0.030 (**tas** –0.258, **zg** –0.124, **zos** –0.094), and the tropics as a whole come out neutral at –0.0002. The cost is real but lands elsewhere — on **thetao**, +0.018 globally and +0.051 in the tropics, i.e. the ocean absorbing the shortwave. That is a spin-up concern rather than a present-day skill one, and it belongs with the standing imbalance rather than with this lever.

49.5 Base → best → this round, at 1850

All three arms at 1850 forcing, so the differences are clean even though the absolute biases against a present-day ERA5 carry a forcing offset of order a kelvin that is *not* model error.

1360–1389	11E base	11N LX4	11Q +RSBLB2	11Q–11N	11Q–11E
DJF T2m 60–90N	–6.149	–5.528	–4.913	+0.614	+1.236
DJF T2m 30–60N	–3.854	–3.757	–2.875	+0.882	+0.979
DJF T2m all land	–3.351	–3.213	–2.647	+0.566	+0.704
ANN T2m all land	–2.509	–2.370	–1.854	+0.516	+0.655
ANN T2m ocean	–1.425	–1.420	–1.266	+0.154	+0.159
JJA T2m Siberia	–2.532	–2.299	–1.532	+0.767	+1.000
DJF inversion 60–90N	+3.271	+3.109	+2.428	–0.681	–0.843
net TOA	+0.778	+0.878	+0.877	–0.001	+0.099
SW CRE SO 45–65S	–62.285	–65.462	–65.828	–0.367	–3.543
NH ice Mar [10 ⁶ km ²]	14.142	13.862	13.697	–0.165	–0.445

11Q gains more over 11N than 11N gained over the base, on every land metric, and the annual land bias falls 26 % from base. The result that was not predicted is **Siberian JJA improving by 1.000 K over base**,

0.767 of it from RSBLB itself: this campaign’s history is levers that buy winter by spending summer or spend energy to buy Siberia, and this is the first time both targets move together. It is free in energy terms, net TOA changing by 0.001 W m^{-2} .

What it does not fix. Net TOA is $+0.88$ in all three arms. That is the campaign’s standing imbalance, far outside the ± 0.3 target and the other half of this report’s title, and RSBLB neither helps nor hurts it.

49.6 Two caveats to carry forward

- **The winter Arctic gain may not be stable.** Restricted to 1380–89 the DJF Arctic screen reads $+0.104$ (11Q) and $+0.639$ (11R), neither significant — but $n=10$ gives a 0.685 K threshold, so that window cannot resolve a 0.6 K signal either way. What is informative is the fitted trend across the 30 years: -0.456 K/decade for 11Q and -0.210 for 11R. The *annual* land and global warming does not decay ($+0.409^*/+0.402^*$ all land in the last decade alone).
- **The ocean is absorbing the shortwave.** `thetao` degrades in `cmpitool`, global T2m rises $0.26\text{--}0.28 \text{ K}$, and net TOA remains $+0.88$. Ocean heat content should be priced before this state is committed to a long PI control.

49.7 One infrastructure note

11Q lost its 1380–89 leg once to LPJ-GUESS reporting `failed to find element to deserialize`. The state was *not* corrupt: all 65 files re-read cleanly at 446 MB/s afterwards, the failing `32.state` is 55.14 MB against 55.04 at 1370, and 11R passed the identical leg minutes later. It was a transient read, consistent with the DKRZ Lustre/Infiniband incident open since 2026–08–17. The leg was re-run and the failed attempt preserved at `run_13800101-13891231_dead_lustre_20260822`. Recorded because the obvious misreading — a genuine LPJ-GUESS restart bug — would have argued for switching `guess` binaries mid-experiment and destroyed the pairing against 11N.

50 Round 33 (2026–08–23): the Southern Ocean has two problems, not one

Round 32 left the model with a -0.74 K global cold bias and a $+0.88 \text{ W m}^{-2}$ imbalance, i.e. still warming toward equilibrium. The question this round answers is where that warming lands, and it turns out the Southern Ocean is not a passive recipient of a global imbalance — it is where the imbalance is generated.

50.1 The drift is in every arm, including the 1850 controls

Drift from 1350–59 to 1380–89, SH ocean south of 60S :

arm	forcing	ΔT2m	$\Delta \text{Sep ice } [10^6 \text{ km}^2]$
11N	1850	$+0.70$	-2.21
11Q	1850	$+0.81$	-1.94
11P	1990	$+0.90$	-2.61
11R	1990	$+0.97$	-2.50

The 1850 arms drift almost as hard as the 1990 arms, so this is **not** the response to present-day forcing — that adds only about $+0.2 \text{ K}$ and $-0.5 \times 10^6 \text{ km}^2$ on top. The PI controls are warming at roughly $+0.8 \text{ K}$ and shedding $2 \times 10^6 \text{ km}^2$ of September ice per 40 years.

Two consequences. First, at 1350–59 the T2m bias south of 60S was -0.06 to -0.65 K , i.e. neutral to slightly cold: **the air was not too warm at the start**, so the ocean cannot have been warmed by sitting under warm air. Both warm together. Second, September ice goes $20 \rightarrow 17.6$ against 16.27 observed, so the model is passing *through* the right answer on its way past it — anything tuned to match the ice now is tuned to a transient.

50.2 The imbalance is generated in the Southern Ocean

Net TOA bias against CERES, 11R, with each band’s share of the global mean:

band	area %	net bias	contribution	SW CRE bias
90–60S	6.7	+5.76	+0.383	+3.65
60–45S	7.9	+4.46	+0.353	−0.53
45–30S	10.3	−3.59	−0.370	−6.82
30S–30N	49.7	−1.75	−0.872	−3.35
30–45N	10.3	+2.41	+0.248	+3.40
45–60N	7.9	+3.75	+0.296	+3.47
60–90N	6.7	+2.05	+0.137	−5.83

Superseded by §50.7, which redoes this over ocean. The two Southern Ocean bands are the largest positive contributors, $+0.74 \text{ W m}^{-2}$ combined, against a tropical -0.87 . The global figure is a small residual of large regional cancellations, so “close the TOA imbalance and the SO follows” is circular: the SO ice loss absorbs the shortwave that *is* the imbalance.

50.3 What the campaign has actually done to those bands

SW CRE bias against CERES; positive means the model cloud reflects too little.

arm	90–60S ANN	90–60S DJF	60–45S ANN	60–45S DJF
11E base	+6.25	+13.63	+4.67	+4.84
11G +S4	+5.99	+12.09	+2.61	+2.39
11I v2soil	+6.33	+13.49	+2.61	+1.55
11N LX4	+4.39	+8.91	+1.25	+0.34
11P LX4 1990	+4.37	+9.05	+0.08	−2.20
11Q +RSBLB	+4.70	+9.85	+0.35	−1.46
11R +RSBLB 1990	+3.65	+6.98	−0.53	−2.97

The polar band has been more than halved, $+13.63 \rightarrow +6.98$ in DJF, so it is *not* an untouched region. What the table actually shows is a **meridional gradient in lever effectiveness**: the same levers drove 60–45S from +4.84 through zero to -2.97 , overcorrected, while 90–60S still reads +6.98. Pushing any of them harder buys the last of the polar band at the cost of driving the sub-Antarctic further negative — and 60–45S is inside the campaign’s own 45–65S SO metric, which is part of why this looked solved.

Only 11P→11R is a clean like-for-like step here; the 1850 arms carry a forcing mismatch against a present-day CERES. The 11N→11Q reading of +0.31 has the wrong sign against the -0.367^* measured for 45–65S over 30 years, and is a 10-year window on a noisy band.

50.4 Amount versus opacity: the two bands need opposite corrections

Superseded by §50.7: over ocean the split is half amount and half opacity, not predominantly opacity.

DJF, SW CRE per unit cloud cover, which isolates opacity from amount:

band		CRE	cover	CRE / cover
90–60S	CERES	−65.99	0.748	−88.2
90–60S	11R	−59.01	0.726	−81.3
60–45S	CERES	−121.01	0.879	−137.7
60–45S	11R	−123.98	0.827	−149.9

At **60–45S** the model has 5 pp too *few* clouds, each 9 % too *bright*; they compensate, which is how the net CRE ended up slightly overcorrected. At **90–60S** the amount is nearly right (-2.2 pp) but each cloud is 8 % too *dim*. Splitting the $+6.98 \text{ W m}^{-2}$: about 5.2 is opacity and 2.0 amount. So the polar residual is an **opacity** problem — the opposite of the 45–65S diagnosis that closed the SO cloud question in round 30.

That also explains the gradient above: every campaign lever works by raising opacity, and at 45–60S opacity was already too high.

The phase measurement agrees, and it flips sign at 60S. Low-level liquid fraction against CALIPSO-GOCCP: 45–65S model 0.873 vs GOCCP 0.858 (too liquid, +0.015); **60–90S model 0.731 vs 0.759 (too icy, -0.028)**; Siberia 0.813 vs 0.818. Icier cloud precipitates out and carries less liquid water, which is how a cloud can be present and optically thin. **The INP family was retired on a phase measurement taken in the one band where the model is too liquid, and that conclusion does not extend poleward.**

50.5 The sea ice itself

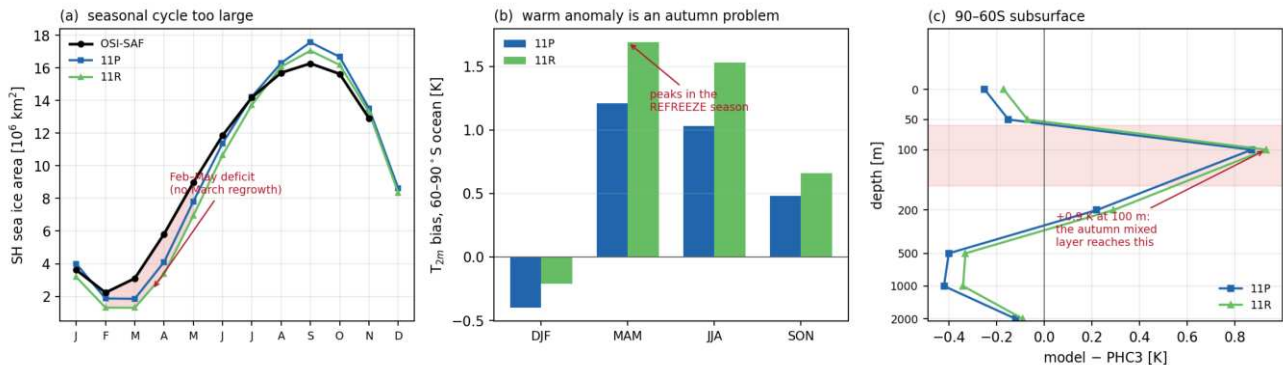


Figure 37: Two Southern Ocean problems in two different seasons. **(a)** The SH ice seasonal cycle is too large: September is $+0.78$ (10^6 km^2 area, $+2.01$ extent) against OSI-SAF, while February–May is far too low and the March regrowth is absent entirely (model 1.30 against 3.10 observed). **(b)** The T_{2m} warm anomaly over the ice zone peaks in *MAM*, the refreeze season, which no shortwave mechanism explains. **(c)** A $+0.9 \text{ K}$ anomaly at exactly 100 m, directly under a neutral surface layer and above *cold* water at 500–1000 m — the depth the autumn mixed layer reaches.

The ice is **not too dark**: September albedo over compact ice is 0.760 at $ci \geq 0.9$ and 0.778 at $ci \geq 0.95$, with snow albedo 0.850, inside the observed range. But in January–March there is **no compact ice anywhere** — every SH cell is below 90% concentration — so the area-mean albedo the sun sees falls to 0.49–0.51. The pack is also too diffuse: mean concentration inside the ice edge is 0.767 in September against OSI-SAF’s 0.804, spread over $2 \times 10^6 \text{ km}^2$ more extent. Diffuse ice has less volume per unit area to lose and melts out rather than retreating to a compact core.

50.6 The subsurface, measured at last

Temperature against PHC3, area-weighted zonal means, 90–60S:

depth [m]	PHC3	11P	11R	11N	11Q
0	−0.52	−0.25	−0.17	−0.17	−0.25
50	−0.69	−0.15	−0.07	−0.07	−0.15
100	−0.78	+0.87	+0.93	+0.76	+0.79
200	+0.50	+0.22	+0.29	+0.09	+0.19
500	+1.14	−0.40	−0.33	−0.54	−0.35
1000	+1.03	−0.42	−0.34	−0.52	−0.37

A sharp $+0.9 \text{ K}$ anomaly at 100 m, beneath a near-neutral surface and *above* cold water at depth. Not a deep warm drift — a shallow one, at the depth the winter mixed layer reaches. It is present in all four arms, so it predates RSBLE, which worsens it by only 0.03–0.06 K.

This closes the melt-from-above versus melt-from-below question: it is both, in different seasons. In DJF, $+6.98 \text{ W m}^{-2}$ of cloud shortwave melts from above. In MAM, autumn cooling deepens the mixed layer into the 100 m warm layer and entrains it, which is why the MAM anomaly is $+4.48 \text{ W m}^{-2}$ of *longwave* with almost no shortwave, why the surface cannot cool, and why March ice never regrows. Each mechanism explains the season the other cannot.

A cloud-opacity lever therefore addresses the summer half only. Even a perfect fix to the $+6.98$ leaves the autumn refreeze failure intact, because that one is fed from 100 m down.

50.7 Corrected: recomputed over ocean, and the numbers above understate it

Everything in §50.2–§50.4 was area-averaged over whole latitude bands. That folds the **Antarctic ice sheet** into the 90–60S band. The ice sheet is brilliant white and radiatively unlike the sea-ice zone, and it cancels part of the signal: over the whole cap the surface net SW bias reads -0.57 W m^{-2} , over ocean alone $+7.99$. The sea-ice question is about the ocean, so the band averages are the wrong domain and the committed numbers are too small.

Two further points of method, both checked rather than assumed. A model's `tsrc` is *total-region* clear sky, so the comparable CERES field is `toa_sw_clr_t`, not `clr_c` which is clear sky over cloud-free scenes only; the choice moves the polar numbers by 1.7 W m^{-2} , not enough to change any conclusion here. And the surface flux is now compared alongside the TOA, because agreement at two semi-independent levels is what distinguishes a real radiative error from a cloud-mask artefact.

SW CRE bias against CERES `clr_t`, OCEAN ONLY:

arm	90–60S DJF	90–60S ANN	60–45S DJF	60–45S ANN
11E base	+24.24	+11.46	+4.20	+4.31
11G +S4	+21.66	+11.01	+1.75	+2.21
11N LX4	+16.50	+8.44	−0.36	+0.85
11P LX4 1990	+16.91	+8.48	−2.91	−0.29
11Q +RSBLB	+18.07	+8.96	−2.17	−0.04
11R +RSBLB 1990	+13.36	+7.26	−3.70	−0.93

The polar ocean deficit is **+13.36 W m^{-2}** in DJF, not the +6.98 reported above — close to double. The campaign's progress is also larger than it looked, $24.24 \rightarrow 13.36$, and the sub-Antarctic overcorrection is deeper, -3.70 rather than -2.97 . The qualitative conclusion — a meridional gradient in lever effectiveness, with 60–45S driven through zero while 90–60S still leaks — is unchanged and in fact sharper.

The surface confirms it, which §50.4 could not show:

ocean band, summer	CRE	cover	CRE/cover	sfc net SW	CERES	bias
60–90S DJF, model	−92.62	0.858	−108.0	139.4	131.4	+7.99
60–90S DJF, CERES	−105.98	0.916	−115.7			
50–70S DJF, model	−116.34	0.851	−136.7	163.0	163.4	−0.40
50–70S DJF, CERES	−120.75	0.912	−132.5			
60–90N JJA, model	−90.00	0.884	−101.8	114.2	127.6	−13.34
60–90N JJA, CERES	−71.78	0.826	−86.9			

The Southern Ocean sea-ice zone absorbs +7.99 W m^{-2} too much shortwave at the surface in summer. That is the direct measurement the melt mechanism needed and did not have: §50.6 inferred the excess from the ice and the 100 m heat, and it is now confirmed at the surface. Note it is confined to the ice zone — 50–70S reads -0.40 , i.e. nothing.

But the amount/opacity split changes. Over ocean the polar deficit is roughly *half amount and half opacity*: cover is 5.8pp too low (0.858 against 0.916) and each cloud is 6.7% too dim (-108.0 against -115.7), each accounting for about 6.5 of the 13.36. The band-averaged version reported $\approx 72\%$ opacity, and that was an artefact of including the ice sheet. A pure opacity lever therefore addresses at most half of this, which is consistent with SC2/SC3 underperforming.

The Arctic, and why it is not yet a target. Over ocean in JJA the model has 5.8pp *more* cloud than CERES and each is 17% brighter, giving TOA CRE -18.22 and a surface shortwave deficit of -13.34 W m^{-2} . TOA and surface agree, so the radiative error is real rather than a cloud-mask artefact. *But the temperature does not corroborate it*: 60–90N ocean T2m is $+0.84 \text{ K}$ **warm** in JJA. A 13 W m^{-2} surface shortwave deficit that leaves the surface warm means something else is compensating, and until that is explained the Arctic cloud error should not be treated as a lever target — even though it is three times the Antarctic one and points toward the campaign's founding cold bias.

50.8 Method notes from this round

- **reval's PHC3 ocean plots are empty for every run in this campaign** — grey maps, `rmsd=nan`, `bias=nan`, byte-identical 153 687-byte files for 11E, 11G, 11N, 11P and 11R. The ocean interior had never been validated. The comparison above was done by hand because of it.
- PHC3 as staged is itself in FESOM format but on a *different* mesh (126 858 nodes against CORE3's 211 567), so it cannot be differenced node-by-node. Both share an identical 47-level vertical grid, so binning onto a common (latitude, depth) grid avoids interpolation entirely. Node coordinates for the PHC3 mesh are in its `fesom.mesh.diag.nc` as `nodes` (2, `nod2`) in *radians*; the CORE3 `diag` instead carries explicit `lat/lon` in degrees.
- Sea ice AREA and EXTENT are different numbers and mixing them inverts the sign of the answer. September: area $+0.78$ but extent $+2.01$.

- **SST is the wrong variable for this question.** Under sea ice it is pinned near freezing and reads -0.9 to $+0.0$ K in the ice zone, which says nothing. T2m is the free variable and reads $+1.5$ to $+1.9$ K at 70–80S. An early reading of this round used SST and concluded the Southern Ocean was too cold and that the ice-albedo feedback would work in the model’s favour as it warms. Both were wrong.

51 Round 34 (2026–09–06→09): four defects in the coupled ice skin

A separate thread from the radiative campaign, recorded here because it changes what the coupled arms mean. The coupled-slab ice surface temperature diverged to $-36,470$ K and overflowed single-precision OpenIFS at `ece_updclicpl.F90:104` after ~ 43 model days. Four independent defects were found; the first is the one that mattered.

51.1 The anchor that was never delivered

`oasis_put` returns `OASIS_Sent` (4) only to the call that completes a coupling group. Every earlier field in the group gets `OASIS_Waitgroup` (14) — the data is accepted and buffered, and transmission happens when the last field puts. FESOM’s `cpl_driver.F90` tested only for 4 and 8, read 14 as “not sent”, and never wrote the received field.

The consequence was silent and specific. `ist_ref`, the reference skin temperature the implicit solve is expanded about, was 0.0 in all 63,923 logged events. `tref` then fell back to the model’s own previous `t`, at which point the ζ_{lam} terms cancel identically:

$$t = \frac{c_p \Delta t^{-1} t + a_{2i} h f + \zeta_{\text{lam}} t_{\text{ref}} + \zeta_{\text{ice}}}{c_p \Delta t^{-1} + \kappa/h + \zeta_{\text{lam}}} \xrightarrow{t_{\text{ref}}=t} t^* = T_{\text{frz}} + \frac{a_{2i} h f}{\kappa/h},$$

an unbounded steady state with no restoring term at all. The fix is one line — `action=(info==OASIS_Sent .OR. info==OASIS_SentOut .OR. info==OASIS_Waitgroup)`.

51.2 The other three, and what a year of running says

- Recovering the advected ice temperature divided by `max(a_ice, 1e-6)`, which injects ≈ 0 K wherever concentration collapses rather than leaving the node alone.
- The fallback anchor had no sanity bound; it now clamps outside 173.15–400 K to 271.35 K.
- The skin is now taken from OpenIFS (`A_Ice_tskin`, new coupling field) instead of FESOM solving a second, independent one. Two models solving the same skin and exchanging only part of the state is the structural version of the same bug.

Verified over a full coupled year: `ist` minimum $-36,470$ K $\rightarrow$ 224.39 K, p1 233.57 K, **zero** points below 200 K, no overflow.

51.3 GSMART is not smart enough for a field that changes sign

`A_Q_ice` carried CONSERV method GSMART, which is GLBPOS (multiplicative, $\Sigma_{\text{src}}/\Sigma_{\text{dst}}$) with a fallback used only when the denominator is *exactly* zero — a condition floating point never meets. The field changes sign between the hemispheres, so the denominator passes through zero and the factor blows up near it. Measured over a coupled year, GSMART delivered **9105 excursions above 1 kW m⁻² onto real ice, maximum 22,550 W m⁻²**; GLOBAL and the new GSSMAR deliver **none**. GSSMAR (OASIS `ip_cgssmar`, method 9) scores the additive, whole-field multiplicative and per-sign multiplicative corrections at each step and applies whichever distorts the field least, with no extra MPI reductions. Note the keyword is six characters because the `namcouple` parser tokenises on that width; a seven-character name shifts the line and aborts with a garbage model name.

51.4 A restart that was never collected

11Y ran its first ten years cleanly and then lost legs 2 and 3. FESOM 2.8 renamed the NetCDF restarts — one directory per restart *instant*, stamped with its date (`fesom.1360-01-01.oce.restart`), where pre-2.8 wrote one directory per year named by that year (`fesom.1359.oce.restart`). The reader still falls back to the old name, so a leg branched off a pre-2.8 restart works and the change is invisible for exactly one leg; the writer only ever

emits the new name. esm-tools’ unchanged patterns therefore collected nothing, `restart/fesom` stayed empty, and the next leg died reading a file that had never been staged — while the date file advanced, so the wreckage looked like progress. Fixed version-bound in the awiesm3 v3.5 arm.

52 Round 35 (2026–09–09): the heat does not go down in the tropics

The campaign’s standing account of the imbalance is that the model removes heat from the surface faster than it should, so the surface never warms enough to radiate the excess away. That account had never been located: “too much mixing” was an inference from a cold surface and a warm interior, not a measurement. This round measures it, on 11X — the 40-year 1850 control on the corrected CORE3 mesh (220509 nodes). The conclusion is that the mechanism is real and the region was wrong.

52.1 Definitions first, because they are not the same quantity

“Mixed layer depth” names four different things here, and a model-minus-obs map is worthless if the two numbers are not the same one.

source	criterion	reference
WOA18 <code>M_an</code>	σ_θ exceeds reference by 0.125 kg m^{-3}	10 m
FESOM <code>MLD2</code>	the same 0.125 kg m^{-3}	top level, 2.5 m
FESOM <code>MLD3</code>	CMOR 0.03 kg m^{-3}	top level
FESOM <code>MLD1</code>	Large et al. (1997) buoyancy gradient	—

WOA’s criterion is stated verbatim in the file’s `summary` attribute rather than inferred; FESOM’s is `oce_ale_pressure_bv.F90` with the threshold set at line 219. Comparing `MLD2` against `M_an` compares a surface-referenced depth with a 10 m-referenced one, and where a warm or fresh skin sits above 10 m the surface-referenced version trips its threshold earlier and reads *shallower* — it flatters the model on precisely the hypothesis under test. So the model MLD is recomputed offline from the model’s own monthly T and S under WOA’s criterion, with EOS-80 σ_θ (the equation of state WOA is built on).

That recompute was validated before it was used. Reproducing FESOM’s *own* convention from the same fields gives $r = 0.99763$ against the written `MLD2`, a global area-weighted 91.08 against 93.33 m, and a median per-node difference of 1.42 m — the residual being FESOM’s Jackett–McDougall EOS against EOS-80. The definitional effect itself is small: switching the reference from 2.5 m to 10 m deepens the global mean by +1.9 m, at most +3.6 m in any band. **So the definition matters at the few-metre level and cannot explain anything below.** That is a measurement, where before it was a worry.

52.2 The mixed layer: right in the tropics, catastrophic in the Southern Ocean

Deepest month of the climatology (winter mixing) -- 11X vs WOA18, same criterion

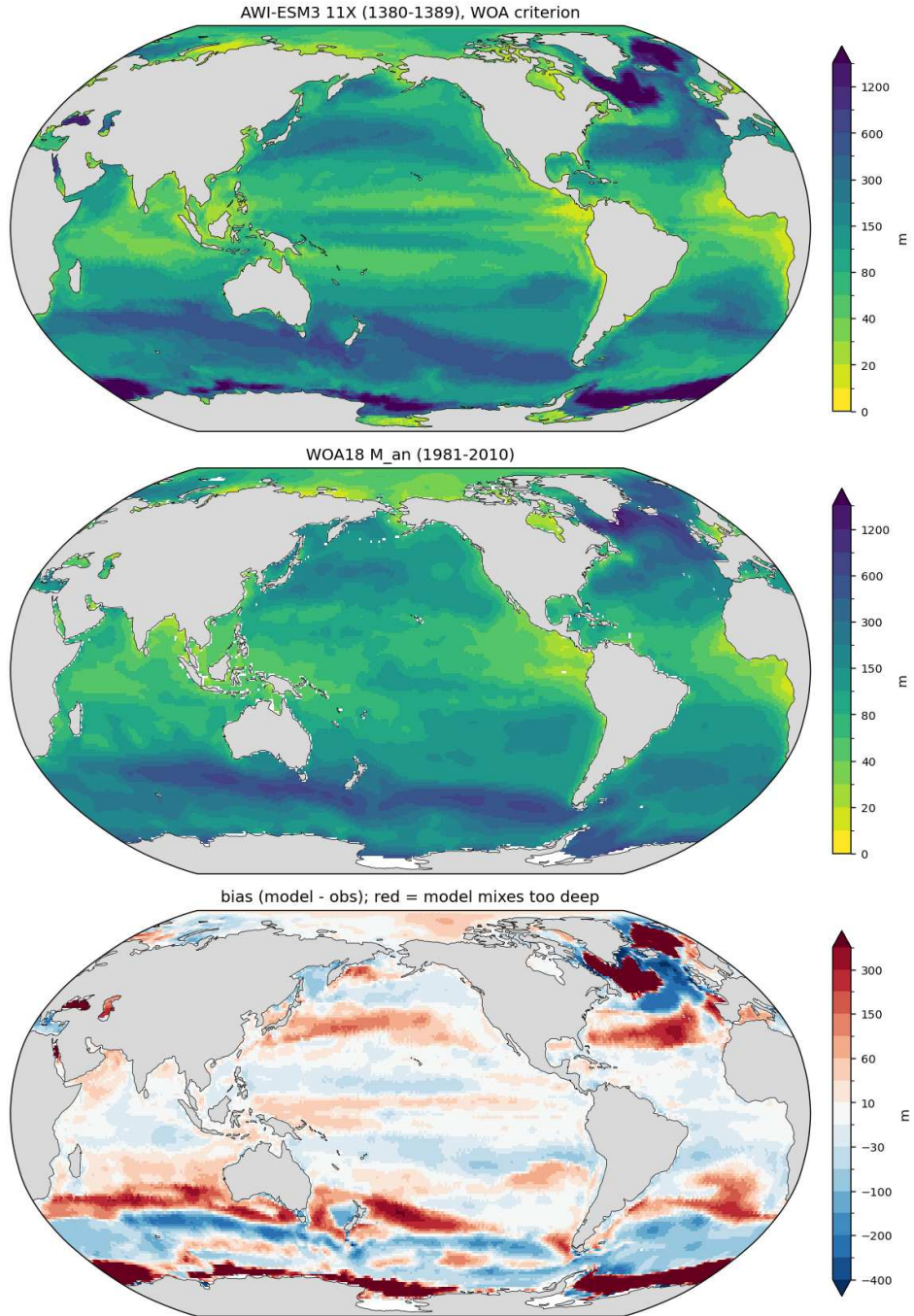


Figure 38: Deepest month of the climatology, 11X (1380–89) against WOA18, **both on WOA’s $0.125 \text{ kg m}^{-3} / 10 \text{ m}$ criterion**. The model’s over-deep mixing is confined to the Southern Ocean, the subpolar North Atlantic and the western boundary currents; the tropical band is unbiased. Red is the model mixing too deep.

band	ann model	ann obs	ann bias	max model	max obs	max bias
90–60S	371.4	123.2	+248.2	840.1	232.7	+607.4
60–45S	140.0	169.8	–29.8	264.1	301.2	–37.1
45–30S	105.0	97.6	+7.5	251.1	202.2	+48.9
30S–30N	43.6	51.6	–8.1	81.0	80.8	+0.3
30–45N	78.5	70.8	+7.6	206.4	157.5	+48.9
45–60N	141.5	104.3	+37.2	389.8	234.0	+155.7
60–90N	118.9	80.2	+38.7	272.5	176.9	+95.5

The tropics are the one band that comes out clean, and the sign is wrong for the mixing story: 8.1 m too *shallow* in the annual mean, and a dead heat at the seasonal maximum (81.0 against 80.8). The seasonal cycle (Fig. 39) shows the model shallower than WOA in all twelve months, not only in the annual mean. The convention also works against the model here: MLD computed from a monthly-*mean* profile is biased shallow relative to WOA’s per-profile-then-analyse, so correcting it moves the tropical model toward the observations, not away.

Where the model does over-mix is entirely extratropical and entirely winter, and 90–60S is not a mixing-coefficient error at all: **18.7 % of model month-nodes there mix to the sea floor** (60–90N 12.5 %, 45–60N 7.9 %, tropics 1.4 %). That is open-ocean deep convection, and no diffusivity knob addresses it.

Two caveats, stated rather than buried. WOA’s Southern Ocean coverage is 13.9 % of analysed cells annually and 6.6 % **in September**, so the +607 leans on a sparsely constrained analysis — though the “reaches the bottom” statistic is a model-side number needing no observations. The Arctic is 18.7%; every other band is 50–73 %. WOA18 is also 1981–2010 against an 1850 control.

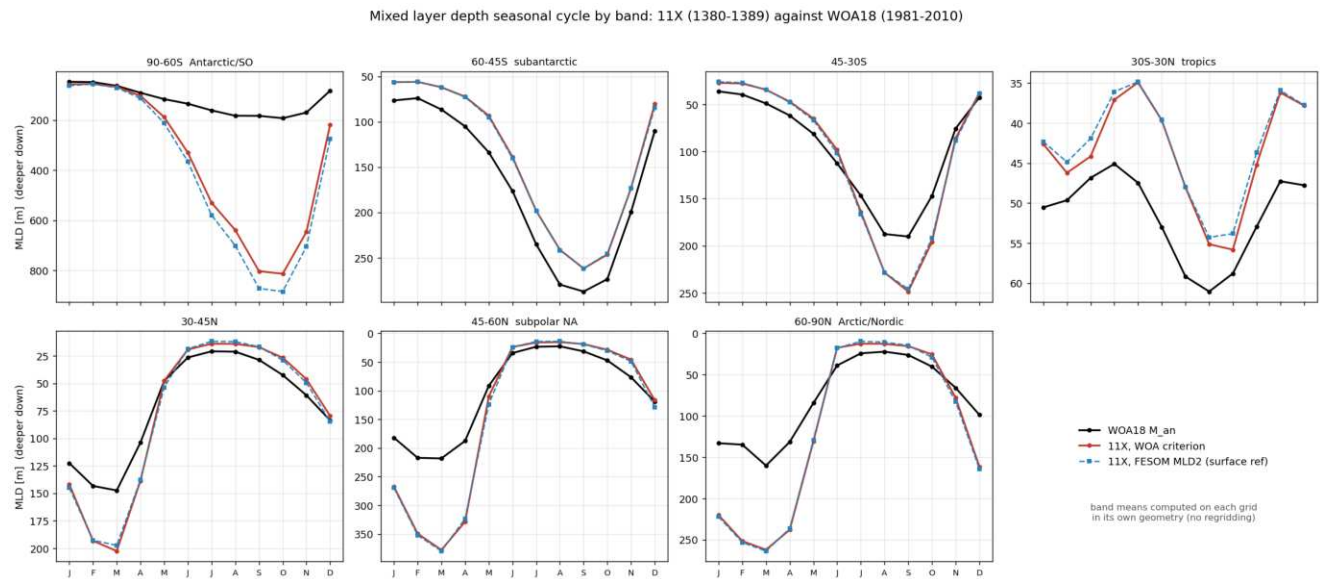


Figure 39: Seasonal cycle by band. The Southern Ocean panel is the failure: model and observations agree from January to March, then the model runs away to 840 m in September–October while WOA stays near 210 m. The tropical panel shows the model shallower than WOA in every month. Band means are computed on each grid in its own geometry, so no regridding enters.

52.3 Where the heat actually lands

The mixed layer is not heat uptake, so the next measurement is the heat itself: the drift of 11X’s own temperature between its first and last decade (1350–59 to 1380–89), which is unambiguously what *this run* put there rather than what it inherited.

The ocean gains $+1.502 \text{ W m}^{-2}$ **per unit ocean area** ($\approx +1.07$ globally), which is the right size to be the TOA imbalance — so this is the missing heat, not a residual. Its depth distribution is the finding:

band (share of global budget)	0–100 m	100–300 m	300–700 m	700–2000 m	>2000 m	total
90–60S	+0.001	−0.004	−0.020	−0.063	+0.009	−0.076
60–45S	+0.016	+0.022	+0.060	+0.007	+0.038	+0.144
45–30S	+0.005	−0.018	+0.157	+0.245	+0.072	+0.462
30S–30N	−0.043	−0.217	+0.373	+0.411	+0.082	+0.607
30–45N	+0.013	+0.035	+0.084	+0.029	+0.031	+0.192
45–60N	+0.015	+0.005	+0.014	+0.039	+0.024	+0.098
60–90N	+0.006	+0.006	+0.010	+0.040	+0.013	+0.075
GLOBAL	+0.013	−0.170	+0.678	+0.710	+0.270	+1.502

The top 100 m gains essentially nothing and the 100–300 m layer cools. Almost all of it lands between 300 and 2000 m, peaking at $+0.50 \text{ K}$ near 580–680 m with the zero crossing at $\sim 340 \text{ m}$. And 90–60S is the one band with a *net heat loss*, cooling at every depth from 70 m to 2400 m and peaking $−0.26 \text{ K}$ near 1000 m — the

runaway convection ventilates the abyss with cold surface water. **The Southern Ocean releases heat; it does not bury it.**

52.4 Heave versus water mass: the discriminator

A cooling-above / warming-below dipole at fixed depth has two possible causes, and they call for opposite fixes: heat crossing density surfaces (diapycnal mixing or a changed source water), or the surfaces themselves moving (circulation). Temperature on isopycnals separates them.

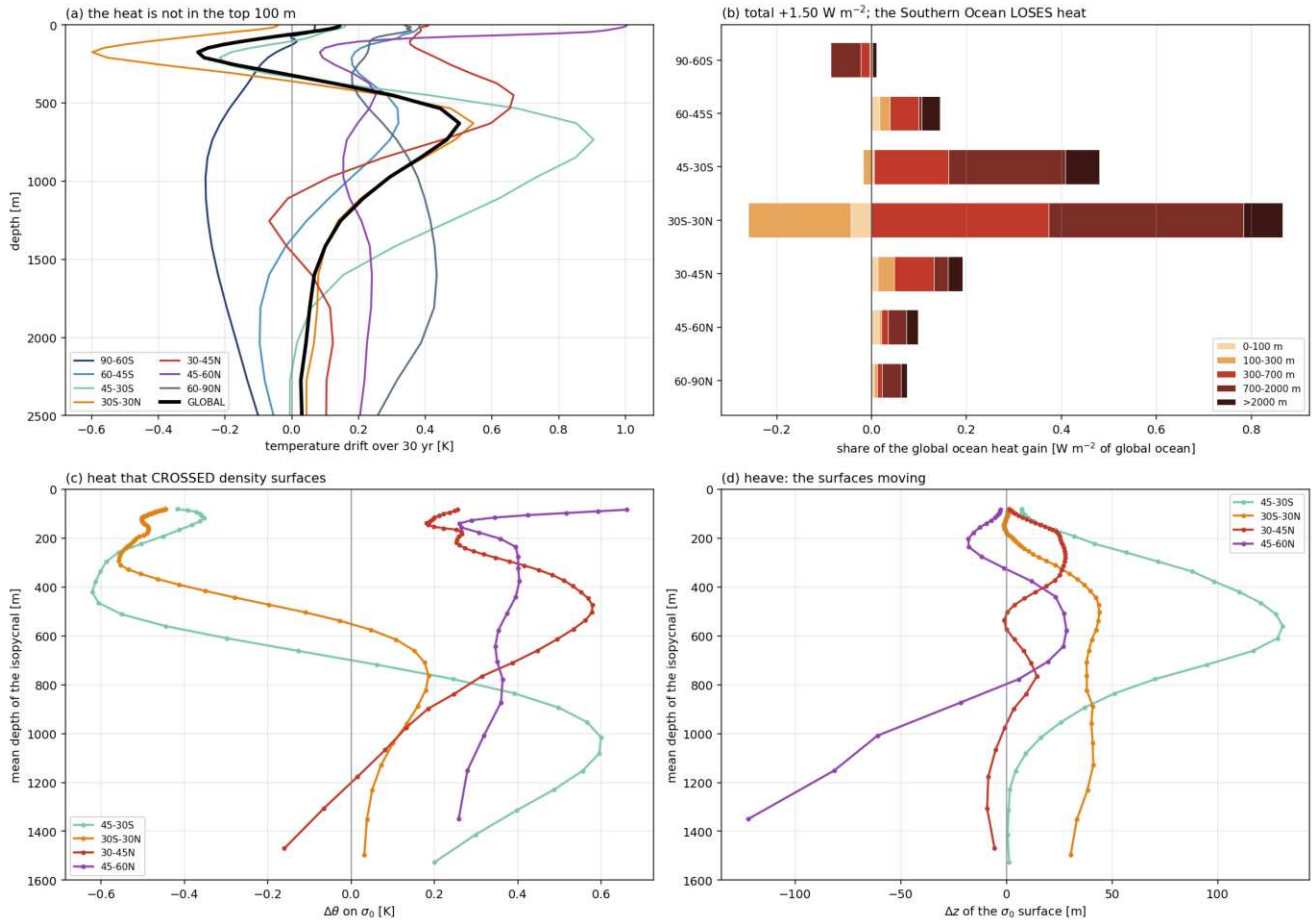


Figure 40: (a) The heat is not in the top 100 m: the global drift is negative from ~ 60 to ~ 340 m and peaks $+0.50$ K at 580–680 m. (b) The tropics take the largest share of the global heat gain and the Southern Ocean is the only band that *loses* heat. (c) Heat genuinely crossing density surfaces is a mid- and high-latitude phenomenon (30–45N $+0.58$ K at 500 m, 45–30S $+0.57$ K at 950 m); the tropical upper thermocline goes *colder* on isopycnals. (d) The tropical deep warming is instead heave — the $\sigma_0 = 27.0$ surface sinks $+42$ m. Only isopycnals present in $\geq 25\%$ of a band’s nodes are drawn; sparser ones give hundreds of metres of spurious heave.

In the tropics at $\sigma_0 = 27.0$ (mean depth 575 m) the water on that surface warmed by $+0.047$ K while the surface itself sank $+42.4$ m. The fixed-depth warming there is $+0.477$ K, and 42 m against the local ~ 0.01 K m⁻¹ gradient accounts for 0.42 of it. **Roughly 90 % of the tropical deep warming is heave.** Above it, the tropical upper thermocline went *colder* on isopycnals, -0.45 to -0.56 K from 83 to 350 m, with salinity down -0.153 psu over 0–100 m — colder and fresher at fixed density is a water-mass change running the opposite way from heat burial.

Heat that genuinely crossed density surfaces is mid- and high-latitude: 30–45N $+0.53$ to $+0.58$ K at 400–500 m, 45–60N $+0.35$ to $+0.52$ K over 100–700 m, 45–30S $+0.57$ K at 950 m, 60–90N $+0.43$ to $+0.72$ K at 280–750 m. **Those are the winter outcrop regions, and they are exactly where the mixed layer is too deep** — 30–45N $+48.9$ m, 45–30S $+48.9$ m, 45–60N $+155.7$ m at the seasonal maximum.

52.5 What this rules out, and what it does not

- **Ruled out: a tropical interior mixing lever.** The tropical mixed layer is not too deep, and 90 % of the tropical deep warming is heave rather than diapycnal flux. A diffusivity knob there has almost nothing to act on.

- **Ruled out: the Southern Ocean as the burial route.** It loses heat at every depth. Its convection is a separate and larger pathology, and it works against the interior warming rather than causing it.
- **Not established: causality at the outcrops.** The subduction warming sits in the same regions as the too-deep winter mixed layer. That is a coincidence of location, not a demonstration; only a lever test settles it.
- **Method limits.** σ_0 is a poor coordinate below ~ 2000 m, so the deepest rows are weak. Heave is a local decomposition and can be driven remotely. And band drift cannot separate local vertical transfer from lateral import — which is precisely why the isopycnal split was needed.

52.6 The free checks, and the two hypotheses they killed

Before pricing any lever, four checks that cost nothing and could each have invalidated the whole framing. Two of them did.

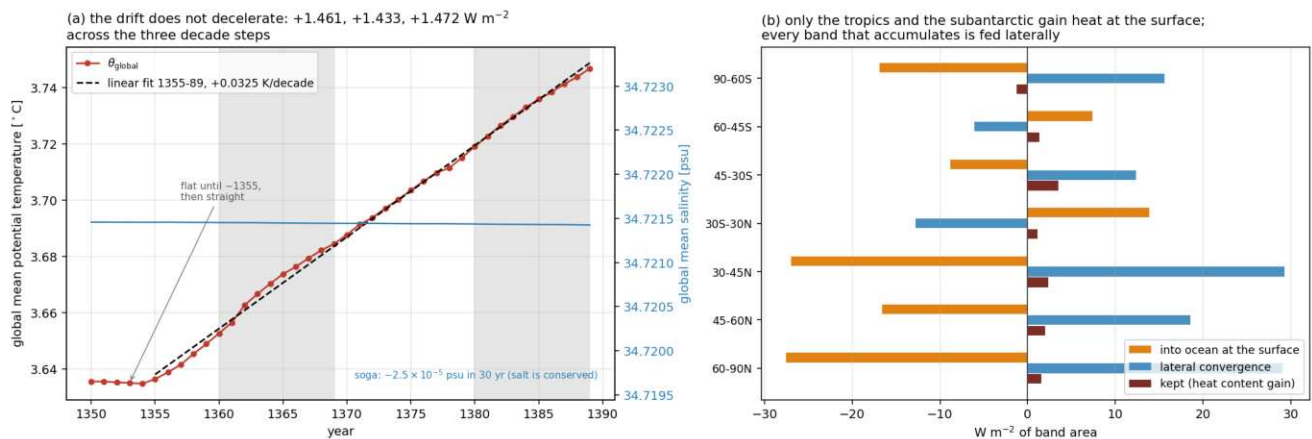


Figure 41: (a) The drift is flat for about five years after the branch and then perfectly straight: the three decade-to-decade steps are $+1.461$, $+1.433$ and $+1.472 W m^{-2}$, so this is a standing bias and not an equilibrating transient. Global mean salinity (blue, right axis) drifts -2.5×10^{-5} psu in 30 years — salt is conserved, which kills the virtual-salt-flux hypothesis. (b) Only the tropics and the subantarctic take heat in through the surface; every band that accumulates strongly (45–30S, 30–45N, 45–60N) *loses* heat at the surface and is fed by lateral convergence.

Check 1: the drift does not decelerate. `thetaoga` over all forty years gives decade-to-decade steps of $+1.461$, $+1.433$, $+1.472 W m^{-2}$ — flat to 3%. There is a short adjustment over roughly the first five years after the branch and then a straight ramp. **So the imbalance is a standing bias, and tuning against it is legitimate** rather than tuning against a spin-up that would have equilibrated anyway.

Check 2: salt is conserved, and the virtual-salt-flux hypothesis is dead. This was the leading candidate: `which_ale = 'linfs'` applies freshwater as a virtual salt flux, and the surface salinity trends at the outcrops ($+0.30$ psu at 45–60N, $+0.37$ psu at 60–90N over 30 years) looked like the numerical signature of it. If salt were accumulating at the outcrops it would deepen the winter mixed layer there, and every mixing coefficient would be the wrong end of the problem.

It is not. `soga` drifts -0.000025 psu over the same 30 years, five orders below the regional trends; `balance_salt_water` closes the budget. And the sign kills it a second time: at 45–60N the virtual salt flux is *freshening* ($-8.3 \times 10^{-7} m s^{-1}$ psu, equivalent to -2.6 psu per decade if it stayed in the top 100 m) while the layer became 0.30 psu *saltier*. **The salinification is advective and convective — a consequence of the too-deep mixing, not its cause.**

Check 3: the winds did not drift. Zonal surface stress changes by at most 4.3% in any band carrying meaningful stress (60–45S -2.4% , 45–30S $+4.3\%$, tropics $+2.2\%$; the two larger percentages are on stresses of $0.01 N m^{-2}$ and are noise). So the 42 m of tropical thermocline heave in §52.4 is **adjustment to a standing bias, not a trend inside the run** — and with check 1 showing no deceleration, that adjustment is still running after forty years.

Check 4: the surface budget closes, and the accumulating bands are fed laterally. FESOM's `fh` is positive *upward*; its 1380–89 global mean is -1.535 , against a storage rate of $+1.455 W m^{-2}$ from `thetaoga`, so the budget closes to $0.08 W m^{-2}$ only with that convention. **An earlier pass read it as positive-downward and produced lateral terms with the wrong sign throughout.** With the sign right, `fh` and the CMOR pair `opottempdiff/opottempmadvect` agree band by band to about $1 W m^{-2}$, which is what licenses the table:

band	into ocean at surface	lateral convergence	kept
90–60S	–16.88	+15.62	–1.260
60–45S	+7.42	–6.09	+1.331
45–30S	–8.83	+12.36	+3.524
30S–30N	+13.91	–12.74	+1.164
30–45N	–26.98	+29.33	+2.356
45–60N	–16.58	+18.57	+1.994
60–90N	–27.54	+29.10	+1.559
GLOBAL	+1.53	—	+1.455

Only the tropics and the subantarctic are net surface heat gainers. Every band that accumulates the most — 45–30S, 30–45N, 45–60N — loses heat at the surface and imports more than it loses. So the burial is not the mid-latitude surface pushing heat down. It is heat entering at the tropical surface, being exported poleward, and being placed into the 300–2000 m layer at the mid-latitudes.

One limitation, and a fix for the next run. `opottempdiff` is written column-integrated, so it cannot separate interior diapycnal mixing from the surface boundary flux — integrated over a column the interior term nearly cancels and what remains is essentially the surface flux, which is why it reproduces `fh`. The 3D field would measure the interior diapycnal heat flux directly and settle §52.4 without the isopycnal inference. **Add `opottempdiff` and `opottemprmadvect` in 3D to the FESOM `io_list` for the next arm.**

What survives. The subduction account survives but its *source* moves: the heat enters in the tropics. That raises the question none of these checks asked — whether $+13.9 \text{ W m}^{-2}$ of net tropical surface uptake is itself too much, which ERA5 and NCEP2 on disk can answer. And the structural candidate is untouched by any of this: **FESOM 2.7 carries no Fox–Kemper mixed-layer-eddy restratification** (nothing in `src/` matches), whose absence is the standard reason coarse models over-deepen winter mixed layers at exactly these outcrops.

52.7 Method notes from this round

- **Read the observational product’s own definition out of the file.** WOA18’s criterion is in the `summary` global attribute. Two products can share a threshold (0.125 kg m^{-3}) and still differ in the reference depth, which is the part nobody writes on the axis label.
- **Reproduce the model’s own diagnostic before trusting a recompute of it.** $r = 0.99763$ against MLD2 is what licenses the WOA-convention number; without that check a difference of the size found here would be indistinguishable from a coding error.
- **Do not mask an objectively analysed field on raw observation counts.** A first pass masked WOA `M_an` on `M_dd` ≥ 3 and deleted most of the ocean — `M_an` is complete over the ocean (41088 of 64800 cells) while `M_dd` is sparse. Coverage belongs in a caveat, not in a mask.
- **Compute nonlinear diagnostics on the native mesh, regrid afterwards.** MLD is a nonlinear functional of the profile; regridding T and S first and computing after gives a different quantity. Band means were additionally computed with no regridding at all as a cross-check.
- **Check a flux variable’s sign convention against a budget, not against its name.** FESOM’s `fh` is positive upward. Taking it as positive-downward inverted every lateral-transport term in §52.6; the error was caught only by requiring the global surface flux to match the `thetaoga` storage rate.
- **Require coverage on isopycnal averages.** Density surfaces that outcrop in a handful of nodes produced –1134 m of “heave” in 45–60N. A 25 % band-coverage floor removes it.

53 Round 36 (2026–09–10→12): mid-depth heat storage sets the TOA plateau

Two things changed underneath the coupled work before this round, and both are documented in `notes/ICE_SKIN_OPTIONS_2026-09-10_14series.md`. The single-precision FESOM lost about 0.5 W m^{-2} of ocean heat: the FCT low-order tendency was rebuilt as $LO \cdot h_{\text{new}} - T \cdot h$ from a float32 provisional solution, losing every per-step increment below half an ulp. It is fixed upstream (FESOM PR #1054). **15F** is the ice-physics configuration of 15C (OpenIFS column-heat flux `ECE_CPL_FESOM_ICE_QOUT`, ungated snow melt, 1 K albedo ramp, ECHAM6 ice albedos) on the fixed ocean. Its ocean heat budget closes at -0.069 W m^{-2} over 10 years (-0.035 over 20), against -0.073 for the double-precision 13A and -0.544 for the unfixed 15C, and Southern Ocean convection returns (2.39 M km² deeper than 1000 m in years 1–10, 5.66 in years 11–20). 15F is the new coupled baseline.

This round asks what sets the TOA imbalance a coupled run settles at, using every coupled run since the 06 series. Annual global net TOA and annual ocean heat content in four depth ranges are kept for 76 runs in one store, `data/coupled_annual_diag.nc`.

53.1 Every run ramps, then settles at a configuration-dependent plateau

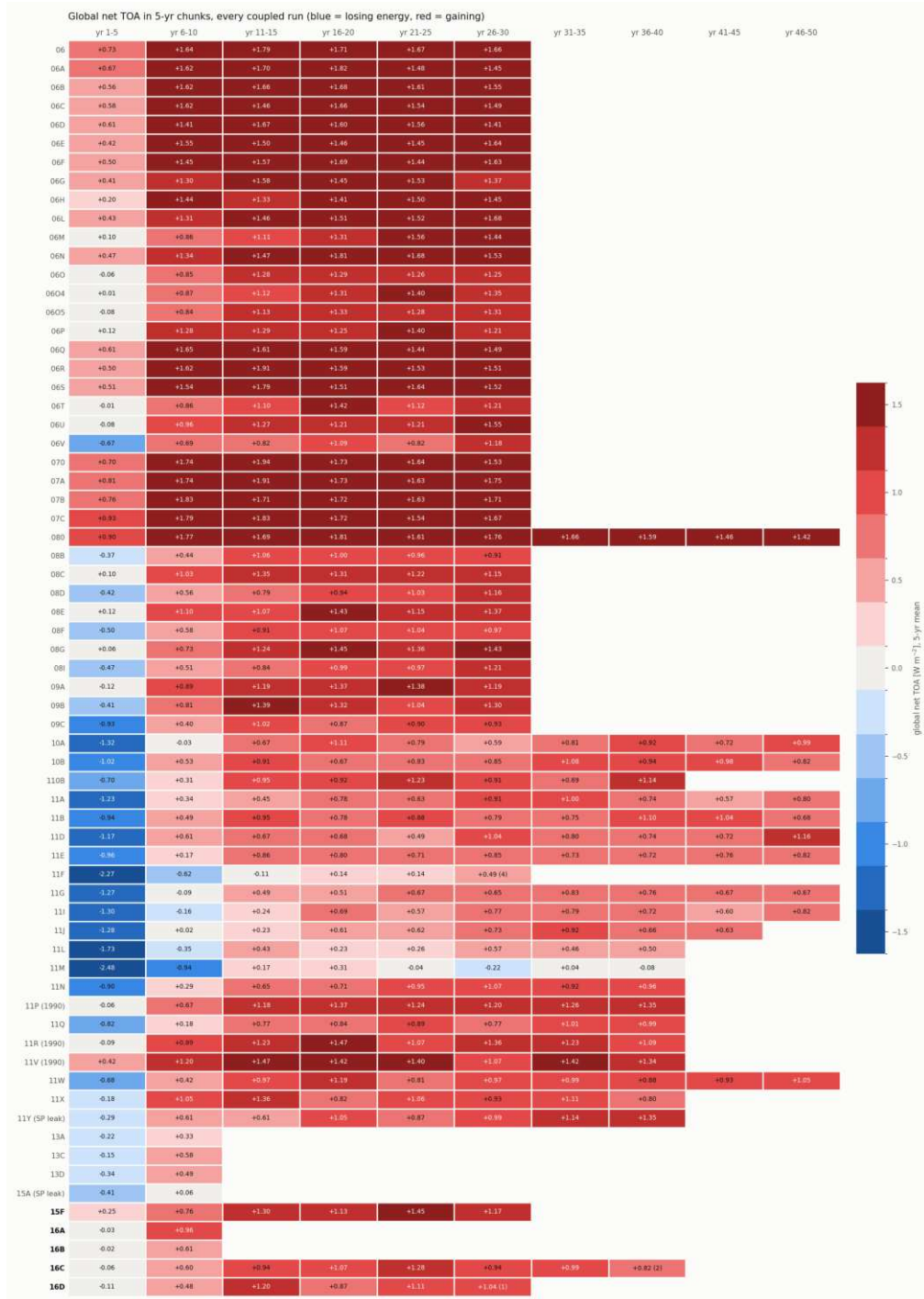


Figure 42: Global net TOA in 5-year chunks counted from each run's start, every coupled run with at least 10 complete years (the CMIP7 spin-ups are excluded: their TOA comes out near $+10 \text{ W m}^{-2}$, a different output convention). The first chunk is the lowest in almost every run; TOA rises through years 6–15 and is flat afterwards. The level it settles at is the configuration's: $+1.4$ to $+1.8$ for the 06/07/08 series, $+0.7$ to $+1.0$ for the new-sea-ice chain 09C–11J, 0.9 – 1.0 for 11W and 11X, 1.2 – 1.4 for the 1990-forced 11P/11R/11V. 15F reads 0.25, 0.76, 1.30, 1.13, 1.45, 1.17, the highest 1850 plateau since the 07/08 series. The darker ice of 15F absorbs about 0.9 W m^{-2} more shortwave than 13A. Of the 40-year 16 arms, 16C and 16D settle at 0.9 – 1.1 and **16E** at 0.66 – 0.83 , the lowest plateau of any run the campaign has kept.

A single year scatters by about $\pm 0.3 \text{ W m}^{-2}$, so a 5-year mean carries about ± 0.13 and differences below ~ 0.35 between two chunks are not resolved. 11E is flat at 0.71–0.86 for 40 years once past year 10; 11X oscillates around 1.0; 11Y, the single-precision run with the heat leak, is the only one that keeps climbing ($0.61 \rightarrow 1.35$).

53.2 Where the heat is stored

Deep-ocean heat uptake in 5-yr chunks, every coupled run (red = the layer is warming; * = chunk shorter than 5 yr)

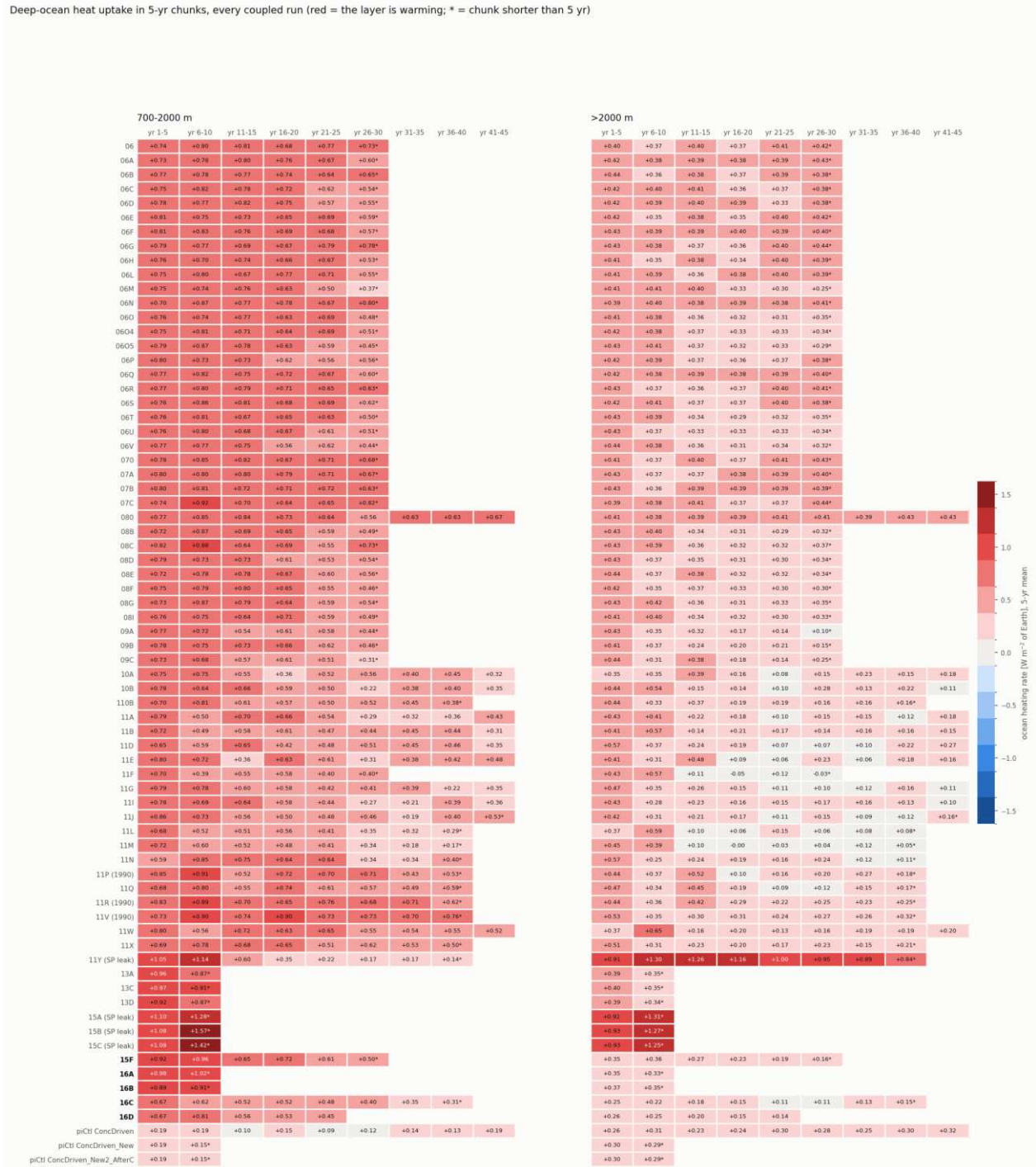


Figure 43: Ocean heating rate in 5-year chunks at 700–2000 m (left) and below 2000 m (right), in W m^{-2} of Earth area so it compares directly with TOA. From annual-mean temperature, $\rho c_p T A \Delta z$ with below-bottom points masked, on each run's own mesh. * marks a chunk shorter than 5 years. Every run starts at 0.7–1.0 at mid-depth; only the new-sea-ice chain lets it decay.

- **The first decade is universal.** Every run puts $0.7\text{--}1.0 \text{ W m}^{-2}$ into 700–2000 m and $0.3\text{--}0.5$ below 2000 m.
- **Only 09C–11Q let the mid-depth uptake decay.** By years 25–45, 11G reads $0.22\text{--}0.41$ and 11I $0.21\text{--}0.44$ at mid-depth, both $0.10\text{--}0.17$ below 2000 m. The 06–08 series never decay ($0.55\text{--}0.8$ and a constant $0.3\text{--}0.4$ at depth).

- **11W and 11X decay only partly**, to 0.50–0.62 at mid-depth and 0.15–0.23 below. 15F tracks 11X after its first decade (0.65, 0.72, 0.61 against 0.68, 0.65, 0.51), and its first decade is the highest of the recent runs (0.92, 0.96).
- **Stronger GM moves the whole column down from the first chunk.** 16C and 16D decay to 0.29–0.37 at mid-depth by years 31–40. **16E starts at 0.37 and stays at 0.24–0.45**, and below 2000 m it is nearly shut: 0.04–0.09 throughout, against 0.16–0.36 in 15F. The heat 16E does take goes into 100–700 m instead (0.57 in years 31–35).
- 11Y's 0.84–1.30 W m^{-2} below 2000 m is the heat-leak artefact, not physics.

53.3 The relationship: lower mid-depth storage, lower decade-4 TOA

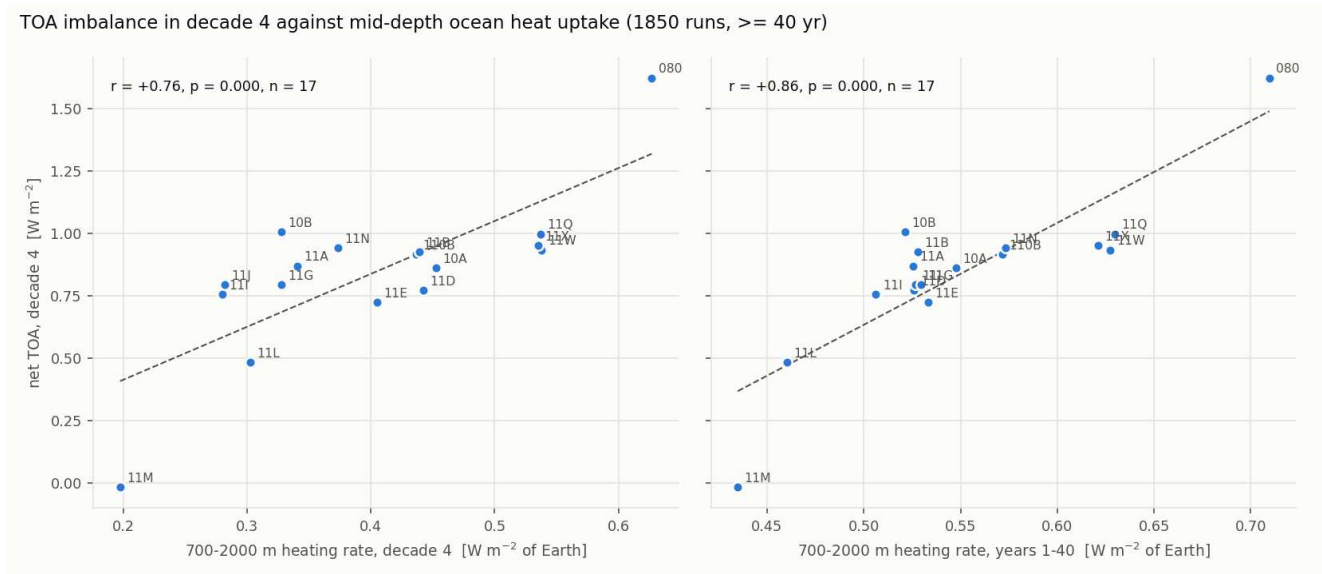


Figure 44: Net TOA in decade 4 (years 31–40) against the 700–2000 m heating rate in decade 4 (a) and averaged over years 1–40 (b), for the 20 runs with 1850 forcing and at least 40 years, now including the 40-year arms 16C, 16D and 16E. The 1990-forced runs and the single-precision heat-leak runs are excluded. Dashed: least-squares fit. 16E has the lowest 40-year mid-depth uptake of any run and sits well above the line.

decade-4 TOA against	all 20	without 080	without 080, 11M	without 080, 11M, 11L
700–2000 m uptake, years 1–40	+0.72	+0.59 ($p = 0.008$)	+0.59 ($p = 0.01$)	+0.58 ($p = 0.015$)
700–2000 m uptake, decade 4	+0.73	+0.61 ($p = 0.006$)	+0.49 ($p = 0.04$)	+0.47 ($p = 0.06$)
total ocean uptake, decade 4	+0.96	+0.94	+0.82	+0.72

The leave-out columns were computed from the tabulated values to two decimals. 080 is the old configuration at the top of both axes, and 11M and 11L are the rejected overlap levers at the bottom. Without them the 40-year relation stays significant (+0.58, $p = 0.015$, 17 runs), and cumulative mid-depth storage still predicts decade-4 TOA better than the decade-4 rate does (+0.58 against +0.47): what the run has already buried sets how far it still is from balance. With the extremes included the two are equal.

The first version of this table (17 runs, $r = +0.86$) overstated the slope. Its fit, 4.1 W m^{-2} of decade-4 TOA per W m^{-2} of 40-year mid-depth uptake, was set by 080 and 11M. It predicted the three GM arms far too low: 16C +0.59 (measured +0.96), 16D +0.67 (+0.91), **16E -0.03 (+0.70, 4.4 residual sd off)**. The fit without the extremes (slope 1.3) predicts 16E at +0.59, close to what it did. So the relation holds, but cutting mid-depth uptake buys about a third of what the headline correlation implied: stronger GM lowers the mid-depth heating rate by more than it lowers TOA, and part of the heat it keeps out of 700–2000 m is stored higher up instead (§53.2). The last row is largely the energy budget itself (slope 0.91) and confirms the bookkeeping rather than adding evidence. From 11X to 11G, about 0.1 W m^{-2} less 40-year mid-depth uptake goes with 0.15–0.2 lower decade-4 TOA (0.95 against 0.80).

53.4 The 16 series: three ocean levers against the mid-depth uptake

Each arm is 15F plus one change, 10 years from the same 1350 restart and libraries (FESOM 540e1029, OpenIFS 3726203165e7; 16B uses the MLE build 79dfae3f). Rates are 5-year chunks, the second ending at 1359 and

therefore 4 years long.

arm	change	700–2000 m [W m ⁻²]	> 2000 m [W m ⁻²]	NH Mar ext. [M km ²]	global T2m bias [K]
15F	baseline	0.92 / 0.96	0.35 / 0.36	18.8	−1.52
16A	Redi_Kmax 300	0.98 / 1.02	0.35 / 0.33	19.9	−1.79
16B	Fox–Kemper MLE	0.89 / 0.91	0.37 / 0.35	19.5	−1.37
16C	K_GM_max 1500, Redi 1000	0.67 / 0.62	0.25 / 0.22	20.2	−1.41

Ice and T2m are 1355–59 means; the 4-year thresholds from 15F’s own detrended scatter are ± 0.64 M km² for NH March extent and ± 0.16 K for global T2m.

- **16C (stronger GM) cuts first-decade mid-depth uptake by about 30 %** and deep uptake by 0.1 W m^{-2} . Nothing else degrades beyond noise. At 60–45S the upper ocean is the closest of any run to PHC3 ($-0.37/-0.16$ K at 0/100 m against 15F’s $-0.62/-0.44$), and the Antarctic convection excess shrinks (+491 against +594 m at the seasonal maximum). **The one cost is NH winter ice:** March extent +1.4 M km², beyond threshold, and trending up at $+1.8 \text{ M km}^2$ per decade where 15F is flat.
- **16A (weaker Redi) goes the wrong way on everything:** more mid-depth uptake, 0.27 K colder, more Southern Ocean convection (3.35 against 2.39 M km²), more ice in both hemispheres. Rejected.
- **16B (MLE) leaves mid-depth uptake unchanged.** It cuts 100–700 m instead, and it fixes the subtropical winter outcrops (45–30S +2 m against +35, 30–45N +28 against +50) and SH summer ice (3.28 against 4.65 M km², obs 3.0). But it over-shoals the subantarctic mixed layer (-103 m at the seasonal maximum).
- **None of the three changes TOA within the first decade beyond noise** (0.60 to 0.96 against 15F’s 0.76 in 1355–59). The first decade is the adjustment in every run (§53.1); the lever’s effect on the plateau shows in decades 2–4.

53.5 The 40-year arms: 16E settles lowest, and pays in NH ice and AMOC

Three arms ran 40 years from the 1350 restart: 16C (K_GM_max 1500, Redi 1000), 16D (16C plus Fox–Kemper MLE) and 16E (K_GM_max 2500, Redi 1000). The 15F control stopped at 1380, so it enters with its third decade. Heating rates are over the decade from annual heat content, in W m^{-2} of Earth.

run	decade	net TOA	100–700 m	700–2000 m	> 2000 m	TOA decade 3
15F	1370–79	1.31	0.36	0.57	0.19	1.31
16C	1380–89	0.96	0.37	0.36	0.15	1.11
16D	1380–89	0.91	0.34	0.31	0.16	0.99
16E	1380–89	0.70	0.40	0.28	0.07	0.77

Clean pairs from the same restart, mean over the common years with the standard error from the scatter of the annual difference:

pair	net TOA	700–2000 m	> 2000 m	total ocean
16C – 15F (1350–79)	-0.21 ± 0.06	-0.19 ± 0.03	-0.09 ± 0.01	-0.24 ± 0.04
16E – 15F (1350–79)	-0.45 ± 0.08	-0.37 ± 0.03	-0.19 ± 0.01	-0.48 ± 0.05
16E – 16C (1350–89)	-0.24 ± 0.06	-0.15 ± 0.02	-0.09 ± 0.01	-0.25 ± 0.05

- **The GM dose response continues to 2500.** 16E takes 0.45 W m^{-2} off 15F’s TOA and another 0.24 off 16C’s, and settles at $+0.66$ to $+0.83$ in every 5-year chunk after year 10 with no upward drift. It is the lowest plateau of any run the campaign has kept.
- **The pairs measure the slope directly:** TOA falls 1.2 (16E – 15F) to 1.6 (16E – 16C) times the mid-depth cut and about one-for-one with the total ocean uptake. That is the shallow slope of the extremes-free fit in §53.3, not the steep one of the first 17-run fit.
- **The deep sink is nearly shut:** below 2000 m 16E takes 0.04–0.09 in every chunk. What 16E still stores goes to 100–700 m (0.40 in decade 4, the most of the four).
- **MLE on top of GM 1500 adds nothing resolvable at 40 years:** 16D minus 16C is -0.05 in decade-4 TOA and -0.05 at mid-depth.

The climate, decade means (15F: 1370–79; others 1380–89):

run	AMOC 26.5N [Sv]	AMOC 40–60N [Sv]	NH Mar ext. [M km ²]	NH Sep ext. [M km ²]	SH Sep ext. [M km ²]	global T2m [K]
15F	12.5	14.9	16.4	7.1	20.9	286.28
16C	14.9	17.3	15.7	7.4	20.6	286.60
16D	15.1	18.1	16.1	7.2	20.3	286.62
16E	13.3	14.5	17.2	6.8	20.2	286.56

Differences below are 16E minus 15F over 1350–79 unless stated. The standard error of a 30-year mean difference is about $\text{sd}\sqrt{2/30}$ from 16E's detrended interannual scatter: 0.24 M km² for NH March extent, 0.15 for SH September, 0.04 K for global T2m, 0.23 K for Arctic winter land.

- **NH winter ice is the main cost:** March extent +1.3 M km² against 15F and +0.7 against 16C over 40 years; in decade 4 16E has 17.2 against 16C's 15.7 and the observed 15.0. It shrinks at -0.8 M km² per decade, as 16C does, but from a higher start. September extent is 0.2 lower.
- **SH winter ice shrinks, as expected with less heat stored:** September extent -0.4 against 15F, and 20.2 in decade 4. The direction holds; the size is small.
- **The surface warms:** global T2m +0.21 K, 90–60S +0.39 K. Arctic winter land +0.20 K is not resolved. Southern Ocean SW cloud radiative effect (45–65S) is 0.76 W m^{-2} more negative.
- **The AMOC does not keep the GM 1500 gain (§53.7):** 16E is 1.0 ± 0.2 Sv weaker than 16C at 26.5N and 1.7 ± 0.2 at 40–60N over 40 years. Against 15F it is level at 26.5N ($+0.04 \pm 0.24$) and 0.67 ± 0.23 weaker at 40–60N. 16E's first decade already showed a subpolar North Atlantic winter mixed layer 144 m shallower than 15F's.

53.6 The stable-BL lever carries part of the late mid-depth uptake, but not the TOA

Within the 11 chain the late mid-depth uptake splits along $\text{RSBLB} = 2$ (the stable boundary-layer stability coefficient, §49.3): 11E/11G/11I read $0.21\text{--}0.48 \text{ W m}^{-2}$ at 700–2000 m in years 26–45, 11Q/11W/11X $0.49\text{--}0.62$. Two pairs differ only in RSBLB ($5 \rightarrow 2$), 40 years each on core3_beta. RSBLB arm minus control, 700–2000 m heating rate [W m^{-2} of Earth], with the standard error of the 40-year mean from the year-to-year scatter:

pair	yrs 1–10	11–20	21–30	31–40	40-yr mean
11Q – 11N (1850)	+0.02	–0.05	+0.10	+0.17	$+0.057 \pm 0.037$
11R – 11P (1990)	–0.02	+0.05	+0.01	+0.19	$+0.054 \pm 0.033$

- **Both pairs agree:** the same sign and nearly the same size, together about $+0.055 \pm 0.025 \text{ W m}^{-2}$. The effect grows with time and in decade 4 is most of the gap between 11N (0.37) and 11Q (0.53).
- Below 2000 m nothing changes ($+0.01$, $+0.03$); total ocean uptake rises by $+0.09 \pm 0.05$ and $+0.13 \pm 0.03$.
- **Net TOA does not move:** -0.003 ± 0.066 and $+0.002 \pm 0.053 \text{ W m}^{-2}$ over 40 years, and decade 4 disagrees in sign between the pairs. The extra ocean heat is drawn from another reservoir, not added at the top of the atmosphere; stronger stable mixing pushes heat from the air into the surface over ice and land. Which reservoir gives it up is not checked.
- **Consequence for §53.3:** part of what separates 11Q/11W/11X from 11G/11I at mid-depth is RSBLB , and that part does not raise TOA. The cross-run correlation is therefore partly carried by other differences between those runs, and $\text{RSBLB} = 2$ stays: switching it off would give up its Arctic winter ($+0.6\text{--}0.7$ K) and Siberian summer ($+0.77$ K) gains for no measured TOA benefit.

53.7 The AMOC: GM 1500 strengthens it, GM 2500 gives it back

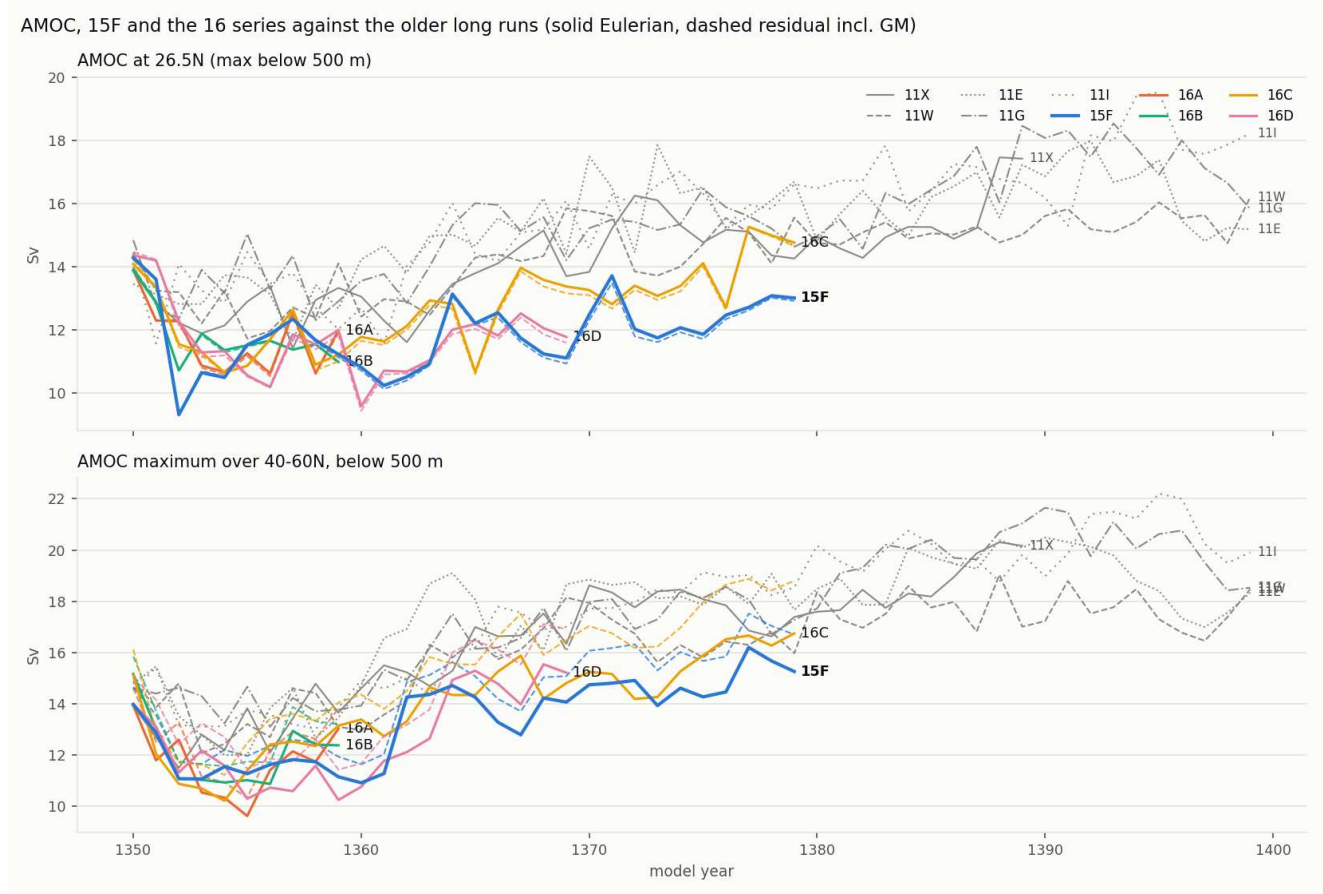


Figure 45: Atlantic overturning maximum below 500 m at 26.5N (a) and over 40–60N (b), annual means. Solid: Eulerian, from w ; dashed: residual, $w + w^*$ with the GM bolus velocity. Older runs in grey. Every run strengthens as it spins up; 15F is the weakest of the older configurations, 16C (GM 1500) and 16D sit above it, and 16E (GM 2500) falls back to 15F's level at 26.5N and below it at 40–60N.

AMOC at 26.5N [Sv]	yrs 1–10	11–20	21–30	31–40	41–50
11X	12.7	13.4	15.0	15.4	
11W	12.9	13.7	14.8	15.0	15.5
11E	13.1	14.9	16.3	16.0	16.4
11G	13.5	14.6	15.4	16.3	17.5
11I	13.2	14.4	16.0	16.8	17.8
15F	11.7	11.4	12.5		
16C	11.8	12.6	13.8	14.9	
16D	12.0	11.4	13.5	15.1	
16E	12.0	11.7	12.0	13.3	

- **15F has the weakest AMOC of all runs**, 11.4–12.5 Sv and 2–3 Sv below 11X over the same years; RAPID observes about 17 Sv today. The likeliest cause is the new ice physics (a warmer, less icy North Atlantic and Nordic Seas forming less dense water); this is not tested.
- **The response depends on the dose.** GM 1500 strengthens it: 16C minus 15F is $+0.83 \pm 0.21$ Sv at 26.5N and $+0.63 \pm 0.19$ at 40–60N over 1350–79. GM 2500 gives that back: 16E minus 16C is -0.99 ± 0.21 and -1.68 ± 0.21 over 40 years, and in decade 4 16E reads 13.3 and 14.5 Sv against 16C's 14.9 and 17.3.
- **MLE does not take the GM 1500 gain back:** 16D dipped in decade 2 (11.4 Sv) and caught up by decade 4 (15.1); 16D minus 16C is -0.25 ± 0.18 Sv over 40 years.
- The GM eddy-induced part is negligible at 26.5N (Eulerian and residual differ by at most 0.2 Sv) and adds 1–2 Sv at 40–60N in every run.
- ~~So the AMOC is no obstacle to a stronger GM dose.~~ 16E killed this: a larger dose than 1500 weakens it again (§53.5). 16C's early NH ice excess ($+1.4 \text{ Mkm}^2$ in 1355–59, gone by 1375–79) is still not an overturning effect.

53.8 Method notes from this round

- `scripts/analysis/coupled_annual_store.py` keeps `data/coupled_annual_diag.nc` (run $\times$ year: toa, ohc_0_100, ohc_100_700, ohc_700_2000, ohc_gt2000) and computes only missing entries on each call, checkpointing every 200. New runs and years are added by running it again.
- Only years with all 12 months count. 15B and 15C wrote to `work` only and are read from there. The mesh is chosen by node count (`core3 220509`, `core3_beta 211567`).
- Figures: `scripts/figures/toa_decadal_heatmap_all_runs.py` (CHUNK=5), `scripts/figures/ohc_rate_heatmap_all_runs.py`, `scripts/analysis/toa_vs_middepth_uptake.py`.
- AMOC: `scripts/analysis/amoc_annual_store.py` keeps `data/amoc_annual_diag.nc` with `pyfesom2 xmoc_data` (as reval part 17), `Atlantic_MOC` mask, and grows incrementally. It needs `fesom.mesh.diag.nc`, which FESOM writes only on a cold start (`fesom_module.F90:698`); none of the restarted coupled runs had one, so it was generated for the corrected CORE3 mesh with FESOM's `fesom_meshtdiag` utility (`scripts/model/comp_fesom_meshtdiag.sh`, `scripts/model/run_fesom_meshtdiag.sh`) and placed in the mesh directory. The RSBLB pairs: `scripts/analysis/rsblb_pair_ohc.py`.
- Clean-pair differences (heat by depth, TOA, AMOC) with standard errors: `scripts/analysis/pair_diff_store.py` (PAIRS=15F:16E, 16C:16E). Per-year T2m, TOA, SW CRE and ice extent for any arm: `scripts/analysis/detection_threshold_4yr.py` (ARM=16E Y1=1389 SERIES=1).

54 Round 37 (2026-09-13→17): five levers at the Southern Ocean, none of them moves it

Round 33 split the Southern Ocean problem in two: a winter over-ice bias and a warm, shallow CDW. This round spends five 20-year branches of **PI200** on the second, and the result is uniformly negative. The value of the round is the exclusion, so the arms are recorded with what each was supposed to do.

Four arms branch at 1560, control **PI200** 1560–1579, everything else identical (v3.5 defaults, INPSEA 0.1 / INPPMIN 50000, FESOM and LPJ-GUESS from PI200's 1560 restarts, atmosphere cold start):

- **PI200_spp** — salt-plume parameterisation on. Brine rejection reaches deeper than the mixed-layer scheme allows, which is the textbook route to dense shelf water.
- **PI200_h0** — lead closing `h0min` 0.5 $\rightarrow$ 0.25 m. The pack is dispersive because FESOM closes leads at the thickness of the ice already there.
- **PI200_mle** — Fox-Kemper mixed-layer eddies, settings as 16B.
- **PI200_all3** — all three together, to catch a combination effect that the single levers miss.

A fifth arm attacks the same loop from the ice shelves. One-year screens **PI200_cav05/06/07** gave cavity basal melt of 1038 / 1190 / 1402 Gt yr⁻¹ against 1893 at `cavity_gamma_scale` 1.0 and 1100–1325 observed, so 0.6 was taken forward as **PI200_cavg06** (1600–1619, control PI200 1600–1619). The hypothesis was a loop: warm CDW drives melt, meltwater freshens the shelf, no dense water forms, the CDW stays warm. Less meltwater should break it.

54.1 The result, and what it costs to state it

None of the five fixes the Southern Ocean in 20 years. The winter over-ice bias is a *freshwater regime*, not a mixing deficiency: the model sits in one of two states, salty and venting or fresh with warm CDW, and none of these levers moves it between them. The basal-melt calibration is worth keeping for its own sake — 1190 Gt yr⁻¹ against 1893 is a defensible number where the uncalibrated one was not — and it became the v3.5 CORE3 default `cavity_gamma_scale`: 0.6 (FESOM PR #1064, merged upstream 2026-09-17). But it is not a Southern Ocean fix.

Recorded as an exclusion list, each with the observation that killed it, in §59.2. **The SSS restoring lever stays excluded**: `surf_relax_s` is 0 and remains 0, even for Weddell convection.

55 Round 38 (2026–09–18→22): the PICAL production line, and the albedo that had to be computed

With the lever search exhausted the campaign switches from screening to producing. The v3.5 TCO95/CORE3 defaults go into `awiesm3.yaml` rather than living in runscripts (2026–09–21), and a single pre-industrial line is carried forward: **PICAL** (1850–1929) → **PICAL_momixoff** (branched 1920, to 2049) → **PICAL_ccnice** (branched 1940).

PICAL_v35def and **PICAL_v35def_darkice** are the paired control for the two namelist values the campaign had retired but the production line still carried (`RCL_INPPMIN` 50000 vs 70000) and for the ice albedos (`albi` 0.70 → 0.66, `albim` 0.68 → 0.64). Both are drawn in the campaign TOA figure as context: they are the counter-example, an Arctic volume runaway that fifty years of correction did not arrest.

55.1 Sea-ice albedo, computed rather than guessed

The ice was too dark and the first instinct — darken the ponds — was wrong: the four `ice_therm` albedos are *pond-free*, `ice_thermo_cpl.F90` builds `alb_noponds` from them and only then applies `albpnd` over the pond fraction, so darkening them to stand in for ponds double-counts. Ponds stay dark; the ice and snow get brighter. The two snow albedos do different jobs, and the split was *measured* on the model’s own melt seasons rather than assumed:

	$\partial\alpha/\partial\text{albsn}$	$\partial\alpha/\partial\text{albsnm}$
NH JJA	0.149	0.504
SH DJF	0.908	0.051

The Antarctic pack surface rarely reaches melting point, so `fmelt` ≈ 0 there and `albsn` carries the southern response; `albsnm` is the Arctic knob. Both stay global: the split is measured surface state, not a hemispheric parameter. `albsn` was raised to 0.80 (2026–09–22).

The sensitivities are state-dependent, which is the part that generalises. As the Arctic cooled, NH $\partial\alpha/\partial\text{albsn}$ moved 0.149 → 0.504, a factor of three. A number measured in one state is not reusable in another, and albedo changes have to be priced in W m^{-2} before they are made: the Arctic runs away above about $+5.5 \text{ W m}^{-2}$ while the Antarctic is stable to at least $+10.4$.

55.2 Lead closing: the right flux response, the wrong state response

`h0min` was tried once more on the production line, 0.5 → 1.0 from 2000. The flux response was exactly as designed — the area-to-volume ratio of freezing fell 1.71 → 1.10 — and the state did not follow: mean thickness 0.61 → 0.62 m, and the 60–90 S bias went $+0.82$ to $+1.45$ K. Reverted. The namelist exposure survives as FESOM branch `feat/leadclose-h0min` and is *not* in `main`, which matters later (§57.1).

56 Round 39 (2026–09–22→24): vertical mixing, and two real bugs in the CVMix path

Four vertical mixing schemes, 20 years each (2020–2039) from one restart, differing only in `mix_scheme`:

arm	scheme	SH Feb volume	SH Feb extent	cavity melt
PICAL_ccnice	KPP (control)	—	—	1255.6
PICAL_cvKPP	<code>cvmix_KPP</code>	—	—	1283.5
PICAL_cvTKE	<code>cvmix_TKE</code>	1.31	1.95	1174.5
PICAL_cvTKEIDEMIX	<code>cvmix_TKE+cvmix_IDEMIX</code>	1.62	2.26	1062.1

Volumes in 10^3 km^3 , melt in Gt yr^{-1} ; observed SH Feb volume about 2.07, extent about 3.0. Both KPP variants lose Antarctic summer ice, both TKE variants grow it, and `cvmix_KPP` reproduces the KPP control closely enough to serve as the null. TKE+IDEMIX comes closest to the observed summer pack and brings cavity melt from 1255.6 to 1062.1 Gt yr^{-1} , towards Adusumilli (2020) ~ 1100 from Rignot (2013) ~ 1500 .

Adopted: `cvmix_TKE` from 2060, `cvmix_TKE+cvmix_IDEMIX` from 2080, and the pair became the v3.5 CORE3 default (`esm_tools 35a0524f7`). The cost was stated at adoption and is northern: NH April volume 38.04 rising $+2.46/\text{decade}$ against about 31.2 observed, where TKE alone sat at 35.17 and was flat.

56.1 Bugs VII and VIII: IDEMIX read garbage, and one bug hid the other

IDEMIX would not initialise. Four hypotheses were wrong before the right one (§60); the two real defects are in the CVMix forcing path and are now FESOM PR #1087.

Bug VII, the implicit netCDF fill value. `read_other_NetCDF` reads `missing_value` into `miss` and fills those points before interpolating. A variable with neither `missing_value` nor `_FillValue` leaves `miss` at whatever it held, in practice 0.0. Every genuine zero is then treated as missing while the real sentinel, netCDF’s implicit float fill 9.96921×10^{36} , is left in; the nearby-point average pulls it in and the field is `Infinity` before it reaches `interp_2d_field`, which documents that it assumes no dummy values exist. On the IDEMIX near-inertial surface forcing this masked 146694 real zeros and gave min/max of -7.6×10^{-4} to `Infinity`.

Bug VIII, an MPI type mismatch. `loc_Etot` and `glb_Etot` are `real(kind=WP)`, but five `MPI_AllREDUCE` calls declared the buffer `MPI_DOUBLE_PRECISION`. In a single-precision build that reads 8 bytes from a 4-byte variable, so half of every energy total is whatever sits next to it in memory. It is not only a diagnostic: `idemix_botforc_Etot` rescales the field by `Etot/glb_Etot`, so the corrupt total goes back into the physics.

With both fixed, `Etot_bot` reads 0.9135067 TW against the forcing file’s own `Etot_in_TW_bin` attribute of 0.91798, and `Etot_srf` 0.3211645 TW. That pair is now the standing gate on any IDEMIX leg: the signature of an unfixed build is `Etot_srf` 6.05×10^{-4} with `Etot_bot` -1.54×10^{16} .

57 Round 40 (2026–09–24→25): PICAL_ccnice reaches 2100, and it is drifting cold

`PICAL_ccnice` completed 1940–2099 and stopped at its configured `final_date`. Throughput was about 77 SYPD on 47 nodes under TKE+IDEMIX (3:18:32 for 2080–89, 2:58:12 for 2090–99), against about 86 for `cvmix_TKE` alone and 75 for KPP.

57.1 Bug IX: the staged library was a symlink, and md5sum follows it

The 2080 leg — the first with IDEMIX on — aborted 100s in with `Etot_bot` = -1.54×10^{16} TW, the exact unfixed-build signature, and NaN’d the ocean. `bin_sources` named the right file and the staged `fesom` executable matched the source md5. The defect was one level down: `<exp>/bin/fesom/lib/fesom/libfesom.so` is a *symlink* into `esm_tools`’ own per-leg snapshots `libfesom.so_<leg>`, and it pointed at a pre-fix one. `md5sum` follows symlinks, so every check passed while the wrong library loaded. The FESOM executable is a driver stub, byte-identical across all these builds, so a matching binary checksum proves nothing: **only the .so distinguishes them, and `ls -l` must come before `md5sum`.**

57.2 The drift: cooling at the surface, flat in the whole ocean

Over 2060–2099, the whole `cvmix` era, every headline metric is outside its confidence interval:

metric	2060–64	2095–99	trend/decade	95 % CI
T2m global	14.439	13.645	−0.214	0.025
T2m 60–90 N	−8.143	−9.842	−0.511	0.118
NH extent, March	16.262	18.188	+0.522	0.103
net TOA	−0.775	+0.236	+0.272	0.079

The run cools 0.21 K/decade globally while the Arctic grows ice, and net TOA over the last two decades is *positive* at $+0.167 \text{ W m}^{-2}$. That is not a contradiction and it is not an energy leak. Total ocean heat content is flat, $+0.04 \text{ ZJ/decade}$, and the cooling is a redistribution inside the ocean:

band	ZJ/decade	W m^{-2} equivalent
0–100 m	−17.8	−0.110
100–700 m	−34.4	−0.214
700–2000 m	+19.4	+0.121
>2000 m	+32.7	+0.203
total	+0.04	0.000

The upper 700 m loses heat to below 700 m at about 0.32 W m^{-2} equivalent, almost exactly balanced. Split by mixing era the mechanism follows the scheme changes: under KPP (2040–59) the upper ocean was *gaining*

(+0.061); `cvmix_TKE` flipped it hard negative (−0.488) and the whole system lost energy (net TOA −0.468); adding IDEMIX halved the upper-ocean loss (−0.201) and drove deep uptake up (+0.176 → +0.452), putting the total back positive. **IDEMIX did what internal-wave-driven diapycnal mixing should do — it moved the burial deeper and stopped the net drain — but the upper ocean is still losing heat, and that is what keeps the surface cooling.**

57.3 Where the state stands against observations

Against GIOMAS 1989–1999 (the early record is the fairer anchor for a PI run; the Arctic thinned sharply across 1989–2014 while the Antarctic has no significant trend):

10^3 km^3	model 2090–99	GIOMAS 1989–99	bias
NH April volume	34.56	31.19	+3.37
NH September volume	13.97	11.96	+2.01
SH September volume	12.33	19.89	− 7.56
SH February volume	0.73	2.07	− 1.34

The Antarctic sits at about 62 % of observed winter volume. NH April volume is still rising at $+1.145 \pm 0.968$ per decade over 2080–2099, which is roughly half the +2.46 priced at adoption but not flat.

CMPI went 0.844 (1980–89) → 0.879 (2000–09) → 0.902 (2080–89) → 0.903 (2090–99). **The degradation is tropical, not polar:** `nino34` +0.155 and `tropics` +0.134 against `arctic` +0.003 and `antarctic` +0.008; by variable `tas` +0.149, `thetao` +0.127, `zg` +0.111.

57.4 Drake Passage, and the price of the GM tuning

Computed with `tripyview`’s model-accurate transect path, which resolves the 105 mesh edges the 66 W line crosses and integrates the flux FESOM’s own discretisation carries. At 2090–99 the net transport is **98.9 Sv** (128.1 eastward, −29.2 westward), against Cunningham (2003) 134 ± 11.2 and Donohue (2016) 173.3 ± 10.7 . It has been declining slowly: 105.1 (2040–49) → 98.9 (2090–99), with no step at either mixing-scheme switch, so the ACC weakness is structural rather than a `cvmix` artefact.

The GM tuning of Round 36 is a different matter. On identical years 1580–1599 from identical restarts:

run	GM	resolved	net
PI200	resolution scaling (before)	110.0	110.0
PI200_gmR1500	Rossby cutoff, K_{GM}^{max} 1500	103.2	103.2
PI200_gmR2500	Rossby cutoff, K_{GM}^{max} 2500	93.3	93.3

The GM tuning cost 16.7 Sv of ACC transport, 15 %, monotonic in K_{GM}^{max} , far outside the ~ 1.5 Sv decadal scatter. The bolus term responds cleanly and monotonically ($\pm 1.8 \rightarrow \pm 2.7 \rightarrow \pm 3.6$ Sv) but its *net* is zero in every arm, because the eddy-induced circulation across the passage is a closed overturning cell. GM 2500 was adopted for the CDW improvement; Drake Passage was not scored at the time, and it is a real cost that belongs on the ledger next to the benefit.

57.5 The boreal forest is gone, and the canopy work needs one

In the Siberian box 55–75 N / 60–180 E at 2099: BNS 0.001, BNE 0.024, TREEFPC 0.050, GRASSFPC 0.216, and grass beats tree on 84.9 % of cells. The reference points are the offline CRUNCEP state it descends from (BNS 0.227, TREE 0.415, grass>tree 29.2 %) and 11E after 50 coupled years (0.017 / 0.083 / 69.6 %). The state is stable between 2090 and 2099, so this is settled, not transient. **The mechanism is competition with C3 grass, not a climate gate:** every BNS establishment gate stays open while the forest disappears, the only temperature gate is a warm one 34 K away, and GDD5 clears on 91 % of cells.

This matters for the LPJ-GUESS canopy work merged on 2026–09–25 (stem area index via `ltos`, shrubs routed to low IFS vegetation, `ifreefracca` 1). Laszlo Hajdu measured +0.46 K global T2m from it, concentrated in the boreal zone, in a run that has a boreal forest — and it improved his ERA5 comparison, bias −0.46 against −0.92 K and RMSD 1.38 against 1.77. **That result is not transferable here:** the change acts on the tree side of the tile and this configuration has 5 % tree cover. **PICAL_crunveg** was built to answer it — identical in every respect except that LPJ-GUESS starts from the CRUNCEP spin-up end state — and was cancelled before starting when the campaign moved to albedo.

57.6 Bug X: the instruction file nothing reads

The canopy parameters were first put in `namelists/lpj_guess/global.ins`, on the evidence that `run_coupled_4_1_2.ins` imports it and that the staged copy matched. That was wrong, and the correct chain has to be traced through the source:

```
framework.cpp:1063    xtring insfilename="guess.ins";          <- hard-wired
awiesm3.yaml:1331     ecearth.ins.j2 -> runPROC_NUM/ecearth.ins  (one per rank)
runN/guess.ins        import "ecearth.ins"
```

`global.ins` and `run_coupled_4_1_2.ins` are staged into the top of the work directory and never reached; `run_coupled_4_1_2.ins` even imports a path that does not exist, which nothing noticed because nothing reads it. The two copies had drifted apart in exactly the way that catches you out: `global.ins` carried `iftreefracca` 1 while the file actually read carried 0. Corrected in `esm_tools cf0a27943`.

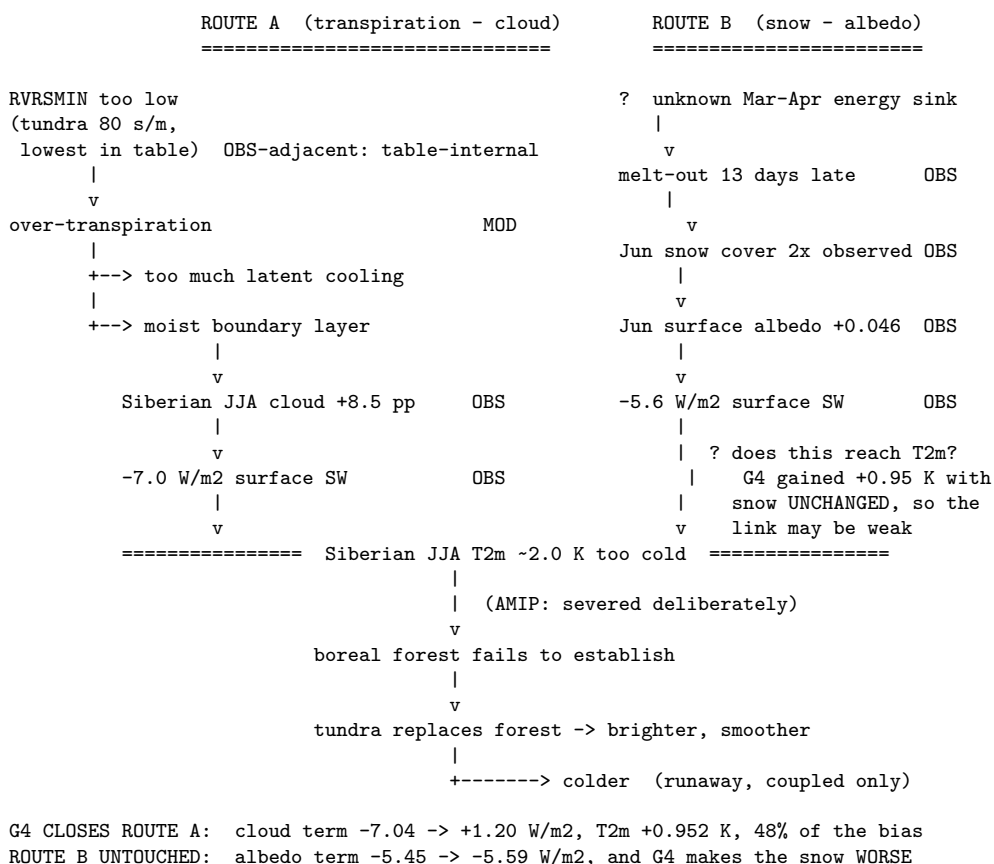
Both 50-year continuations queued on 2026-09-25 were cancelled before starting, partly for this: they would have run with the canopy work compiled and inert.

58 Synthesis: the causal web, and what is still dark

This section is the campaign's current best account of *why* the boreal cold bias exists, assembled from every round. It is written to be falsifiable: each link carries a status marker, and the unknowns are listed rather than glossed.

✓ OBS	established against observation (CERES, Rutgers/IMS satellite)
● MOD	established only model-vs-model (ERA5 = HTESSEL sibling); suggestive, not authoritative
✗ DEAD	tested and falsified
?	unknown — no evidence either way

58.1 The picture



58.2 Route A — transpiration to cloud: closed

Status: the strongest *measured* result of the campaign — but the mechanism is only partly established. The chain as proposed is: excess stomatal conductance cools the surface latently *and moistens the*

boundary layer; the moisture becomes low cloud; the cloud removes shortwave; the surface cools.

The moisture step is contradicted by this report's own profiles. §21.1 finds the excess boreal cloud is *thermally* driven, not moisture-driven: Siberian RH is +5.6 to +8.2% too high while q is *too low* (−0.2 to −0.3 g kg^{−1}). What was measured is the *correlation* F4 → less cloud → warmer; the moisture pathway is [?], not ✓OBS.

- ✓OBS Siberian JJA cloud fraction was 78.1% against CERES 69.6%, and the cloud share of the SW deficit was -7.04 W m^{-2} (§24.6).
- ✓OBS G4 removes it: cloud term $-7.04 \rightarrow +1.20 \text{ W m}^{-2}$, worth +0.952 K.
- The lever is defensible on *table-internal* grounds for tundra (RVRSMIN(9)= 80, the lowest of any vegetated type, §24.4) but **not** for needleleaf, where 250 → 1000 remains an unanchored 4× excursion that does not saturate (§24.6).
- ? G4 slightly *overshoots* in August ($+3.19 \text{ W m}^{-2}$ cloud term). A new bias in the making; watch it.

58.3 Route B — snow to albedo: open, and the interesting one

- ✓OBS June surface albedo +0.046 too high (CERES, §25.1); June snow cover 0.380 vs satellite 0.203; melt-out 13 days late (§25.2).
- ●MOD Snow *accumulation* is right; the loss term is mistimed — too weak in Mar–Apr, too strong in May (§24.2).
- ✗DEAD It is not the snow-*cover formulation*: round 13 changed RQSNCR and moved the albedo response into Sep–Nov instead (§23.2).
- ✗DEAD It is not an energy *supply* problem: the model absorbs *more* net SW than CERES in Mar–Apr ($+2.5, +1.2$), because its spring albedo is too *low* (§25.1).
- ✗DEAD It is not snow *brightness*: model snow is *darker* than both ERA5 and CERES. Brightening snow would worsen June.
- ●MOD It is probably not cold content: τ_{sn} is within a few tenths of a K of ERA5 — but this is model-vs-model (§25.3).
- ✓RESOLVED for the cover half — see §26: HTESSEL has no sub-grid snow depletion, so cover stays pinned near 1 while the median cell still holds 27 cm in May. That explains the late melt-out and the June albedo, and explains why round 13 could not have worked.
- ? The March–April mass deficit remains open. ERA5-only, The absorbed shortwave is present and the snow does not melt. Candidates: turbulent flux to the atmosphere (*unconstrained by any observation*); conduction/refreeze within the pack; or the canopy-shaded snow tile (7) behaving differently from exposed snow — the round-13 residual showed canopy masking already dominates in the melt months (§24.1).

58.4 The link that may not exist

The most consequential uncertainty in this report. Route B is a well-measured *radiative* bias, but its effect on *temperature* is unproven:

G4 raised Siberian JJA T2m by +0.952 K while May–June snow mass did *not* change (§24.6), and while snow cover and melt date got *worse* (§25.2).

Temperature moved a full kelvin with the snow term stationary or degrading. Either the June albedo error is weakly coupled to T2m — a real radiative bias that the atmosphere largely exports — or the two routes offset each other in a way we have not separated. **Until this is resolved, the size of the prize in fixing Route B is unknown**, and it is possible that closing it buys far less than its -5.6 W m^{-2} suggests. A clean test would be a run that changes the snow term alone with Route A held fixed; round 13 attempted exactly that and failed for an unrelated reason (it moved the wrong season).

58.5 The coupled amplifier, deliberately severed

The AMIP configuration exists to cut the vegetation–albedo feedback so the root cause can be hunted with the loop open. In coupled mode with LPJ-GUESS the chain closes: cold summers → boreal forest fails to establish

→ tundra replaces forest → higher albedo and lower roughness → colder. That runaway is why a 2K AMIP bias becomes forest loss in the coupled model, and it is why 48 % closure may or may not be sufficient — **a question AMIP structurally cannot answer.**

A new risk sits precisely here. G4 delays melt-out from 24 to 28 May (§25.2). Later melt is later growing-season onset for LPJ-GUESS. G4 may therefore improve AMIP T2m while *harming* forest establishment through a pathway AMIP cannot see. This should be weighed before G4 is committed to the PI spin-up, and it is an argument for running coupled with G4 sooner rather than later.

58.6 The energy target, now decomposed PER RUN

Where the imbalance lives was settled long ago: §11.2.2 showed the global budget is right *by cancellation*, with the Southern Ocean carrying $+5.8 \text{ W m}^{-2}$ and contributing $+0.57$ on its own, from a cloud deficit — the Hyder et al. signature. Nothing below revises that.

What was missing is that the decomposition existed **only for the control**. `amip_toa_decomp.py` never used the RUNS list, so across the tuning runs the energy target was tracked by a single global number with no idea which band any lever was moving. That is now part of the standard table and produced for every run automatically (§58.7). The new information is the **response**:

Δ net TOA vs control [W m^{-2}]	60–90S	30–60S	tropics	30–60N	60–90N	global
<i>control, absolute</i>	−90.44	−22.24	+42.61	−21.78	−97.74	+0.64
<i>CERES</i>	−98.82	−21.63	+45.11	−24.29	−97.98	—
<i>gap to close</i>	+8.38	−0.61	−2.50	+2.51	+0.24	
A1a ovl=0.10	−3.33	−2.80	−1.06	−2.44	−2.75	−1.90
D2a inpsea.2	−2.83	−1.91	−0.38	−0.79	−0.81	−0.93
D2b inp+p700	−1.76	−0.83	+0.09	−0.19	−0.34	−0.28
B3 clldiff	−0.11	+0.89	+2.02	+0.86	+0.25	+1.34
B4 entshalp3	−0.56	−2.00	−1.55	−0.75	−0.24	−1.33
B5 capdcycl0	−0.47	+0.06	−1.93	−0.14	+0.18	−1.00
F4 rsmn1000	−0.40	+0.04	+0.07	+0.30	+0.38	+0.10
G1 = F4+D2b	−2.03	−0.83	+0.10	+0.26	−0.02	−0.20
G4 = G1+tundra	−1.98	−0.65	+0.12	+0.28	+0.13	−0.13

Three things this view gives that the global number could not:

(i) **The 60–90S band is the most responsive in the model**, moving $−0.4$ to $−3.3$ under levers that were designed for entirely different purposes. It is not inert; it has simply never been *targeted*.

(ii) **The G-series buys about $−2.0$ of the $+8.4$ needed** there, all of it from D2b, and D2a alone buys $−2.8$ — but D2a costs $−0.93$ globally and was rejected on other grounds. *Roughly a quarter of the Southern Ocean TOA gap is already closed as a by-product*, and the remainder is the same ~ 6 pp cloud-area deficit that no lever has moved (§58.8).

(iii) **The surface levers are confirmed regionally clean.** F4 moves the tropics by $+0.07$ and G4 by $+0.12$, against B3’s $+2.02$ and B5’s $−1.93$. The tropical guardrail was being checked as a single number; it is now visible as a band, and it shows the F/G family does what it was designed to do and nothing else.

One trap the guardrail table invited. The PI target is **net TOA** ≈ 0 , *not* the CERES column. CERES is present-day (2005–2015) and carries the real planetary warming imbalance, so it reads $\sim +1.1 \text{ W m}^{-2}$ globally: the control’s $+0.64$ is *too positive* even though it sits *below* CERES. Per-band CERES differences are the right way to use it — for locating error, never as a global tuning target.

58.7 One command: `evaluate.sh`

The protocol had grown to five scripts run from memory, and skipping one has repeatedly produced wrong conclusions here — a lever promoted on 4-year noise (B8), a winter penalty unnoticed for eleven rounds (B5), and a TOA decomposition done only for the control. `scripts/analysis/evaluate.sh` now runs the whole protocol, with the TOA decomposition included in the main table rather than as a separate invocation. The way to make a guardrail effective is to make skipping it require effort.

58.8 The open list

1. ? Where the March–April melt energy goes. Unconstrained; no satellite product measures turbulent flux.

2. ? Why F4/G1 *increases* May–June snow mass (+5.58, +5.68 mm, both significant). Sublimation was the suspect and is *not* implicated in Mar–Apr; May snowfall rises significantly instead, for reasons unknown.
3. ? Whether the June albedo bias reaches T2m at all (§58.4).
4. ? Why G4 *repairs* G1’s Nordic Seas SW RMSE damage (+0.300 $\rightarrow$ +0.089). Adding tundra resistance should not act on a Nordic ocean box. An unexplained improvement is as much a loose end as an unexplained degradation.
5. ? The Southern Ocean cloud-*area* deficit (~ 6 pp), untouched by every lever in all 34 evaluated runs.
6. ? Tropics net TOA 2.4 W m^{-2} too low.
7. ✓ **but unaddressed** Global net TOA $+0.52 \text{ W m}^{-2}$ against a ~ 0 target — *one of the two original problems*, barely moved by the whole campaign (control +0.64). For a PI spin-up this drift matters more than the last tenths of Siberian warming.

59 State of the campaign (2026–09–25)

A snapshot, not a story. Numbers are 44-year means over 1872–1915 unless stated.

59.1 Proven

Added 2026–09–25 (rounds 37–40, §54–§57).

- **The Southern Ocean winter bias resists every lever tried.** Salt plumes, Fox-Kemper MLE, a deeper lead-closing floor, all three together, and a calibrated ice-shelf basal melt all fail to move it in 20 years (§54.1). It is a freshwater regime, not a mixing deficiency.
- **Ice-shelf basal melt is calibrated.** `cavity_gamma_scale` 0.6 gives 1190 Gt yr^{-1} against 1893 uncalibrated and 1100–1325 observed; upstream as FESOM PR #1064.
- **The two snow albedos act in different hemispheres, by measured surface state.** $\partial\alpha/\partial\text{albsn}$ is 0.908 in SH DJF against 0.149 in NH JJA; `albsnm` is the reverse. Both stay global. The sensitivities are state-dependent and moved by a factor of three as the Arctic cooled (§55.1).
- **`cvmix_TKE+cvmix_IDEMIX` is the adopted mixing scheme**, on a four-way 20-year comparison from one restart. Both TKE variants grow the Antarctic summer pack where both KPP variants lose it; cavity melt falls $1255.6 \rightarrow 1062.1 \text{ Gt yr}^{-1}$ towards Adusumilli (2020). The cost is northern and was stated at adoption (§56).
- **Two real bugs in the CVMix forcing path**, an implicit netCDF fill value and an `MPI_DOUBLE_PRECISION` buffer on a `real(WP)` variable, both in FESOM PR #1087. The gate on any IDEMIX leg is `Etot_bot` $0.9135067 \rightarrow 0.9179869 \text{ TW}$ (§56.1).
- **The cold drift is an ocean redistribution, not an energy leak.** Total ocean heat content is flat at $+0.04 \text{ ZJ/decade}$ while the upper 700 m loses to below 700 m at 0.32 W m^{-2} equivalent, and the surface cools 0.21 K/decade with net TOA positive (§57.2).
- **The GM tuning cost 16.7 Sv of Drake Passage transport**, $110.0 \rightarrow 93.3 \text{ Sv}$ on identical years, monotonic in K_{GM}^{max} (§57.4).
- **The boreal forest is effectively gone** at 2099: BNS 0.001, TREEFPC 0.050, grass beating tree on 84.9% of the Siberian box, stable across 2090–99. By competition with C3 grass, with every establishment gate open (§57.5).
- **The model skill degrades in the tropics, not the poles.** CMPI $0.844 \rightarrow 0.903$ across 1980–2099, driven by `nino34` $+0.155$ and tropics $+0.134$ against arctic $+0.003$ (§57.3).

Added 2026–09–11 (round 36, §53).

- **The single-precision ocean heat leak is fixed.** 15F closes at -0.069 W m^{-2} over 10 years and -0.035 over 20, like the double-precision 13A (-0.073); the unfixed 15C reads -0.544 (FESOM PR #1054).
- **Decade-4 TOA follows mid-depth heat storage, with a shallow slope.** Across 20 runs with 1850 forcing and at least 40 years (now including 16C, 16D, 16E), decade-4 TOA against the 700–2000 m heating rate averaged over years 1–40 gives $r = +0.72$, and $+0.58$ ($p = 0.015$) without 080, 11M and 11L. The slope without the extremes is about 1.3 W m^{-2} per W m^{-2} , not the 4.1 of the first 17-run fit (§53.3).

- **Only the new-sea-ice chain lets mid-depth uptake decay**, to $0.21\text{--}0.44\text{ W m}^{-2}$ by years 25–45 (11G, 11I). 11X levels off at $0.50\text{--}0.62$, and 15F tracks it (§53.2).
- **Stronger GM is the lever that reaches mid-depth**. 16C (K_{GM_max} 1500) cuts first-decade 700–2000 m uptake by about 30 % ($0.67/0.62$ against $0.92/0.96$). Weaker Redi raises it and MLE leaves it unchanged (§53.4).
- **16E (K_{GM_max} 2500, Redi 1000) settles at the lowest TOA plateau the campaign has kept:** $+0.70\text{ W m}^{-2}$ in decade 4, -0.45 ± 0.08 against 15F and -0.24 ± 0.06 against 16C in clean pairs. Its deep uptake below 2000 m is nearly shut ($0.04\text{--}0.09$). The costs: NH March ice $+1.3\text{ M km}^2$ against 15F (17.2 in decade 4, obs 15.0), global T2m $+0.21\text{ K}$ (§53.5).
- **The AMOC response to GM depends on the dose**. GM 1500 strengthens it ($+0.83 \pm 0.21\text{ Sv}$ at 26.5N against 15F); GM 2500 is $1.0 \pm 0.2\text{ Sv}$ weaker than GM 1500 at 26.5N and 1.7 ± 0.2 at $40\text{--}60\text{N}$. MLE changes neither (-0.25 ± 0.18 , §53.7).
- **RSBLB = 2 adds late mid-depth uptake but not TOA:** $+0.17/+0.19\text{ W m}^{-2}$ at 700–2000 m in decade 4 of both clean pairs, net TOA unchanged within ± 0.07 (§53.6).

Added 2026–09–09 (rounds 34–35, §51, §52).

- **The ocean heat gain accounts for the imbalance, and it lands below 300 m**. 11X gains $+1.502\text{ W m}^{-2}$ per ocean area over 30 years; the top 100 m gains $+0.013$ and the 100–300 m layer *loses* 0.170 , while 300–2000 m gains $+1.388$ (§52.3).
- **The tropical mixed layer is not too deep**. 43.6 against WOA18’s 51.6 m annually and 81.0 against 80.8 at the seasonal maximum, both on WOA’s criterion. The model is shallower than observed in all twelve months (§52.2).
- **The tropical deep warming is heave, not diapycnal flux**. At $\sigma_0 = 27.0$ (575 m) the water warmed $+0.047\text{ K}$ while the surface sank $+42.4\text{ m}$, accounting for 0.42 of the $+0.477\text{ K}$ seen at fixed depth (§52.4).
- **Heat crosses density surfaces at the mid- and high-latitude winter outcrops**, where the mixed layer is $+48.9$ to $+155.7\text{ m}$ too deep: $30\text{--}45\text{N}$ $+0.58\text{ K}$ at 500 m , $45\text{--}30\text{S}$ $+0.57\text{ K}$ at 950 m (§52.4).
- **The Southern Ocean loses heat rather than burying it**. -0.076 W m^{-2} of the global budget, cooling at every depth from 70 to 2400 m , with 18.7% of month-nodes mixing to the sea floor (§52.3, §52.2).
- **The coupled ice skin was diverging because oasis_put returns OASIS_Waitgroup for every field in a group but the last**. The anchor was 0.0 in all $63,923$ logged events; the skin reached $-36,470\text{ K}$. After four fixes the minimum is 224.39 K with zero points below 200 K (§51).
- **OASIS GSMART on a sign-changing field is unsafe**. 9105 excursions above 1 kW m^{-2} onto real ice in one coupled year, maximum $22,550$; GSSMAR and GLOBAL deliver none (§51.3).

Added 2026–08–23 (round 33, §50).

- **The Southern Ocean generates the TOA imbalance rather than responding to it**. The $90\text{--}60\text{S}$ and $60\text{--}45\text{S}$ bands contribute $+0.383$ and $+0.353\text{ W m}^{-2}$ to the global mean, the two largest positive terms, against a tropical -0.872 (§50.2).
- **Every arm drifts, including the 1850 controls**: $+0.8\text{ K}$ and $-2 \times 10^6\text{ km}^2$ of September ice per 40 years. The air was NOT warm at the branch point (-0.06 to -0.65 K), so the ocean was not warmed by the atmosphere — both warm together (§50.1).
- **The polar cloud residual is an OPACITY problem, the sub-Antarctic one an AMOUNT problem**. At $90\text{--}60\text{S}$ cover is right to 2 pp but each cloud is 8% too dim; at $60\text{--}45\text{S}$ there are 5 pp too few clouds each 9% too bright. Every campaign lever raises opacity, which is why $60\text{--}45\text{S}$ is now overcorrected at -2.97 while $90\text{--}60\text{S}$ still reads $+6.98$ (§50.4).
- **Cloud phase flips sign at 60S**: $+0.015$ too liquid at $45\text{--}65\text{S}$, -0.028 too icy poleward. The INP family was retired on a measurement taken in the one band where the model is too liquid (§50.4).
- **The ice melts from above in summer and fails to refreeze from below in autumn**. $+6.98\text{ W m}^{-2}$ of DJF cloud shortwave, and a $+0.9\text{ K}$ anomaly at exactly 100 m that autumn mixed-layer deepening entrains — which is why the MAM anomaly is longwave with no shortwave component (§50.6, Fig. 37).
- **The sea ice is not too dark** (compact-ice albedo $0.760\text{--}0.778$); it is too diffuse. No cell exceeds 90% concentration in January–March, so the area-mean albedo falls to $0.49\text{--}0.51$ (§50.5).

Added 2026-08-23 (round 32, §49).

- **Stable-BL mixing is the winter land lever, and it works coupled.** 11Q/11R = 11N/11P + RSBLB 2.0 give Arctic DJF screen $+0.614^*/+0.674^*$ against AMIP's non-significant $+0.285$, with T925/T850 null — coupled, the inversion weakens by warming the screen, not by chilling the air above (§49.3, Fig. 36b).
- **It is free in energy terms coupled.** Global net TOA $-0.001/-0.027$, both null, against AMIP's $+0.175^*$: the tropical shortwave gain is absorbed as ocean warming and cancelled by the longwave response (§49.3).
- **Winter and summer move together for the first time.** Siberian JJA improves 1.000 K over base, 0.767 of it from RSBLB, against a campaign history of levers that buy one by spending the other (§49.5).
- **11R is the best-scoring run in the campaign,** CMPI 0.7577 against 11E's 0.8237 , and it reverses 11G's tas/zg regression (§49.4).
- **The stable-mixing trade is linear in dose and cannot be dodged.** Tropical SW CRE per K of Arctic inversion is $0.88/0.87$ for RSBLB $b=3/b=2$ and 3.43 for the mixing-length route, so the shortwave cost is a property of the scheme rather than of the knob (§49.2, Fig. 36a).
- **Best coupled land state so far:** annual all-land bias $-2.509 \rightarrow -1.854\text{ K}$ from base at 1850, a 26 % reduction, at unchanged net TOA (§49.5).

Added 2026-08-21 (rounds 30-31, §46-§48).

- **The Southern Ocean bias is an AMIP problem, not a coupling problem.** 84 % of the coupled SO SW CRE error is already present in AMIP over a prescribed, correct ocean; cloud cover matches to 0.001 and CRE to 0.91 W m^{-2} (§46.1).
- **Tuning net TOA to zero does stop the ocean drift.** The apparent “ -1.0 W m^{-2} structural leak” was the omitted snow-enthalpy term; with it the column conserves to $0.02\text{--}0.04\text{ W m}^{-2}$ and 11M (TOA -0.016) drifts -0.02 K/century (§46.3).
- **The winter land bias sits between the screen and 925 hPa.** In 11P at 60-90N DJF the air-side inversion bias is $+2.81\text{ K}$ while the skin-side bias is $+0.11\text{ K}$, and the profile decays to -0.42 K by 850 hPa (§48.3, Fig. 35).
- **The surface-side levers are retired as a class, by measurement.** Skin conductivity ($+0.125\text{ K}$ against a 0.41 threshold, §48.1) and Raupach z_0 (DJF land-ocean gap -0.017 and -0.003 in two windows, §48.2) both score null, for the reason the decomposition gives: they adjust a part of the column that carries no error.
- **Only about half the residual land bias is reachable by any mixing lever.** Of the -2.23 K all-land DJF bias, $\sim 1.0\text{ K}$ is excess inversion and $\sim 1.2\text{ K}$ is a deep cold bias still present at 850 hPa (§48.3).
- **The LX4 stack is the best coupled state so far,** all-land annual $-2.79 \rightarrow -2.11\text{ K}$ at fixed forcing basis and 60-90N DJF $-6.35 \rightarrow -4.49\text{ K}$ (§47.3). Note 11N→11P is a change of comparison basis, not a further gain.

Added 2026-08-16 (round 29, §45).

- **Raupach canopy roughness is dead, structurally rather than statistically.** Its winter mechanism needs a canopy; the model's Siberian trees are 2.89 m tall with 2 cm stems, giving $\lambda_{\text{stem}} = 0.0092$ and $z_0 = 0.0132\text{ m}$ against IFS's 0.058 m. It makes the winter surface $4.4\times$ smoother. The lever cannot bootstrap the forest it requires in order to work.
- **The land-surface repair is a trade.** Paired, it warms Siberian DJF soil $+0.308\text{ K}$ ($t = 2.49$, still growing) and recovers TREEFPC $+0.0094$ of what S4 cost — but it moves the Siberian JJA bias against ERA5 from -0.673 to -1.123 , worse than the 11E baseline.
- **This design cannot resolve anything below $\approx 0.35\text{ K}$ of Siberian JJA soil at 50 coupled years.** The interannual noise floor on the paired difference is 1.2 K ; the measured $+0.312\text{ K}$ would need 57 years. Future land levers must be screened on a lower-noise diagnostic — the surface momentum budget resolved $\text{skt}-2t$ at $p = 3 \times 10^{-5}$ where soil temperature saw nothing.
- **Branched coupled arms must be scored paired.** All arms share the 1350 state, so the difference series starts at zero and grows; the unpaired $1.96\text{ sd}\sqrt{2/n}$ threshold discards the pairing and hid a resolved result.

Added 2026-08-11 (round 28, §44).

- **S4 (RCL_INPPMIN 70000 → 50000) is the best configuration the campaign has.** Southern Ocean surface bias $+3.66 \rightarrow -0.96$, global net TOA $+0.64 \rightarrow +0.25$ — nearest the piControl target of anything run — Siberia JJA $+1.03$ and DJF $+0.05$, tropics $+0.26$. One namelist number, and it had sat unevaluated since 2026–08–09.
- **The round-26/27 levers are global dimming knobs, not Southern Ocean levers.** LX3 moves the tropics -4.05 against the Southern Ocean’s -5.61 , near-uniformly across every basin. DMS’s one-year seasonal selectivity does not become spatial selectivity at 44 years.
- **RSNOWLIN2 is confirmed at full length on both counts:** it delivers the tropical longwave gain ($+2.06$ global LW CRE) and warms the Southern Ocean ($+0.61$ net) while worsening global energy ($+0.42$). Not adoptable; still the only lever that moves the tropical longwave at all.
- **A1a’s boreal damage reproduces exactly** — -0.67 K Siberian JJA on the new base against -0.749 on the bare base, both $\approx 3\times$ threshold.
- **The campaign’s SO “band mean” and “SW RMSE” are different fields:** TOA cloud forcing and surface net shortwave. They disagree about which arm is best, and the surface error exceeds the TOA error everywhere — an atmospheric-absorption signature.
- **Krakatoa is visible in the AMIP runs** (1883–86, all arms), independently confirming that these “pre-industrial” runs read transient forcing.

Added 2026–08–10 (round 27, vegetation, §43.12).

- **Warming restores the boreal forest, and the transfer function is now measured.** Offline 2000-year LPJ-GUESS equilibria at TCO95: AMIP forcing gives TREE 0.202 and BNS 0.034; $+1$ K gives 0.261/0.058; $+2$ K gives 0.327/0.099 against the observed-forcing 0.415/0.227. Monotonic, not saturating, and $+2$ K recovers 59 % of the gap. **With the AMIP cold bias at ≈ 2 K period-clean, closing it should recover most of the forest** — the campaign’s founding premise, previously assumed and now calibrated.
- **The coupled model dismantles an inherited forest in 50 years.** 11E starts at BNS 0.213 / TREE 0.394 (the CRUNCEP restart) and ends at 0.017/0.083 — a 92 % loss of larch — with grass beating trees on 70 % of cells against 28 % at the start. It is a transient toward a much poorer equilibrium, not an endpoint.
- **Warming does not dissolve the competition.** grass>tree sits at 43–45 % across AMIP, $+1$ K and $+2$ K against CRUNCEP’s 29 %. Forcing is the driver; grass-versus-larch remains the mechanism (§31.2).
- **Siberia is NOT solved.** 11E fixes the *atmospheric* half — summer warmth and the winter penalty every previous snow lever carried — and leaves the vegetation untouched.

Added 2026–08–10 (round 27, §43).

- **RSNOWLIN2 0.030 → 0.04 closes the tropical longwave deficit.** $+2.581 \text{ W m}^{-2}$ alone and $+2.740$ on the adopted stack, against a ± 0.279 threshold — 120–127 % of the -2.16 deficit and five times the largest response among the other fifty arms. It is the temperature exponent of ice→snow autoconversion, so it builds cold thin cirrus rather than thickening anvils, and thin cirrus is longwave-dominant.
- **It also warms the Southern Ocean.** On the adopted stack its SO longwave $+2.626$ swamps its shortwave -0.507 , giving net $+2.294$, resolved. On the bare base it was neutral: the sign reversal is produced by the base. **The campaign’s two remaining regional targets are in tension through this one parameter.**
- **Southern Ocean levers must be scored on NET, not SW CRE.** Longwave cancels a very consistent 25–29 % across the whole 44-year archive, so SO SW $\times 0.72$ is the rule for ordinary cloud levers — and the new levers break it. Best net lever in the archive is A1a at -4.567 , ahead of D2a’s -3.602 .
- **DMS is the cleanest of the new Southern Ocean levers on net forcing** (-1.976 , only 17 % longwave cancellation) and the only one that reaches DJF, where the deficit is $+10.86$ and austral-summer insolation is 460 W m^{-2} against 87 in JJA. Its annual mean coincides with RCL_INPPMIN’s; its seasons do not.
- **A jointly-tuned eight-parameter cloud stack for TCO95L91 existed unused** in the v3.4.2 DARS2 runscripts, with no `fort.4` on disk setting any of it.
- **The AMIP runs are not at 1850 forcing** — NCMIPFIXYR sits in the dead &NAMECECMIP6 block, runtime is -1 , GHG runs 1869→1911. Deltas are unaffected; the absolute energy baseline is offset $\approx 0.15 \text{ W m}^{-2}$. The coupled runs *are* correctly pinned. Z1 verified a one-line fix.
- **S4 – P5:** SO SW CRE -1.98 , global net TOA -0.41 , tropics -0.10 , Siberia within noise — but SO SW RMSE $+0.255$ against ± 0.019 . Best mean, worse pattern.

Added 2026-08-10 (round 26, §42).

- **Marine biogenic CCN is a real Southern Ocean lever, and the best one found.** At $S = 166$ it moves SO SW CRE -7.25 W m^{-2} in DJF against a ± 3.82 threshold — 67% of the $+10.86$ DJF deficit, in the season the deficit is worst, from the bloom rather than from a tuned constant.
- **It cannot be adopted as a tuning knob at any published coefficient.** The tropical cost is 0.0129 W m^{-2} per unit S , so the ± 0.5 tolerance binds at $S \approx 39$, below Woodhouse et al.'s floor of 43; and Twomey's $\Delta \ln N_d$ scaling makes the bound conservative. Scaling S cannot separate the two bands because tropical background CCN is *lower* than the SO's.
- **Detection thresholds for a 1-year pair, per band:** SO SW CRE annual ± 1.97 , SO DJF ± 3.82 , tropics annual ± 0.99 , tropics DJF ± 1.22 , global ± 1.11 . Two of the three bracket arms fall below their own thresholds and carry no information.
- **The tropical longwave deficit gates the Southern Ocean fix.** The tropical cost of the DMS lever is shortwave; the tropical bias is longwave. They are independent terms, so closing the LW deficit makes the SO lever affordable rather than competing with it — and the two are complementary in the global budget (-1.72 against $\approx +1$).

Added 2026-08-09 (round 25, §41).

- **The tropopause is not displaced.** Model and ERA5 put the WMO lapse-rate tropopause on the same level in all five bands. The 200/100 hPa sign reversal is a *sharper* transition, not a moved one — the model cools $+1.8 \text{ K km}^{-1}$ faster than ERA5 through 300–250 at the poles and -1.5 to -1.9 slower above 200.
- **Cold and dry share one entry point.** A -2.07 K tropical cold point predicts -28.5% of entry-air drying by Clausius–Clapeyron against a measured -46.9% at 100 hPa, i.e. **61%**. They are not the independent compensating pair §38.4 recorded, so a lever that warms the tropical cold point should move both the right way (§41.3).

Added 2026-08-09 (round 24, §40).

- **Southern Ocean droplet number is correct in the mean** — $N_d \approx 73 \text{ cm}^{-3}$ against an observed 60–120, evaluated on 6-hourly wind. The marine CCN path is mapped end to end and RCCNSEA is inert because LCCNO=.TRUE.
- **But the seasonal phase is inverted:** model N_d is lowest in DJF (68.5) and highest in JJA (76.1), because it is wind-driven, where observations peak in DJF from biogenic sulphate. The measured SO CRE error is worst in exactly those months.
- **The cloud-lifetime CCN route would divide by zero** at NAERCLD= 0, so the amount two-thirds is unreachable from the CCN path (§40.3).
- **A DMS emission climatology already ships at TCO95 and nothing reads it** — emdms, paramId 210043, on the model's own N96 grid, with ICMCL having headroom for an eighth field (§40.4).

Added 2026-08-09 (round 23, §39).

- **The Southern Ocean amount deficit is confirmed against the right reference.** CERES and MODIS put SO cloud area at 89.5 and 89.3% against the model's 83.1; ERA5 reads 81.6 but reflects 6.31 W m^{-2} less than CERES against the model's 7.45, i.e. it has low area *and* low reflection in proportion. **ERA5 reproduces the model's bias because ERA5 is IFS**, so it is not an independent cloud reference (§39.3).
- **The Southern Ocean and the tropics fail through different cloud types.** SO loses energy via SW cloud ($+7.45$); the tropics via LW cloud (-2.16). No global microphysics knob can serve both, which is why RVICE trades one for the other (§39.2).
- **Nothing in 50 runs warms the tropics through the longwave**, the term that is actually deficient there. The only tropics-warming levers on record act in the shortwave.
- **CERES cloud area is valid over dark ocean, not over ice** — CERES and MODIS disagree by 7.7 pp at 90S–65S.

Added 2026-08-09 (round 22, §38).

- **The period-clean radiative error is $+1.23 \text{ W m}^{-2}$, about twice what the PI framing reports.** amip_presentday reads net TOA $+2.20$ against CERES's $+0.97$ (§38.2).

- **The tropical TOA deficit is mostly the reference period** — -2.51 PI-vs-CERES against -0.67 period-clean. The Southern Ocean is the opposite: robust to the epoch and *larger* period-clean, $+7.46$ W m^{-2} and $+0.735$ of the global mean, now unambiguously the largest single regional term.
- **The SO deficit is cloud AMOUNT, 65.5 % of it**, zonally uniform to 1.42 pp and worst in austral spring and summer — and **no lever in the campaign moves cloud area**: D2a $+0.63$ pp and D2b $+0.43$ pp of a 6.43 pp gap, against $+0.48$ and $+0.45$ pp from two boreal-*land* levers that cannot physically act there (§38.3).
- **The global cold bias is extratropical and lives at the tropopause**, peaking at 200 hPa over both poles (-4.20 , -4.45) and reversing sign at 100 hPa, with the tropics the *least*-biased band. The convective-heating hypothesis is excluded (§38.4).
- **The atmosphere is cold AND dry, and the two cancel in OLR**. -1.62 K and -10.1 % over 700–300 hPa; a Planck response to the cold alone predicts -6.10 W m^{-2} of OLR against -2.53 measured. Fixing either alone moves net TOA several W m^{-2} the wrong way.
- **The tanh falsification reproduces coupled on a controlled pair**: 11B minus 11A is -18.99 K of Siberian DJF soil (§38.1).
- **The coupled branch already carries D2b**, not just G4 — 11A/11B/11D all set RCL_INPSEA with the 700 hPa gate. K1 and P5 are what remain missing.

Added 2026–08–07.

- **The tanh depletion is falsified on observations, not on preference**. Its range is open at 1, so complete snow cover is unrepresentable at any parameters, while Russian snow courses report exactly 10/10 in 99.74 % of 14 369 Siberian DJF surveys. Cost: -20.2 K against 174 676 station soil observations where the scheme *off* is right to $+0.8$ K (§33.3).
- **The damage was distributional, not a matter of mean cover**. DJF cover is identical to 0.002 across N1/N2/O1/O2 while the soil differs by 23 K; the clipping ramp leaves 93.5 % of the box perfectly insulated, the *tanh* smears the same deficit into every cell (§33.2).
- **P3 is adopted** — both falsifiers passed, and it beats the scheme-off baseline on bias, RMSE and JJA against 105 RIHMI stations (§34).
- **Aerosol is ~ 2 % of the Southern Ocean clear-sky excess, not 13 %**, measured rather than inferred, and the two reflection columns are not separable (§35).
- **K1 is the carried-forward configuration (adopted 2026–08–05)** = G4 + the snow-free land albedo correction. Siberian JJA $+1.036$ K ($t=8.29$), global T2m RMSE $1.579 \rightarrow 1.524$ (the campaign’s best bar K2), every season inside its own threshold, and no Arctic energy penalty — 60–90N net TOA -97.463 against a control -97.737 and CERES -97.98 (§30.2). Namelist-only on top of G4.
- **Land surface albedo is CLOSED as an explanation for the universal bias**. Removing 74 % of a directly measured bias bought 0.089 K of ~ 0.7 K; a perfect fix buys ~ 0.12 K, one sixth of the target (§30.3).
- **G4 is the base K1 is built on**: Siberian JJA T2m $+0.952$ K ($t=7.68$). *Closure depends on a denominator this report does not pin down*: against the ≈ 2.0 K working figure it is 48 %, but §17.2 gives an honest range of -1.3 to -2.0 K and §27.3 measures -2.58 K against ERA5, so the defensible span is **37–73 %**, SO SW RMSE $6.88 \rightarrow 4.80$, subpolar N Atl $5.01 \rightarrow 4.74$, tropics untouched ($+0.124$), every season inside its own threshold. Namelist-only (§24.7).
- **The boreal budget is nearly spent**. Siberia’s JJA -2.58 K is ~ 0.7 universal (present over prescribed-SST ocean too), ~ 0.3 orographic and 1.3–1.6 boreal-specific; G4 has taken ~ 0.95 of the last (§21.3). The remainder is not reachable by a boreal-indexed lever, and this — not a parameter limit — is why $F5 = F4$.
- **The boreal forest fails by TWO routes, not one** (§31.1). BNE/BINE are *gate-excluded* — 0.7 % of NE-Siberian cells clear `tcmn_est` = -30 and their LAI is exactly 0.000. BNS (larch) has *no* cold gate, clears its GDD gate on 84 % of cells, and is *outcompeted*: grass carries $5.8\times$ its LAI. Since NE Siberia is ~ 80 % larch, the winter-gate mechanism does not explain the region that matters most.
- **The near-surface cold is global, and the soil is not the cause**. All 20 surface types are cold in every season, and over low-lying land the soil and skin are the *least*-biased parts of the column (§21.2, §21.3).
- **The boreal cold bias had two routes**. Route A (over-transpiration $\rightarrow$ low cloud $\rightarrow$ SW loss) is **closed**: the Siberian JJA cloud term went $-7.04 \rightarrow +1.20$ W m^{-2} . Route B (late melt $\rightarrow$ June albedo) is **diagnosed and fixable** but carries almost no temperature leverage (§27.2).

- **The snow field can be made correct, but not at this price:** melt-out 12 May against a satellite 11 May (§27.1) — and the scheme that does it costs -16.2 K of coupled DJF soil for $+1.2\text{ K}$ of summer (§32). **Rejected**; round 19 tests a re-parameterisation (§32.2).
- **The winter damage is seeded in AUTUMN, not winter.** Midwinter snow cover is identical between the schemes ($\Delta\text{ZCVS} = -0.0004$ in January); the deficit peaks in October (-0.075) exactly as the soil begins to diverge, then vanishes while the soil runs to -24 K (§32.1).
- **The control soil is right, verified against direct observations.** RIHMI-WDC station soil temperature under natural cover puts the control within $\sim 1\text{ K}$ at every depth the Russian network reports in winter, and its thermal offset within $0.7\text{--}2.3\text{ K}$ of the observed $+17.8$ to $+24.2\text{ K}$ (§27.8). The campaign no longer depends on ERA5 — an HTESSEL sibling — for any soil claim.
- **I3 is the strongest summer lever built:** Siberian JJA $+1.322\text{ K}$ ($t=10.61$), and it closes the Siberian surface SW deficit to 166.58 against CERES 166.26 — the deficit §22 opened the albedo thread with. It is **not adopted**: it buys that with $+1.72\text{ W m}^{-2}$ at $60\text{--}90\text{ N}$, the largest NH–SH albedo shift of any run, and the campaign’s only significant Nordic Seas RMSE degradation (§27.6).
- **The Siberian winter penalty is not radiative.** DJF snow albedo, SWE and absorbed SW are identical across control/G4/I1/I2/I3 to three decimals, on an 8 W m^{-2} midwinter budget that cannot build -2.4 K (§27.7). The route is thermal.
- **Half the land albedo excess is the snow tile.** Of $+0.0154$ against CERES, $+0.0074$ is snow and $+0.0080$ the snow-free surface; every forest type is inside $+0.006$ once snow is masked per cell per month (§29.1).
- **RVRSMIN does not saturate;** 1000 is a cap set by winter damage, not a knee (§24.6).
- **The energy target is Antarctic/Southern Ocean,** and the two campaign targets share that root cause (§11.2.2, §58.6).
- **Tuning is namelist-driven and mergeable with EC-Earth** — one added line in an ECMWF file (§24.7).

59.2 Open, in priority order

Added 2026–09–25 (rounds 37–40).

1. **The upper-ocean heat drain.** 0.32 W m^{-2} equivalent out of the top 700 m is the mechanism behind the cold drift and nobody has tried to stop it. Whether it is the mixing scheme, the Southern Ocean freshwater regime, or both, is open.
2. **NH April volume** decides whether `cvmix_TKE+cvmix_IDEMIX` keeps its place: $+1.145 \pm 0.968$ per decade over 2080–2099, sitting 3.37 above the early GIOMAS record.
3. **The Antarctic at 62 % of observed winter volume,** unchanged in character by any Round 37 lever and not a vertical-mixing problem.
4. **The ACC is 30 % weak** and slowly weakening, and the GM setting that improved the CDW made it worse. The two constraints have not been reconciled.
5. **Does the canopy work do anything without trees?** `PICAL_crunveg` was built to answer it and cancelled before starting.
6. **Will the CRUNCEP state load on TC095-land?** It was written on an 11538-cell L096 gridlist; the old respin handoff recorded a missing-cell case as fatal with peatland on, and the code has changed substantially since. Untested.
7. **The tropical CMPI degradation** has no mechanism attached to it yet.

Re-ranked 2026–09–11 after round 36 (§53). The goal is now to use the relation between mid-depth heat storage and TOA to bring TOA close to zero without degrading the rest of the climate; then retune the melt-pond albedo for NH sea ice.

1. **Bring TOA to about zero.** The three 40-year GM arms are done (§53.5). 16E is the best, at $+0.70$ in decade 4, and GM alone will not close the rest: the pairs give $1.2\text{--}1.6\text{ W m}^{-2}$ of TOA per W m^{-2} of mid-depth uptake, 16E’s mid-depth rate is already 0.28, and the step from GM 1500 to 2500 cost 1.7 Sv of AMOC at $40\text{--}60\text{ N}$. Open: which second lever takes the remaining 0.7.

2. **Then retune the melt-pond albedo for NH sea ice.** 16E has NH March extent 17.2 M km² in decade 4, against 16C's 15.7 and the observed 15.0. This comes after TOA, because the ice responds to the ocean state.
3. **SH sea ice should improve by itself** once the ocean holds less heat and warms the ice less from below. Verify this rather than tune it. 16E: September extent -0.4 M km² against 15F over 30 years; the direction holds, the size is small.
4. ~~Check the AMOC for the GM arms.~~ Done 2026-09-11: GM strengthens it (§53.7). Open instead: why 15F has the weakest AMOC of the older configurations, and why GM 2500 gives back the gain GM 1500 made.
5. **16E_1990** (constant 1990 forcing as 11P/11R/11V, with 1850 land use; running, 38 years at 1387) gives the present-day pair for period-clean scoring against observations. 16C_1990 failed: LPJ-GUESS refused the 1990 land use on the 1850 state.

Previous ranking, 2026-09-09:

Re-ranked 2026-09-09 after round 35 (§52). The ocean side now has a located mechanism, and it is not where the plan assumed.

1. **Price a winter mixed-layer lever at the subtropical and subpolar outcrops.** That is where heat crosses density surfaces, and where the model over-mixes by +48.9 (30-45N, 45-30S) to +155.7 m (45-60N) at the seasonal maximum. The coincidence of location is established; the causal link is not, so the first arm is a measurement, not an adoption.
2. **The Southern Ocean convection, still unfixed and now quantified.** 18.7 % of month-nodes reach the sea floor and the band cools at every depth. The corrected CORE3 mesh did not fix it. This is a stratification and salinity problem, not a diffusivity one, and it is the largest single ocean defect in the model.
3. **Re-baseline the energy target** against 11X now that the ocean heat gain ($+1.502$ W m⁻² of ocean area) can be set against a TOA number directly.
4. **Is the tropical surface uptake itself too large?** $+13.9$ W m⁻² net into the ocean at 30S-30N is the source of the heat that the mid-latitudes then bury (§52.6). ERA5 and NCEP2 are on disk and this has never been scored.
5. **Implement Fox-Kemper mixed-layer-eddy restratification.** FESOM 2.7 has none, and its absence is the standard reason coarse models over-deepen winter mixed layers at the outcrops. Implementation work rather than a knob, but it acts exactly where the measurement points.
6. **Write opottempdiff and opottemprmadvect in 3D** in the next arm's io_list. The column-integrated versions cannot separate interior diapycnal mixing from the surface flux.
7. **Dropped: tropical interior ocean mixing, and dropped: the 11f's virtual salt flux** — both falsified in round 35.
8. **Excluded by instruction: sea surface salinity restoring.** surf_relax_s stays 0 in every arm, including as a suppressant for the Weddell convection. Restoring hides the model's error rather than fixing it and makes a coupled PI spin-up non-conservative.
9. **A PHC3 climatology on the corrected CORE3.** Only core3_beta (211567 nodes) exists, which forces every obs comparison onto a regular grid or onto drift rather than bias.

Campaign plan, set 2026-08-10 (evening): finish the Southern Ocean and the tropics, then one final land round. The vegetation result (§43.12) shows Siberian tuning is not finished, but it also shows what finishing it requires — closing the ≈ 2 K cold bias, for which the offline +1/ + 2 K series is now the calibration. That is a separate campaign from the radiative one, and mixing them is what made round 27 sprawl.

1. ~~Finish SO and tropics. LX1/LX3/LY2 completed 48 years on 2026-08-10 and await evaluate.sh.~~ **Done 2026-08-11** (data/evaluate_L.out); scored in §44.3. RSNOWLIN2's tropical gain does *not* survive: LX3 and LY2 reach the Southern Ocean by dimming the planet (-0.88 and -0.58 global TOA), and LX1 pays 0.83 W m⁻² of imbalance for 0.4 of SO surface. **S4 wins on all three constraints at once.** The one decision still genuinely open from this item is **A1b**, credited here with solving the energy target while appearing in no run of the adopted stack.
2. **Then the final land round**, with three things it did not have before: the +1/ + 2 K transfer function, the three unread LPJG_COMP_EXP* competition experiments, and the knowledge that 50 coupled years shows a transient, not an equilibrium, so vegetation must be judged offline or over centuries.

3. **Re-baseline the energy target** with one fixed-forcing control (§43.9); Z1 proved the one-line fix works.
4. **11E** is complete at 50 coupled years and is the best coupled configuration on Siberian T2m, but net TOA is still +0.765 and SO CRE 5.65 short of CERES.

Re-ranked 2026–08–10 after round 27 (§43). The tropical longwave deficit now has a lever that closes it, and the lever costs the Southern Ocean. Everything below follows from that.

1. **Price RSNOWLIN2 at 44 years.** LX1/LX3/LY2 are in flight against P5. The tropical gain and the Southern Ocean penalty are both established at one year; what is completely unmeasured is **Siberia**, and there is no headroom — the land stack recovers 1.15 K of a ≈ 2 K period-clean bias, leaving ≈ 0.9 K cold. LY2 carries A1a's value, rejected in round 3 *for boreal damage*.
2. **Resolve the tropical-LW / Southern-Ocean tension.** They are coupled through one parameter. If RSNOWLIN2 survives Siberia, the question becomes whether DMS or A1a can pay back the SO longwave it adds.
3. **Decide A1b.** RCL_OVERLAPLIQICE $0.65 \rightarrow 0.35$ is credited in this report with solving the energy target and appears in *no* run of the adopted stack. Y1 shows it is essentially energy-neutral on the new base (global -0.005) and reaches only -1.96 of SO SW. The inconsistency should be closed explicitly.
4. **Re-baseline the energy target** with one fixed-forcing control run (§43.9). Do *not* mix fixed and transient arms in one comparison.
5. **11E coupled**, resumed at year 1370 with 20 of 50 banked; still the gate on the PI spin-up.
6. **The tropical cold point**, -2.07 K (§41), untouched.

Re-ranked 2026–08–10 after round 26 (§42). The Southern Ocean lever now exists and is priced; what blocks it is the tropics. The list collapses accordingly:

1. **The tropical longwave cloud deficit**, -2.16 W m $^{-2}$. Promoted to first because it is no longer a parallel item: it *gates* the Southern Ocean fix (§42.5). The lever must be keyed on **deep convection**, not on cloud ice in general — RVICE is the counter-example. *Do the free check first*: no arm of ~ 52 has ever been scored on this term, and `tropical_lw_scan.py` is written to answer it from data already on disk.
2. **11E coupled**, running since 2026–08–10. Decides whether the coupled configuration is level with AMIP, and gates the PI spin-up. **11F** is running alongside it but has been **reclassified as a mechanism probe rather than a tuning arm** (§42.6): its $S = 166$ is not adoptable, and S4 delivers the same Southern Ocean forcing at twenty times the tropical selectivity. It is kept only for the generic question — what ≈ 2 W m $^{-2}$ of SO brightening does with free SST and sea ice — which transfers to S4 and which AMIP cannot answer at any length. First to cut if compute tightens.
3. **The tropical cold point**, -2.07 K, the shared entry point for the cold bias and 61 % of the stratospheric dry bias (§41).
4. **S4** (RCL_INPPMIN $70000 \rightarrow 50000$) completed 48 years on 2026–08–09 and is **unevaluated**.
5. **The SO cloud-amount two-thirds** stays open with no identified lever; the CCN path reaches the opacity third only.
6. **Dropped:** the Blomquist et al. (2017) transfer velocity. It is less steep in wind than the Nightingale quadratic, so it moves flux away from the windy Southern Ocean toward the calm tropics — worse selectivity, not better (§42.5).

Re-ranked after round 24 (§40), which answers round 23's question. Round 23 asked for a *regionally selective* Southern Ocean reflection lever. Round 24 found one and priced it: marine CCN is selective by construction, but it reaches only the **opacity third** — 2.57 W m $^{-2}$ in-band, ≈ 0.25 global — and the amount two-thirds is unreachable from the CCN path because the lifetime route divides by zero at NAERCLD = 0. The list is therefore:

1. **11E coupled** (11D at SWEMIN = 15 plus K1). A namelist edit on a lever already validated over 44 AMIP years, and the only route to three questions AMIP cannot price. Cheapest decisive run available.
2. **The tropical longwave cloud deficit**, -2.16 W m $^{-2}$, untouched by any lever in 50 runs and the one term that is deficient in the direction the tropics need.
3. **The tropical cold point**, -2.07 K, which §41 shows is the shared entry point for the cold bias and 61 % of the stratospheric dry bias. The tropopause *height* is correct, so this is a cold-point temperature problem, not a displacement one, and one lever should move both errors.

4. **Marine biogenic CCN via emdms** — real missing physics with the input already on disk, but $\approx 0.25 \text{ W m}^{-2}$ global and it needs a published flux $\rightarrow$ CCN relation, so it follows the cheaper items.
5. **The SO cloud-amount two-thirds** stays open with no identified lever.

Re-ranked again after round 23 (§39). The Southern Ocean stays first but the requirement has sharpened: it needs a *regime-selective* reflection lever, because every global low-cloud knob overshoots a tropical band that is only -0.67 W m^{-2} from CERES. Second is the tropical *longwave* cloud deficit, untouched by any lever. Third is the extratropical tropopause. Fourth is 11E coupled.

Re-ranked 2026–08–09 after round 22 (§38). The top of this list is now: (i) **the Southern Ocean cloud-amount deficit**, 65.5 % of a $+7.46 \text{ W m}^{-2}$ band error, untouched by every lever and *not* reachable by the INP branch (§38.3); (ii) **the extratropical tropopause cold-and-dry pair**, which is the dominant land-T2m term and simultaneously the reason the global energy budget cancels (§38.4); (iii) **11D re-run at SWEMIN = 15 plus K1**, the first coupled configuration level with the AMIP one.

Two entries below are **superseded** and struck rather than deleted: ~~The snow-free land albedo residual $+0.0080$ is untouched by all 41 runs.~~ K1 is exactly that lever and removed 74 % of it for $+0.089 \text{ K}$; the entry predates K1's adoption and survived unedited (§30.3, which also retracted K3 before it ran). ~~Tropics net TOA 2.4 W m^{-2} too low.~~ Period-clean it is 0.67 (§38.2).

Added 2026–08–07. (i) **The autumn cover excess P3 introduced** — $+0.259$ in September against Rutgers, where as-released was $+0.107$. P5/P6 test SWEMIN = 15 and 30; the offline optimum is 30 (§34.1). (ii) **The Southern Ocean unexplained residual has grown to $\sim 1.9 \text{ W m}^{-2}$** now that the aerosol term is measured (§35). (iii) **Scale awareness is a two-anchor interpolation, not a law** — a new resolution needs validation (§34.2). (iv) The M1 clear-sky sign anomaly is significant and untraced.

1. **The blocker: a -4.7 K Siberian winter bias** accompanying the snow fix (§27.3). Not a radiation-supply problem (§27.4) and now positively excluded as radiative at all (§27.7), so the route is thermal. Round 16 tests skin conductivity; if that fails, boundary-layer mixing. *J1/J2 are built on I1, not I3* — diagnostic either way, but the production pair needs re-running on the better base (§28).
2. **Decide the coupled base: I3 or I1.** I3 gives $+0.072 \text{ K}$ more JJA and a better global T2m RMSE, but carries the Arctic energy bill of §27.6, which **AMIP structurally cannot price** — prescribed SST and ice absorb 1.7 W m^{-2} without responding, and the coupled spin-up will not. Open question: whether a recalibrated mode 2 keeps the $+1.3 \text{ K}$ without the 60–90N surplus, since the SDOR scaling does most of its work outside the box it was tuned in (§29.2).
3. **The snow-free land albedo residual $+0.0080$** — crops $+0.0204$, semidesert $+0.0169$, bare soil $+0.0145$, tundra $+0.0105$ — is untouched by all 41 runs and is the one lever aimed at the $\sim 0.7 \text{ K}$ *universal* bias rather than the boreal one. It **must not simply be stacked on I3** (§29.3): the mid-latitude entries are safe, the tundra entry competes for the same Arctic energy budget.
4. **Global net TOA $+0.52 \text{ W m}^{-2}$** against a PI target of ~ 0 — one of the two original faults, barely moved by the whole campaign. Note this sits oddly beside a cold troposphere: in AMIP the prescribed SST absorbs the imbalance without warming, so the two are not contradictory, but the pairing is unexplained.
5. **The global tropospheric cold bias** of §21.1 (-0.9 K at 700–850 hPa, -2.8 to -3.0 K at 200 hPa) is now the dominant remaining term for land 2 m temperature and has never been worked on. It is out of reach of every lever this campaign has built.
6. **SO cloud-area deficit $\sim 6 \text{ pp}$** , untouched by all 34 evaluated levers, and now known to drive the energy target too.
7. **-12 W m^{-2} downward-LW deficit in Oct–Nov**, unexplained.
8. Tropics net TOA 2.4 W m^{-2} too low; Nordic Seas SW RMSE moved by G1 and partly repaired by G4, both unexplained.
9. Whether 48 % of the boreal bias suffices for forest survival — **AMIP structurally cannot answer this**; it needs the coupled tests.

60 Falsified claims, kept

Every mechanism this campaign has killed, with the measurement that killed it. Retained because the history has to stay honest, and because three of them were killed only after being built on.

claim	what killed it	where
Decade-4 TOA falls about 4 W m^{-2} per W m^{-2} of 40-year mid-depth uptake removed ($r = +0.86$, 17 runs), so a GM arm that cuts the uptake by 0.2–0.3 reaches TOA balance	16E cut the 40-year 700–2000 m uptake to 0.34, the lowest of any run; the fit predicted decade-4 TOA -0.03 , and it measured $+0.70$ (4.4 residual sd). The slope was set by 080 and 11M; without them it is about 1.3.	§53.3
Stronger GM strengthens the AMOC, so it is no obstacle to a larger GM dose	True for GM 1500 only. GM 2500 (16E) is $1.0 \pm 0.2 \text{ Sv}$ weaker than GM 1500 at 26.5N and 1.7 ± 0.2 at 40–60N over 40 years, and below 15F at 40–60N.	§53.7
Fox-Kemper MLE takes back the AMOC gain of GM 1500	Read off 16D's second decade (11.4 Sv). By decade 4 16D is at 15.1, and 16D minus 16C is $-0.25 \pm 0.18 \text{ Sv}$ over 40 years.	§53.7
The links virtual salt flux is accumulating salt at the mid- and high-latitude outcrops, which deepens the winter mixed layer there and is the real cause of the heat burial	Global mean salinity drifts $-2.5 \times 10^{-5} \text{ psu}$ in 30 years, so salt is conserved. And at 45–60N the virtual salt flux is <i>freshening</i> while the layer becomes 0.30 psu saltier — the salinification is advective, a consequence of the deep mixing rather than its cause.	§52.6
The ocean heat uptake is spin-up from the initial condition and will equilibrate, so it should not be tuned against	The three decade-to-decade steps of thetaoga are $+1.461$, $+1.433$ and $+1.472 \text{ W m}^{-2}$ — flat to 3% over forty years, with no deceleration after the first five.	§52.6
The model buries surface heat by mixing it down in the tropics, which is why the surface stays cold while TOA stays positive — so tropical ocean mixing is the lever	The tropical mixed layer is 8.1 m too <i>shallow</i> annually and level at the seasonal maximum (81.0 against 80.8), and 90% of the tropical deep warming is heave ($\sigma_0 = 27.0$ sinks $+42.4 \text{ m}$ while the water on it warms $+0.047 \text{ K}$). Heat crosses density surfaces at the mid-latitude outcrops instead. The burial itself is real: the ocean gains $+1.502 \text{ W m}^{-2}$ and none of it stays in the top 100 m.	§52.2, §52.4
The Southern Ocean convection is a route by which surface heat reaches the deep ocean	90–60S is the only band with a net heat <i>loss</i> , -0.076 W m^{-2} of the global budget, cooling at every depth from 70 to 2400 m and peaking -0.26 K near 1000 m. Convection ventilates: it connects the abyss to a cold surface.	§52.3
force_flux_consv in FESOM's coupling receive is amplifying A_q_ice into the kilowatt excursions that reach the ice skin solve	A probe comparing the field immediately before and after the two calls found raw_max = corrected_max in all 20,760 samples of a coupled year. The path is inert in this configuration; the excursions came from OASIS's GSMART conservation instead.	§51.3
The Southern Ocean surface is too cold, so the ice albedo feedback works in the model's favour as it warms toward equilibrium	Measured with SST, which under sea ice is pinned near freezing and cannot show the error. T2m is the free variable and reads $+1.5$ to $+1.9 \text{ K}$ at 70–80S, peaking at $+4.5 \text{ K}$ in MAM. The feedback works against us.	§50.8
The residual Southern Ocean cloud deficit is in a band the campaign has never worked on	It has: 90–60S DJF SW CRE has gone $+13.63 \rightarrow +6.98$ across $11\text{E} \rightarrow 11\text{R}$. The real structure is a meridional gradient in lever effectiveness, not an untouched region.	§50.3
RSBLLMIN spares the tropics: the 30 m floor binds only where the stable layer is shallower than 300 m, so a shallow Arctic winter SBL feels it and a deep subtropical Sc boundary layer does not	It also sets the length in free-shear layers, so it enhances entrainment across the Sc capping inversion. Same Arctic gain as SB2 at four times the tropical cost ($+2.106$ against $+0.541$ of SW CRE) and $+1.673 \text{ W m}^{-2}$ of net TOA. The risk was recorded in the runscript before the run.	§49.2
The tropical low-cloud loss will degrade nino34 , where 11G already regressed	cmpitool on 11R: nino34 is the MOST improved region, -0.030 (tas -0.258), and the tropics are neutral at -0.0002 . The cost is real but lands on thetao ($+0.018$ global), the ocean absorbing the shortwave.	§49.4
Snow skin conductivity is the winter lever: λ_{sk} over exposed snow is too small, letting the skin radiatively decouple in polar night, so raising ECE-LAMSK_SN from 7 to 15–25 should warm DJF by $+1$ to $+3 \text{ K}$	Measured null at 48 yr, paired: $+0.125 \text{ K}$ at 60–90N DJF against a 0.41 threshold (J1) and -0.039 against 0.45 (J2). Verified live first — the symbol is in libsurf.SP.so , fort.4 carries the values, and the arms differ locally by up to 21 K. The decomposition then gave the reason: the skin-side bias is already only $+0.11 \text{ K}$.	§48.1
Raupach roughness will move the coupled winter screen temperature	Scored on the 11J–11I run-time-gated pair: DJF land-ocean gap -0.017 (1380–89) and -0.003 (1360–79), sign flipping at 60–90N between windows. Confirms by measurement the structural argument already on this list.	§48.2
Raupach (1994) canopy roughness is a WINTER lever: a leafless boreal canopy is the roughest state, so DJF grid-box z_0 rises $0.058 \rightarrow 0.140 \text{ m}$ and warms Siberian winter soil	The winter mechanism requires a canopy the model does not have. Measured in 11J: $h = 2.89 \text{ m}$, stem diameter 0.0198 m, so $\lambda_{stem} = 0.0092$ — an order of magnitude below the peak of the Raupach z_0/h curve. Its own constants then give $z_0 = 0.0132 \text{ m}$, the bare-soil value, against IFS's 0.058 m: $4.4\times$ <i>smoother</i> , which is the sign the momentum budget measures (DJF $ \tau $ -4.1% , 10 m wind $+3.0\%$) DJF soil $+0.054 \text{ K}$, $p = 0.670$.	§45.5
The land surface repair is free — nothing resolves against its threshold, so adopt	Scored with the wrong statistic and against no target. Paired (the arms branch from one 1350 state) Siberian DJF soil $+0.822 \text{ K}$ in 1348, which is the highest	§45.4

Two further items are *provisional* rather than dead: the March–April mass deficit rests on ERA5 SWE (forest-biased), and mode 2’s scale-awareness carries only $\sim 16\%$ of the length scale over this box (§26.4).

.1 Round 37–40 (2026–09–13→25)

- ~~The Southern Ocean winter over-ice bias is a vertical mixing deficiency, fixable with a salt-plume scheme, mixed-layer eddies, or a deeper lead-closing floor.~~
Killed by **PI200_spp**, **PI200_mle**, **PI200_h0** and **PI200_all3**, four 20-year branches from one restart: none moves it. It is a freshwater regime, salty-and-venting or fresh-with-warm-CDW, and these levers do not cross between them (§54.1).
- ~~Reducing ice-shelf basal melt breaks the loop warm CDW → melt → fresh shelf → no dense water → warmer CDW.~~
Killed by **PI200_cavg06**: the calibration is right on its own terms (1190 against 1893 Gt yr^{−1}, observed 1100–1325) and the Southern Ocean does not respond in 20 years.
- ~~Raising the lead-closing floor h0min 0.5 → 1.0 thickens the Antarctic pack.~~
The flux response was exactly as designed, area-to-volume of freezing 1.71 → 1.10, and the state did not follow: thickness 0.61 → 0.62 m and the 60–90 S bias +0.82 → +1.45 K. Reverted (§55.2).
- ~~The ice is too dark because the melt ponds are too bright; darken albpnd.~~
Wrong direction and double-counting: the four **ice_therm** albedos are pond-free and **albpnd** is applied over them afterwards. Ponds stayed dark and the ice and snow were brightened instead (§55.1).
- ~~albsn 0.75 and albsnm 0.65 are FESOM’s own defaults.~~
They are not: FESOM ships 0.81 and 0.77, so those values were deliberate darkenings, not inherited. Corrected before the numbers were committed.
- ~~The albedo sensitivities are fixed properties of the configuration.~~
They are state-dependent: NH $\partial\alpha/\partial\text{albsn}$ moved 0.149 → 0.504 as the Arctic cooled. Re-measure rather than reuse.
- ~~IDEMIX fails because the bottom forcing file is wrong, or unnormalised, or carries NaNs that the 3-D interpolator mishandles, or NaNs that the 2-D reader mishandles.~~
Four wrong hypotheses in a row. The file has *zero* NaNs, proven by a 30-line offline Fortran harness. The real defects were an implicit netCDF fill value and an MPI type mismatch (§56.1, FESOM PR #1087).
- ~~A matching fesom executable checksum proves the right FESOM is running.~~
It proves nothing: the executable is a driver stub, byte-identical across these builds. Only **libfesom.so** distinguishes them, and it can be a symlink that **md5sum** silently follows (§57.1).
- ~~Net TOA is positive, so the run is warming.~~
Net TOA is +0.167 W m^{−2} over 2080–2099 and the surface cools at 0.21 K/decade. Total ocean heat content is flat; the upper 700 m is losing heat to below 700 m at 0.32 W m^{−2} equivalent (§57.2).
- ~~GM tuning is free at the Southern Ocean.~~
It cost 16.7 Sv of Drake Passage transport, 110.0 → 93.3 Sv on identical years, monotonic in K_{GM}^{max} (§57.4).
- ~~The coupled run reads namelists/lpj_guess/global.ins; ecearth.ins.j2 is unused by this setup.~~
Exactly backwards. **guess.ins** is hard-wired in **framework.cpp:1063**, rendered per rank, and imports **ecearth.ins**; **global.ins** is staged and never reached. The two had drifted apart, **ifreefracca** 1 against 0 (§57.6).
- ~~tripyview is not installed on this system.~~
It is, in `~/.conda/envs/esm-tools_auto_tripyview`; its editable-install **egg-link** points at a path that no longer exists, so the import fails misleadingly. `PYTHONPATH=work/ab0246/a270092/software/tripyview` fixes it.

A Parameter catalogue (distinguishing change per run)

Run	Change (file/namelist)	Baseline → New	Process
06A	albpnd (namelist.ice)	0.2 → 0.28	melt-pond albedo
06B	+ rfracmax	1.0 → 0.75	pond water retention
06C	+ pndaspect	0.8 → 1.3	pond depth:area (=common pond)
06D <i>HRlike</i>	(only common pond shift)	–	no other diff found
06G	albpnd/rfracmax/pndaspect	0.35/0.6/1.6	stronger meltpond
06L	albpnd/rfracmax/pndaspect	0.4/0.55/1.8	extra-strong meltpond
06H	strong pond + RVICE (&NAMCLDP)	0.16 → 0.18	meltpond + cloud-ice fall speed
06Q	albw/albocn	0.1 → 0.045	open-water/ocean albedo
06R	albw/albocn	0.1 → 0.075	open-water/ocean albedo
06S	whichevp (&ice_dyn)	1 → 0	sea-ice rheology mEVP→EVP
06E	RVICE+GGAUSSB (&NAMGWMS)	0.18; –0.5 → –0.6	cloud-ice + gravity-wave
06F	LRDALB (NALBEDOScheme=3)	F → T	diagnosed surface albedo
06P	ENTSTPC3 (&NAMCUMF)	absent → 1	deep-convection entrainment
06N	use_momix (namelist.tra)	F → T	near-surface ocean mixing (mospp)
06M	OASIS coupling period + LAG	7200 → 3600 s	atmos↔ocean 2h→1h (1hcpl)
06O	1hcpl + mospp	–	coupling + mixing
06O4	O + momix_kv	0.01 → 0.012	+ larger M-O mixing
06O5	O + momix_kv	0.01 → 0.02	+ even larger mixing
06T	O + KPP-low (kpp_av0/kv0 0.005 → 0.003, bckg ↓ 10×)	–	+ weaker KPP mixing
06U	T + open-water albedo 0.075	–	combo
06V	T + ENTSTPC3=1	–	combo (most aggressive)

B Implementation & lab-notebook detail

Operational detail moved here to keep the work-plan science legible. Retained for reproducibility.

A.1 esm-tools fixes for the AMIP–noLPJG generator (§10.2). Model tree oifsamip-cy48 git-aligned to awiesm3-develop (oifs local_combined_fixes @ 1303a90, oasis afac6a3, xios 2.5.3) + bundle/field patches. Three bugs fixed, committed & pushed (commit 9921d95b, branch development_interactiv_mesh_awiesm3, and a follow-up for the WAM recipe):

- xios.yaml dereferenced the fesom namespace via choose_fesom.* and crashed *any* FESOM-less setup — now gated on "\$(('fesom' in config))".
- oifsamip.yaml never overrode the OIFS comp_command, so the binary compiled *without* XIOS or coupling despite LXIOS:TRUE — now adds -with-cplng2 -with-oifs-xios -build-type=release (like awicm3/awiesm3).
- WAM aborted (CDATEA≠CDATEF) at any start year ≠1990 because the OIFS preprocess step (change_icm_date.sh, which re-dates ICMGG/ICMSH/wave init files) never ran: oifsamip.yaml lacked prepcompute_recipe: \${oifs.preproc} and fell back to the default recipe with no preprocess step. Added (matches awicm3.yaml).

Runscript oifsamip-cy48-levante-TC095L91_lpjgforcing.yaml overrides XIOS output to file_def_lpjg_spinup + field_def_lpjg_safe, emitting daily tas, tasmin, tasmax, pr, rsns, rlms, sfcWind + tdps, ps, psl, clt, prsn, evspsbl, p... *esm_master gotcha*: the single-component target is recomp-<setup>/oifs-48r1 (the *component* name); the alias /oifs silently falls back to the setup and doubles the path (oifsamip-cy48/oifsamip-cy48).

A.2 Flux-accumulation fix (§10.2), at root. The de-accumulation divisor in field_def_lpjg_safe.xml was the hard-coded /21600 (6 h). It is now templated: field_def_lpjg_safe.xml.j2 uses /{{oifs.flux_accumulation_period}} with oifs.flux_accumulation_period defaulting to \${oifs.time_step} (=NFRHIS×TSTEP; with NFRHIS=1 this is the timestep). A validated ×6 scalar patch (AMIP_noLPJG_1d_1870–1879_TC095_PI_fluxfix.nc) corrected the existing forcing without a rerun.

Controlled temperature test now ready. Because the AMIP–CRUNCEP comparison shows meaningful forcing sensitivity but the CERES replacement rules out net radiation as its dominant source, two additional AMIP forcing cycles were created for the next attribution step. They add exactly +1 or +2 K to tas, tasmin and tasmax at all TCO95 cells north of or at 45°N; every field south of 45°N and every non-temperature variable is unchanged. Their NH-land forcing-cycle means are –3.57 and –2.57 °C, compared with –4.57 °C in the source AMIP file. These are *prepared inputs, not spin-up results*. Running them with unchanged LPJG parameters will quantify whether a uniform boreal temperature increment crosses a vegetation threshold or merely produces a modest continuous response.

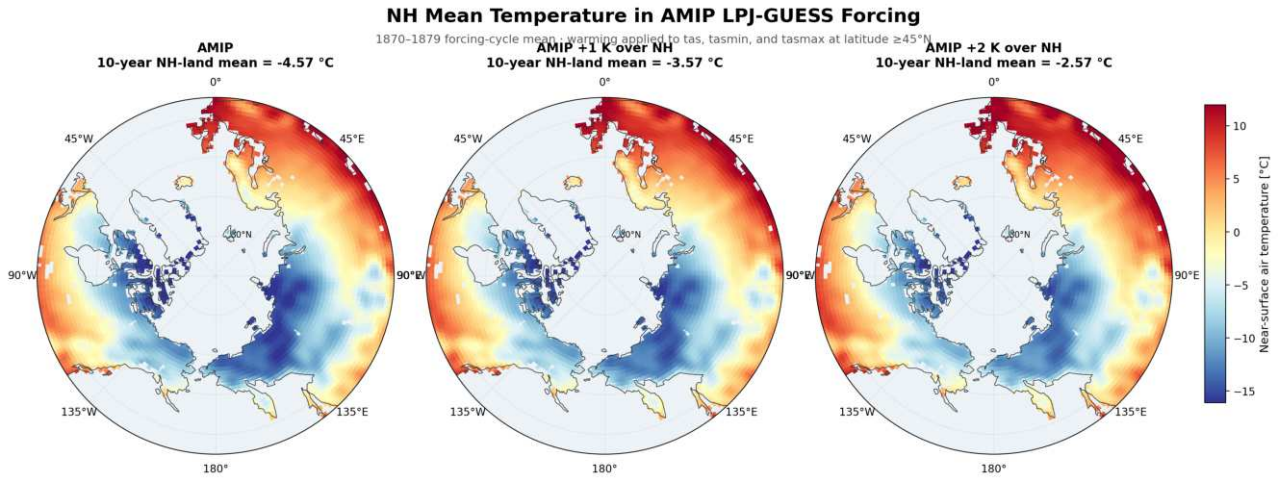


Figure 46: Input check for the planned AMIP temperature sensitivity. Ten-year NH-land mean near-surface temperature in the original, +1 K and +2 K forcing cycles. The increments were applied consistently to `tas`, `tasmin` and `tasmax` at latitude $\geq 45^\circ\text{N}$; no vegetation response is shown because these two spin-ups have not yet been run.

A.3 CERES forcing construction and validation (§7). The direct source is `/work/ab0246/a270092/obs/CERES/CERES_EBAF_Ed4.1_Subset_200003-202106.nc`; the climatological source is `/work/ab0246/a270092/obs/CERES/CERES_EBAF_Ed4.1_Subset_CLIM01-CLIM12.nc`. The output files are `/work/ab0995/a270270/input/cruncep_v7/CRUNCEP_noLPJG_1d_1901-1910_TCO95_CERES_rsns_rlms.nc` and `/work/ab0995/a270270/input/cruncep_v7/CRUNCEP_noLPJG_1d_1901-1910_TCO95_CERESclim_daily_variability.nc`. Both contain 3652 daily timesteps and 40 320 TCO95 cells. Coordinates and all non-radiation fields are unchanged from CRUNCEP v3. The direct file has `rsns` range 0–358.23 and `rlms` range -197.30 – 34.76 W m^{-2} . The daily-variability file has ranges 0–455.61 and -206.92 – 93.85 W m^{-2} ; every mapped CERES cell-month target is reproduced within $8 \times 10^{-6}\text{ W m}^{-2}$. Construction and preanalysis scripts, cached intermediates and the detailed provenance document are retained under `/work/ab0995/a270270/analysis/Offline_LPJ_GUESS_analysis/`.

A.4 Phase-1B OpenIFS source edits (§10.3). Paths under `oifs-48r1/ifs-source/arpifs/phys_ec/`:

knob	file:line	current	proposed	effect
ZLMIN	<code>vdfexcu.F90:201</code>	30 m	40–50 m	min stable mixing length
ZCB	<code>vdfexcu.F90:190</code>	5.0	3.0–4.0	LTG <i>b</i> : lower $\Rightarrow$ longer tail $\Rightarrow$ more <i>K</i>

Secondary companions if needed: the 0.1 / 300 m stable mixing-length limits (`vdfexcu.F90:374,376`) and the surface-layer constants `RCHETA/RCHETB/RCHBHD` (`surf/module/suvexcs_mod.F90`, mirror in `suvdfs.F90`).

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Citations added in response to internal review (rev. 2026-07-06); bibliographic details for Beljaars & Viterbo 1998, Jiang et al. 2012, Medeiros et al. 2011 and Zhang et al. 2011 to be page-verified. Snow-cover references added 2026-08-03: volume/page numbers for Essery et al. 2013, Helfrich et al. 2007 and Thackeray et al. 2015 are from memory and should be page-verified before external use; the Niu & Yang 2007, Swenson & Lawrence 2012, Roesch et al. 2001, Liston 2004, Dutra et al. 2010 and Betts & Ball 1997 entries are the load-bearing ones for §26.